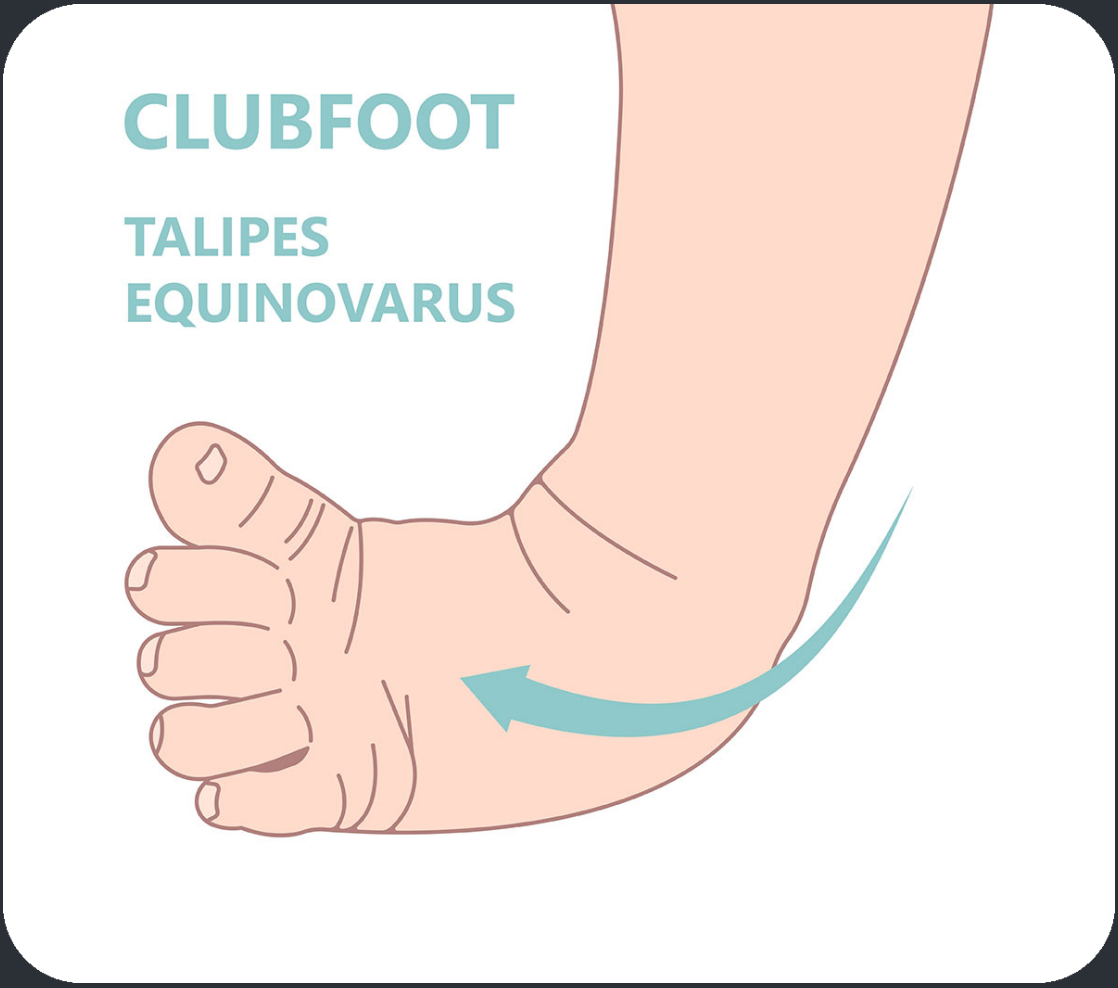
COURSE

Pediatric Orthopaedics



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14 lectures

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22 October 2026

TRANS PRO

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Lecturers

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Prof Dr Hisham Abdel-Ghani

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This work was created using a computer program developed by us through Vibe Coding, together with several other AI-assisted tools. Using these tools, we were able to create transcripts of lectures, including lectures delivered in Egyptian Arabic, and organize them into a more academic, book-like format.

The aim is to help the listener focus on listening without having to write notes, while having an organized transcript they can easily return to for quick revision.

This transcript is not a replacement for the original lecture or the relevant textbooks and scientific references. The accuracy is generally very good, but AI-assisted transcription can still contain errors. If you are unsure about any point, please check the original lecture or a reliable reference.

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- Click any line in the Contents to jump to that lecture.
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- "(see Fig.)" jumps to the figure; "↑ back to text" under the figure brings you back to where it was cited.
- "Ask AI" beside each heading opens a page that turns that section into a ready question for an AI: simplify it, clarify what the recording left unclear, or check for newer guidelines.
- Orange notes such as [C1] and [Q1] are AI corrections and doubts - mostly where the recording was hard to transcribe. Each lecture lists them after its index of figures.

Contents

1. Management of Congenital Clubfoot 6
2. Relapsed and Neglected Clubfoot 34
3. Developmental Dysplasia of the Hip: Pathology 61
4. Developmental Dysplasia of the Hip: Diagnosis 78
5. Developmental Dysplasia of the Hip: Treatment 94
6. Perthes Disease: Pathology 138
7. Perthes Disease: Treatment 171
8. Lower Limb Deformity Planning - Part 1 196
9. Lower Limb Deformity Planning - Part 2 215
10. Lower Limb Deformity Planning - Part 3 236
11. Frontal Plane Malalignment of the Knee 256
12. Management of Rickets 288
13. Supracondylar and Lateral Condyle Fractures of the Humerus 317
14. Traumatic Radial Head Dislocation 346

Management of Congenital Clubfoot

(Principles and Ponseti-based management of idiopathic clubfoot)

Dr Mohamed Fathey

Watch Original Lecture

Lecture transcript processed with AI assistance.

Contents

↑ Book contents

Foot Biomechanics . . . 11
Acetabulum Pedis . . . 11
Inversion and Eversion . . . 11
Abduction, Adduction and Flexion . . . 12
Supination and Pronation . . . 12
Heel: Varus and Valgus . . . 12
Forefoot Terms, Equinus and Cavus . . . 12
Subtalar Joint: Mitred Hinge . . . 12
Locking and Unlocking During Gait . . . 13
Clubfoot: Definition, Epidemiology and Etiology . . . 13
Definition and Epidemiology . . . 14
Associated Anomalies . . . 14
Prenatal Diagnosis . . . 14
Etiology . . . 14
Collagen Synthesis: Clinical Implications . . . 15
Deformities of Clubfoot . . . 15
Four Components . . . 15
Tibial Torsion . . . 16
Summary . . . 16
Pathological Anatomy . . . 16
Navicular and Talonavicular Joint . . . 16
Tibialis Posterior . . . 16
Talus . . . 16
Calcaneus, Cuboid and Cuneiforms . . . 17
Tendons . . . 17
Ligaments . . . 17
Summary Picture . . . 17
Muscle and Tendon Changes . . . 18
Crimp: Histological Basis of Manipulation . . . 18
Clinical Examination and Classification . . . 18
Clinical, Not Radiological . . . 18
Palpation: Talar Head vs Calcaneus . . . 19
Foot Size and Leg . . . 19
Radiology and Correction . . . 19
Types of Clubfoot . . . 20

Pirani Score 20
Midfoot Items 21
Hindfoot Items 21
Significance of the Score 22
Principles of Ponseti Treatment 22
Scope 22
Aim 22
Guidelines of Treatment 23
Ponseti Casting Technique 23
Preparation 23
First Cast: Cavus 23
Abduction and Serial Casting 24
Padding and Cast Application 24
Moulding: Four Key Areas 24
Above-knee Cast and Trimming 25
Progression and Cast Removal 25
Casting Complications and Errors 25
Complications of Casting 25
Common Errors in Management 26
Achilles Tenotomy 26
Indications and Criteria 26
Technique 27
Post-tenotomy Cast 27
Bracing, Follow-up and Relapse 27
Denis Browne Abduction Brace 27
Shoes of the Brace 28
Bracing Protocol 28
Non-compliance 28
Follow-up Schedule 28
Relapse Patterns 29
Atypical Clubfoot and Recurrence 29
Tibialis Anterior Transfer 29
Rationale 29
Timing and Site 30
Technique 30
Pre-transfer Casting 30
Take-home Message and Questions 30
Take-home Message 30

Brace Timing and Medical Shoes 30
Casting, Tenotomy and Stretching 31
Transfer Incisions and Mixed Brace 31
Positional Clubfoot and Achilles 32
Practical Casting Points 32
Neutral Brace Position and DDH 32
Transfer Site and Number of Casts 32
Remaining Questions 33

AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

- **C1** (p. 12) “heel varus: calcaneus first abduction” → “heel varus: calcaneus first adduction”
- **C2** (p. 14) “after the third trimester” → “after the first trimester”
- **C3** (p. 15) “calcaneus internal rotation + abduction” → “calcaneus internal rotation + adduction”
- **C4** (p. 21) “2-3 deep creases scored 0” → “2-3 deep creases scored 0.5”
- **C5** (p. 21) “reaching only neutral scored 0” → “reaching only neutral scored 0.5”

Foot Biomechanics Ask AI

- Pediatric foot ≠ adult foot; large subject → one topic per session
  - Each topic → relevant **biomechanics** + principles of deformity correction
  - Today: clubfoot → short biomechanics → treatment principles

Acetabulum Pedis Ask AI

- Foot likened to hip joint → term **acetabulum pedis**
  - Hip: femoral head moves inside acetabulum
  - Foot: reverse arrangement
- "Acetabulum" of the foot
  - Navicular
  - Anterior + middle facets of calcaneus
  - **Spring ligament**
- "Femoral head" of the foot → **head of the talus**
- Talar head fixed → whole foot moves around it as one block
  - Hip: head moves in socket; foot: socket moves around head
- Talar head lies opposite to foot direction
  - Foot **abducts** (lateral) → talar head in internal rotation
  - Foot **adducts** (inversion, medial) → talar head in external rotation
- Correction principle → **stabilise the talar head** → move whole foot around it
  - Any deformity built on fixed talar head + foot moving around it
- Foot: about `26` bones, many mobile joints → moves as a block
- Concept underlies much of later foot work

Inversion and Eversion Ask AI

- **Inversion** → foot turns inward; three movements together
  - Plantar flexion
  - Forefoot adduction (internal rotation)
  - Supination
- Level: foot + ankle joint; forefoot adduction within the foot
- Eversion (right foot) → lesser toes up, big toe down
- External rotation occurs at ankle + **subtalar joint**
- Eversion ≈ **pronation**

	Inversion	Eversion
Direction	Whole foot inward	Whole foot outward
Movements	Plantar flexion → internal rotation → supination	Dorsiflexion → external rotation → pronation
Mnemonic	"Down and in"	"Up and out"

Abduction, Adduction and Flexion Ask AI

- **Adduction** → tarsal + midtarsal bones turn toward midline
  - Term applies mainly at midfoot + forefoot
- **Abduction** → tarsals, metatarsals, toes turn away from midline
- **Hindfoot** → different terms (varus/valgus)
- Plantar flexion → foot down; dorsiflexion → foot up

Supination and Pronation Ask AI

- **Supination** ≈ inversion; pronation ≈ eversion
  - Inversion → more correct term: includes all three movements
  - Supinated foot: plantar flexion → internal rotation → turns upward
  - Pronation: dorsiflexion → external rotation → moves up toward lateral side

Heel: Varus and Valgus Ask AI

- Heel → **varus** or valgus, never abduction/adduction
  - Forefoot/midfoot: toward midline = adduction; away = abduction
- Heel toward midline → heel varus; away → **heel valgus**
- Any heel deformity → either varus or valgus

	Heel varus	Heel valgus
Direction	Toward midline	Away from midline
Calcaneus	Adduction (medial) → inversion (internal rotation)	Abduction (lateral) → eversion/pronation (external rotation) *[C1 · transcript: "heel varus: calcaneus first abduction"]*

Forefoot Terms, Equinus and Cavus Ask AI

- **Forefoot supination/pronation** → inversion/eversion components at forefoot level
  - Revisited later
- **Equinus** = plantar flexion
- **Cavus** → dropped first metatarsal → exaggerated arch
  - ↑ Height of vault (apex) of foot

Subtalar Joint: Mitred Hinge Ask AI

- Foot never moves in a single axis
- **Subtalar joint** → anterior, middle, posterior facets
  - Each articular surface in a different plane
  - → Three facets cannot move together the same way
- Ponseti → **mitred hinge**
  - Subtalar movement plane ~4°–40° → several axes
- Right foot: calcaneal tuberosity points outward; talar head points medially
- Movement axis → **anteromedial to posterolateral**
  - Anterior → middle → posterior facet → axis shifts → greater ROM
- → **Complex joint:** >1 movement axis
  - Know its motion before correcting its deformity

Locking and Unlocking During Gait Ask AI

- Subtalar **unlocking and locking mechanism** during gait
  - Phases: swing, stance
  - Key events: heel strike → mid-stance → push-off
- **Heel strike** → first gait stage: heel touches ground
  - Tibia, fibula, talus → internal rotation
  - Talus moves opposite to foot (acetabulum pedis)
  - → Foot + subtalar joint externally rotate → **unlocked** (supple)
  - Purpose → foot as shock absorber → protects subtalar joint
- Foot settles on ground → talus externally rotates
  - Foot internally rotates; subtalar inverts
- **Push-off** → heel rises off ground
  - Inversion of foot → **locking mechanism**
  - Subtalar + foot + ankle → one segment
  - → **Triceps surae** on calcaneus lifts body → next step

Event	Talus (+ tibia, fibula)	Foot / subtalar	State	Purpose
Heel strike	Internal rotation	External rotation	Unlocked (supple)	Shock absorber
Push-off	External rotation	Internal rotation (inversion)	Locked	One segment → triceps surae lift

High Yield

- Acetabulum pedis: navicular, anterior + middle facets, spring ligament
- Talar head fixed → foot moves around it, opposite direction
- Mitred hinge: subtalar plane ~4°–40°, several axes
- Heel strike → unlocked (shock absorber); push-off → locked

Clubfoot: Definition, Epidemiology and Etiology Ask AI

- **Ponseti** → Ponseti technique for clubfoot
  - Modified earlier manipulation methods; not original discovery
  - → Most successful treatment method of clubfoot

Definition and Epidemiology Ask AI

- Clubfoot ("claw-like foot") → **four deformities**
  - Forefoot adducted + supinated
  - Equinus
  - Cavus

Parameter	Value
Incidence, Caucasians	~1/1000
Incidence, South Africans	~3/1000 (higher)
Male : female	3:1
Bilateral	~40%

Associated Anomalies Ask AI

- **Metatarsus adductus**
- Congenital hip dislocation (DDH)
- Ring constriction
- **Bilateral clubfoot** → always examine hip
  - DDH incidence in bilateral cases: not high, but considerable

Prenatal Diagnosis Ask AI

- Diagnosable in utero after first trimester *[C2 · transcript: "after the third trimester"]*
- **Ultrasound** → not visible before week 16 → scan at week 16

Etiology Ask AI

- Proposed: small uterus, oligohydramnios, **intrauterine compression** → not accurate
  - Compression deformities → calcaneovalgus, **positional clubfoot**
  - These correct spontaneously in first month with stretching
  - → Compression/oligohydramnios theory of clubfoot not valid
- **Neuromuscular defects** → clubfoot, but not idiopathic
- Idiopathic etiology → still unknown; many theories
- Basis → **contracture** + shortening of tendons, muscles, ligaments
  - Most important: ligaments between joints
  - Joint/ligament contracture → deformity begins

Collagen Synthesis: Clinical Implications Ask AI

- Ligaments → **collagen fibres**
- Synthesis ↑ markedly first 6 months after birth → continues to 3–4 years
- Deformity left after birth → worsens
  - → Ponseti: start treatment within ~1–2 weeks of birth
- Synthesis to 4 years → drives **recurrence**
  - → Follow up until 3–4 years
  - → Denis Browne brace kept 3–4 years
  - Active synthesis → recurrence risk stays active
- **Posteromedial release** after first 1–2 months → very severe fibrosis
  - Release in first 6 months → extensive scarring, post-op fibrosis
  - → Surgical release preferably after 6 months–1 year
- Synthesis ↓ at ~4–5 years → **brace discontinuation** age

Clinical Pearl

- Bilateral clubfoot → examine hip for DDH
- Ultrasound diagnosis possible from week 16
- Collagen surge first 6 months → start treatment within 1–2 weeks
- Surgical release preferably after 6 months–1 year

Deformities of Clubfoot Ask AI

- Four deformities + **tibial torsion**

Four Components Ask AI

- Cavus, forefoot adduction, equinus, heel varus
- **Cavus** → dropped first metatarsal → exaggerated arch
  - Calcaneus → adduction + varus (internal turn)
  - Heel varus (supinates) → forefoot opposite → pronates
  - → **Twisting mechanism** → first metatarsal drops → ↑ cavus
  - Deformity on one side of foot → opposite on other side
  - Model: heel into varus + forefoot pronated
- **Forefoot adduction** → midtarsal level (possibly also tarsometatarsal) toward midline
- **Equinus** → foot down shortened triceps surae/Achilles
- **Heel varus** → calcaneus internal rotation + adduction *[C3 · transcript: "calcaneus internal rotation + abduction"]*
  - Normal right foot: calcaneus points laterally, talus medially
  - → Calcaneus nearly beneath talus, almost **parallel** to it

Tibial Torsion Ask AI

- "Internal tibial torsion with clubfoot" → not accurate
- Actual change → ↓ **normal external rotation**, not internal rotation

Leg	External tibial rotation
Normal child	~40°
Clubfoot side	~17–20°
Normal side, unilateral cases	~30°

Summary Ask AI

- Front → back: forefoot adduction → cavus → heel varus + **equinus**

Pathological Anatomy Ask AI

- Bone by bone, joint by joint

Navicular and Talonavicular Joint Ask AI

- **Navicular** → displaced medially
  - Forefoot adduction at **midtarsal joint** (navicular + cuboid)
  - Tuberosity reaches / abuts medial malleolus
- → **Talonavicular subluxation** + uncoverage of talus
  - X-ray: navicular shifts medially → part of talus uncovered
- Navicular shape → **wedged**
  - Narrow laterally, wide medially medial movement

Tibialis Posterior Ask AI

TP

- Inserts on navicular
- Main cause of navicular medial displacement
- Shortened, thickened, very broad insertion → **traction** → navicular pulled medially
- Insertion markedly widened
  - Slips → cuneiforms + undersurface of **cuboid**

Talus Ask AI

- Navicular medial shift → talus uncovered
- Follows calcaneus (medial displacement + inversion)
  - → **Flexion with inversion** inside ankle mortise
- Large part of head uncovered
- Talar head shape → **wedge**
  - Medial part → covered, articulates with navicular
  - Lateral part → no articulation (uncovered) → palpable
- Talar flexion → only small posterior part of **trochlea** in ankle
  - Non-cartilaginous posterior talus → intra-articular in mortise

Calcaneus, Cuboid and Cuneiforms Ask AI

- **Calcaneus** → internal rotation + adduction
  - Achilles insertion medially displaced → ↑ heel varus
- **Cuboid** → midtarsal joint shifted medially
  - → Calcaneocuboid subluxation; cuboid medially displaced
- Cuneiforms + metatarsals → adducted, **normal shape**

Tendons Ask AI

- Anterior + medial compartment tendons → **shortened** + medially displaced
  - Tibialis anterior, extensor digitorum, extensor hallucis
- Lateral compartment (**peroneals**) → little affected
- Clubfoot: three tendons aligned **in front of medial malleolus**
  - Normal foot: normal anatomical position
- Medial displacement → contributes to, increases, maintains deformity
  - → Among causes of later **recurrence**
- **Tibialis posterior** → enlarged insertion, shortened, thickened

Ligaments Ask AI

- All medial + plantar ligaments → **tight**
- **Spring ligament** → undersurface supporting talar head
  - Calcaneus (posterior) → navicular (anterior)
- Lateral/posterior ligaments → **stretched**, thin, not thickened
  - Probably bifurcate ligament; calcaneocuboid ligament

Ligament	State	Reason
Tibionavicular	Tight, shortened	Navicular medial → tibia–navicular distance ↓
Calcaneonavicular (spring)	Shortened, thickened	Talar head flexion + internal rotation → navicular–calcaneus distance ↓
Lateral/posterior (bifurcate, calcaneocuboid)	Stretched, thin	Lateral = stretched side of foot

Summary Picture Ask AI

- Navicular → wedged: **wide medially, narrow laterally**
- Calcaneocuboid joint subluxated; cuboid medially displaced
- Pathology → basis of later treatment
  - Talus → **severe equinus**
  - Calcaneus → under talus, rotated, inverted
- Cuneiforms + metatarsals → adducted, near-normal shape
- Medial tibial tendons → shortened, in front of medial malleolus
- **First metatarsal** dropped → cavus

Muscle and Tendon Changes Ask AI

- Clubfoot muscles + tendons → **smaller and shorter** than normal
  - Gastrocnemius at 6 months: clubfoot side shortened
  - Shortened gastrocnemius muscle-tendon unit → **equinus**
- Medial tendons → shortened + thickened; peroneals → near normal
- **Tibialis posterior** tendon → 2–3× normal size; very wide insertion
  - Expansion → cuboid + cuneiforms
- **Achilles tendon** → thickened, shortened
  - Medial insertion → ↑ heel varus

Crimp: Histological Basis of Manipulation Ask AI

- **Crimp** → basis of manipulation + serial casting
- Ligament collagen fibrils → wavy arrangement → stretchable
  - Stretch to maximal length → hold → crimp **restored** → loosens again
- **Type 1 collagen** ≈ 90% of ligament collagen fibres
- Calcaneus rotating under talus → detailed later
- Spring ligament → markedly shortened in clubfoot
  - Manipulation → stretches only **up to a limit**
  - Limit = maximal crimp

High Yield

- Navicular: medial, wedged, tuberosity abuts medial malleolus
- Talar head: lateral part uncovered → palpable
- Tibialis posterior: 2–3× size, slips to cuneiforms + cuboid
- Medial/plantar ligaments tight; lateral ligaments stretched
- Crimp: stretched collagen regains waves → basis of serial casting

Clinical Examination and Classification Ask AI

Clinical, Not Radiological Ask AI

- Examination → **mainly clinical**
  - Late ossification centres → navicular on X-ray at 3 years
  - X-ray at 6–12 months → only small calcaneus, talus, cuboid centres
  - Cuneiforms may not yet appear
- → Assessment **purely clinical** → requires bone palpation

Palpation: Talar Head vs Calcaneus Ask AI

- Most palpable → **medial + lateral malleoli** → palpate first
- Below + anterior to lateral malleolus → two bony prominences
  - Anterior process of calcaneus vs **talar head**
  - Hand on calcaneal process instead → all manipulation wrong
- Lateral malleolus → first distal prominence = anterior process of calcaneus
  - First anterior prominence = talar head
- Still unclear → stabilise it + **abduct the foot**
  - Prominence disappears / less prominent → talus
  - Unchanged → **calcaneus**
  - Talar head diminishes, no longer well felt

Foot Size and Leg Ask AI

- Parents' question → feet never identical after correction
- New shoes → clubfoot side **one size smaller**
  - Measure on normal foot
- Smaller anteromedial muscles → **smaller leg circumference**
- No **limb length discrepancy**

Parameter	Clubfoot side vs normal
Foot length	Shorter by ~1–1.5 cm
Foot width	Narrower by ~0.5 cm
Leg circumference	Smaller
Leg length	Equal

Radiology and Correction Ask AI

- Radiology → little role in assessment or treatment
- **Never assess correction by radiograph**
- Clinical end point → **plantigrade**, normal-shaped, functional foot
  - 100% normal X-ray → never in clubfoot
  - Normal radiographic alignment → not an indicator of success
  - Not required for full/successful correction

Types of Clubfoot Ask AI

- **Classic clubfoot** → four components
- **Atypical** → rigid: syndrome or rigid foot
  - "Fluffy" foot: full, fatty leg; bony prominences hardly felt; severe
- **Syndromic** → rigid/resistant group

Type	Features	Course
Classic	Cavus, forefoot adduction, heel varus, equinus	Standard Ponseti
Positional	Intrauterine compression	Usually resolves spontaneously
True	Presents soon after birth or delayed	Treated as classic
Atypical	Rigid, fluffy, severe	Rigid/resistant
With associated anomalies	Myelomeningocele, other neurological anomaly	—
Syndromic	Syndrome-related	Rigid/resistant

Pitfall

- Never judge correction by X-ray → clinical plantigrade foot only
- Hand on calcaneal process instead of talar head → wrong manipulation
- Abduction test: talar head disappears; calcaneus stays unchanged

Pirani Score Ask AI

- First step for each patient → **scoring** → baseline for follow-up
- Systems: **Pirani**, Dimeglio
  - Pirani → simple, quick in clinic, repeatable each visit
  - Great functional importance
- Two regions × three items
  - **Midfoot contracture**: lateral curvature, medial crease, talar head coverage
  - **Hindfoot contracture**: posterior crease, empty heel, equinus
- Uses → assessment, follow-up, **Achilles tenotomy** decision
- Each item → `0` normal, `0.5` moderate, `1` severe
- Total → most **severe** `6`; moderate `~3`; mild `<3`
- Score each item in **corrected position**, never resting position
  - Medially deviated resting foot → not automatic score `1`
  - Correct until it stops → then assess

Midfoot Items Ask AI

Pirani

- **Lateral border** → abduct foot first → then assess
  - Normal: straight from heel to fifth metatarsal head
- **Medial crease** of midfoot
- Uncovered **talar head** → palpable anterior to fibula
  - Thumb on talar head → abduct (reduction) → judge coverage
  - Fully covered → no talar head abnormality

Item	0	0.5	1
Lateral border curvature	Straight to fifth MT head	Curve begins at fifth MT head	Curve begins at calcaneus
Medial crease	Multiple fine creases	2–3 deep creases	Single deep crease *[C4 · transcript: "2-3 deep creases scored 0"]*
Talar head coverage	Fully covered with abduction	Partly covered	Still totally uncovered

Hindfoot Items Ask AI

Pirani

- **Posterior crease** → above calcaneus; graded like medial crease
  - Multiple → 0; 2–3 deep → 0.5; single deep → 1
- **Empty heel sign** → normally calcaneus felt at heel
  - Shortened Achilles → calcaneus pulled → inversion + plantar flexion
  - → Calcaneus not felt from heel
- Severe → only heel pad felt; finger cannot reach calcaneus
- **Equinus** (rigidity) → how plantar flexed; assessed in corrected position
  - Equinus of 10–15° → severe

Item	0	0.5	1
Posterior crease	Multiple creases	2–3 deep creases	Single deep crease
Empty heel	Felt immediately, no effort	Felt with mild/moderate pressure	Needs deep pressure, or not felt
Equinus	Dorsiflexion beyond neutral (90°)	Barely reaches 90°	Stays below 90° *[C5 · transcript: "reaching only neutral scored 0"]*

Significance of the Score Ask AI

- Evaluation → >3.5–4 severe; ≤3 moderate
  - → Estimates **treatment duration**
- Score >4 → usually need **tenotomy;** high recurrence
  - → Strict with family at follow-up
- Score <3 → may not need tenotomy
  - Now **routine tenotomy** in all cases high recurrence
  - Even when manipulation achieves dorsiflexion
- **Every visit** → score (6 → 5 → 4)
  - Evaluates patient + treatment method
- Tenotomy timing → not with score above 2–2.5
  - Last cast still 3–4 → midfoot deformity persists → no tenotomy
  - Target 2–2.5 → e.g. equinus 1 + posterior crease 1 only
- → **Pre-operative criterion** for Achilles tenotomy

Must-Know

- Six items: three midfoot, three hindfoot; each 0 / 0.5 / 1
- Maximum 6; moderate ~3; mild <3
- Always score in the corrected position
- Score >4 → tenotomy usually needed, high recurrence
- Tenotomy when score has fallen to 2–2.5

Principles of Ponseti Treatment Ask AI

Scope Ask AI

- Today → Ponseti **manipulation and casting** only
- Separate lecture → resistant + atypical clubfoot
  - Surgery, medial releases, talectomies, Ilizarov

Aim Ask AI

Ponseti

- **Functional,** pain-free, normal-looking foot
  - Good mobility + function; no callosities
  - No shoe modification afterwards
- **100% normal foot** → unattainable: bone size never equalised
  - Counsel family → near-normal, well functional; no comparison with normal side
- Manipulation → stretches **ligaments + capsules** (wavy, stretchable collagen)
  - → Joints realigned → deformity corrected

Guidelines of Treatment Ask AI

- All components corrected **simultaneously**, except equinus
  - Old: forefoot adduction → heel varus → equinus, separately → wrong
  - Subtalar + foot → **mitred joint:** >1 axis, multiple joints
  - One joint moves → next moves opposite; block one → next blocked
  - → Forefoot adduction + heel varus together; equinus last
- **Cavus** dropped/pronated first MT → supinate forefoot → align with hindfoot
  - Twisted foot: calcaneus inverted, forefoot pronated
  - → Untwist → forefoot + hindfoot in one plane → foot moves as **one block**
  - → Cavus corrected first
- Foot abducted in **supination + plantar flexion**
  - Never pronate, or correct before abducting → **subtalar blocking**
  - Stabilise **talar head** → pivot of whole foot
- Equinus never corrected before **heel varus**
  - → No tenotomy until all else corrected
- Stretched collagen regains crimp in ~5 days
  - → **Casting period** never <5 days

Exam Pearl

- All components together except equinus → equinus last
- Cavus first → supinate forefoot to align with hindfoot
- Abduct in supination + flexion; never pronate
- Talar head = pivot → always stabilised
- Each cast ≥5 days → collagen re-crimp

Ponseti Casting Technique Ask AI

Preparation Ask AI

- After Pirani evaluation → **calming the child**
  - Breastfeed before visit, or bring a bottle
  - 2–3-month-olds → very fussy, move during casting
  - Calm only when fed
- **Locate talar head** → in front of lateral malleolus
  - Finger on it + abduct → covered / disappears gradually → confirmed
  - Palpate medial + lateral malleoli, navicular, talar head

First Cast: Cavus Ask AI

Cast 1

- First cast → **reduction of cavus** (untwist foot)
- Elevate **first metatarsal head**
  - → Forefoot supinates, as if deformity increases
- Grip → behind at malleoli, finger on talar head
  - **No hand under calcaneus**
  - Right hand corrects; left holds malleoli + talar head
  - Right raises first MT head → forefoot aligned with hindfoot

Abduction and Serial Casting Ask AI

- Second stage → **forefoot adduction** + varus
  - Forefoot + hindfoot already one block, one plane
  - **Abduct in equinus**, talar head stabilised
  - Thumb of one hand on talar head
  - Other hand → forefoot / medial first MT → abduct
- Serial casting → **every week**
- Manipulation → ~5–10 min
- Grip → four fingers behind malleoli (or Achilles), **thumb on talar head**
  - Other hand → forefoot → abduct
- Thumb → stops talar rotation inside **ankle mortise**
  - Abduct gradually → maximal stretch limit

Padding and Cast Application Ask AI

- **Padding** → minimal: 1 roll (max 2) on forefoot, up to thigh
  - → Talar head still palpable for stabilisation
- **Two people** cast
  - Assistant rolls cotton once maximal stretch reached
- Wrap opposite to deformity → heel varus → **medial to lateral**
  - Never lateral to medial
- Cast **from below upward** → two stages
  - 3–4 plaster turns on forefoot (metatarsals) → upper tibia
  - Foot held in maximal corrected position throughout
- Foot–tibia part **moulded** + dried → then thigh

Moulding: Four Key Areas Ask AI

- Continuous thumb pressure → **pressure sore** → press and release
  - Talar head scars → among commonest post-Ponseti findings
- Index finger above heel; thumb on talar head
  - → Prevents **flat heel** + upward calcaneal migration in cast
- Arch moulding → **maintains arch**
- Medial border moulding → straightens medial border

Area	Technique	Purpose
Talar head	Thumb press and release	Avoid pressure sore
Above calcaneus	Index finger just above heel	Prevent flat heel + calcaneal migration
Arch	Mould cavus	Maintain arch
Medial border	Mould	Straighten medial border

Above-knee Cast and Trimming Ask AI

- Thigh cast → **knee flexion**
- Holds correction → below-knee alone lets leg + foot rotate
- Prevents **slippage** → small legs; below-knee cast falls off
- Trimming → plantar side to **toe tips**; nails + toe bases visible dorsally
- Thigh trimmed → no proximal compression
- Final cast → knee flexed, medial border straight, toes exposed

Progression and Cast Removal Ask AI

- Gradual progression → final **abduction of 60°**
- Last cast → ≥40–50° abduction in equinus → ready for tenotomy
- Weekly change; same technique
  - Abduction ↑ per stretch obtained
- Removal → **knife**, or warm-water soak ½–1 h → soften → cut
- Trick → free plaster end left twisted → parent pulls → unrolls gradually
- Never **oscillating saw** → burn / skin injury

Casting Complications and Errors Ask AI

Complications of Casting Ask AI

- **Rocker-bottom deformity** → premature equinus correction
  - Dorsiflexion before forefoot adduction + heel varus corrected
  - → Midfoot breakage, talonavicular dislocation, talus drops
- **Crowded toes** → assistant holds toes during casting
- **Flat heel** → no moulding behind heel → abnormal / incompletely corrected foot
- **Pressure sore** → commonest site: talar head
- **Deep sore** → tight cast, or full-extension cast then knee bent
  - → Popliteal fossa pressure ± neurovascular pressure
  - → Knee at 90° before casting; never bend after

Complication	Cause	Prevention
Rocker-bottom	Premature dorsiflexion	Equinus corrected last
Crowded toes	Toes pressed in cast	Assistant holds toes
Flat heel	No heel moulding	Mould above calcaneus
Pressure sore	Talar head pressure	Thumb press and release
Deep (popliteal) sore	Tight cast; knee bent after casting	Knee at 90° first

Common Errors in Management Ask AI

- **Pronating** instead of supinating
  - Correct: supinate (↑ deformity) → cavus corrected → foot moves as block
- Not stabilising talar head → **Kite** error: calcaneocuboid joint stabilised
  - Talus → external rotation in mortise → fibula displaced → ankle deformity
  - Finger on calcaneus / CC joint → blocks calcaneal eversion
  - → **Persistent heel varus**
  - → Feel talar head; counter-pressure during casting
- **Casting error** → below-knee cast; poor manipulation / moulding around key areas
  - Knee extended/flexed after plaster → popliteal pressure
- Premature dorsiflexion → **rocker-bottom**
  - X-ray: talonavicular dislocation, talus pointing downward
  - Navicular articulating with (probably) talar neck
- No proper **brace** → detailed later

Pitfall

- Early dorsiflexion → rocker-bottom foot
- Kite's calcaneocuboid fixation → persistent heel varus + ankle deformity
- Pronating the foot instead of supinating
- Below-knee cast; knee bent after casting

Achilles Tenotomy Ask AI

Indications and Criteria Ask AI

- All components corrected except equinus → **tenotomy** stage
  - Indication → equinus, after varus + adduction corrected
- **Pirani** <3 (2–2.5) → mainly hindfoot signs (equinus) left
- Anterior process of calcaneus → palpable **lateral to talus**
  - Deformity: calcaneus + talus nearly parallel
  - Full abduction + calcaneal **eversion** → process moves away from talus
  - **Sinus tarsi** palpable: calcaneal process lateral, talar head medial
  - → Near-anatomical reduction of (presumably) talocalcaneonavicular joint
- Abduction → 60° or more
- Early tenotomy → surprise **recurrence**
- **Heel varus** fully corrected

High Yield

- Pirani <3
- Anterior process of calcaneus palpable lateral to talus
- Abduction ≥60°
- Heel varus fully corrected

Technique Ask AI

Tenotomy

- Blade → No. 11 or No. 15; lecturer prefers curved **15 blade**
- Skin draping → local infiltration or general anaesthesia
- **Local** not preferred
  - Awake child → painful injection
  - Injection bump → tendon poorly felt → less accurate
  - Last cast before brace → must be in full correction, child calm
- → **General anaesthesia**, mild (~10–15 min)
  - Tenotomy + cast in fully relaxed, corrected position
- **Neurovascular structures** medial to tendon
  - → Enter immediately medial → direction medial to lateral
  - Lateral entry → blade may cut neurovascular structures
- Knee full extension + foot **full dorsiflexion**
  - → Gastrocnemius tensed, Achilles tight → palpable
- Entry → 1.5–2 cm above insertion (**calcaneal tuberosity**)
- Blade **parallel to tendon** (pointing up or down); never transverse
- Rotate blade 90° → cut with **pendulum-like movement**
- **Pop** or snap → complete release
- Success → 10–15° dorsiflexion

Post-tenotomy Cast Ask AI

- **Fully abducted** (~60–70°)
- Neutral or slight dorsiflexion
- Duration → 3–4 weeks
- Errors → **premature equinus correction**; incomplete tenotomy (no pop)

Bracing, Follow-up and Relapse Ask AI

Denis Browne Abduction Brace Ask AI

DB brace

- After 3–4-week post-tenotomy cast → **Denis Browne** brace
- Abduction brace → metal **bar** mandatory → maintains abduction
- **Bar width** = distance between child's shoulders
  - Comfortable; legs move up/down together → less brace stress
- Bar **convex downward** → maintains dorsiflexion
  - Straight bar → neutral foot
- Foot bases → **abduction** according to deformity

Shoes of the Brace Ask AI

- Foot base external rotation → **angle** per foot
- **Heel opening** under shoe → heel confirmed fully down
  - → No unnoticed slip into equinus

Foot	External rotation of foot base
Normal	40–45°
Corrected clubfoot	~70°

Bracing Protocol Ask AI

- First 3 months → **full time:** 23 of 24 h
  - 1 h for changing, shower
- Then → 12 h night + 3–4 h day ≈ 16 h
- Until → age 3–4 years
  - Collagen synthesis ↑ → high **recurrence rate**
  - Most foot bones **ossified**, normal position → ↓ recurrence
- After 3–4 years → bracing stopped

Non-compliance Ask AI

- **Non-compliance** → commonest cause of relapse; biggest Ponseti problem
- Parents fed up → child grows, wants to walk, refuses brace
  - Tired of repeated on/off → recurrence
- Stress at every cast, from the start, before bracing
  - "Until the operation I am responsible; after it, you are"
- ↑ Compliance → **continuous education**
  - Small written material; repetition every visit
  - Bracing preparation; good instructions
  - Teach brace application in clinic

Follow-up Schedule Ask AI

- Close follow-up → confirms correct brace use + **compliance**
- Relapses occur over time → regular follow-up → **detect relapse early**

Period	Visits
First 1–1.5 months of bracing	Every 2 weeks
Then until age 2	Every 3 months
Then	Every 6 months
After age 4 (brace removed)	Every 1–2 years

Relapse Patterns Ask AI

- Frequency → **equinus** → varus → dynamic supination
- Dynamic forefoot supination → after brace removal at 3–4 years
  - **Overactive tibialis anterior**
- Varus → usually **non-compliance** with bracing

Relapse	Cause	Management
Equinus (commonest)	Brace worn improperly → heel raised inside brace	Repeat Ponseti → full correction → brace
Varus	Recurrent medial tightness/contracture; brace removed (non-compliance)	—
Dynamic forefoot supination	Overactive tibialis anterior after brace stopped	Must be dealt with

Atypical Clubfoot and Recurrence Ask AI

Atypical

- Atypical (full, fatty) → usually severe → **very high recurrence**
- Usually standard Ponseti technique
- Sometimes **forefoot adduction before cavus**
  - Fluffy foot; bony prominences (first MT) hardly felt
- Atypical + syndromic → detailed separately

Must-Know

- Denis Browne bar: shoulder width, convex downward
- Foot base: normal 40–45°, clubfoot ~70°
- 23 h/day for 3 months → ~16 h/day → until 3–4 years
- Non-compliance = commonest cause of relapse
- Commonest relapse: equinus

Tibialis Anterior Transfer Ask AI

Rationale Ask AI

- Brace off at 3–4 years → **dynamic supination**
  - Corrects when held, recurs when released → flexible deformity
  - Overactive tibialis anterior
- Insertion → base of first MT + first cuneiform
  - Action → **dorsiflexion + inversion** (= supination)
  - Overactivity → foot inverts + supinates
- Aim → move **deforming force** to neutral site
  - Midfoot reinsertion → inverting force → dorsiflexion force
  - → Maintains correction

Timing and Site Ask AI

- Never before 2.5 years
  - Cuneiform still **cartilaginous,** not ossified → weak fixation + action
- Site → middle or lateral cuneiform
  - Lecturer prefers **lateral (third) cuneiform**

Technique Ask AI

TA transfer

- Identify tendon → **prominent** medially in dynamic supination, like EHL
- Incision → 2 cm at insertion
- **Maximal tendon length** → detach from insertion → tag with suture
- Free tendon up to **inferior limb of extensor retinaculum**
  - Retinaculum never cut → avoids **bowstringing** (small foot)
  - Superior limb preserved; part of inferior limb may be freed
- **Image intensifier** → identify lateral cuneiform
  - Hole → 2.5 mm drill bit
  - Widened gradually → 3.5 or 4.5 mm → tendon passes
- **Suture passer** → tendon to sole
  - Tied over button, gauze, or antibiotic vial cap
- Tension in **maximum dorsiflexion** tendon slackens over time
- Close → cast 1.5 months
  - Below-knee or above-knee; lecturer → **below-knee**
  - Until **tendon–bone incorporation** → cast removed

Pre-transfer Casting Ask AI

- 3–4 serial manipulations + casts before transfer
  - → **Stretch medial structures**
  - → Easier correction + faster post-op corrective action of tibialis anterior

Take-home Message and Questions Ask AI

Take-home Message Ask AI

- Proper **assessment** + scoring
- Gradual casting per Ponseti steps
- Tenotomy timing → after successful abduction
- **Follow-up of bracing** → key to recurrence
  - Pick up relapse early → deal promptly

Brace Timing and Medical Shoes Ask AI

- Tenotomy cast removal → brace interval ≤3–4 days
  - Brace measured same day; applied immediately
- Brace hard to wear → replace with shoes?
  - **Medical shoes** → no proven evidence in clubfoot
  - Brace at night + special shoes by day → orthotics money-maker
  - Ordinary shoes; only Denis Browne maintains correction

Casting, Tenotomy and Stretching Ask AI

- Big toe deformity from casting → no
  - Only dropped first MT (cavus) → corrected in **one cast**
  - Raise first MT head → forefoot aligned → no deformity risk
- **Tenotomy healing** → 100% guaranteed
  - Sheath intact in 90% (at least 2–3 sides)
  - Gap **bridging** → complete healing, faint scar
  - Some → open **Z-plasty** after 1–1.5 years
  - Ponseti → percutaneous safe up to 4 years; after 4 → open Z-plasty
- Physical stretching instead of Ponseti?
  - Individual efforts, personal papers; exception → **French method**
  - Not applicable in Egypt → visits every 3 days + continuous stretching
  - Tenotomy still needed; greater effort
  - No evidence superior to Ponseti
  - Worldwide agreed treatment → **Ponseti**
- Remaining clubfoot topics → relapse, foot **osteotomies**
- Each cast → 5–7 days
  - Collagen fibrils **heal** + regain wavy appearance
- First cast for cavus only?
  - Cavus → like a twisted **towel**
  - Both hands back to one plane → turn foot as block → abduct
  - Abduction before cavus correction → forefoot still pronated → not allowed
  - First cast → cavus supination only; no cavus → straight to **forefoot adduction**
- Achilles release age → percutaneous to 4 years; then **open**
- Why equinus last → need a **fulcrum** = Achilles
  - Foot held in plantar flexion → rest of deformity corrected

Transfer Incisions and Mixed Brace Ask AI

- Tendon transfer → **two incisions:** insertion + lateral cuneiform
- Clubfoot one side + **vertical talus** other → each corrected separately (common)
  - Vertical talus → reverse Ponseti (later lecture)
  - Clubfoot → Ponseti
- **Mixed brace** → Denis Browne, foot set per deformity
  - Clubfoot side → 70° abduction
  - Vertical talus side → neutral position

Positional Clubfoot and Achilles Ask AI

- "Physiological" clubfoot → meant **positional**
  - No intervention → corrects in first month on follow-up
  - Not corrected → classic clubfoot → standard treatment
  - Mother → stretching exercises first month
  - Fully corrected within 3–4 weeks
- **Over-lengthening** with tenotomy → no
  - Only with open Z-plasty set in excessive dorsiflexion (10–15°)
  - → **Calcaneus deformity** → very difficult condition
- Ponseti effective → up to 4 years; then foot **stiff**
- Open vs percutaneous → open at older age (>2–4 years)
  - Below 2 years → **percutaneous tenotomy** sufficient

Practical Casting Points Ask AI

- Treatment start → ~1 week, not day after birth
  - **Physiological jaundice** → incubator → cast removed
  - → Wait 1 week–10 days → then cast
- **Plaster of Paris** over Dynacast → mouldable; Dynacast not mouldable
- Padding kept **thin** → talar head felt through plaster
  - → Stabilisation + correct casting
- Denis Browne → until 3–4 years

Neutral Brace Position and DDH Ask AI

- **Neutral position** = 0° rotation (vertical talus side)
- Clubfoot + **DDH** → correct foot first, then hip
  - Young age → foot surgery much less; older → foot surgery exceeds hip
  - Presents at 1 month → foot corrected by ~3–4 months
  - → Hip **closed reduction under GA**
  - Waited beyond 1 year → **open reduction**
  - Hip first, foot last → surgical release + post-op foot problems
- After full, stable foot correction → DDH
  - Younger → closed reduction; older → **open reduction**

Transfer Site and Number of Casts Ask AI

- **Tibialis anterior transfer to cuboid** → no
  - Too lateral → evertor → deformity reversed
  - Keep it central → neutral
- Total casts → ~6–8, patient-dependent
  - >12 casts, still uncorrected → **technique error**
- Toe flexion during serial casting
  - Plaster to toe tips (sole); nails visible dorsally → no **flexion deformity**

Remaining Questions Ask AI

- Positional clubfoot → no cast; **follow-up only** + exercises first month
- Rocker-bottom = **vertical talus** → entirely different topic
  - Complication questions + resistant clubfoot → separate lecture
- Tenotomy must reach 15° dorsiflexion? → yes
  - 10–15° **dorsiflexion** required; otherwise no tenotomy done
- Pirani >2.5 from hindfoot signs + abduction to 60° → tenotomy?
  - Full abduction + calcaneal eversion → part of **empty heel** corrected
  - Crease 1 + equinus 1 + empty heel 0.5 = 2.5 → proceed
- Denis Browne brace in all cases → **mandatory**

Relapsed and Neglected Clubfoot

(Recognising and managing recurrent and neglected deformity)

Dr Mohamed Fathey

Watch Original Lecture

Lecture transcript processed with AI assistance.

Contents

↑ Book contents

Principles of Surgery in Relapsed and Neglected Clubfoot . . . 39
Two Entities . . . 39
À la Carte Surgery . . . 39
Timing of Surgery . . . 39
Relapsed versus Neglected Clubfoot . . . 40
Causes of Relapse . . . 40
Clinical Differences . . . 40
Timing of Relapse . . . 41
Pathoanatomy . . . 41
Problems of Neglected Clubfoot . . . 41
Aim of Treatment . . . 41
Soft Tissue Release . . . 42
Indications . . . 42
Timing and Preoperative Casting . . . 42
Three Main Incisions . . . 43
Posteromedial Release (Turco) . . . 43
Principle . . . 43
Medial Steps . . . 44
Summary . . . 44
Plantar Release Illustrated . . . 44
Adequacy of Medial Release . . . 45
Subtalar Release . . . 45
Lateral and Posterior Release . . . 45
Lateral Release . . . 45
Provisional K-wire Fixation . . . 45
Joint Releases Illustrated . . . 46
Posterior Release . . . 46
Intraoperative Check of Reduction . . . 46
Norris Carroll Modification . . . 46
Residual Deformities . . . 47
Residual Adductus . . . 47
Lateral Column Shortening: Rationale . . . 47
Four Procedures . . . 47
Residual Equinus . . . 48
Residual Heel Varus . . . 49

Dwyer's Lateral Closing Wedge . . . 49
Medial Opening Wedge . . . 49
Sliding Osteotomy . . . 49
Forefoot Supination . . . 49
Dynamic vs Fixed . . . 50
Tibialis Anterior Transfer . . . 50
Rigid Supination: Osteotomies . . . 50
Salvage Procedures . . . 51
Triple Arthrodesis: Indications . . . 51
Technique . . . 51
Wedge Resection and Lambrinudi . . . 51
Achilles and Plantar Fascia . . . 52
Talectomy . . . 52
Internal Tibial Torsion . . . 52
Treatment Algorithm . . . 52
Ilizarov . . . 53
Indications . . . 53
Ilizarov versus Triple Arthrodesis . . . 53
Principle and Methods . . . 54
Complications of Surgery . . . 54
Recurrence . . . 54
Dorsal Navicular Subluxation . . . 54
Valgus Overcorrection . . . 55
Triple Osteotomy . . . 55
Dorsal Bunion . . . 55
Calcaneus Deformity . . . 55
Skin Complications . . . 56
Take-Home Message . . . 56
Questions and Answers . . . 56
Ponseti First . . . 56
Ilizarov in Scarred Feet . . . 56
Casting Before Release . . . 56
Ponseti in Relapsed Clubfoot . . . 56
Complications of Calcaneal Varus Correction . . . 57
Tibial Torsion and Definition of Neglected . . . 57
Posteromedial Release Tips . . . 57
Split versus Total Tibialis Anterior Transfer . . . 57
One Stage or Two . . . 58

Achilles: Formal Lengthening vs Ponseti Tenotomy 58
Z-plasty Technique 58
Dorsal Bunion Surgery 58
Residual Metatarsus Adductus 58
Master Knot of Henry 58
Plantar Fascia, Release, Calcaneal Shortening 59
Preferred Approach: Norris Carroll 59
Soft Tissue Swelling 59
Triple Arthrodesis Techniques 59
Ilizarov Rate of Distraction 59
Diabetic and CP Children 59
Multiple Metatarsal Osteotomy 59
Cause of Calcaneus Deformity 60

AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

- **C1** (p. 44) “increases forefoot abduction and supination” → “increases forefoot adduction and supination”
- **C2** (p. 45) “calcaneus returns inverted” → “calcaneus returns to eversion”
- **C3** (p. 51) “midfoot taken apart” → “hindfoot taken apart”
- **C4** (p. 55) “over-lengthening of tibialis anterior” → “over-lengthening of the Achilles tendon”
- **C5** (p. 59) “tibialis anterior leads to TN joint” → “tibialis posterior leads to TN joint”

Principles of Surgery in Relapsed and Neglected Clubfoot Ask AI

Two Entities Ask AI

- Two entities → **relapsed clubfoot** vs neglected clubfoot → different management
  - Relapsed → previous procedures; not a primary case
    - Scars, previous operations; prior foot surgery unknown
  - Neglected → usually severe; never approached; no previous operation

À la Carte Surgery Ask AI

- Surgery apart from **Ponseti** → three scenarios
  - Already treated with Ponseti
  - Never received Ponseti → neglected → late presentation
  - Surgery decided (basis of decision covered later)
- Clubfoot surgery is **"à la carte"** → key statement
  - Defined set of surgical steps
- Not all steps in every patient
  - Some patients → some steps; others → all steps
  - Only what the patient needs → not every memorised step

Timing of Surgery Ask AI

- Applies to **posteromedial release** and osteotomy alike
- Preferably **not before first year** of life
  - Foot > 8–9 cm → clearer pathoanatomy, anatomy, neurovascular structures → easier
  - First 6 months → increased **collagen** production
    - Collagen → **recurrence** in clubfoot
    - Surgery at 2–3 months → severe post-op fibrosis → recurrence expected
    - Later operation → better
  - After 1 year → **weight bearing** begins
    - Weight bearing maintains correction → **plantigrade position**
- Rule → after 1 year, or foot length > 8–9 cm

High Yield

- Clubfoot surgery is **"à la carte"** → only the steps the patient needs
- Surgery preferably after 1 year, or foot > 8–9 cm
- First 6 months → high collagen → post-op fibrosis → **recurrence**
- Weight bearing after 1 year → maintains **plantigrade position**

Relapsed versus Neglected Clubfoot Ask AI

Causes of Relapse Ask AI

- Causes → revision and secondary procedures
  - **Faulty techniques** and faulty manipulations
  - Parental **non-compliance** with Denis Browne splint
  - Older age at presentation
  - Recurrence after previous procedures
  - **Early surgery** within first 6 months → post-op fibrosis
- All → recurrence

Clinical Differences Ask AI

- **Relapsed** → previously treated foot that recurs
  - Any intervention counts → even Ponseti alone
- **Neglected** → primary patient, never treated
  - No operation, no surgical scars, no Ponseti, no **percutaneous tenotomy**
- Relapsed → usually **less severe**
  - Previous Ponseti or surgical release → reduced deformity magnitude
- Neglected → **severe deformity**
  - Untouched foot → deformed bones, contracted soft tissues
- Relapsed gait → on sole, but on **lateral border**
  - Forefoot supination, adduction, heel varus
- Neglected gait → on **dorsum** of foot (left picture)
- Large dorsal **callosity**, not plantar
  - Untreated severe clubfoot → weight on dorsal callosity, never on sole
- Relapsed → bony deformities partly **remodel** → less severe
  - Untreated → all foot bones deformed
- Neglected → most have dorsal **bursa**
- Relapsed (lower picture) → weight on sole; some supination, **forefoot adduction**

Feature	Relapsed	Neglected
Prior treatment	Yes — even Ponseti alone	None — no surgery, Ponseti or tenotomy
Scars	Present	None
Severity	Less severe	Severe
Weight bearing	Sole, lateral border	Dorsum
Foot position	Forefoot supination, adduction, heel varus	—
Bones	Partial remodelling	All deformed
Dorsum	—	Callosity, bursa

Timing of Relapse Ask AI

- Usually 2–3 years; wider range 10 months–5 years
- After 5 years → uncommon
  - Almost all bones ossified by 5 years
  - Last foot bone to ossify → **navicular**, at 3–4 years
  - Bones then remodel to new position
- No relapse by 5 years → usually none; few exceptions
- Most often → 1–3 years

Pathoanatomy Ask AI

- Same as primary clubfoot → deformed bones and soft tissues
  - **Talus** → plantar flexed, inverted
  - Calcaneus → plantar flexed, inverted
  - **Navicular** → medially displaced → close to medial malleolus
  - Medial joints tight; lateral joints loose
  - **Calcaneocuboid joint** → medial subluxation → deforming force
  - Tight soft tissues
    - Medial ligaments, especially plantar → **spring ligament**
    - Posterior tendons → **tibialis posterior**, Achilles
- Neglected → full pathology, more severe
- Relapsed → only some elements

Problems of Neglected Clubfoot Ask AI

- **Pain** → walking on dorsum
- **Awkward gait** → no long distances; pain with long distances
- Dorsum unsuited to weight bearing → **skin breakdown** and ulceration
  - Damage on dorsum and on dorsal bursa
- Severe pain + **severe disability** → intervention necessary

Aim of Treatment Ask AI

Neglected

- Success → two requirements
  - **Weight bearing on sole**
  - Normal shoes
- Not a fully anatomically reduced or normal-looking foot
  - Shoes + walking on sole, not dorsum (pain, **ulceration**, skin breakdown)
- Aim → **plantigrade foot:** sole bearing, shoe wearing, walking in shoes
  - Patient satisfied even if foot stiff or movement limited

Soft Tissue Release Ask AI

Indications Ask AI

STR

- Age limit → up to 4 years
- **Failed Ponseti**
  - Or second Ponseti failed from stiffness
    - Syndromic patient
    - Old surgical scars → prevent complete or repeat Ponseti
- Now rarely in **idiopathic** cases
- Current main indications
  - Previous release + recurrence → extensive scarring → Ponseti fails
  - **Syndromic** → myelomeningocele, arthrogryposis
    - Respond poorly to Ponseti, despite many papers on Ponseti in them
    - Not all respond → treated by release

Timing and Preoperative Casting Ask AI

- Best time → after 1 year, before 4 years
  - Foot still **mobile** → responds to surgical correction
- **Casting** first, before any surgical correction
  - Reduces surgical correction required
  - Stretches soft tissues, ligaments, muscles → easier release

Three Main Incisions Ask AI

- Three most common of many described
  - **Turco** → posteromedial + lateral
  - Norris Carroll → reverse: medial + posterolateral
  - **Cincinnati** → circumferential behind heel; one incision
- Turco posteromedial incision
  - Starts at **metatarsal** → medial side → beneath medial malleolus → medial side of Achilles
  - May stop at Achilles midpoint
  - + lateral incision → **posterior release**
- Norris Carroll medial incision
  - Starts at **base of first metatarsal**
  - Curved up toward medial malleolus → down to middle of heel
  - + posterolateral incision on lateral side of Achilles
- Cincinnati (yellow line) → medial side → behind heel → lateral side
- Turco lateral incision → lateral border of Achilles → **Achilles lengthening** + posterior release
- Norris Carroll posterolateral → posterolateral release on lateral border of Achilles

Approach	Incisions	Course
Turco	Two: posteromedial + lateral	Metatarsal → beneath medial malleolus → medial to Achilles; lateral → lateral border of Achilles
Norris Carroll	Two: medial + posterolateral	Base of 1st metatarsal → curved to medial malleolus → heel; posterolateral → lateral to Achilles
Cincinnati	One, circumferential	Medial side → behind Achilles → lateral side

- Order of discussion → **posteromedial release** first

Posteromedial Release (Turco) Ask AI

Principle Ask AI

Turco

- Two incisions → posteromedial + lateral
- Specific steps, **à la carte**
  - One step → check correction → achieved: stop; not: continue

Medial Steps Ask AI

Turco

- Through posteromedial incision
- Release **abductor hallucis** from origin on calcaneus
  - Deforming force → ↑ forefoot adduction and supination *[C1 · transcript: "increases forefoot abduction and supination"]*
  - Work **above** it, not beneath
    - Upper border → away from neurovascular structures → no injury
- Expose **plantar fascia**
  - Plantar fascia → cause of **cavus**
  - Not in Turco → described by Norris Carroll; included for complete steps
  - Release near **calcaneal insertion**
  - Cut with it → **abductor digiti minimi**, flexor digitorum brevis
- **Neurovascular structures** appear → retract
- Joints
  - Two most important → **talonavicular** and calcaneocuboid
  - Talonavicular → opened **dorsally, plantarly, medially**
  - Calcaneocuboid → only **medial capsule** reachable medially → opened
  - **Medial subtalar** joint → opened → subtalar release begins
- Tight medial ligaments
  - **Deltoid ligament**
  - **Spring ligament** = plantar calcaneonavicular ligament
- Last → lengthen **tibialis posterior tendon** → facilitates navicular reduction

Summary Ask AI

Turco

- Sequence → abductor hallucis elevated from calcaneus → plantar fascia (cavus) → joints → ligaments → tibialis posterior
- Talonavicular release dorsal + plantar + medial → navicular onto **head of talus**
- **Deltoid ligament** → very tight medially (picture) → must release
- Calcaneonavicular ligament → better seen from lateral side
- **Tibialis posterior** → deforming force → lengthened → talonavicular reduction

High Yield

- Medial order → abductor hallucis → plantar fascia → joints → ligaments → tibialis posterior
- Three joints from medial side → **talonavicular**, calcaneocuboid, medial subtalar
- Talonavicular released dorsally, plantarly, medially → navicular reduces onto talar head
- Key medial ligaments → **spring** (calcaneonavicular) and deltoid

Plantar Release Illustrated Ask AI

- Structures of the **plantar release**
- **Spring ligament** → cut
  - Connects calcaneus and navicular
  - Very tight → causes or contributes to navicular medial subluxation
  - Incised → talonavicular reduction
- **Talonavicular joint** → fully incised, dorsum → plantar surface
- **Subtalar joint** → medial capsule completely incised
- **Plantar fascia** → released from calcaneal attachment

Adequacy of Medial Release Ask AI

- Sign → **navicular easily reducible** onto talar head
  - Navicular still tight → incomplete → complete the release
- Whole foot rotates around **head of talus**
  - Talus and calcaneus together inverted → talus externally rotated, calcaneus inverted
  - Freed from each other → calcaneus to **eversion** → normal talocalcaneal angle *[C2 · transcript: "calcaneus returns inverted"]*

Subtalar Release Ask AI

Turco

- Talus–calcaneus held by **capsule** + interosseous ligament
- Release **subtalar joint capsule**; leave interosseous ligament intact
  - Interosseous cut → calcaneal **lateral translation** → later iatrogenic valgus
- **Complete release** → medial, posterior, lateral sides
  - Capsule only → no translation, no iatrogenic calcaneal deformity

> **Pitfall**
> - Cutting talocalcaneal **interosseous ligament** → calcaneal lateral translation
> - Result → later iatrogenic valgus deformity
> - Subtalar release → capsule only, all sides

Lateral and Posterior Release Ask AI

Lateral Release Ask AI

- Few structures; key → **lateral subtalar joint** capsule
- **Calcaneocuboid joint** → not opened laterally; medial side only
  - Pathology → CC medial subluxation → tight medial capsule
  - Medial CC capsule released → reduction
- Joint released laterally → **subtalar joint**
- **Peroneal tendon sheath** → released; tendons not lengthened
  - Sheath → **tether** against reduction
- **Calcaneofibular ligament** → released
  - Very tight; displaced somewhat posteriorly
  - Blocks calcaneal reduction and eversion
- Remaining → **posterior release** of calcaneus → easier

Provisional K-wire Fixation Ask AI

- Navicular and cuboid reduced onto talus and calcaneus → **provisional K-wire fixation**
  - Done **before posterior release** → key tip
- Wire `1` → talus, posterior → anterior, up to head
  - Used as **joystick** → reduce talonavicular joint
- Wire `2` → across **calcaneocuboid** joint
  - After medial CC release, calcaneofibular ligament and subtalar release
  - CC reduced → fixed
- Sequence → posteromedial release → lateral release (calcaneofibular ligament, subtalar) → K-wires

Joint Releases Illustrated Ask AI

Cincinnati

- Picture via **Cincinnati incision** → shows joint freeing
- Subtalar → **circumferential** release
- Ankle → mostly posterior → for **equinus**
- Subtalar circumferential **without cutting interosseous ligament** → talocalcaneal rotation restored
- Calcaneofibular ligament shown
- Anatomy of ligaments and capsules to release
  - Calcaneus, talus (capsule-covered), navicular, cuneiforms, metatarsals
- Joints opened → **talonavicular**, subtalar (medial, lateral, posterior), ankle posteriorly
- Most important ligaments → **spring ligament**; calcaneofibular laterally
- Medial and lateral sides complete

Posterior Release Ask AI

- After **temporary stabilisation** of TN and CC → remaining: equinus deformity
- Corrected by **posterior release**
  - Turco → straight longitudinal incision on lateral border
  - Norris Carroll → posterolateral incision
- Two targets → **Achilles tendon** + joints
  - Joints → **posterior subtalar** and ankle
- Achilles → **formal lengthening**, not percutaneous as in Ponseti
  - Incision → **Z-plasty**
  - **Distal cut medially**, not laterally
    - Achilles inserts medially medial deviation of calcaneus
    - Medial attachment released → **line of pull** more lateral → aids calcaneal reduction

Intraoperative Check of Reduction Ask AI

- K-wire sequence (left picture)
  - Wire into talus → reduction → advanced into navicular → **talonavicular joint** reduced first
  - TN reduced → **spontaneous reduction** of CC
  - Second wire → CC fixation → **stabilised foot**
- Lateral view before talar reduction → **fibula** posteriorly displaced
  - Talus and calcaneus rotated in **ankle mortise** → fibula posterior
- TN reduced, talus back → fibula returns to normal lateral view
  - Intraoperative check of correct reduction

Clinical Pearl

- Posteriorly displaced **fibula** on lateral view → talus rotated in mortise
- TN reduced → normal fibular lateral view → reduction confirmed
- TN reduced → CC reduces spontaneously

Norris Carroll Modification Ask AI

- Same steps as Turco + two additions
  - **Plantar fascia release** → not done by Turco
  - **Capsulotomy of calcaneocuboid joint** → important addition
- Rationale → many relapses after posteromedial release uncorrected **cavus**

  - Cavus correction → release plantar structures, mainly plantar fascia
- Technique → **one scissor**, one cut
  - Plantar fascia + flexor digitorum brevis + abductor digiti minimi
  - → relieves **plantar tethering** → reduces cavus
- Otherwise same structures as Turco
- Scar → Norris Carroll somewhat better than Turco
  - Posteromedial scar → deforming or conspicuous, especially in relapsed children

Residual Deformities Ask AI

- Up to 4 years → soft tissue release possible
- Beyond 4 years → bones deformed → release and reduction harder
- **Residual deformities** → more advanced, bigger operations
  - Each deformity addressed per se; e.g. residual adductus, residual equinus
- Procedures → **separate** or combined in same child
- **À la carte** → several deformities → combined procedures; one deformity → one correction

Residual Adductus Ask AI

- **Residual adductus** → forefoot adduction, usually with supination
- Corrected by **lateral column shortening**

Lateral Column Shortening: Rationale Ask AI

- Foot deviated medially → **lateral column** longer than medial
  - Lateral column → calcaneus, cuboid, fifth metatarsal
- Shortening → theoretically lengthens or realigns medial column → **straight lateral border**
- Preferred age → 4–8 years
  - Below → soft tissue release usually enough
  - Above 8 → more severe → may need more than lateral column shortening
- Origin → Norris Carroll study
  - Main cause → **obliquity of calcaneocuboid joint** → residual adductus
  - Shortening corrects CC obliquity + shortens lateral column

Four Procedures Ask AI

Adductus

- Closing wedge osteotomy of calcaneus
- Excision of anterior process of calcaneus (**Lichtblau**)
- Calcaneocuboid wedge resection (Evans)
- Cuboid osteotomy / double column osteotomy
- Calcaneal or cuboid osteotomy → preceded by **posteromedial release**
  - Separate first stage, or all in one stage
- Target → **calcaneocuboid articulation**
  - Primary pathology → **apex** of adductus deformity
- Incision → lateral surface of calcaneus, **centred on calcaneocuboid joint**
- Cuboid closing wedge → less preferred
  - Primary pathology in joint more than cuboid → cuboid away from **apex**
- Calcaneal closing wedge → wedge **just proximal to CC articulation**

  - Joint stays mobile; no fusion
  - Lateral column shortened; joint **preserved**
- Anterior process excision → **resection arthroplasty**
  - Anterior articular end of calcaneus removed → lateral column shortened
  - **Pseudoarthrosis** → mobility kept; not stiff, not fused
- **Evans** → complete CC resection + CC fusion
  - Very good stability, satisfactory results
  - Low recurrence among lateral column shortening osteotomies
- Cuboid → severe deformity
  - **Cuboid closing wedge osteotomy** or cuboid enucleation (removal)
  - More severe → + medial column osteotomy
    - Cuboid wedge → into **medial cuneiform** → double column osteotomy
    - Severe cases only
- Residual forefoot adduction → four main operations

Procedure	Action	CC joint	Note
Calcaneal lateral closing wedge	Wedge just proximal to CC joint	Preserved, mobile	Straight lateral border
Anterior process excision (Lichtblau)	Anterior articular end of calcaneus removed	Pseudoarthrosis, some motion	Resection arthroplasty
Evans	Complete CC resection	Fused	Good stability, low recurrence
Cuboid / double column	Cuboid wedge or enucleation; wedge → medial cuneiform	—	Severe; lengthens medial, shortens lateral column

Residual Equinus Ask AI

- **Residual equinus** → corrected patient re-presents with equinus ± forefoot adduction
  - One element or more → one or more procedures
- Cause → soft tissue only, or bone
- Stepwise
  - Standard **soft tissue release** → evaluate
  - No improvement → **lateral column lengthening**
    - Theoretically reduces equinus at **midtarsal joint** level
  - Still no improvement → bony correction; three operations
    - **Naviculectomy**
      - Navicular acts as **hinge** blocking reduction → excised
      - Talus then articulates with cuneiforms → some motion kept
    - Partial excision of **head of talus**
      - Marked foot stiffness → lecturer dislikes it
    - **Distal tibial dorsiflexion osteotomy** → anterior wedge from distal tibia → dorsiflexion
- Preferred → **Lichtblau** + naviculectomy → best results and satisfaction

Exam Pearl

- Residual equinus, preferred → **Lichtblau** + naviculectomy
- Talar head excision → marked stiffness
- Distal tibia → anterior wedge → dorsiflexion

Residual Heel Varus Ask AI

- **Residual varus** → calcaneal level; internally deviated or inverted, uncorrected
- Three options only
  - Lateral closing wedge = **Dwyer's operation**
  - Medial opening wedge
  - Sliding osteotomy

Dwyer's Lateral Closing Wedge Ask AI

- Simple lateral incision over calcaneus
- Two osteotomy lines
  - One **parallel to subtalar joint**
  - One parallel to posterior surface of calcaneus
- Wedge removed completely → gap closed → fixed (**staple**, screw or wires)
- Very good results

Medial Opening Wedge Ask AI

- Requires **graft** → another procedure
  - Distal tibia or iliac crest → ↑ morbidity
- Lengthens calcaneus → ↑ **Achilles tightness** → needs Achilles tendon lengthening

Sliding Osteotomy Ask AI

- Very simple; lateral approach
- **Oblique osteotomy** → posterior to subtalar joint, anterior to Achilles insertion
- Posterior segment translated **laterally** (medial → lateral)
  - Corrects calcaneal varus

Forefoot Supination Ask AI

- Fourth residual deformity → after adduction, heel varus, equinus
- **Supination** → first metatarsal elevated off ground
- Dynamic or fixed (rigid)

Dynamic vs Fixed Ask AI

- Dynamic → correctable
  - **Coleman test** → corrected
  - Talar head stabilised + forefoot abducted → reduces; TN joint adjustable
  - Cause → **overacting tibialis anterior**
  - Treatment → tibialis anterior transfer
- Rigid → non-correctable; options
  - **Plantar closing wedge osteotomy** of medial cuneiform → not very common
  - **Midtarsal osteotomy**
    - Complete → cuneiform → cuboid
    - Or **double column** → cuboid wedge → into cuneiform
- Dynamic transfer target → **lateral cuneiform**

	Dynamic	Fixed (rigid)
Correctable	Yes — Coleman test; abduction with talar head stabilised	No
Cause	Overacting tibialis anterior	—
Treatment	Tibialis anterior transfer → lateral cuneiform	Plantar closing wedge of medial cuneiform; double column (cuneiform + cuboid); complete midtarsal osteotomy (medial cuneiform → cuboid)

Tibialis Anterior Transfer Ask AI

- Very simple operation
- **Medial incision** at insertion (cuneiform, base of first metatarsal) → detach → free to inferior extensor retinaculum
- Second incision over **third cuneiform** and third metatarsal → drill hole → pass tendon
- Now 2 incisions, formerly 3
  - Preserves **superior extensor retinaculum** → no bowstringing of tendon
- Tied over **button** on sole
- **Cast** → 6 weeks

Rigid Supination: Osteotomies Ask AI

- Complete **midtarsal osteotomy** → medial cuneiform → cuboid; complete wedge removal
- **Double column** → cuboid wedge (shortens lateral column) → into opened medial cuneiform (lengthens medial column)
  - Restores normal foot shape
- Complete midtarsal wedge, cuneiform → cuboid → large wedge (clinical picture)

Salvage Procedures Ask AI

Triple Arthrodesis: Indications Ask AI

- Child > 8–9 years, nearing **adolescence** (10–12 years), severe deformity
  - Previous procedures → no full correction
  - Joints contracted, bones deformed, tendons and muscles shortened → osteotomies insufficient
- Solution → bones taken apart and reassembled → **triple arthrodesis**
  - Hindfoot disassembled → reassembled in **corrected position** *[C3 · transcript: "midfoot taken apart"]*
- Not before 12 years → avoids **growth retardation**
- Three joints fused → **talonavicular**, calcaneocuboid, subtalar
  - All articular surfaces resected → talus, navicular, cuboid, calcaneus
- Talus → minimal bone removal
  - Tenuous blood supply + small bone volume → **AVN** risk
  - Calcaneus → rich supply, large volume → freer resection

Technique Ask AI

Triple

- Standard **Ollier** incision
  - Lateral, centred on **sinus tarsi**
  - Sinus tarsi **soft spot** → from dorsum → curved, below lateral malleolus → ends over calcaneus
  - Good bone exposure
- Skin opened → sinus tarsi released and cleared → calcaneus exposed
- CC joint covered by **extensor digitorum brevis**
  - Sharp dissection → CC joint exposed
- **Denude** CC articular surfaces → calcaneal + cuboid cartilage
- Subtalar → all articular surface removed from **undersurface of talus** and calcaneus
  - **All cartilage** removed → good fusion
- **Hohmann** retractor on neck of talus → denude talonavicular joint
- Three joints → CC, TN, subtalar
- Foot reduced to normal **plantigrade** position → fixed

Wedge Resection and Lambrinudi Ask AI

Triple

- Denudation not always enough → **wedges** in severe deformity
  - Mostly lateral → CC level; up to 1.5–2 cm
- TN irreducible → resect TN by **Lambrinudi** technique
  - Bone removed from talar head down to posterior facet distally → talar head articulated with navicular
- Less severe → simple **denudation** allows reduction
- More severe → large lateral CC wedge ± large talar wedge → TN reduction

Achilles and Plantar Fascia Ask AI

Triple

- Check **Achilles** after triple + fixation
- Usually no lengthening needed
  - Subtalar release or denudation → some flexibility and length to Achilles
- Achilles lengthening → last; re-evaluate after triple and fixation
- **Residual cavus** → plantar fasciotomy
- TN and CC wedge resection → shorter foot → laxer plantar fascia
  - Plantar fascia resection or recession → may be unnecessary

Talectomy Ask AI

- Other **salvage procedure** at older age → talectomy
  - Above 10–12 years + severe deformity → talectomy or triple arthrodesis
- Triple → marked **superiority** over talectomy
- Reserved for specific cases
  - **Rigid equinovarus** in myelodysplasia or arthrogryposis
    - Arthrogryposis → very high recurrence, severe deformity → talectomy preferred
  - Neglected clubfoot > 10–12 years not amenable to triple arthrodesis
- Disadvantages
  - **Incongruent joint** → talus removed → tibia-to-calcaneus articulation
  - Pseudoarthrosis in many cases
- Advantage → **plantigrade foot**, though possibly incongruent or unstable
- Triple preferred whenever feasible

Internal Tibial Torsion Ask AI

- Point within residual deformity
- Some children → **internal tibial torsion**
- After complete foot correction → **derotation osteotomy**
  - **Supramalleolar osteotomy** → cut above malleoli
- Rotation > 10–15 degrees → fibula cut too
  - Fibula intact + large rotation → fibula becomes rotation axis → severe stress at osteotomy
  - **Combined tibial and fibular osteotomy** → derotation about tibial axis

Treatment Algorithm Ask AI

- First line → release; corrected → done
- Persistent adductus → lecturer prefers **Evans**
- Salvage → above 10–12 years, severe, not amenable to other treatment

Situation	Procedure
> 1 year: neglected, after or failed Ponseti, arthrogrypotic, syndromic	Posteromedial or Norris Carroll release first
Uncorrected, > 4 years, persistent adductus	Calcaneal closing wedge, Lichtblau, or CC resection (Evans — preferred)

Situation	Procedure
Persistent equinus	Naviculectomy (preferred), talar head excision, distal tibial osteotomy
Persistent supination — dynamic	Tibialis anterior transfer
Persistent supination — fixed	Cuneiform osteotomy: complete midtarsal (medial cuneiform → cuboid) or double column (medial cuneiform + cuboid)
> 10–12 years, severe, not amenable	Salvage procedures

Ilizarov Ask AI

Indications Ask AI

Ilizarov

- Most foot deformities → correctable without Ilizarov
- **Severe scars** → no safe incision; scalpel → skin problem risk
- Additional **lengthening procedure** needed (tibia or foot) → combined via Ilizarov
- Preserves foot **length** → no shortening, unlike triple

Ilizarov versus Triple Arthrodesis Ask AI

	Triple arthrodesis	Ilizarov
Foot	Stiff, shorter	Stiff, long
Function	Short stiff foot better → movement, pain, walking	Inferior to short stiff foot
Recurrence	—	Somewhat low

- Triple → advantage over Ilizarov in function
- Ilizarov → after soft tissue release, or primary without release
- Low recurrence; good **plantigrade foot**; key in extensive scarring

Principle and Methods Ask AI

Ilizarov

- **Distraction histiogenesis**, not osteogenesis → soft tissue level, not bone
- Combined with prior soft tissue release and osteotomies
- Or primary closed Ilizarov → **closed release**, no opening
- Three methods in foot → open; closed constrained; closed non-constrained
- Frame → **tibial frame** + foot frame
- Open → frames separate; forefoot may be separate from hindfoot → all free
  - No control of correction → subluxation or incomplete correction
- **Closed** → tibial and foot frames connected
  - Constrained → hinges respected, at deformity apex
  - Non-constrained → hinges anywhere, away from deformity
- Non-constrained → **subluxation** → correction problems
- Constrained → hinge exactly at **head of talus** → perfect correction

Method	Frames	Hinges	Outcome
Open	Separate; forefoot may be separate from hindfoot	—	Uncontrolled → subluxation, incomplete correction
Closed constrained	Connected	At deformity apex (head of talus)	Perfect correction
Closed non-constrained	Connected	Away from deformity	Subluxation, correction problems

- Large topic; key point → constrained vs non-constrained

Exam Pearl

- Ilizarov in foot → **distraction histiogenesis**, not osteogenesis
- Constrained → hinge at deformity apex (head of talus) → perfect correction
- Non-constrained → hinges away from apex → subluxation
- Ilizarov → stiff, long foot; triple → stiff, short foot (functionally better)

Complications of Surgery Ask AI

- After posteromedial release or foot osteotomies

Recurrence Ask AI

- Even after soft tissue release
  - Inadequate correction
  - **K-wires** not left after posteromedial release
  - Wires removed too early

Dorsal Navicular Subluxation Ask AI

- **Dorsal subluxation of navicular** → rocker-bottom deformity
- Cause → incomplete plantar and dorsal release
  - Navicular subluxes dorsally on talar head → rocker-bottom
  - **Tibionavicular ligament** not released → rocker-bottom

Valgus Overcorrection Ask AI

- Important cause → cut **talocalcaneal interosseous ligament**
  - Calcaneus translates laterally on talus → valgus deformity
- Severe pain → probably from weight bearing on **medial arch**
- Treatment → lateral column lengthening, triple arthrodesis, or triple osteotomy
  - Details → valgus foot topic
- **Iatrogenic valgus** → > 12 years → triple arthrodesis; or triple osteotomy

Triple Osteotomy Ask AI

- **Midfoot osteotomy** + calcaneal osteotomy
- Midfoot → cuboid and cuneiform, reversed direction
- Calcaneus → **translation (sliding) osteotomy**, reversed direction

Component	Varus foot correction	Triple osteotomy (valgus)
Midfoot	Cuboid wedge → into cuneiform	Medial cuneiform wedge → into cuboid
Calcaneal sliding	Posterior segment medial → lateral	Posterior segment lateral → medial

- **Triple osteotomy** ≠ triple arthrodesis

Dorsal Bunion Ask AI

- **Dorsal bunion** → prominent first metatarsal head + hallux flexus
  - **Hallux flexus** → flexion at first MTP joint
- Cause → **flexor hallucis longus** overpull
  - Pulls hallux toward ground
  - + **tibialis anterior** overpull on base of first metatarsal
- Treatment
  - Release **plantar plate** (volar/plantar capsule)
  - **Plantar closing wedge osteotomy** of first metatarsal
  - ± tibialis anterior tendon lengthening
- Occurs with overcorrection or after clubfoot operations

Calcaneus Deformity Ask AI

Complic.

- **Calcaneus deformity** Achilles over-lengthening *[C4 · transcript: "over-lengthening of tibialis anterior"]*
  - Dorsiflexion + loss of Achilles plantar flexion → walks on calcaneus
- Very difficult to treat
- **Double tendon transfer** → medial + lateral
  - Lateral → **peroneus brevis** → attached to calcaneus
  - Medial → tibialis posterior → woven or sutured into Achilles
  - → medial and lateral **augmentation** → ↑ plantar flexion
- Drawback → **drop foot**
  - Transfer + tibialis anterior lengthening (tight, overpulling first metatarsal) → dorsiflexion power lost
  - Must be **balanced**

Skin Complications Ask AI

- **Skin complications** → very common with Turco or medial release
  - Complete medial release + correction → tight medial skin lengthened medial column
- Solution → **connect the two incisions** across dorsum
  - Lateral redundant skin → brought medially → coverage without medial tension

Take-Home Message Ask AI

Must-Know

- Neglected clubfoot → no **perfection**; normal foot unachievable
- Goal → **plantigrade foot**
- Weight bearing on sole + easy shoe wear

Questions and Answers Ask AI

Ponseti First Ask AI

- Flexible foot → always start with **Ponseti**, especially young child
- Release or other procedures → recurrent, or neglected at older age
- Ponseti → up to 4 years
- **Syndromic** → primary surgery preferred from outset
  - Very stiff; very high recurrence
- **Paralytic** foot → recurs unbalanced muscles
  - Start Ponseti → follow protocol until recurrence → then soft tissue and muscle balancing

Ilizarov in Scarred Feet Ask AI

- Allowed → advantage; closed, no scalpel
- **Severe scarring** → gradual stretching of scar → improved foot shape

Casting Before Release Ask AI

- **Serial casting** + manipulation before soft tissue release
  - Full correction not reached → otherwise no release needed
- Continue until no further improvement → then release
  - Initial casting → reduces extent of release
- No fixed position or timing → e.g. 4 or 6 casts

Ponseti in Relapsed Clubfoot Ask AI

- Young relapsed child → Ponseti again
  - E.g. treated at 1–2 months → returns at 6–7 months no **Denis Browne** splint
- Any case suitable for Ponseti → start Ponseti, even if relapsed
- Release or osteotomies → older patients or **failed Ponseti**
- Repeat Ponseti once, twice or three times while responsive
  - Recurrence cause → parental **non-compliance**, not stiff or severe foot
- TN and CC K-wire duration → 6–8 weeks (question; audio unclear)

Complications of Calcaneal Varus Correction Ask AI

- Corrected by **Dwyer's** or sliding osteotomy
- Complications → very few in lecturer's experience
  - Unite well; smooth; easy operation
- Lateral skin normally **redundant** → no skin complications
- Lateral wedge closed → no foot lengthening → no heavy tension on neurovascular structures
- Dwyer's and sliding osteotomy → low-risk

Tibial Torsion and Definition of Neglected Ask AI

- **Internal tibial torsion** → never corrected before foot release
  - Foot first → then assess and correct torsion
- Neglected → no previous operations, severe
  - Age → above 1 year, even above 30
  - Clubfoot → treated immediately after birth, from first month
  - Any late presentation (after 6 months, 1 year, years) → **neglected**

Posteromedial Release Tips Ask AI

- Skin and subcutaneous tissue opened → **abductor hallucis**
- Abductor hallucis released from calcaneal attachment → neurovascular structures → plantar fascia beneath
- Neurovascular structures → enter **groove** toward lateral border
  - Scissors in groove → one blade above plantar fascia, one beneath
  - Plantar fascia + abductor digiti minimi + flexor digitorum brevis → cut as one component
  - → releases **cavus**
- Three joints
  - **Talonavicular** → dorsally, medially, plantarly
  - Calcaneocuboid → medially
  - Subtalar → **circumferentially,** medial round to lateral
- Ligaments
  - **Deltoid** → between medial malleolus and calcaneus
  - Spring (calcaneonavicular) → TN reduction
  - **Tibiotalar ligament** → usually cut with capsule
- Last medial structure → **tibialis posterior** → lengthened → TN reduction

Split versus Total Tibialis Anterior Transfer Ask AI

- **Rocker-bottom deformity** → next week's lecture
- Split → **neuromuscular** (CP) or muscle imbalance
  - Whole transfer → reverses deformity
- Idiopathic → **total transfer**
  - Tibialis anterior → balancing **dorsiflexor**; neither invertor nor evertor

	Split	Total
Indication	Neuromuscular (CP), muscle imbalance	Idiopathic clubfoot
Reason	Total transfer → reverses deformity	Balancing dorsiflexor; neither invertor nor evertor

One Stage or Two Ask AI

- Soft tissue release + osteotomy → one or two stages, by deformity
- Severe + risk of **neurovascular** or skin complications → two stages
- Mild but stiff → mild release (joints, abductor hallucis) → osteotomy same sitting
  - Provided no neurovascular effect or **skin tension**

Achilles: Formal Lengthening vs Ponseti Tenotomy Ask AI

- Formal lengthening **scarring** in Achilles (previous procedure, deformity stiffness)

	Ponseti tenotomy	Formal lengthening
Technique	Percutaneous complete cut	Complete Z-plasty
Age	< 2 years	—
Healing	Regenerates spontaneously	—
Setting	Ponseti	Posteromedial or lateral release

Z-plasty Technique Ask AI

- Tendon cut **sagittally**, lengthwise, into two halves
- Distal segment → medial side; proximal segment → lateral side
  - Cancels **medial insertion** → line of pull lateral → aids correction
- Foot → 90 degrees (neutral) or slight plantar flexion
  - Fear of **calcaneus deformity**
- Sutured in 5 degrees plantar flexion
- **Cast** → 6 weeks

Dorsal Bunion Surgery Ask AI

- **Jones procedure** → original description → flexor hallucis transfer to first metatarsal head
- Many no longer do it
  - **Plantar closing wedge osteotomy** + plantar capsule release → sufficient
  - Flexor hallucis transfer → fear of losing flexion power
  - Tibialis anterior released in same operation → flexor transfer usually unnecessary
- Lecturer's practice → plantar wedge + capsular release + tibialis anterior lengthening

Residual Metatarsus Adductus Ask AI

- Residual **metatarsus adductus** → casting first
- Recurrence or resistance → **medial soft tissue release**

Master Knot of Henry Ask AI

- Omitted earlier → part of medial release
- **Master knot of Henry** → fibrotic area at decussation of FHL and FDL
- Released in posteromedial release
- Formerly considered deforming force → medial surface tightness → not certain

Plantar Fascia, Release, Calcaneal Shortening Ask AI

- Plantar fascia → **complete release**, not 50%
- **Release** → sliding or cutting, not reattaching
- **Shortening of calcaneus** after osteotomy → small; foot size unaffected
  - Wedge ~1–1.5 cm
  - Deformed calcaneus → axis correction gains some length
- Upper age limit for Ponseti → 4 years

Preferred Approach: Norris Carroll Ask AI

- Lecturer prefers **Norris Carroll**
  - Medial scar much better than Turco posteromedial
  - **Step by step** → each structure leads to the next
    1. Abductor hallucis → neurovascular structures → plantar fascia → joints → master knot of Henry → tibialis posterior → TN joint → medial CC joint *[C5 · transcript: "tibialis anterior leads to TN joint"]*

Soft Tissue Swelling Ask AI

- **Soft tissue swelling** after release → possible complication; skin tension too
- Swelling present or expected → **under-correct** in cast
  - 2 weeks → return → complete correction

Triple Arthrodesis Techniques Ask AI

- Several techniques → differ only in simple **denudation** vs wedge resection, by severity
- Decided intraoperatively → wedge size unknown until deformity and stiffness seen

Ilizarov Rate of Distraction Ask AI

- Very large subject in clubfoot
- Like Ponseti → **hinges at head of talus** → then distract
- 1 mm per day → half morning, half night (every 12 hours)
- One side distracted; other moves **passively**, constrained
  - Rod moves along its own fixed path
  - Concave side distracted; convex side moves by itself
  - Foot follows defined path → no stress on lateral surface

Diabetic and CP Children Ask AI

- **CP** → completely different algorithm
  - Foot deformity more often valgus than varus
  - Varus in CP → different treatment
- Diabetes → **systemic disease**
  - Affected skin or wound healing → like any operation; nothing specific

Multiple Metatarsal Osteotomy Ask AI

- Residual adductus → debate
  - Capsulotomies and soft tissue release vs metatarsal osteotomy
- **Ponseti's book** → multiple metatarsal osteotomy over joint capsulotomy
  - Capsulotomy → later subluxation + joint incongruity
- Stiff foot + severe residual adductus → lecturer prefers **multiple metatarsal osteotomies**

Cause of Calcaneus Deformity Ask AI

- Cause → same as clubfoot pathology; review from the beginning
- Calcaneus deformity → dorsiflexion **Achilles over-lengthening**; plantar flexion power lost
- Tibialis posterior + peroneus brevis transfer → risk of drop foot
  - ↑ plantar flexion, ↓ dorsiflexion power

Developmental Dysplasia of the Hip: Pathology

(Anatomy, pathogenesis and natural history)

Prof Dr Hisham Abdel-Ghani

Watch Original Lecture

Lecture transcript processed with AI assistance.

Contents

↑ Book contents

Introduction and Agenda . . . 65
Terminology and the DDH Spectrum . . . 65
Spectrum Entities . . . 66
Neonatal Examination: Barlow and Ortolani . . . 66
Teratogenic Dislocation . . . 67
Incidence and Natural History . . . 67
Incidence . . . 67
Natural History of Unstable Hips . . . 68
Timing of Ultrasonography . . . 68
Case: Newborn Hips . . . 68
Treatment Threshold . . . 68
Normal Hip Formation . . . 69
Role of Movement in Joint Development . . . 69
Cartilage Model Study . . . 70
Growth of the Acetabulum . . . 70
Five Growth Centers . . . 70
Triradiate Cartilage . . . 71
Acetabular Cartilage . . . 71
Labrum . . . 71
Cartilage Disorders and Growth Mode . . . 72
Periosteum of the Pelvic Bones . . . 72
Injury to the Growth Centers . . . 72
Secondary Ossification Centers . . . 72
Summary of Acetabular Growth . . . 73
Growth of the Proximal Femur . . . 73
Pathological Anatomy of DDH . . . 74
The Neolimbus . . . 74
Growth and Development After Treatment . . . 75
Accessory Ossification Centers . . . 75
Resumption of Acetabular Growth . . . 75
Case: Closed Reduction Only . . . 76
Obstacles to Reduction . . . 76
Hourglass Capsule . . . 76
Extra-articular Obstacles . . . 76
Intra-articular Obstacles . . . 76

Take-Home Message 77

AI detected corrections (Mostly bad transcripts)

Queried - kept as spoken, but looks wrong

Q1 (p. 67) As spoken; lecture later says Barlow-positive instability is 1% (10 per 1000); 1 in 1000 is persistent dislocation.

Q2 (p. 68) As spoken; same lecture earlier gives ultrasonographic incidence as 34 per 1000, not 34%.

Q3 (p. 73) As spoken; triradiate cartilage grows interstitially, as the lecture itself said earlier (a055).

Introduction and Agenda Ask AI

- **DDH** → core topic of pediatric orthopedics; very large subject
- Two sessions
  - Session `1` → **pathology** and diagnosis
  - Session `2` → treatment, late diagnosis, **pelvic osteotomies**
  - Session `2` timing → set by Dr. Mohamed Hany
- Pathology lecture → concepts recurring throughout later lectures
- Pathology agenda
  - **Terminology** → meaning of DDH
  - Incidence of DDH
  - **Normal development** → acetabulum and femoral head
  - **Pathological anatomy** of DDH
  - **Obstacles to reduction**
  - Growth and development after treatment

Terminology and the DDH Spectrum Ask AI

- Former term: **CDH** = congenital dislocation of the hip
- "Congenital" = dating from birth → not all dislocations date from birth → term abandoned
- **DDH** = Developmental Dysplasia of the Hip; used since 1989
  - Not all cases discoverable at birth
  - Not all cases dislocated; not all congenital dislocations
  - "Developmental" → process developing over time
- DDH = **spectrum of abnormalities** → not a focal / point diagnosis

Spectrum Entities Ask AI

DDH

- **Acetabular dysplasia** → primary pathology in DDH
  - Hip dislocation without acetabular dysplasia → not DDH
  - Example: **septic arthritis** → dislocation
    - Acetabulum normal → not DDH
  - Emphasis → always on the **acetabulum**
  - Dysplastic acetabulum on X-ray / any modality, no clinical manifestation → **dysplasia**
    - = acetabular dysplasia / hip dysplasia
- **Instability** → hip in place but dislocatable
  - = positive Barlow
  - Located hip very easily dislocated → unstable hip
- **Subluxation** → partial loss of contact of articular surfaces
  - = loss of joint congruity
- **Dislocation** → total loss of contact of articular surfaces

Entity	Finding	Label
Dislocation	Secondary to established dysplasia	DDH
Subluxation	Partial loss of articular contact	DDH
Instability	Dislocatable hip	DDH
Dysplasia	Radiological picture of established dysplasia only	DDH

- Diagnosis "DDH" → specify type
  - Dislocated, subluxed, unstable, or radiological finding only

Neonatal Examination: Barlow and Ortolani Ask AI

- Neonatal dislocation → state **reducibility**

Test	Manoeuvre	Result
Barlow	Dislocate the located hip	Positive → instability (dislocatable hip)
Ortolani positive	Reduce the dislocated hip	Relocates → reducible
Ortolani negative	Reduce the dislocated hip	Does not relocate → irreducible

- **Ortolani test** → neonatal period, performed with abduction
  - Relocation or **clunk** → hip was dislocated and reduced
  - Alternatively → subluxed and reduced
- **Ortolani negative** → completely irreducible by neonatal examination
- Ortolani-negative types → very severe, very **rigid**

Teratogenic Dislocation Ask AI

- **Teratogenic dislocation** → associated with paralytic / neuromuscular disorders
  - **Myelomeningocele**
  - **Arthrogryposis**
- **Long-standing intrauterine dislocation** → closed reduction at examination very difficult
- Ortolani → always negative → cannot be reduced
- **Ortolani-negative hip** → two possibilities
  - Ordinary DDH, severe → irreducible
  - Teratogenic dislocation
- High, long-standing intrauterine dislocation → not called DDH
  - Arthrogryposis
  - Myelomeningocele
  - Other neuromuscular disorders
- Term → **teratogenic dislocation of the hip**

Exam Pearl

- Barlow → dislocates a located hip → instability
- Ortolani → reduces a dislocated hip, with abduction
- Ortolani positive → reducible; Ortolani negative → irreducible
- Ortolani negative → severe DDH or teratogenic dislocation
- Teratogenic dislocation → not called DDH

Incidence and Natural History Ask AI

Incidence Ask AI

DDH

- **Instability** incidence → 1 in 1000 *[Q1 · As spoken; lecture later says Barlow-positive instability is 1% (10 per 1000); 1 in 1000 is persistent dislocation.]*
  - Includes dislocation and dislocatable hip
- **Dislocatable hip** → born with dysplasia, not dislocation
  - Dislocated on examination
- Dislocatable hip rate in normal children → 1 in 1000
  - Child has it or progresses to dislocation
- **Ultrasonography** diagnosis → 34 per 1000
  - Detects dysplasia without dislocation → higher incidence

Natural History of Unstable Hips Ask AI

Barlow

- **Barlow's study** → Barlow-positive neonates
  - Barlow positive → 1% in neonatal period
  - Persistent dislocation → 1 in 1000
  - Untreated → 12% of the 1% remain dislocated / progress to dislocation
  - Untreated → 60% stable in first week
  - Untreated → 88% stable by second month
- Roughly 1% born with **hip instability** → 1 in every 100 children
  - Hip easily dislocated by Barlow test
  - Untreated → 10%, precisely 12% → dislocation
  - = 1 in 1000 of all children
- Remaining 999 → **spontaneous resolution**
  - Instability does not persist, does not progress to dislocation
- All of them → DDH

Timing of Ultrasonography Ask AI

US

- Ultrasonography of all neonates before first week → many DDH → splinting
  - → **Over-treatment**
- No dislocation → wait → ultrasound at 1.5–2 months → most children improved
- → defined **timing for ultrasonography** (discussed later)
- **Positive Barlow test** → dislocated until proven otherwise
  - Untreated → 12% dislocate → needs attention

Case: Newborn Hips Ask AI

Case

- Right → good **acetabular index**
- Left → dysplastic acetabular index
  - Dysplasia → more frequent on left than right
- Later, no treatment → very normal hips, spontaneously

Treatment Threshold Ask AI

Barlow

- **Acetabular dysplasia** → very large proportion of children
  - Ultrasonography → diagnosed in 34% of cases *[Q2 · As spoken; same lecture earlier gives ultrasonographic incidence as 34 per 1000, not 34%.]*
- Only 1% → **positive Barlow test** → may need treatment
- Of the 1%, untreated → 12% or 10% dislocate; rest resolve spontaneously
- 12% dislocation → **every Barlow-positive hip** needs treatment
- 33% with acetabular dysplasia on early ultrasonography → not treated
- → no very early ultrasonography → avoids **over-treatment**

High Yield

- Barlow positive → 1% of neonates; persistent dislocation → 1 in 1000
- Untreated → 12% dislocate; 60% stable at week 1, 88% at month 2
- Positive Barlow → dislocated until proven otherwise
- Every Barlow-positive hip → needs treatment
- Early neonatal ultrasonography → over-treatment

Normal Hip Formation Ask AI

- Hip formation begins → 7th week, like all joints
  - ≈ 1.5 months from missed period
- Large **cartilaginous anlage**
- → **Cleft** in mesenchyme
  - Defines acetabulum vs femoral head
- 11 weeks (near 3rd month of pregnancy) → hip formed
  - By 3rd month → nearly all structures formed
  - After 3rd month → growth and final development
- **Normal neonatal hip** → cannot be dislocated
  - Attempt → femur may fracture; hip still not dislocated
- Femoral head at birth → **deeply seated** in acetabulum
  - Even after capsule incision → difficult to displace
- Hip development → balance
  - **Genetic factor**
  - Acetabulum–femoral head relationship
- Best stimulus for acetabular growth → **well-seated, reduced hip**
- Best stimulus for femoral head → **good, containing acetabulum**
- **Dislocated hip** → neither femoral nor acetabular growth normal
- Reduction → head and acetabulum develop normally → **rule**

Role of Movement in Joint Development Ask AI

- Genetics → cleft → femoral head + acetabulum → not sufficient
- Joints need **movement** to finalize development
- No intrauterine movement → abnormal joint development
- Causes of absent **active movement**
  - **Intrauterine paralysis**, any cause
  - Limited movement → uterine abnormality
    - **Bicornuate uterus**
    - Abnormally shaped uterus
  - → incomplete joint development with limited movement
- DDH expected with **joint stiffness**
  - Arthrogryposis
  - Myelomeningocele
  - Sacral agenesis
- Arthrogryposis → **anterior horn cell** problem → intrauterine paralysis
- Mother → "baby does not move well"
  - Especially **multiparous**, with experience of fetal movement
- Absent movement → abnormal joint
  - **Adhesions** in joint
  - Synovial membrane → not formed properly
  - Capsule → adhesions
  - Ligaments → attenuated

- Final joint appearance → modulated by **intrauterine active movement**

Cartilage Model Study Ask AI

Movement

- In-vitro **cartilage model** study, 2014
  - Models placed facing each other
  - 3 models → growth only, compression, movement
- **Form follows function** → shape determined by type of movement

Model	Result
Growth only	Haphazard growth → no defined shape
Compression / static loading	Flat articular surfaces
Multi-axial motion	≈ ball-and-socket joint

- Growth only → explains abnormal articular surfaces in **paralysis**
- Cartilage shape → physically determined mainly by movement type
- **Biological effect of movement** → development of
  - Synovial cavity
  - Synovial membrane
  - Ligaments
  - Capsule
  - Joint shape
- **Arthrogryposis** → knee, ankle, hip affected; possibly also wrist, elbow
  - Imaging / surgery → abnormal articular surfaces and bony ends
- First 3 months → base formation normal
  - Cartilage, capsules, articular surfaces → as in any normal fetus
- Later stage → **active motion** needed → absent → abnormal
  - → normal intrauterine joint development fails
- Movement essential → **pre-cavitation and post-cavitation** stages

Growth of the Acetabulum Ask AI

- Femoral and acetabular growth → independent
  - Acetabulum → best stimulus: **located femoral head**
  - Femoral head → best stimulus: head deeply seated in acetabulum

Five Growth Centers Ask AI

Acetabulum

- Acetabular growth → **five centers**
  - Triradiate cartilage
  - Acetabular cartilage
  - Labrum
  - Secondary ossification centers
  - Periosteum of the pelvic bone

Triradiate Cartilage Ask AI

Triradiate

- Three bones → ilium, pubis, ischium
- Division into three → human naming only
- Actual divider → cartilage / **growth plates** between bones = triradiate cartilage
- Anatomists → upper part separate from two lower parts
  - Two lower parts → separate from each other
  - → three bones: **ilium**, pubis, ischium, with cartilage between
- **Y-shaped cartilage** → growth plate on each side
  - Ilium–ischium → plate facing ischium + plate facing ilium
  - Ischium–pubis → plate facing pubis + plate facing ischium
- Growth plate → works in **two directions**
- **Interstitial growth** → increasing depth and width of acetabulum
  - = increased acetabular diameter

Acetabular Cartilage Ask AI

Acetabulum

- Pediatric joint cartilage layers
  - Articular → **hyaline cartilage**
  - Beneath: mature → bone directly
  - Beneath: child → **epiphyseal cartilage**
  - Beneath epiphyseal cartilage → **ossific nucleus**
    - Keeps enlarging; probably by enchondral ossification
- Acetabulum → same: cartilage → epiphyseal cartilage → growth plate
- X-ray analogy → femur, **capital femoral epiphysis**, epiphyseal cartilage, hyaline articular cartilage
  - Epiphyseal cartilage → between articular cartilage and ossific nucleus
  - Growth plate present → ossific nucleus grows until maturity
- Maturity → epiphyseal cartilage **replaced by bone**
  - Only hyaline cartilage remains → covers bone, forms joint
- Thick epiphyseal cartilage → growth plate beneath → **labrum** at its end

Labrum Ask AI

Labrum

- Labrum → acts as **distal-end growth plate**
  - Growth plate beneath labrum, on epiphyseal cartilage
  - Growth beneath labrum → increased **acetabular depth**
- Labrum → very important structure for acetabular growth
- Removing / cutting labrum for reduction → **labral growth** stops
  - → decreased acetabular depth

Cartilage Disorders and Growth Mode Ask AI

Acetabulum

- Large part of acetabular growth → cartilage-dependent
- Any **abnormal cartilage formation** disease, any type
  - → abnormal acetabulum + abnormal femoral head
- Ossific nucleus abnormality + cartilage / articular cartilage abnormality
  - Usually thought of → long bones only
  - Also → **abnormally shaped acetabulum**
  - Not only abnormal ends of long bones (distal / proximal)
- Acetabular cartilage growth
  - **Interstitial growth** → width; beneath labrum
  - **Appositional growth** → under perichondrium

Periosteum of the Pelvic Bones Ask AI

Periosteum

- Acetabulum grows → ilium, pubis, ischium must grow with it
- **Flat bones** → growth via subperiosteal new bone formation
- Good acetabular growth requires
  - Acetabular cartilage
  - Labrum
  - Triradiate cartilage
  - **Periosteum** of pelvic bones
- Periosteum needed → acetabulum enlarges → bones must enlarge

Injury to the Growth Centers Ask AI

Acetabulum

- Severe periosteal injury → abnormal acetabular growth → **acetabular dysplasia**
- **Labral injury** → affected acetabular growth
- **Septic hip** → damage to articular cartilage + underlying physeal plate
  - → abnormal acetabular growth
- Infection → **triradiate cartilage** damage → same result
- Infection → labral damage → same result
- **Careless surgery** → labrum removed
  - → abnormal acetabular growth

Secondary Ossification Centers Ask AI

Acetabulum

- **Extra centers** → exist in all; not visible in all / not always on X-ray
  - **Os acetabuli** → pubis
  - **Acetabular epiphysis** → ilium; also anterior inferior iliac spine
  - **Os marginalis** → ischium
- Sometimes visible on X-ray
- Appear in 2–3% of **normal hips**
- Appear **near maturity** → after 11–12 years
  - Never in young children
- Assist growth → like any **long-bone** ossification center

Summary of Acetabular Growth Ask AI

Acetabulum

- Acetabular growth depends on
  - **Triradiate cartilage** → appositional growth *[Q3 · As spoken; triradiate cartilage grows interstitially, as the lecture itself said earlier (a055).]*
  - Acetabular cartilage → normal interstitial + appositional growth
    - Acts like **physeal cartilage** of any long bone
  - Secondary ossification centers
  - Periosteal new bone formation
  - **Labral growth**

Must-Know

- Five acetabular growth centers → memorize
- Triradiate cartilage → Y-shaped; growth in depth and width
- Acetabular cartilage → interstitial + appositional growth
- Labrum → growth plate beneath → acetabular depth
- Secondary ossification centers + pelvic periosteum

Growth of the Proximal Femur Ask AI

- Acetabulum best stimulus → **deeply seated, located, mobile femoral head**
- Proximal femur components
  - **Capital femoral epiphysis** → ossifies at 4–7 months
  - **Greater trochanteric epiphysis**
- Neonatal / early childhood → linked by **physeal cartilage**
  - Capital femoral epiphysis ↔ trochanter
  - → femoral neck growth in width = **neck isthmus**
- **Hyperemia** → may increase growth in head and neck
  - → very wide neck
- **Femoral growth** depends on
  - Pressure and forces
  - Normal circulation
  - Normal muscle tone
  - Normal movement
  - Normal development of the baby
- Delayed sitting / walking / crawling → **abnormal proximal femur shape**
  - → possibly abnormal acetabular shape
- Any **paralytic problem** → same

Pathological Anatomy of DDH Ask AI

Feature	Normal hip	DDH hip
Femoral head	Deeply seated inside acetabulum	Not seated inside acetabulum
Acetabular cavity	Good, very deep; not superficial	Shallow; not concave
Cartilage	Thick; concave; no ridges	Ridge inside acetabulum
Labrum	Covers femoral head	Opened out; not covering femoral head

The Neolimbus Ask AI

Neolimbus

- Ridge inside acetabulum → **neolimbus**
- Form follows function → no pressure → **haphazard growth** of cartilage
  - → cavity not concave → ridge remains
- At surgery → not seen
  - Seen → shallow acetabulum with apparent superior edge
- Named by **Ortolani**
- Head reduced → head pressure on acetabulum → **neolimbus disappears**
  - → normal concave acetabulum
- Not seen intraoperatively → needs **cut through acetabulum**
- Ortolani test → head slides over neolimbus → **false impression of reduction**
  - → only a click
- Ortolani test → **clunk, not click**
  - Click source → sometimes the neolimbus
- Cartilage → not visible; after reduction → head feels **inferiorly seated**
  - Cause → abnormal bulge = neolimbus
  - With time → **remodels** → head returns to normal position

Pitfall

- Neolimbus → false impression of reduction in Ortolani test
- Ortolani → clunk, not click
- After reduction → head inferiorly seated → remodels with time

Growth and Development After Treatment Ask AI

Accessory Ossification Centers Ask AI

DDH

- **Accessory centers** → distinct from normal secondary ossification centers
  - Appear in 60% of hips
  - Appear **after treatment**
  - Indicate acetabular growth
- Misread by beginners → "irregular articular surface, **arthritis** starting"
  - Child → fully mobile, painless
- Actual → **accessory ossification centers;** very nice appearance
- More common after surgery or **closed reduction**
- Absent → child without acetabular dysplasia
- Present in 60% of cases
- Irregularity → happy news → acetabulum **still active,** growing
  - → **deepening,** good development
- Seen structure → **bone underneath** the articular surface, not articular surface

Pitfall

- Accessory centers → irregular look, not arthritis / osteoarthritis
- Seen structure → bone underneath the articular surface
- Accessory centers → good sign → acetabulum still growing

Resumption of Acetabular Growth Ask AI

Acetabulum

- **Acetabulum** → potential for recovery
  - → resumption of growth for many years after maintained concentric reduction
- After open or closed reduction → acetabulum grows, remodels, changes shape
- Timing of acetabular growth cessation → **no agreement**
- Usually, probably after age 5 → **acetabular depth** not improved → probably will not complete development
  - Specific parameters → discussed in diagnosis
- **Accessory ossification centers** → good news → acetabulum likely to grow
- Resumption of acetabular development depends on
  - **Well-seated femoral head** → most important
    - Earlier reduction → better acetabular growth → **better results**
    - **Concentric reduction** → best stimulus
  - **Active mobility** and normal stresses
    - Normal tone + normal development → acetabulum improves → normal hip shape
    - **Paralytic problem** → even good / high tone, e.g. CP
      - Delayed normal development → abnormal hip loading
      - → abnormal development persists → **recurrent dislocation** possible
  - **Normal growth potential**
    - Labrum removed + periosteum stripped → **centers of growth** lost
    - → no acetabular growth to expect

Case: Closed Reduction Only Ask AI

Case

- **Dislocated hip** + very dysplastic acetabulum
  - Contralateral → normal hip, deep acetabulum
- **Closed reduction** only; no pelvic osteotomies
- Over time → acetabulum improves; initially very shallow
- At 13 years, after maturity → **normal hip**
- No remaining sign of acetabular dysplasia

Obstacles to Reduction Ask AI

- Apply to **closed and open reduction**

Hourglass Capsule Ask AI

Obstacles

- Dislocated head + **iliopsoas tendon** inserted into lesser trochanter
  - → constriction ring → head cannot descend
- Capsule → **hourglass appearance**
  - Upper balloon → contains head; large, stretched
  - Middle → waist
  - Lower balloon → shrunk, adherent to acetabulum = **inferomedial capsule**
- Inferomedial capsule → must be widened / opened
  - Otherwise → adherent capsule pushes head out again
- **Transverse acetabular ligament** → at end of inferomedial capsule → must be opened
- Iliopsoas tendon present → no reduction → must be **released**

Extra-articular Obstacles Ask AI

Obstacles

- **Adductor longus tightness**
  - Head restricted high → femur proximally migrated → short adductors
  - → reduction blocked → lengthening
  - DDH → lengthening of adductor longus only
- **Iliopsoas tendon** → prevents head descent
- **Inferomedial capsule** → adherent to acetabulum → prevents reduction

Intra-articular Obstacles Ask AI

Obstacles

- **Hypertrophied ligamentum teres**
  - Huge → fills acetabulum → pushes head out
  - Well-seated reduction → requires **excision**
- **Transverse acetabular ligament**
  - Constant ligamentum teres pull, especially **walking child** → proximally convex
  - → fills this part → prevents reduction
- **Inferomedial capsule** → prevents reduction
- **Hypertrophied / inverted labrum**
  - Natural pathology → labrum **everted**, not inverted
  - Inversion cause → repeated **trials of closed reduction**, casting
  - Head not seated → labrum pushed repeatedly → inverted
  - Inverted into acetabulum → obstacle to reduction
  - **Inverted labrum** → always iatrogenic; not natural pathology

- **Pulvinar** → fibrofatty tissue inside acetabulum
  - Very scanty → no effect
  - Very thick fat lobule → blocks reduction; likely needs **removal**

Extra-articular	Intra-articular
Adductor longus tightness	Hypertrophied ligamentum teres
Iliopsoas tendon	Transverse acetabular ligament
Inferomedial capsule	Inferomedial capsule
—	Hypertrophied / inverted labrum
—	Pulvinar

High Yield
- Iliopsoas → constriction ring → hourglass capsule
- Inferomedial capsule → adherent → pushes head out
- DDH → adductor longus lengthening only
- Inverted labrum → always iatrogenic (closed reduction trials)
- Hypertrophied ligamentum teres → excision for reduction

Take-Home Message Ask AI

- Acetabulum → best stimulus: **concentrically reduced femoral head**
- Femoral head → best stimulus: **located femoral head**
  - Plus normal stresses, development, tone, hip movement
- **Early reduction** → resumption of acetabular growth
- Lecture aim → understand where growth comes from
- Surgery / reduction → **respect the biology**
  - Respect labrum
  - Respect periosteum
  - Avoid **over-dissection**
  - All → growth plates for normal development
- Reduction only performed → later course depends on **biology** and growth
- → respect the **growth centers**

Clinical Pearl
- Concentric reduction → best stimulus for acetabular growth
- Earlier reduction → better results
- Labrum excision / periosteal stripping → acetabular growth lost
- Avoid over-dissection → respect the growth centers

Developmental Dysplasia of the Hip: Diagnosis

(Clinical and imaging diagnosis across age groups)

Prof Dr Hisham Abdel-Ghani

Watch Original Lecture

Lecture transcript processed with AI assistance.

Contents

↑ Book contents

Risk Factors for DDH . . . 82
Postnatal Factor: Swaddling . . . 82
Neonatal Examination: Barlow and Ortolani . . . 83
The DDH Iceberg and Ultrasonographic Screening . . . 83
The Iceberg of DDH . . . 83
Why Ultrasonography . . . 84
Universal versus Selective Screening . . . 84
Timing of Screening . . . 85
Clinical Presentation of the Missed Dislocated Hip . . . 85
Unilateral versus Bilateral . . . 85
Residual Acetabular Dysplasia . . . 86
Ultrasonography . . . 86
Role and Timing . . . 86
Ultrasound Anatomy and Angles . . . 87
Graf Classification . . . 87
Radiographic Assessment . . . 88
X-ray Lines . . . 88
Tönnis Classification . . . 88
Acetabular Index . . . 88
Center-Edge Angle . . . 89
DDH Spectrum and Post-Septic Dislocation . . . 89
Screening Policy . . . 89
The DDH Spectrum . . . 89
Post-Septic Hip Dislocation . . . 89
Diagnostic Pitfalls . . . 90
Congenital Femoral Deficiency . . . 90
Femoral Anteversion Mimicking Dislocation . . . 90
Early Ultrasonography: Ossific Nucleus and Screening Timing . . . 90
Delayed Ossific Nucleus . . . 90
Ultrasound Parameters Before 6 Months . . . 90
Screening and the Barlow–Ortolani Tests . . . 91
Hip Dislocation in Cerebral Palsy versus DDH . . . 91
Frog Lateral View, MRI and CT . . . 92
Frog Lateral View . . . 92
MRI and CT . . . 92

The 1-Year-Old with Delayed Ossific Nucleus 92
Signs the Head Is Located 93
Barlow and Ortolani Beyond the Neonate 93
After 3 Months 93
Repeated Testing 93

AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

- **C1** (p. 82) “legs tied together in abduction” → “hips adducted”
- **C2** (p. 82) “legs tied together in abduction” → “hips adducted”
- **C3** (p. 89) “CEA must exceed 15 degrees” → “CEA must exceed 25°”
- **C4** (p. 89) “CEA must exceed 15 degrees” → “more than 25°”

Risk Factors for DDH Ask AI

- Session focus → **diagnosis of DDH**
- Many risk factors; most important → **first born**
  - Highest incidence in the first-born baby
  - Many explanations; none change practice
    - Tighter uterus → **limited movement**
    - Limited movement → abnormal acetabular growth and development → joint pathology
- **Female** sex → more affected
  - Many factors; among them **hyperlaxity** → risk factor for DDH
- **Breech with knee extension** — not any breech
  - Knee extended → severe **hamstring** stretch
  - Hamstrings span ischium → knee; knee extension → further ↑ tension
  - Hamstring stretch → hip stress → femoral head **dislocatable**
- **Oligohydramnios** (term uncertain in recording) → abnormal hip stresses → dislocatable
- **Family history**
  - Baby with DDH → next sibling also at risk
- Anything reducing space and movement around the baby
  - **Torticollis** or foot deformities → markers of overcrowding
  - Limited movement → abnormal hip position, ↓ active hip movement → DDH

Postnatal Factor: Swaddling Ask AI

Swaddling

- Normally born baby with **unstable hip** = positive Barlow
  - Positive Barlow → 1% of babies
  - Left alone → 88% improve; 12% → dislocation
  - **Swaddling** → lowers the 88%, pushes dislocation above 12%
- Swaddling → community habit aiming for a tall child with straight legs
  - Both legs tied together → hips **adducted** and extended *[C1 · transcript: "legs tied together in abduction"]*
  - Extension → further hamstring stretch
  - Mild acetabular dysplasia or positive Barlow → **completely dislocated hip**
  - Swaddling → forbidden
- Protective practices → hips **abducted**
  - Baby nursed and sleeping on the belly
  - Mother carries baby on belly or back, legs spread around her
  - These communities → **low incidence** of hip dislocation vs swaddling communities
- Swaddling formerly common in Egypt; practised in Saudi Arabia, many countries → higher dislocation incidence

High Yield

- Most important risk factor → first born
- Breech with knee extension → hamstring stretch → dislocatable hip
- Positive Barlow → 88% improve, 12% dislocate
- Swaddling → hips adducted and extended → dislocation; forbidden *[C2 · transcript: "legs tied together in abduction"]*

Neonatal Examination: Barlow and Ortolani Ask AI

- Every **newborn** → hip examination, whatever the complaint
  - Not examining the hips → mistake
  - Applies equally to orthopaedic surgeon and paediatrician
- Two key tests → **Barlow and Ortolani**

	Barlow	Ortolani
Starting hip	Located (reduced)	Dislocated
Manoeuvre	Attempt to dislocate	Attempt to reduce
Old name	"Click of exit"	"Click of entry"
Positive	Clunk → hip dislocates	Hip reduces
Negative	—	Hip does not reduce → Ortolani negative
Indicates	Dislocatable	Dislocated

- Forget "click" → it is a **clunk**
- Both tests → **instability**
  - "Instability" → ask: Barlow or Ortolani? dislocatable or dislocated?
- Ortolani → reduction succeeds = positive; fails = negative
- Clunk (model demonstration) → not a click or snap
  - Felt as **translation** → a step
  - Head outside → brought back → returns to place
  - Must actually feel the head returning and dislocating
- Click alone → **means nothing**

Exam Pearl

- Barlow → dislocatable hip; Ortolani → dislocated hip
- Clunk = translation of the head; click = nothing
- Instability → always specify Barlow or Ortolani

The DDH Iceberg and Ultrasonographic Screening Ask AI

The Iceberg of DDH Ask AI

Iceberg

- Clinically detected
  - Dislocation → 1 per 1000
  - Instability → 1 per 100
  - Not looked for → not found
- **Acetabular dysplasia** → common; behaves like an iceberg
  - Surface → dislocation
  - Below → instability; untreated → 12% progress → the 1 per 1000
  - Base → dysplasia seen **radiologically** only; no clinical manifestation

Why Ultrasonography Ask AI

Screening

- **Ultrasonographic screening** in neonates → detects what clinical exam cannot
  - Needs a radiological method
  - X-ray for all babies → radiation → no; ultrasonography instead
- Technique → **very easy to learn**
- Advantages
  - Assesses **cartilage** → three quarters of region; invisible on X-ray
  - **Dynamic examination** → head comes out or not → instability

Universal versus Selective Screening Ask AI

Screening

- Two types worldwide
  - **Universal** → all babies, regardless of Barlow or Ortolani
  - **Selective**
- Screening → babies with **no clinical manifestation**
  - Normal hip exam → neither Ortolani nor Barlow
  - Positive Barlow or Ortolani, or visibly dislocated → **clinically diagnosed** → not screening
  - Screening → non-clinically diagnosed cases only
- Universal drawback → **over-treatment**
  - 88% even of positive Barlow improve spontaneously; more so for dysplasia alone
  - Ultrasound for all babies → 34% acetabular dysplasia vs real problem 1 per 1000
  - Small wealthy countries afford it → non-surgical treatment instead of late surgery
- **Selective** → babies with risk factors
  - Positive family history
  - Breech presentation
  - Foot deformity or torticollis → overcrowding or **intrauterine over-moulding**
- **Sign of instability** → not screening
  - Barlow or Ortolani → imaged straight away; investigate early, act early

	Universal	Selective
Who	All babies	Risk factors only
Indications	—	Family history; breech; foot deformity or torticollis
Drawback	34% dysplasia vs 1 per 1000 → over-treatment	—
Users	Small wealthy countries	High-risk patients

Timing of Screening Ask AI

Screening

- No instability sign → when to screen?
  - 88%, even positive Barlow, improve by 2 months
- Ideal screening → **6–8 weeks**
  - Reassurance without early treatment → avoids over-treatment

High Yield

- Ideal screening age → 6–8 weeks
- Universal ultrasound → 34% dysplasia → over-treatment
- Instability sign → immediate imaging, not screening
- Ultrasound → sees cartilage, dynamic

Clinical Presentation of the Missed Dislocated Hip Ask AI

- Persistent undetected dislocation → examination findings
  - **Limitation of abduction** (left dislocated side in example)
  - Lower limb shortening → specifically **femoral**
- **Galeazzi test** → demonstrates femoral shortening
  - Supine, feet side by side → one side lower
  - Femur or knee displaced posteriorly → shortening femoral, not tibial
- Before walking → noticed by mother or paediatrician
  - Shortening
  - **Asymmetric folds** → gluteal, thigh or labial
- After walking → **limp**
  - Most children diagnosed at this stage

Unilateral versus Bilateral Ask AI

Gait

- **Unilateral** → easy diagnosis
  - Shortening; possible toe-walking
  - Excessive knee flexion on the right (example) → compensates **leg length discrepancy**
  - → classic appearance of DDH
- **Bilateral** → sometimes harder
  - **Waddling gait**
  - Hyperlordosis
  - Bilateral limitation of abduction
  - Waddling child → not **calcium deficiency**
  - Lower limbs equal → no discrepancy
  - Abduction symmetric but limited
  - Hip **laxity** → unstable

Pitfall

- Bilateral DDH → waddling, equal limbs, symmetric limited abduction
- Waddling child → think DDH, not calcium deficiency

Residual Acetabular Dysplasia Ask AI

- **Acetabular dysplasia** may present after normal life
  - Or after DDH surgery and treatment with normal walking
- At maturity or adolescence → **residual acetabular dysplasia**
  - Family reassured after surgery → complaints years later
  - Age ~11–12 years; girls after 10
- **Secondary acetabular dysplasia** → problem recurs; head migrates outward
- Complaints
  - **Abductor fatigue**
  - Activity-related **groin pain** or labral pain
  - Many varied complaints
- Child or adolescent with such complaints → acetabular dysplasia may be discovered
  - With or without **history of DDH treatment**

Ultrasonography Ask AI

Role and Timing Ask AI

US

- **Unstable hip** (positive Ortolani or Barlow) → ultrasonography first
- Useful in the first **6 months**; after 6 months → no sense relying on it
  - Ultrasound to diagnose DDH at 1–2 years → wrong; ultrasonography has its time
- Purpose → avoid repeated X-ray exposure
- Ultrasound follow-up
  - Acetabular development
  - **Pavlik harness** management (possibly other closed treatment)
- Older than 6 months or 1 year → not the **primary diagnostic tool**

Ultrasound Anatomy and Angles Ask AI

US

- **Standard coronal plane** → most used; equivalent to AP hip X-ray
- Structures
  - **Iliac bone** → white line
  - Bony roof of acetabulum
  - Capsule
  - Femoral head
- Alpha and beta angles → opposite behaviour

	Alpha angle	Beta angle
Lines	Iliac bone line + bony roof line	Along the capsule
Larger	Deeper acetabulum	Head more outside
Smaller	Shallower acetabulum	Head more inside → better
Emphasis	Focus on it	Leave aside

- **Alpha angle** → normal more than 60°
  - Less than 60° → acetabular dysplasia
- Ultrasound for all newborns → 34% alpha below 60° → reason not done early

Graf Classification Ask AI

Graf

- Classic alpha-angle classification → **Graf**

Alpha angle	Graf type	Meaning
More than 60°	—	Normal
50°–60°	Type 2	Good stage → conservative treatment
Less than 50° (or 49°)	Type 2c	Bad type

- Below 50° (or 47°) → conservative success falls very sharply
  - **Pavlik harness** success falls very sharply → treatment lecture
- Same degree, subtypes **type 2a, 2a+, 2a–** → age-related
  - Older child, same degree → worse
  - Younger child, same degree → better → hip improves with growth and development

High Yield

- Ultrasound → useful in first 6 months only
- Alpha angle → normal more than 60°
- Graf 50°–60° → type 2 → conservative treatment
- Graf type 2c (below 50°) → Pavlik success falls sharply
- Same angle, older child → worse

Radiographic Assessment Ask AI

X-ray Lines Ask AI

X-ray

- X-ray → standard measurements
- First line → **Hilgenreiner's line**
  - Joins the triradiate cartilages, one each side
- Triradiate cartilages not visible → line along **inferior border of pubic rami**
  - Parallel line drawn → parallel to Hilgenreiner's line
- **Perkins' line** → longitudinal, along lateral edge of acetabulum
  - Drawn perpendicular to Hilgenreiner's line
- Two lines → four quadrants
  - Well-located head → **inferomedial quadrant**
- **Shenton's line** → must be intact
  - Inferior edge of neck → continuous with inferior border of pubic ramus
  - Interrupted → head dislocated
- Head in superior outer quadrant → **fully dislocated**

Tönnis Classification Ask AI

Tönnis

- **Descriptive** → communication; little value in management planning
- Purpose → degree of dislocation
- Based on the four quadrants

Grade	Femoral head position
1	Inferomedial quadrant
2	Inferolateral quadrant
3	Superolateral quadrant
4	Very high in superolateral quadrant

Acetabular Index Ask AI

AI

- **Acetabular index** → very important; first thing to look at
  - Hilgenreiner's line + line from triradiate cartilage → outer edge of acetabulum
  - Smaller → deeper acetabulum, better-inclined roof
  - Larger → shallow (superficial) acetabulum
- Remember 20 → normal ≤ 20° from age 2 years; above → abnormal
- Before 2 years → higher values accepted

Age	Accepted acetabular index
Newborn	27.5°
First 6 months	23.5°
From 2 years	≤ 20°

Center-Edge Angle Ask AI

CEA

- **Center-edge angle**
  - Vertical line through centre of femoral head
  - Second line from same point → outer edge of acetabulum
- Remember 25 → beyond 13–14 years, CEA must exceed 25° *[C3 · transcript: "CEA must exceed 15 degrees"]*
  - Less → head **not well covered**
  - Much more → **over-coverage** of femoral head
- Hip → **deep joint**, unlike knee, elbow, hand → actively look for DDH in neonate

Must-Know

- Located head → inferomedial quadrant
- Interrupted Shenton's line → dislocated head
- Acetabular index ≤ 20° from 2 years
- Newborn 27.5°; first 6 months 23.5°
- CEA beyond 13–14 years → more than 25° *[C4 · transcript: "CEA must exceed 15 degrees"]*

DDH Spectrum and Post-Septic Dislocation Ask AI

Screening Policy Ask AI

Screening

- Worldwide trend → **ultrasonographic screening**
  - Universal or selective; selective → high-risk patients
- **Egypt** → selective screening + National Program for Universal Screening
- Critics → universal screening **over-diagnoses** self-resolving cases → over-treatment
- Neonate presenting for any reason → hip examination mandatory

The DDH Spectrum Ask AI

Spectrum

- **DDH = spectrum**
  - Unstable → positive Barlow, positive Ortolani
  - Dislocated → reducible or **irreducible** (negative Ortolani)
  - Acetabular dysplasia → best diagnosed by radiology
- Common to all → **dysplastic acetabulum**

Post-Septic Hip Dislocation Ask AI

Septic

- Dislocation from **septic arthritis**, not DDH
  - Early → acetabulum normal
- Later → acetabulum becomes dysplastic
  - Sepsis injures cartilage and **labrum** → abnormal acetabular growth and development
  - Hip left dislocated, unreduced → acetabulum does not grow normally
- Best acetabular growth stimulus → **well-seated femoral head**
- Calling it DDH → **acetabular dysplasia** required
- Ultrasonography → high-risk patients

Diagnostic Pitfalls Ask AI

Congenital Femoral Deficiency Ask AI

PFFD

- Looks like DDH → shallow, near-absent acetabulum; high femoral head
- Clue → **subtrochanteric varus**
  - → **congenital femoral deficiency,** not DDH
  - Former name → proximal focal femoral deficiency (PFFD)

Femoral Anteversion Mimicking Dislocation Ask AI

Anteversion

- Referred case → quadrants drawn
  - **Hilgenreiner's and Perkins' lines** drawn
  - Left → head in inner inferior quadrant; located; Shenton's line intact
  - Right → head in **inferior outer quadrant** → called dislocation; surgery scheduled
- Right side findings
  - Acetabular dysplasia → **acetabular index**; left roof much better than right
  - **Shenton's line** → interrupted
  - Prominent **lesser trochanter** → marked anteversion; head faces forward, not toward acetabulum
- Imaged in **internal rotation** → perfectly located; no treatment needed
- Pitfall → **femoral anteversion** misdiagnosed as dislocation

Pitfall

- Subtrochanteric varus → congenital femoral deficiency, not DDH
- Prominent lesser trochanter → anteversion → apparent dislocation
- Internal rotation view → head located

Early Ultrasonography: Ossific Nucleus and Screening Timing Ask AI

Delayed Ossific Nucleus Ask AI

Q&A

- Question → delayed appearance of femoral head ossification centre
- Normal ossification of capital femoral epiphysis → starts at 4–6 months
- Dislocated side → always **delayed ossification centre**
  - Abnormal loading → affects dislocated hip
- Best femoral head growth stimulus → **location inside acetabulum**
  - → head shape change and delayed ossification → expected

Ultrasound Parameters Before 6 Months Ask AI

US

- Before 6 months → **ultrasonography** preferred
  - Unossified head → invisible on X-ray, seen on ultrasound
  - Shows how much of the head is inside the acetabulum
- Parameters beyond the alpha angle
  - **Dynamic examination** → head comes out or not
  - **Head coverage** → half inside better; more than half outside → treatment
  - **Alpha angle** → among most important; largely decides intervention vs waiting

Screening and the Barlow–Ortolani Tests Ask AI

Q&A

- Screening reduces importance of Barlow and Ortolani? → **No**
- Universal screening → not first week; at ~6 weeks
- Early ultrasound → only positive **Barlow or Ortolani**
- Clinically normal babies → wait until 6 weeks
  - Self-correcting cases resolve → only persistent acetabular dysplasia diagnosed
- Ultrasound for all in first week → 34 of every 100 babies in **Pavlik harness** → done nowhere

Hip Dislocation in Cerebral Palsy versus DDH Ask AI

	DDH	CP spastic dislocation
At birth	Acetabular dysplasia	Normal hip
Acetabulum	Dysplastic from birth	Abnormal development secondary to spastic pathology
Treatment	Different	Completely different

- Different pathology → may look alike when both dislocated
- CP patient born with DDH → possible
  - Born with DDH → later **brain ischaemia** → CP
- CP + **early dislocation** → think DDH with CP on top
  - Among the most difficult patients
  - Prognosis of surgical intervention → very poor
- **Bilateral DDH** in CP with non-walker prognosis → contraindication to hip reduction
  - Only contraindication to reduction in CP hip dislocation

Clinical Pearl

- DDH → born with dysplasia; CP → born with normal hip
- CP + early dislocation → DDH with CP on top
- Bilateral DDH, CP, non-walker → only contraindication to reduction

Frog Lateral View, MRI and CT Ask AI

Frog Lateral View Ask AI

Frog

- Lateral X-ray in internal rotation → no such view
  - Standard → **AP and frog lateral**
- Late-presenting cases (older children) → other views
  - Help **planning of osteotomy** → pelvic osteotomy type
- Frog lateral → **role in diagnosis**
  - Apparently dislocated head → frog lateral → inside acetabulum
  - Head still high → **pure dislocation**
- Difficult cases → **arthrography** (later)
- Never rely on a single view
  - Earlier anteversion case → frog lateral → head deeply seated
  - Head high on frog lateral → dislocated, not anteversion with **apparent dislocation**
- Young-age DDH → **AP and frog lateral** both
  - Frog lateral → may show hip not dislocated, severe anteversion

Exam Pearl

- Young-age DDH → AP and frog lateral, never AP alone
- Head high on frog lateral → true dislocation
- Head seated on frog lateral → anteversion, apparent dislocation

MRI and CT Ask AI

MRI

- Question → role of MRI or CT, besides arthrogram
- **Late-developing acetabular dysplasia** → different planning and thinking
  - Child with pain or **Trendelenburg** at late age
  - With treated DDH history or none
  - X-rays → standing views; **false profile view**
- Pain with suspected **labral tear** → MR arthrography
  - Plain MRI → not indicative
- MRI otherwise mostly not informative → except **AVN** assessment
- Cartilaginous femoral head seen only by → ultrasonography or **MRI**
  - Follow-up after **closed reduction** → MRI; patient in cast → ultrasound difficult
- Each modality → discussed in its time; MRI role → with closed reduction

The 1-Year-Old with Delayed Ossific Nucleus Ask AI

- Question → 1-year-old with delayed ossific nucleus; rely on **inner upper margin of proximal femoral metaphysis**?
- Possibly → but absent capital femoral epiphysis at 1 year → very unusual
- Suspect an unknown **septic hip** → dig into history
  - Incubator (NICU) admission
  - After discharge → hard crying at nappy change; difficult changing; continuous crying
  - Swelling of thigh and hip region

- May prove **post-septic**, not DDH
- Metabolically normal child, no rickets → **Shenton's line** → very good indicator
- Assess acetabulum shape and **acetabular index**
  - At 1 year → not above 23–27° at most
  - Higher → significant acetabular dysplasia

Signs the Head Is Located Ask AI

Located

- **Internal rotation** → Shenton's line continuous with metaphysis → no problem
- Well-developed **teardrop** → head not dislocated
- Good ossification of **sourcil** (acetabular roof) → dense white line
  - → good loading → head inside acetabulum
- Osteopenic roof without the line → head not inside acetabulum

Exam Pearl

- Capital femoral epiphysis absent at 1 year → suspect post-septic hip
- Acetabular index at 1 year → 23–27° at most
- Dense sourcil line → head loaded inside acetabulum

Barlow and Ortolani Beyond the Neonate Ask AI

After 3 Months Ask AI

Q&A

- Question → role of Barlow or Ortolani after 3 months?
  - Very difficult → Barlow-positive hip has **normalised** or dislocated
  - Most cases → hip hard to dislocate or relocate
- Even at 1–2 years → **clunk** may be felt on examination
  - Clunk ≠ reduction
  - Worst mistake → declaring the hip reduced
  - May only slide over the **neolimbus**, not into the acetabulum

Repeated Testing Ask AI

Q&A

- Repeated testing → may **harm** the hip
  - Positive finding once → enough; no repeating for the parents
- Test → **painful** for the baby (Barlow or relocation)
  - Baby cries, tightens legs → repeat test may become impossible
- End of part one → pathology and diagnosis

Pitfall

- Clunk at 1–2 years ≠ reduction → may be neolimbus
- Barlow or Ortolani after 3 months → very difficult
- Repeated testing → harmful and painful; do it once

Developmental Dysplasia of the Hip: Treatment

(Age-based treatment from harness to osteotomy)

Prof Dr Hisham Abdel-Ghani

Watch Original Lecture

Lecture transcript processed with AI assistance.

Contents

↑ Book contents

Principles and Natural History of DDH . . . 99
Why the Principles Matter . . . 99
Natural History . . . 99
Indications, Aims and Age-Based Strategy . . . 100
Indication for Treatment in the Neonate . . . 100
Aim of Treatment . . . 100
Treatment by Age at Presentation . . . 100
Pavlik Method: Under 6 Months . . . 101
Indications . . . 101
Mechanism . . . 101
Types of Splints . . . 101
Correct Application . . . 101
Inducing Reduction . . . 102
Contraindications . . . 103
Duration and Failure . . . 103
Success and Failure . . . 104
Femoral Nerve Palsy . . . 104
Summary . . . 105
Closed Reduction . . . 105
Rationale . . . 105
Correct Technique . . . 105
Arthrography: Case Example . . . 106
Steps of Correct Closed Reduction . . . 106
Post-Reduction Imaging and Casting . . . 107
Outcome . . . 107
Pre-Reduction Traction . . . 108
Problems . . . 108
Open Reduction: Medial Approach . . . 109
Approaches for Open Reduction . . . 109
Technique . . . 110
Limitations and Indication . . . 110
Open Reduction: Anterior Approach, Capsulorrhaphy and Dega Osteotomy . . . 111
Approach . . . 111
Capsulotomy and Obstacles . . . 112
Capsulorrhaphy . . . 113

Dega Osteotomy Technique . . . 113
Stability Testing After Opening . . . 113
Graft Technique in Young Children . . . 114
Closure and Hip Spica . . . 114
Case Examples and Results . . . 115
Preferred Strategy for 6–12 Months . . . 116
Femoral Osteotomy in DDH . . . 116
Intraoperative Decision . . . 116
Amount of Shortening . . . 117
No Varus Osteotomy in DDH . . . 117
No K-Wires to Hold Reduction . . . 117
Open Versus Closed Reduction . . . 118
Comparison . . . 118
Complications After Reduction . . . 119
General Complications . . . 119
Post-Operative Care: No Physiotherapy . . . 119
Residual Acetabular Dysplasia . . . 120
Late-Developing Acetabular Dysplasia . . . 120
Proximal Femoral Growth Disturbance . . . 121
Kalamchi Classification . . . 121
Summary of DDH Treatment . . . 122
Late Acetabular Dysplasia: Indications and Imaging . . . 123
Indications for Pelvic Osteotomy . . . 123
Clinical Presentation . . . 123
Prerequisites and Radiological Indications . . . 123
Imaging Array . . . 124
AP View . . . 124
Standing AP View . . . 125
Frog Lateral View . . . 125
Abduction–Internal Rotation View . . . 125
False Profile View . . . 125
3D CT . . . 125
Aim of Pelvic Osteotomy . . . 126
Pelvic Osteotomy: Principles and Types . . . 126
Classification . . . 126
Basic Principles . . . 126
Fragment Mobility: Salter (Innominate) Osteotomy . . . 126
Double Pelvic Osteotomy . . . 127

Triple Pelvic Osteotomy . . . 127
Pemberton Osteotomy . . . 128
Dega Osteotomy . . . 128
Dega Case Examples . . . 128
Triple Osteotomy Case . . . 129
Shelf Osteotomies . . . 130
Fragment Attachments: Summary . . . 130
Ganz and Bernese Osteotomies . . . 131
Freeing the Fragment Completely . . . 131
Ganz (Bernese Periacetabular) Osteotomy . . . 131
Ganz Technique . . . 132
Ganz Case Example . . . 133
Salvage Use of the Ganz . . . 133
Paralytic Hips . . . 133
Summary of Pelvic Osteotomies . . . 134
Questions and Answers . . . 134
Rotation in the Cast . . . 134
Closed Reduction After Pavlik Failure . . . 134
Pavlik Follow-Up . . . 135
Neonatal Diagnosis and Screening . . . 135
Bilateral Cases: Timing . . . 135
Upper Age Limit for Open Reduction . . . 136
Capsule Suturing . . . 136
Consensus: Open Reduction + Pelvic Osteotomy ± Femoral Shortening . . . 136
K-Wire Fixation After Closed Reduction . . . 136
Waiting for the Ossific Nucleus . . . 137
Post-Cast Stiffness . . . 137
Inverted Labrum . . . 137

AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

- **C1** (p. 102) “posterior strap prevents abduction” → “posterior strap prevents adduction”
- **C2** (p. 102) “Pavlik blocks abduction; abductor stretch” → “adduction blocked; adductor stretch”
- **C3** (p. 119) “bilateral cases move faster” → “unilateral cases move faster”
- **C4** (p. 135) “two 2-month casts = about three months” → “about four months in casts”

Queried - kept as spoken, but looks wrong

- **Q1** (p. 104) As spoken; femoral nerve palsy with Pavlik is reported in roughly 2-3%, not 70%.
- **Q2** (p. 121) As spoken; Kalamchi-MacEwen type IV is total head and physeal damage, not medial physis/coxa vara.

Principles and Natural History of DDH Ask AI

Why the Principles Matter Ask AI

- **Treatment of DDH** → frequently examined; essential in practice
  - Non-operators, general orthopaedics → must know principles → correct patient information
- Knowing everything ≠ performing everything → perform only what you master
  - Worst situation → contradictory opinions from two doctors
    - Close opinions on one operative step → acceptable
    - An opinion and its opposite → never both acceptable
  - Extremes → confused patient; patient excused
- Understanding **principles** → correct information when asked

Natural History Ask AI

DDH

- **DDH spectrum** → acetabular dysplasia → subluxation → dislocation
  - Dislocation → fixed or reducible
  - i.e. **Ortolani negative** vs Ortolani positive
- Neonatal **acetabular dysplasia** → possible courses
  - Spontaneous reduction and development
  - Persistent dysplasia
  - Persistent subluxation
  - Persistent dislocation
- Most dangerous course → **subluxation**
  - Reduced but dysplastic hip → not a major problem
  - Fully dislocated hip → may last long without pain or arthritis
  - Subluxation → very important **predictor of osteoarthritis**
- Subluxation → **prevents acetabular development,** even with growth potential
  - Shearing stresses on labrum; labrum = growth centre
  - Labral shear → **arrest and inhibition** of acetabular growth, not stimulation
- **Concentrically reduced hip** → most important stimulus for acetabular growth
  - Not subluxated, not partially reduced → no labral shear
  - → normal acetabular growth + normal labral function

Indications, Aims and Age-Based Strategy Ask AI

Indication for Treatment in the Neonate Ask AI

- **Unstable hip** (Barlow positive, dislocatable) → 12% persistent dislocation
  - 88% improve within first year → specifically first two months
- 12% risk → any neonatal unstable hip treated as dislocation → **intervention**
  - Dislocated, or dislocatable (subluxatable)
  - i.e. **positive Barlow**
- Ultrasound-screened dysplasia → treated accordingly; separate issue
  - Here → **clinical** criterion: any unstable hip, dislocated or dislocatable
- = **indication for treatment** → untreated Barlow positive → 12% persistent dislocation
  - Treat 100% for the sake of 12%

Aim of Treatment Ask AI

- **Concentric reduction** → aim in dislocation, subluxation and dysplasia alike
  - Not partial, not approximate
- Not achieved → nothing achieved → hip will not develop normally
- Obtained **as early as possible**, then maintained → normal development of proximal femur and acetabulum
- Reduction changes no anatomy; concentric reduction + **muscle balancing**
  - + growth → femur stimulates acetabulum; acetabulum stimulates femur
- Younger age → greater **remodelling potential** of acetabulum and femoral head → longer time available
- **Fastest development** → first ~3 years of life
  - Second rapid phase → at maturity
    - First 3 years → reduction enhances acetabular development and growth
    - At maturity → previously fine hip may develop **late acetabular dysplasia** → remember

Treatment by Age at Presentation Ask AI

- Age of presentation / diagnosis → three groups
  - Less than 6 months
    - Neonate; precisely **infant younger than 6 months**
  - 6 to 18 months → pre-walking to early walking
  - More than 18 months → after walking
- Under 6 months → **harness** or brace
- Over 18 months (walking age)
  - → **open reduction**
- 6–18 months → **problematic group** → open vs closed reduction; pros and cons of each

Age at presentation	Stage	Treatment
< 6 months	Infant	Harness / brace (Pavlik method)
6–18 months	Pre-walking to early walking	Closed or open reduction
> 18 months	Walking	Open reduction

Pavlik Method: Under 6 Months Ask AI

Indications Ask AI

Pavlik

- Birth to 6 months
  - Ortolani positive, Ortolani negative, **Barlow positive** (subluxatable)
  - Reduced hip with dysplasia only → treat if still present after 6 weeks–2 months
  - After 2 months → 88% already improved
- Not improved by 1.5–2 months → treat → **acetabular development**
- **Pavlik** → most famous method
  - Pavlik method ≠ Pavlik harness only → any brace with **same mechanism**
  - Hence "Pavlik method"

Mechanism Ask AI

Pavlik

- Pathology addressed
  - Dislocated hip in flexion → **displaces posteriorly**
  - Limited abduction
  - **Hamstring shortening** → causes dislocation, prevents reduction
- Brace → knee flexion
  - → **relaxes hamstrings**
  - + abduction → reduces hip
  - + hip flexion
- **Movement allowed** → abduction, adduction, flexion, extension
  - Limited → adduction
  - Limited → **extension**
  - → maintains hip position
- Uses → dysplasia, subluxation, **dislocation**
- Irreducible (Ortolani-negative) hip → applied as is
  - No prior reduction → **effective under 6 months**
  - Beyond 6 months → no paper showing efficacy

Types of Splints Ask AI

Pavlik

- Several splint types
  - **Von Rosen splint** → fixed position, like a cast; no movement
  - Tübingen brace → similar to Pavlik; some movement; widely used, especially some Arab countries
  - **Pavlik harness** → classic; still top in non-invasive / non-surgical DDH management

Splint	Movement	Note
Von Rosen splint	None — fixed, like a cast	Clear contrast with Pavlik
Tübingen brace	Some movement	Widely used, some Arab countries
Pavlik harness	Controlled movement	Classic; still at the top

Correct Application Ask AI

Pavlik

- Know what you apply and its goal
- **Chest strap** → at nipple line, not on abdomen

  - Often wrongly placed at abdominal level
  - Lower → less lower-limb control
  - Higher → better control
- All tension control → at knee, not foot → **proximal tibial point**
  - Foot piece + straps
- Two straps
  - Medial strap → anterior
  - Lateral strap → posterior
- **Anterior strap** attachment
  - At anterior axillary line, not more anterior
  - Purely anterior → flexion only, without abduction
  - Correct → flexion + abduction
  - Both straps together → **flexion with abduction**
- **Posterior strap** → prevents extension and adduction *[C1 · transcript: "posterior strap prevents abduction"]*
- Harnesses in Egypt → **strap clips** too anterior → ineffective
  - Fix → re-attach at anterior axillary line on the chest strap → true flexion with abduction
- Manufacturers recently corrected → clips moved laterally
  - → anterior axillary line → **effective hip positioning**

Component	Position	Action
Chest strap	Nipple line	Lower-limb control
Anterior (medial) strap	Anterior axillary line	Flexion + abduction
Posterior (lateral) strap	Tension at proximal tibia	Prevents extension + adduction

Pitfall

- Chest strap on abdomen → lost lower-limb control
- Anterior strap too anterior → flexion only, no abduction
- Local harness clips too anterior → re-attach at anterior axillary line
- Tension at knee (proximal tibia), not foot

Inducing Reduction Ask AI

Pavlik

- Pavlik → may **induce reduction** in dislocated hip → no prior reduction
- Parents instructed → child mostly
  - **Supine**
- Adduction blocked → repeated gradual abduction → **adductor stretch** → hip reduces *[C2 · transcript: "Pavlik blocks abduction; abductor stretch"]*
- Reduction during relaxed sleep → child **wakes crying** → warn parents
  - Startles at each reduction attempt → until reduced
- Then → gradual acetabular and femoral head growth → **normal hip development**

Contraindications Ask AI

Pavlik

- Contraindicated in
  - **Muscle imbalance**
  - Ligamentous laxity
  - Joint stiffness
- Examples → **arthrogryposis**, myelomeningocele, paralytic conditions
- **Severe lower-limb deformities** → no Pavlik

Duration and Failure Ask AI

Pavlik

- Inducing reduction in **complete dislocation** → maximum 3 weeks
- No reduction by 3 weeks → **stop Pavlik**
  - Abduction + failed reduction
  - → femoral head deformity, **avascular necrosis**, major problems
- Rule → no reduction within 2–3 weeks → Pavlik failed → stop
- Left longer → **Pavlik harness disease**
  - Femoral head damage
  - Acetabular cartilage injury
  - Head pushed against acetabulum → **labrum crushed** → impaired acetabular growth
- Inducing reduction → trial 2–3 weeks; beyond → stop

Success and Failure Ask AI

Pavlik

- Success by hip type
  - **Acetabular dysplasia only,** head reduced → very high
  - Dislocatable (Barlow positive) → excellent → head reducible and reduced
  - Dislocated → Ortolani positive > Ortolani negative (irreducible on examination)
- **High failure** → completely dislocated hips
- Failed Pavlik → then open / closed reduction → higher **proximal femoral growth disturbance**
  - Due to Pavlik use?
  - Or **severe case** from start → failure reflects severity → problems expected after open reduction
  - → unresolved question
- **Failure factors**
  - Start after 3–4 months → use
    - Very early
  - Males > females
  - Bilateral > unilateral
  - **Alpha angle** < 47° → higher failure; > 47° or > 50° → higher success
  - Dislocated non-reducible
    - (**Ortolani-negative**) → 60% failure
  - Reduced dislocatable hip → 40% failure

Hip	Pavlik outcome
Acetabular dysplasia only, head reduced	Very high success
Dislocatable (Barlow positive)	Excellent success
Dislocated, Ortolani positive	Better than Ortolani negative
Reduced dislocatable hip	40% failure
Dislocated non-reducible (Ortolani negative)	60% failure

Femoral Nerve Palsy Ask AI

Pavlik

- **Femoral nerve palsy** → approximately 70% (figure uncertain); ↑ with dislocation severity *[Q1 · As spoken; femoral nerve palsy with Pavlik is reported in roughly 2-3%, not 70%.]*
  - Attributed to flexion
  - True palsy disputed
  - Flexion > 90° → child cannot move hip → interpreted as palsy
- Sign → **no knee extension** in flexion position
  - Release harness → ~1 week → full knee extension recovers → reapply
- More common with
  - **Heavier patient,** higher BMI

Summary Ask AI

Pavlik

- Place → **first 6 months** of life; no role afterwards
- Earlier → better; more reducible → better
- No reduction within first 2–3 weeks → stop
- Position → hip flexion > 90°, **abduction** > 60°

High Yield

- Pavlik only in first 6 months; no role after
- Inducing reduction → max 2–3 weeks, then stop
- Longer → Pavlik harness disease: head damage, crushed labrum
- Position → flexion > 90°, abduction > 60°
- Failure → Ortolani negative 60%, alpha < 47°, males, bilateral, late start

Closed Reduction Ask AI

Rationale Ask AI

Closed

- 6 months–2 years → 7 months, 8 months, 1 year, 2 years
  - Just walking; no major femoral head changes
  - Options → **closed** or open reduction
- Closed first → avoid surgery → less invasive reduction
  - Same logic as Pavlik → **non-invasive**, high success → no need for more invasion
- **Theme** → minimise surgical intervention
  - + low morbidity → no wound, no operation
  - Keep goal in mind → does closed reduction achieve it?
- Common misconception → "Abduct
  - → click → cast → X-ray → reduced"
  - = **forced / haphazard reduction**, not closed reduction

Correct Technique Ask AI

Closed

- Count trips under
  - **Anaesthesia** → goal = reduce anaesthetic rate, avoid theatre
- Step 1 → **examination under anaesthesia (EUA)** → is head reducible?
  - Patient and muscles relaxed
  - Not reducible → closed reduction impossible
- Step 2 → **arthrogram** → assess reduction → mandatory
- **Neolimbus** → false sense of reduction
  - Head glides over acetabular edge
  - Or over neolimbus
  - Clunk or click ≠ reduced → head still out, dislocated

Arthrography: Case Example Ask AI

Closed

- Left dislocation → well-formed **pseudo-acetabulum**; head articulates with it
  - Wide acetabulum → reduced, reducible or not? confusing
- Arthrogram → outlines **true head shape**
  - Plain film → ossific nucleus only; arthrogram → true size, normal shape
- Dynamic → dislocation vs reduction → out → **medial dye pooling**
- Wide acetabulum → head appears in, not truly reduced
  - → moves around inside
- **Arthrography data** → true head shape, true acetabular shape, reduced or not
  - Apparent reduction + medial pooling → not reduced → obstacle
  - e.g. large **ligamentum teres**
  - Click felt, head not deeply seated, not concentric
- Basics → aim = **concentric reduction** of femoral head

Steps of Correct Closed Reduction Ask AI

Closed

- EUA
- Arthrogram
- Abduction < 60° (Pavlik position), or difficult reduction → + **adductor tenotomy**
  - Closed reduction + adductor tenotomy → still closed reduction
- **Perfect reduction** only
  - No "acceptable", no "better than before surgery"
  - Perfect or nothing
  - No partial reduction → subluxation worse than leaving dislocated

Must-Know

- Click or clunk ≠ reduction → neolimbus deceives
- Medial dye pooling → not reduced → obstacle (e.g. ligamentum teres)
- Perfect reduction or nothing
- Subluxation worse than leaving dislocated

Post-Reduction Imaging and Casting Ask AI

Closed

- After reduction + cast → X-ray
  - Not informative → head may be **subluxed posteriorly**
- Reputable centres → **post-reduction MRI** mandatory → shows cartilage
  - Otherwise difficult; if reduced → arthrography
  - Arthrography → largely dynamic examination
- Success, MRI confirms → **cast change** at 6 weeks → another general anaesthetic ± repeat arthrogram
- Total cast → not less than 3 months
  - Remove at 3 months → examine → lax, unstable → re-cast 3–4 months
- **Sequence**
  - One theatre trip → arthrography
  - Reduction ± adductor tenotomy
  - Cast → MRI
  - Satisfactory → 3 months cast
  - Removal → possible further 3–4 months (up to 6)
- Total → 3–6 months in cast

Outcome Ask AI

Closed

- High rate of **later operative management**
  - Relies on natural history → head correctly in → continuous acetabular development
  - Development continues to ~5–6 years
  - Dysplasia after 5–6 years → most likely surgery
- Very high incidence of **residual acetabular dysplasia** management after reduction
- Repeated anaesthetics + cast changes
  - + long casting → still dysplasia → surgery
  - → where is **decreased morbidity**, reduced anaesthesia rate?

Pre-Reduction Traction Ask AI

Closed

- ↑ success, ↓ complications → **pre-reduction traction** = most important factor
- Needs paediatric **traction equipment** → not every centre; must be equipped
- Good-result centres → pre-operative pre-reduction traction
- Gradual → longitudinal → **overhead with abduction**
  - Longitudinal alone
    - Almost abandoned → high complications; success 57–58%
  - **Overhead traction** with gradual abduction
    - Success 74–92%
    - Very low avascular necrosis, late growth disturbance
- 6–18 months closed reduction → feasible with correct **equipment**
  - + overhead abduction traction + correct closed reduction
- Incorrect ("heard a clunk", "a bit better") → complications
  - High **avascular necrosis**
  - High proximal femoral growth disturbance
  - High late acetabular dysplasia
- Correct traction → success ↑ greatly; but **cost** may far exceed open reduction
  - Traction + hospital stay + closed reduction + MRI → cost > open reduction

Traction	Success	Complications
Longitudinal alone	57–58%	High; almost abandoned
Overhead with gradual abduction	74–92%	Very low AVN, late growth disturbance

Problems Ask AI

Closed

- **Early and late failures**
  - Early → redislocation; or still subluxatable after 3 months cast
  - 5–15% dislocated at **post-reduction imaging**
    - Despite arthrogram, reduction, reassurance
    - Cast + waking + movement → enough to redislocate
- Next
  - **Open reduction**; then open vs closed for 8 months–2 years

Open Reduction: Medial Approach Ask AI

Approaches for Open Reduction Ask AI

- Two approaches → medial, anterolateral
- **Medial approach** → open reduction only
- **Anterolateral approach** → open reduction
  - + capsulorrhaphy + pelvic osteotomy → multipurpose
- Medial → no **pelvic osteotomy**, no capsulorrhaphy
  - Opens inferior tight capsule → left wide
  - Superior part to tighten → unreachable medially
- Additional procedure
  - **Femoral shortening and derotation osteotomy**
  - No varus osteotomy

	Medial	Anterolateral
Open reduction	Yes	Yes
Capsulorrhaphy	No	Yes
Pelvic osteotomy	No	Yes
Capsule	Inferior tight part opened	Inferior opened; superior tightened

Technique Ask AI

Medial

- Quick, little bleeding, very easy
- **Transverse incision** over adductor longus
  - Longitudinal incision for adductor tenotomy → pity for the scar
  - Transverse → along skin lines → cosmetic
  - Longitudinal → perpendicular to skin tension lines → poor scar
  - Never longitudinal; transverse → full exposure
- **Adductor longus** → divided completely
- Then **pectineus** + anterior nerve branch (likely obturator); beneath → adductor brevis
- Intervals
  - Pectineus / adductor brevis
  - Posterior to adductor brevis (brevis / **adductor magnus**) → preferred → directly onto acetabulum
- Lesser trochanter + **iliopsoas tendon** → divide
  - One of most important obstacles → no reduction without division
- **Medial circumflex femoral artery** division here
  - Main proximal femoral blood supply
  - Injury → avascular necrosis of capital femoral epiphysis
- Adductor magnus retracted posteriorly, brevis anteriorly → onto capsule
- Capsule incised → **transverse acetabular ligament** → divided
  - Opens narrowest capsule part → blocks head reduction
  - Superior part untouched → capsulorrhaphy (plication) site
- Head seen → reduced ± **ligamentum teres** division
  - = open reduction only → obstacles removed; no pelvic osteotomy
- Relies on natural acetabular remodelling and growth
  - Lax capsule → high incidence of **subluxation**
  - Acetabular development failure → future pelvic osteotomy, as with closed reduction
- **Modification** (Dennis Wenger, San Diego, main advocate)
  - Ligamentum teres kept on head → shortened → sutured to transverse acetabular ligament → prevents subluxation

Limitations and Indication Ask AI

Medial

- No pelvic osteotomy → high rate of **later pelvic osteotomy**
  - No capsulorrhaphy
  - No pelvic osteotomy
  - **Very high dislocation** (walking for some time, no comfortable laxity) → head cannot be brought down
    - → no reduction medially
  - Risk to medial circumflex femoral artery
    - → **avascular necrosis** of femoral head
- **Indication** → pre-walking young children
  - ~6–8 months to 1 year, up to 1.5 years / 18 months (possibly 2 years)
- High dislocation in walking child, or pelvic osteotomy needed → medial approach very difficult

Open Reduction: Anterior Approach, Capsulorrhaphy and Dega Osteotomy Ask AI

Approach Ask AI

Anterior

- Preferred, most used worldwide
  - Open reduction via **Smith-Petersen approach**
  - Classic Smith-Petersen incision → bad, ugly scar → not used
  - Skin incision modification
  - → **bikini incision**
- Step 1 → **adductor tenotomy**
  - Adductor longus only
  - Cerebral palsy → adductor longus + gracilis
  - Novice → open tenotomy
  - Experienced → **percutaneous adductor tenotomy** → blade in, lifted upward
  - → divides adductor longus
- Step 2 → **incision**
  - Head to screen right, legs left → incision along iliac crest
  - Assistant tensions skin toward himself → released → scar below crest, not on bony prominence → not painful
  - Along iliac crest → distally to approximately mid-inguinal point
- Entry → interval between **tensor fasciae latae** and sartorius
  - Option 1 → interval found distally → then iliac apophysis split (senior professor's preference)
  - Option 2 (from Catterall) → identify **external oblique** → split apophysis → guides into plane automatically
- Apophysis split + elevated → automatically between sartorius and tensor fasciae latae
- **Periosteal elevator** → always directed medially and inferiorly
  - Purely transverse → enters, injures sacroiliac joint
  - Aim → greater sciatic notch (inferior)
- **Packing** after periosteal elevation → two benefits
  - Haemostasis
  - Periosteal dissection
  - Automatic; long exposure once apophysis opened
- In plane → **lateral cutaneous nerve of the thigh** appears
- **Iliopsoas tendon** → release (tenotomy) → tendinous part only, not muscular
  - Assistant "rolling"
  - Tendon medial → muscle rolled until tendon delivered, lifted on angled instrument, divided
  - Very crucial → one of most important obstacles to reduction
- **Rectus femoris** at anterior inferior iliac spine (AIIS) → divided
  - Personally not reattached; professor reattaches
- Rectus divided →
  - **Hip capsule** exposed

Capsulotomy and Obstacles Ask AI

Anterior

- Capsule opened **parallel to superior pubic ramus**
  - Diathermy → superficial part only
  - Completed with scalpel → protects femoral head
- Capsule open → **ligamentum teres** (not head) seen
  - Opening extended as distally as possible
  - Divided by scalpel → left attached to transverse acetabular ligament
  - = guide to **transverse acetabular ligament** → held with Kocher clamp → followed → ligament divided
- After ligament division → **finger** into acetabulum → passes into depth
  - Finger does not pass → inferior capsule not opened, ligament not divided → unstable reduction → head comes out
- **Pulvinar** removed; sometimes trivial → left
- Acetabulum cleaned → reduction **smooth**, at ease, no tension
  - Not forced
- Transverse ligament revisited → **finger manoeuvre** → keep going behind until ligament opens
  - Definite sound and sensation as ligament opens
  - Video → no narration; scissors sound only
- Reduction → easy, **without force**
  - Reduction under tension, with difficulty = avascular necrosis

Clinical Pearl

- Iliopsoas tendon → key obstacle → always divide
- Finger must pass into acetabular depth → else ligament intact → unstable reduction
- Forced reduction under tension = avascular necrosis
- Periosteal elevator → medially and inferiorly → spares sacroiliac joint

Capsulorrhaphy Ask AI

Anterior

- Capsule = **ring** → cannot be advanced → must be opened
- Capsule cleaned from attached muscles
- Advancement blocked at two points
- **Transverse incision** → diathermy, then scalpel
  - Complete cut → capsule opens → advances
  - Advancement clearly visible
  - Before opening → no take-up, no tightening, no push
- **Capsulorrhaphy** = inferomedial advancement of capsule
  - Ring abuts AIIS → opened into "V" embracing AIIS
- Repeat of steps
  - Capsule cleaned; clamp → cannot advance
  - Incision in superior part → **parallel to superior neck**
  - Capsule → **V-shape** embracing AIIS → advances
  - Redundancy pulled forward → sutured to superior pubic ramus
- Sutures → `3`, Prolene `1` or PDS `1`
  - **Non-absorbable** or delayed-absorbable
  - Superior pubic ramus → capsule
- **Stay sutures** left untied → capsulorrhaphy completed later
  - Head and acetabulum visible → begin pelvic osteotomy
- Pelvic osteotomy in **all cases** → primary pathology = acetabular dysplasia

Dega Osteotomy Technique Ask AI

Dega

- **Dega osteotomy** → current worldwide standard
- Hohmann retractors → **greater sciatic notch** + other side
  - Osteotomy with head inside
- Cut → between ASIS and AIIS, slightly toward AIIS
  - **Oblique** at approximately
  - `30°` → not reaching greater sciatic notch
- After cut → lever
  - → **osteoclasis** of posterior column; column attached, uncut
  - Salter → complete iliac osteotomy

Stability Testing After Opening Ask AI

Dega

- Head reduced → hip extended → may come out
  - External rotation → comes out
  - Wrong conclusion → needs internal rotation → **derotation osteotomy** → done in all such cases
- Correct → assess stability with **osteotomy opened**, before capsulorrhaphy
  - Osteotome in → opened as if pelvic osteotomy done → test
  - Result → no derotation, no shortening → head stable after pelvic osteotomy
- Step spares **unneeded femoral osteotomy** → seen very often

Graft Technique in Young Children Ask AI

Dega

- Younger child → harder osteotomy
  - Iliac graft keeps osteotomy open; child's ilium small, thin
- **Sandwich graft** → iliac piece cut into two halves
  - Placed transversely, not on edge → no slipping
  - Held doubled → impacted
  - **No fixation** → no K-wires
- Standard → open reduction + capsulorrhaphy + **Dega osteotomy**
- Common belief → delay to ≥ 1.5 years for strong bone
  - Actually → **Dega feasible** from 11–12 months
  - Especially with sandwich graft, transverse not longitudinal
  - Longitudinal → needs wide bone → older child → forced wait, or open reduction only
  - Open reduction only → later pelvic osteotomy if poor **acetabular remodelling**
- Open reduction + capsulorrhaphy + pelvic osteotomy
  - → best success, least **later pelvic osteotomy**

Closure and Hip Spica Ask AI

Spica

- Closure in layers; **percutaneous tenotomy** → small stitch
- **Hip spica**
  - Flexion 90° or slightly more
  - Abduction 60°
  - Perineum widely open → hygiene, cleaning
- Cast → 2 months, not 3
  - **Worldwide trend** toward 2 months
  - English colleague convinced; local recommendation 3 months → liability
- Clean cast → mother brings
  - **Opsite** adhesive film
  - Stuck on cast and skin → no stool or urine under cast
  - + cleanliness, frequent nappy changes

Case Examples and Results Ask AI

Dega

- Dislocated hip + pelvic osteotomy → **transverse graft**, two layers, folded iliac crest
  - Acetabular index → markedly improved
- Another → dislocated, reduced, osteotomy, graft
- Old case → **open reduction only**; never done alone nowadays
- Open reduction alone → did well
  - Many do well → but higher **later pelvic osteotomy** need if poor acetabular development
- Current literature → **open reduction + pelvic osteotomy**
  - Highest good radiological and clinical results, fewest complications
  - Very low re-operation for residual acetabular dysplasia
- **Re-dislocation rate**
  - Open reduction + pelvic osteotomy → ~2% → 98% success
  - Open reduction alone → 5.5%
- ~3.5% fewer re-dislocations + less re-operation → **no justification** for open reduction alone
  - Already open → iliac bone
  - In view → osteotomy < 5 extra minutes
  - Best chance → avoid another operation, avoid re-dislocation
- Only difficulty → child < 1 year or 11 months
  - Very **soft bone**, thin iliac crest → graft placement hard

Procedure	Re-dislocation	Re-operation for residual dysplasia
Open reduction + pelvic osteotomy	~2% (98% success)	Very low
Open reduction alone	5.5%	Higher

Preferred Strategy for 6–12 Months Ask AI

- Presentation > 6 months, or **failed Pavlik** stopped
  - No closed-reduction centre
  - Tools, traction lacking → wait to 11–12 months
  - **Single procedure** → open reduction, capsulorrhaphy, osteotomy
  - Best success, least complications, least re-operation
- No need to operate at 6–7 or 8 months
  - Wait a couple of months
  - Tell parents → other options → lower results than open reduction + **pelvic osteotomy** at 11–12 months
- Case → very nice reduction; **acetabular index** improved, perhaps better than other side (original dysplasia)
- Another → acetabular index greatly improved
- Further cases → acetabular development; capsulorrhaphy case with ~10-year follow-up

High Yield

- Standard → open reduction + capsulorrhaphy + Dega osteotomy
- Dega feasible from 11–12 months → transverse sandwich graft, no fixation
- Re-dislocation → ~2% with osteotomy vs 5.5% without
- Spica → 90° flexion, 60° abduction, perineum open, 2 months

Femoral Osteotomy in DDH Ask AI

Intraoperative Decision Ask AI

Femoral

- **Femoral osteotomy** → intraoperative decision
  - Planned open reduction + capsulorrhaphy + Dega → plate and screws ready
  - Cannot reduce, or reduction under severe tension → femoral osteotomy
- Primary aim → **release tension** → shortening, nothing else
- No preoperative promise of femoral osteotomy to parents
  - "Needs derotation"
  - → reduce + open pelvic osteotomy → head **stable** → no derotation
- Beware **excessive derotation** → iatrogenic posterior dislocation
  - Even with shortening + derotation → derotate little
- After derotation → head still tends to dislocate **anteriorly** → pelvic osteotomy stabilises
- Head deep and stable after derotation → = **posterior dislocation**
  - Hidden, deep, tucked posteriorly, never coming out = posteriorly dislocated head
- Difficult reduction → primary aim **shortening**; very small derotation allowed
- **No proximal femoral varus osteotomy** in DDH → no coxa valga

Amount of Shortening Ask AI

Femoral

- Sequence → plate + two screws → osteotomy → reduce
  - **Fragment overlap** after reduction = shortening amount
- Resected bone → **graft** for pelvic osteotomy → better than iliac graft
- Avoid **excessive rotation**
- Case → 6-year-old, very deep **pseudo-acetabulum**
  - Head brought down; femoral bone → pelvic osteotomy graft
  - Femoral osteotomy with untouched pelvis → unacceptable → primary pathology = **acetabular dysplasia**
  - Very good acetabular development; small 6-year-old from Yemen

No Varus Osteotomy in DDH Ask AI

- Apparent **coxa valga** on dysplasia film
  - "Needs varus osteotomy" → wrong → check internal rotation
- External rotation film → anteverted head, visible **lesser trochanter** → neck seen end-on
- **Internal rotation** → neck in profile → normal neck-shaft angle
- No valgus in DDH
- Tragedy → DDH with femoral varus, shortening, derotation, no **pelvic osteotomy**
- Case → open reduction + capsulorrhaphy, no pelvic osteotomy, dysplastic acetabulum, **varus osteotomy**
  - → functional coxa vara
  - → disturbed hip tensions and muscle action
- Open reduction alone with normal **muscle tone**
  - → acetabular development possible; after varus → none
- **Coxa vara** → abnormal hip stress → acetabular dysplasia
  - Lack of abductor tension, tone and function

> **Pitfall**
> - No coxa valga in DDH → internal-rotation view → normal neck-shaft angle
> - Prominent lesser trochanter → anteversion imaged in external rotation
> - Varus osteotomy → functional coxa vara → no acetabular development
> - Excessive derotation → iatrogenic posterior dislocation
> - Femoral osteotomy aim → shortening only

No K-Wires to Hold Reduction Ask AI

- **K-wires** to hold reduction → something wrong
- End of procedure → head reduced, leg neutral on bed, no manipulation
  - Manipulation needed → something wrong → **revise steps**
  - Do not force and insert a wire
- 26–27 years in paediatric orthopaedics → wire used once, **revision case**
  - Correct technique → never needed

Open Versus Closed Reduction Ask AI

Comparison Ask AI

OR vs CR

- More than 1.5–2 years → **open reduction** automatically
- 6 months–1.5 years → open vs closed
- Closed → **early and late failures**
  - Aim → reduce surgery; yet 5–15% dislocated on post-reduction imaging
- Open → re-dislocation 5.5% alone; 2% with pelvic osteotomy (98% success)
- Closed reduction aim → less surgery, less **anaesthesia exposure** → achieved?
- After successful closed reduction → future **pelvic osteotomy** in 66–85%
  - Despite 3 months cast
  - + possible 3 more + MRI + cast changes under anaesthesia
- Re-operation or anaesthesia rate truly reduced? → **no**
- What really reduces it
  - Preoperative **flexion–abduction traction** + closed reduction → comparable to open
  - Not in all centres; needs equipment; extra cost; may exceed open reduction cost
- Open → **single operative intervention**; no post-reduction traction; no repeated procedures
  - Highest revision ~19%
  - → without pelvic osteotomy
  - With pelvic osteotomy → later pelvic osteotomy need very low
- Closed → repeated **general anaesthesia**
  - Long post-operative casting
  - Traction needed
  - 5–15% dislocation
  - Second surgical procedure 66–85%
  - → open reduction preferred
- Delay to ~1 year for open reduction + osteotomy → **no problem**
  - If Pavlik failed, or presentation after Pavlik age (6 months)
- Personal → no closed reduction → wait to 1 year → open reduction + capsulorrhaphy + **Dega** → best chance
- Closed → 66–85% later **pelvic osteotomy**
- Open reduction + pelvic osteotomy → advantages
  - Lower **revision** → vs 5–19% (open alone)
  - And vs 66–85% (closed)
  - Lower avascular necrosis
  - Only 2 months cast
- → worldwide trend **away from closed reduction**
  - Some centres with good traction equipment → very good success

	Closed reduction	Open reduction + pelvic osteotomy
Procedures	Repeated, under GA	Single
Traction	Needed for good results	None
Cast	3–6 months	2 months

	Closed reduction	Open reduction + pelvic osteotomy
Early failure	5–15% dislocated on post-reduction imaging	Re-dislocation ~2%
Later pelvic osteotomy	66–85%	Very low
Avascular necrosis	Higher	Lower
Cost	May exceed open	Lower

Exam Pearl

- Closed reduction → 66–85% need later pelvic osteotomy
- 5–15% dislocated on post-reduction imaging
- Open alone → revision up to 19%; with pelvic osteotomy very low
- Overhead flexion–abduction traction → closed comparable to open

Complications After Reduction Ask AI

General Complications Ask AI

- Any DDH reduction, closed or open
  - **Recurrent dislocation**
    - Very low → open reduction + capsulorrhaphy + Dega
    - Very high → closed reduction
    - High → open reduction without pelvic osteotomy
  - **Residual acetabular dysplasia** → hip reduced, dysplasia remains
    - Very high → closed, or open without pelvic osteotomy
  - **Proximal femoral growth disturbance** → head shape, ossific nucleus shape change
    - Formerly "avascular necrosis" → not AVN
- Causes of recurrent dislocation → technical error (stressed repeatedly)
- Immediate post-op redislocation → **no immediate reoperation**
  - Wait until hip mobilises → don't work on a stiff hip

Post-Operative Care: No Physiotherapy Ask AI

- After cast removal (open reduction) → **physiotherapy forbidden**
  - Child left alone → self-mobilises
- Hip → stiff in **abduction** → adduction range returns spontaneously
- Bilateral → slower
  - Unilateral → faster *[C3 · transcript: "bilateral cases move faster"]*
- Physiotherapy / aggressive mobilisation ~2 months after removal
  - Bone very **osteopenic**, fragile
  - Adduction exercises → subtrochanteric fracture
  - → never physiotherapy

Residual Acetabular Dysplasia Ask AI

Residual

- More after **closed reduction** and open reduction without pelvic osteotomy
  - Very low after open reduction + pelvic osteotomy
- Follow yearly → 1, 2, 3, 4 years
  - Improving → **acetabular index** decreasing → observe
- Beyond 5–6 years → most likely no correction
  - "Most likely" → **no cut-off point** → keep following
  - With subluxation → certainly no correction
- Concentrically reduced, no subluxation → may correct
- **Plateau** → intervene → pelvic osteotomy

Late-Developing Acetabular Dysplasia Ask AI

- Operated at 1–1.5 years → normal hip, walking excellently
- **Adolescent growth spurt** → dysplasia reappears after good development
- Good remodelling → first 3 years (high growth); at maturity → dysplasia may recur
  - Names → **adolescent acetabular dysplasia**, late-developing acetabular dysplasia
- No discharge at 5 years → **follow to end of growth**, yearly X-ray

Must-Know

- Follow to end of growth → yearly X-ray
- Residual dysplasia beyond 5–6 years → most likely no correction
- Subluxation → dysplasia never corrects
- No physiotherapy → osteopenic bone → subtrochanteric fracture
- Early redislocation → no immediate reoperation on a stiff hip

Proximal Femoral Growth Disturbance Ask AI

PFGD

- **PFGD** → formerly avascular necrosis
  - Renamed → no specimen proving AVN worldwide; belief no longer convincing
- Incidence → 20–70%
  - "Never see AVN / PFGD" → does not operate on DDH
- Occurs on **treated sides** and untreated / normal sides
- Features
  - Abnormal **proximal femoral shape**
  - Abnormal growth → capital femoral epiphysis, physis, or both
  - **Enlarged head** → hyperaemia of surgery and casting → stimulates growth
- **Unpredictable**
  - Young child with problem → future appearance unknown
  - No problem young → head problem may appear later
- Causes
  - **Vascular abnormality** → may be AVN; unproven
  - Pressure
  - Hyperaemia
- Contralateral hip → cause unknown → possibly abnormal **position and casting**
- Pavlik / failed Pavlik → open reduction at 1 year → **flattened femoral head**
  - Even after Pavlik in first 3 months, stopped
- Risk factors
  - Head forced in under **tension**
  - Extreme position → **frog position**
  - Post-op position → **human position**, not frog; no 90° abduction
  - → all ↑ PFGD incidence

Kalamchi Classification Ask AI

PFGD

- **Kalamchi classification**
  - Aim → affected physis part, not memorising types
  - Lateral, central or medial physis → very easy
- **Type I** → epiphysis only → small, hypoplastic capital femoral epiphysis
- **Type II** → lateral physis → coxa valga
  - Often well → problems begin at adolescence
  - Conforming and growing well until maturity → appears late
  - Little seen early
  - Case → pelvic osteotomy → epiphysis, physis working; acetabulum growing → then **coxa valga** appearance
  - Later appearance → acetabulum still not bad → head long **well centred**
- **Type III** → central → impaired longitudinal neck growth → short neck
  - Early → affects acetabulum
  - Acetabulum = **counterpart** (mirror) of head
    - Head shape changes → acetabulum changes
    - Round, concentric head → concave acetabulum embracing it
  - Case → operated, nice result → then central affection → **neck shortening**
- **Type IV** → medial physis → coxa vara *[Q2 · As spoken; Kalamchi-MacEwen type IV is total head and physeal damage, not medial physis/coxa vara.]*

- **Early appearance** ≠ final outcome → key message
  - Not every abnormality = AVN
  - Case → open reduction only, no osteotomy → good acetabular growth
    1. Epiphysis looks like AVN / **Perthes disease**
    2. Panic → "reaming for Perthes" → wrong; needs nothing
    3. With growth → very normal hip
- Early PFGD appearance → does not predict later or **final outcome**

Kalamchi type	Site	Result
I	Epiphysis only	Small, hypoplastic capital femoral epiphysis
II	Lateral physis	Coxa valga; appears late (adolescence)
III	Central physis	Short neck; early; affects acetabulum
IV	Medial physis	Coxa vara

Exam Pearl

- PFGD 20–70%; renamed → no specimen proving AVN
- Occurs on treated and untreated sides
- Causes → vascular, pressure, hyperaemia
- Human position, not frog; no 90° abduction
- Early appearance does not predict final outcome

Summary of DDH Treatment Ask AI

- Earlier treatment → better results
- **Open reduction + pelvic osteotomy** → highest success, least complications
- After surgery → **follow to maturity** → late adolescent acetabular dysplasia
- Before 6 months → **Pavlik harness** to induce reduction → success: continue; failure: stop
- 6 months–1.5 years → closed or open
  - Preference → open reduction + capsulorrhaphy + **Dega** at 1 year → wait until 1 year
- More than 1.5 years → always open reduction + capsulorrhaphy + **Dega**
- Moderator → thanks Dr. Hisham for clarity and simplicity

Late Acetabular Dysplasia: Indications and Imaging Ask AI

Indications for Pelvic Osteotomy Ask AI

- Moderator → time for pelvic osteotomies; lecturer continues
- Indications
  - Combined with **open reduction**
  - Previously treated (surgery, closed reduction, Pavlik, any) → **residual dysplasia** not improving, plateau, early subluxation
  - Treated, doing well → **late-presenting dysplasia**
  - **Primary presentation** at adolescence → never treated, never seen
- Late-presenting dysplasia → **mature vs immature**
  - Mature → no growth plates
  - Immature → growth plates
  - Considered in osteotomy choice

Clinical Presentation Ask AI

Late AD

- Not covered in DDH diagnosis lecture → different complaints
  - **Pain and fatigue on walking** → unstable hip fulcrum → excessive muscle forces → fatigue → pain
  - **Shear stresses** on labrum
    - And capsule → pain on prolonged standing or running
  - **Impingement pain** → especially labral injury → sitting, forced flexion / adduction → labral pain
  - **Abnormal gait** → limp; abductor insufficiency → Trendelenburg; bilateral → waddling

Prerequisites and Radiological Indications Ask AI

Late AD

- **Reconstructive pelvic osteotomy** → hip concentrically reduced; not dislocated, not subluxated
  - Even with simultaneous reduction → head reduced, not outside
- Radiological indications
  - **Centre-edge angle** < 20°
  - Or acetabular index > 15° or 20°
- Valid only in **symptomatic patient**
  - Radiological dysplasia only, no complaints (no pain, limp, Trendelenburg) → nothing
  - Operate for limp, pain, fatigue
  - **Abductor fatigue** → real problem to solve
  - Do not operate to treat an X-ray
- Universal rule → no risk for a textbook X-ray
  - e.g. infected, or walking normally → now limping
- Judge indication and success by **function**
  - Not by X-ray adequacy

Must-Know

- Do not operate to treat an X-ray
- Symptomatic patient only → pain, limp, abductor fatigue
- Radiological → centre-edge angle < 20°, acetabular index > 15–20°
- Reconstructive osteotomy → hip must be concentrically reduced

Imaging Array Ask AI

Late AD

- Late dysplasia → **array of radiographs**
  - AP supine
  - AP standing
  - AP
    - In **abduction and internal rotation**
  - Frog lateral
  - False profile view
  - **3D CT** of the hips → one of most useful nowadays

View	Shows
AP supine	Shenton line, acetabular index, centre-edge angle (lateral coverage)
AP standing	Subluxation on standing
AP abduction–internal rotation	Final post-osteotomy position; reducible vs fixed subluxation
Frog lateral	Femoral head shape — round vs flattened
False profile	Anterior coverage (anterior centre-edge angle)
3D CT	Head and acetabular shape; head subtraction

AP View Ask AI

AP

- **Shenton line** → inferior neck border + inferior border of superior pubic ramus → continuous curve
  - Interrupted → dislocated or subluxated hip
- **Acetabular index** → measured from
  - Hilgenreiner's line → through triradiate cartilage, same point both sides
  - Not superior on one side, inferior on other
  - Normal < 20°; more → dysplasia
- **Centre-edge angle** → on Hilgenreiner
  - Line → perpendicular from head centre + line from head centre to acetabular edge
  - Normal side > 20°; affected ~0 → head poorly covered
  - By the acetabulum
- AP → interrupted **Shenton line**
- Triradiate cartilage unclear, sides asymmetric → **symmetric pelvic points**
  - e.g. inferior ischial border or inferior pubic ramus → line
  - Parallel line through visible triradiate cartilage → = Hilgenreiner's line → measure index
- **Normal acetabular index** → older children < 15°; younger < 20°
  - This case → > 30°
- Centre-edge angle → ~30° normal side
  - Affected → 3–5° → **uncovered**; Shenton disturbed

Standing AP View Ask AI

- **Standing view** → stable vs subluxing on standing → distance increase
- Case → incidental acetabular landmark spot
  - Supine → head close; standing → distance ↑ → **subluxation on standing**
  - → very high shear, not concentric, Trendelenburg, major problems

Frog Lateral View Ask AI

- **Frog lateral** → not for acetabulum
  - For femoral head → front and side views (like prisoner photos)
  - AP diameter = frog-lateral diameter → **round head**; otherwise flattened

Abduction–Internal Rotation View Ask AI

- Places head in **final post-osteotomy position**
  - Reducible and reduced → descends into acetabulum
  - Or subluxated and fixed
- Previous case → abduction + internal rotation → Shenton restored → **spherical congruent hip** → reconstructive osteotomy feasible
- Apparent dislocation / subluxation
  - Acetabular index better than other side (previous operation)
  - Do not be deceived → third mention → **internal rotation**
  - Head seated in acetabulum → simply **increased femoral anteversion**
  - Prominent **lesser trochanter** → anteversion imaged in external rotation → head end-on, not in profile

False Profile View Ask AI

- **False profile** → anterior acetabular coverage
  - AP → lateral coverage
  - Planning → lateral only, anterior only, or both
- **Technique**
  - Patient dead lateral → steps back with opposite leg → rotates ~65°
  - Other hip → reverse position
  - Beam ~25–30° from body → complements 65°
- Memorise → describe to **radiographer** → only big centres know
- **Anterior centre-edge angle** → same as centre-edge angle on false profile; taken standing
  - Vertical line from head centre + line to anterior acetabular edge
  - Covered hip → ~45°
  - Poor coverage → < 5°
- → this case needs lateral + anterior coverage

3D CT Ask AI

- **3D CT** → very informative
  - Femoral head + acetabular shape
  - **Head subtraction** → exact acetabular shape → useful in some cases

Aim of Pelvic Osteotomy Ask AI

- Pre-op → disturbed Shenton, ↑ acetabular index, ↓ centre-edge angle
- Goals
  - **Concentric reduction** in acetabulum
  - Normal Shenton line
  - ↑ centre-edge angle
  - ↓ acetabular index
- Post-osteotomy → index improved, Shenton restored, centre-edge angle ↑
- Aim → **normalise hip appearance** + good femoral head coverage

Pelvic Osteotomy: Principles and Types Ask AI

Classification Ask AI

Pelvic Ost

- Types (theory) → **reshaping**, redirectional, shelf
- Or → reconstructive vs salvage
  - **Reconstructive** → congruity achievable
    - Assessed by abduction–internal rotation view (~40°)
    - Head seated, more congruent → reconstructive feasible
    - = rebuilding hip; restoring hip longevity
  - **Salvage** → congruent hip not achievable

Basic Principles Ask AI

Pelvic Ost

- **Head inside acetabulum** during osteotomy
  - Head outside → acetabulum may shrink → head will not fit later
  - Osteotomy is opened
  - Femoral head = **mould** of acetabulum
  - "Dough" takes head's round shape → never osteotomise with head outside
- Rotation → **closer to acetabulum** → less displacement; farther → more displacement
- Open **triradiate cartilage** → never cross; all cuts away from it
- Choice of osteotomy → variable, according to
  - **Degree of dysplasia**
  - Congruity → not achievable → salvage, not reconstructive
  - **Maturity**
    - Mature → stiffer, stronger bone → simple osteotomy insufficient → complex osteotomy
    - Mature → no triradiate cartilage → osteotomies impossible in immature allowed
- Mild dysplasia → **Dega** alone may suffice
- Severe → triple pelvic, **Ganz**, Bernese triple or quadruple

Fragment Mobility: Salter (Innominate) Osteotomy Ask AI

Salter

- Memorise diagram → step by step
  - Pubic ramus cut alone → no movement
  - Superior pubic ramus → no movement
  - Ischium → no movement
  - **Iliac cut** → movement possible
- Ischium + pubis held by joints, capsule, ligaments → acetabular fragment moves around them

- **Salter osteotomy** = innominate = iliac bone osteotomy
- Fragment (true pelvis part) held by
  - **Sacrospinous ligament**
  - Sacrotuberous ligament
  - Symphysis pubis
- Full freedom
  - → separate from all → progressive in other osteotomies
  - Salter → all still hold fragment
- Rotation around **symphysis pubis** only → wrong
  - Other ligaments also hold; tensioned ligament controls direction
- **Axis of rotation** of Salter → really unknown
  - Controlled by symphysis pubis, sacrotuberous, sacrospinous, all tissues
- Salter → historically famous for **acetabular dysplasia**; indications decreasing worldwide
  - Last pelvic osteotomy meeting (Switzerland) → perhaps only current indication → **bladder exstrophy** (ectopia vesicae) → close pelvis
  - Rarely done for acetabular dysplasia
- **Figure-of-four** position → tensions, opens osteotomy
  - Also used in other osteotomies → open, rotate acetabular fragment
- **Iliac crest graft** → inserted → wire fixation
- Disadvantages
  - **Wire fixation** required
  - More retroversion
  - Fragment moves poorly
  - → tension of attached structures
- Possible in **mature** patients → less movement → stiff ligaments and pubic rami; less so in children
- Main disadvantage → **fixation**, unneeded in easier types

Double Pelvic Osteotomy Ask AI

- Aim → ↑ fragment movement
- Innominate cut + **pubic cut** → double pelvic osteotomy
  - Salter concept → rotation around symphysis → pubic cut → more freedom
- Cut far from acetabulum → few degrees rotation → **off-ending** → non-union → abandoned

Triple Pelvic Osteotomy Ask AI

Triple

- Cuts **close to acetabulum** → avoids large displacement
  - Superior pubic ramus + ischium together → better movement
- Still held by **sacrospinous** and sacrotuberous; freed only from symphysis pubis
- Salter fragment → held by sacrotuberous, sacrospinous, **symphysis pubis**
- Triple → freed from symphysis only → movement still **limited**
  - Lecturer's later correction → ischial cut close to acetabulum also frees sacrotuberous

Pemberton Osteotomy Ask AI

Pemberton

- Like Salter, but posterior column **not reached or cut**
  - No exit into greater sciatic notch
- Misconception → lateral cut beside acetabulum (X-ray look)
  - Actual path → backwards like Salter → curves inferiorly → **triradiate cartilage** → rotates around it
- **Rule**
  - Cut closer to acetabular roof → more reshaping → may shrink acetabulum
  - Farther → more reorientation (redirection) → shape unchanged
- **Pemberton** → essentially reshaping
  - Starts at AIIS
  - Curved osteotome backwards
  - X-ray → hinge appears at triradiate cartilage; path runs behind acetabulum

Dega Osteotomy Ask AI

Dega

- Polish surgeon → **Dega osteotomy**
  - Pemberton problem → may injure **triradiate cartilage** → avoid rotating around it
  - Path backwards → stops short → no cut into greater sciatic notch → **hinges on posterior column**
- Pemberton + Dega → like Salter → fragment **still held**
  - Symphysis pubis, sacroiliac joint, sacrotuberous, sacrospinous → limited
- → mostly **reshaping** osteotomies
- Terminology confused → Dega called Pemberton and vice versa
  - Best description → 2020 **Instructional Course Lectures**
  - Original in Polish → poorly translated
- Start more **anteriorly** → more reshaping → smaller acetabulum
- Start slightly more **superiorly** → more redirection
- Graft → Dega's original: two **wedge** grafts
  - Very young → thin ilium → unstable
  - → sandwich graft, transverse not longitudinal
- Opens, covers femoral head → **hinge on posterior column**
  - Posterior column → osteoclasis, not osteotomy → elastic recoil
- **Elastic recoil** + graft → no fixation; cast required
  - Older children → fixation → avoids cast

Dega Case Examples Ask AI

Dega

- 8-year-old girl → dysplasia, poor **anterior coverage** → Dega + screws
  - → no cast; head well-seated, congruent, covered
- 7-year-old girl → open reduction only → **residual acetabular dysplasia**
  - Followed → no improvement → subluxation, instability → Dega + 2 wires → no cast
  - Cast, if applied → 3 weeks, not 6
  - Very nice coverage, well-seated head → Dega very easy, simple
- Another → Dega + **screws** (older age)
  - Avoiding cast at 8–10 years → achievement → cast torments child and parents

Triple Osteotomy Case Ask AI

Triple

- More **redirection** → older child, severe dysplasia
  - Pubic ramus, ischium, ilium → fragment rotated → more mobility
- Case → obvious dysplasia + **large head**
  - Dega unsuitable → reshaping shrinks acetabulum → mismatch with large head (AP, frog lateral)
  - **False profile** → anterior coverage respectable vs healthy side
  - Need → lateral + anterior rotation → **redirection, not reshaping**
  - Head uncovered
  - But large → rotate acetabulum around it → **triple pelvic osteotomy**
- Superior pubic ramus cut
- **Ischium** → Ganz approach, cut completely (not Ganz path)
  - Ilium cut → fragment displaced, rotated → covers head
- False profile → completely changed → **full head coverage**

High Yield

- Closer to acetabulum → reshaping; farther → redirection
- Pemberton hinges on triradiate cartilage; Dega on posterior column
- Dega → elastic recoil → no fixation needed in young
- Large head → no Dega → redirection (triple)
- Salter → only current indication perhaps bladder exstrophy

Shelf Osteotomies Ask AI

Shelf

- Redirectional + reshaping → mostly reconstructive → only with **congruity**
- **Shelf osteotomy** → unstable head, not reducible, older patient → stabilise, improve
  - **Chiari osteotomy**
  - Others → Staheli, Catterall, Spitzy → 20–30 types, essentially same
- Mostly not for acetabular dysplasia → used in **Perthes disease**
  - Dysplasia → some still use Spitzy
  - Included
  - → comprehensive **pelvic osteotomy** coverage
- **Chiari** → cut above hip in ilium → entire distal fragment displaced medially
  - Superior part → shelf → ↑ head coverage
  - Used more in Perthes
- Chiari vs other shelves → **medialization** of femoral head
  - Others → head stays in place
  - Medialization → hip centre closer to centre of gravity → ↓ hip forces
- Other shelves → **iliac bone** piece above femoral head
  - Some claim very good head development, especially late-developing dysplasia
  - Not done much in dysplasia

Shelf	Technique	Head medialization
Chiari	Iliac cut above hip; distal fragment displaced medially	Yes
Other shelves (Staheli, Catterall, Spitzy)	Iliac bone piece above head	No

Fragment Attachments: Summary Ask AI

- Osteotomies covered → **redirectional, reshaping, combined**
- Salter, Dega, Pemberton → fragment held by sacrotuberous, **sacrospinous**, symphysis pubis
- Triple → freed from symphysis pubis only → still attached to sacrospinous ± sacrotuberous → **limited**
- Correction → triple with ischial cut close to acetabulum → frees **sacrotuberous**; sacrospinous remains

Ganz and Bernese Osteotomies Ask AI

Freeing the Fragment Completely Ask AI

- Mature, older, **severe** dysplasia → fragment moves freely, "like jelly"
  - Pubic cut
  - Ischial cut → frees sacrotuberous → still held by **sacrospinous**
  - Free from sacrospinous too → completely free
- Pubis + ilium cuts → fragment held by sacrotuberous + **sacrospinous**
  - Transverse ischial cut (ordinary triple) → frees sacrotuberous; sacrospinous remains
  - Cut farther away → both remain
- 3 cuts = **triple** → may still be held by both
  - Free from sacrotuberous → ischial cut very close to acetabulum, proximal to ligament insertion
- Free from **sacrospinous** too → cut in other direction → more free mobility
  - Or join cuts → = **Ganz**

Ganz (Bernese Periacetabular) Osteotomy Ask AI

Ganz

- Idea only → pubic cut; ischial + iliac cuts joined **within posterior column**
  - Posterior column intact
  - Fragment moves freely
- Disadvantage → crosses **triradiate cartilage** → not in immature
- Blue line osteotomy = Ganz
- **Alternative** with same mobility → ilium + pubis + ischium cuts
  - Ischial cut exits proximal to ischial spine
  - → into greater sciatic notch
- Previous triple → frees from **sacrotuberous**
  - Cut in inferior ischiopubic ramus → also triple → less mobility → still sacrotuberous
- Both free-mobility osteotomies → very difficult
  - Complete separation → **Bernese procedures** → Ganz periacetabular osteotomy + Bernese-type triple
- Ganz → through pubis, posterior column, exits in ischium + **pubic osteotomy**
- Carried to greater sciatic notch + complete iliac osteotomy → same mobility → separated from sacrospinous
  - = **Bernese-type triple pelvic osteotomy**
- "Triple" → **three cuts**; fragment may be
  - Completely free
  - Held by sacrospinous only
  - Held by sacrospinous + sacrotuberous → cut in ischiopubic ramus, not ischium
- Maturity
  - Triple → always feasible in **immature**
  - Ganz → mature only → red limb crosses triradiate cartilage

Osteotomy	Cuts	Fragment held by	Note
Salter (innominate)	Complete iliac	Symphysis pubis, sacrotuberous, sacrospinous	Wire fixation; axis unknown

Osteotomy	Cuts	Fragment held by	Note
Double pelvic	Ilium + pubis	Sacrotuberous, sacrospinous	Off-ending, non-union; abandoned
Pemberton	Iliac, curving to triradiate cartilage	Symphysis, sacroiliac, sacrotuberous, sacrospinous	Reshaping; hinge on triradiate cartilage
Dega	Iliac, stops short of sciatic notch	Same as Pemberton	Reshaping; hinge on posterior column
Triple (ischiopubic ramus cut)	Ilium + pubis + ischiopubic ramus	Sacrospinous + sacrotuberous	Less mobility; immature OK
Triple (ischium close to acetabulum)	Ilium + pubis + ischium	Sacrospinous only	High mobility; immature OK
Bernese-type triple	Complete iliac to sciatic notch + pubis + ischium proximal to spine	Free	Same mobility as Ganz
Ganz periacetabular	Pubis + ischium + ilium joined in posterior column	Free; posterior column intact	Mature only

Ganz Technique Ask AI

Ganz

- Ordinary **Smith-Petersen approach**
- Rectus femoris taken down, reflected → capsule
- Below capsule → **curved Ganz osteotome** → ischium
  - Notched → embraces ischium; follows direction by itself
- Stop → continue in **posterior column** upward → exit ilium
  - Continue up → Bernese-type triple
  - Down → ordinary triple → same osteotome below hip capsule
  - → ischium cut
- Ischial cut at this level → freed from **sacrotuberous** → held by sacrospinous only → high mobility
- Cut into **greater sciatic notch** → difficult → many dangerous structures
- **Pubic ramus** → osteotome or saw
  - Thick periosteum → divide well → else osteotomy does not move
- Ischium, pubis, ilium cut → completed in **posterior column**
  - → acetabular fragment
- Bernese type → **line of osteotomy** matters
  - Adductor approach or described way
  - Same effect

Ganz Case Example Ask AI

Ganz

- 19-year-old girl → needed **rotation** + reshaping / redirection
  - Not bedbound → posterior column stable → sitting from first post-op day
- Pre-op → **Shenton disrupted**
  - False profile → very dysplastic, deficient anteriorly
- Acetabular cut; **ischial cut** to half distance
- Ischial cut first → partial iliac osteotomy from above → joined via osteotome
  - Fragment → very free → any direction
- Main Ganz concern → **overcorrection**
  - After triple and Ganz → examine → full flexion
  - → no excessive head coverage
  - Fixation → screws
- Result → **excellent coverage**, Shenton intact, head seated

Salvage Use of the Ganz Ask AI

Ganz

- Aged 13 years → head articulates with **pseudo-acetabulum**; true acetabulum empty
- Abduction
  - View → predicts post-osteotomy head
  - Triple or Ganz options → no triradiate cartilage → **Ganz**
  - Head → lateralized → pseudo-acetabulum turned over to cover it
- Not reconstructive → **salvage**
  - No congruent hip
- Very uncovered → completely outside; true acetabulum lower
- Ganz performed
- Head covered, much improved → but not true acetabulum
  - Head **lateralized** → never reduced → not congruent from start
- Ganz, but not reconstructive → **congruity** not achieved

Paralytic Hips Ask AI

Paralytic

- Adult (reported ~32 years) → **poliomyelitis**
  - Very thin posterior column → Ganz path impossible
  - Head completely outside → needs roof
  - Paralytic → very high mobility → triple or **Bernese-type triple** → frees from sacrospinous
- Internal rotation → marked **anteversion**; head covered
- Polio / paralytic → **roof beyond normal** → otherwise nothing achieved
- Last patient (possibly myelomeningocele)
  - Same procedure → **sitting** from first post-op day

Clinical Pearl

- Congruity achieved → reconstructive; not → salvage, whatever the osteotomy
- Ganz → mature only; crosses triradiate cartilage
- Main Ganz concern → overcorrection → check full flexion
- Paralytic hips → roof beyond normal

Summary of Pelvic Osteotomies Ask AI

- Classification → not by name → by **congruity**
  - Achieved → reconstructive; not → salvage
- Reshaping vs redirectional
  - Closer to acetabulum → more **reshaping**; farther → more redirectional
  - More cuts (triple) → more redirectional
  - More fixed fragment (single osteotomy) → more reshaping
- Variable correction per osteotomy
- **Maturity** → plan cut direction
  - → protect triradiate cartilage
- Ganz + Bernese-modified triple → very difficult → **experienced surgeons** only → very serious complications
- Lecture duration → two hours twenty minutes

Questions and Answers Ask AI

Rotation in the Cast Ask AI

- Post-op cast → **neither internal nor external rotation**
  - Rotating feet → rotates knees, not hip

Closed Reduction After Pavlik Failure Ask AI

- Q → Pavlik failed in 3 weeks → stop → wait 6 weeks for closed reduction?
- Closed reduction acceptable → after 1, 2 or 3 months → **done correctly**
- Common errors
  - "Felt it reduce" → cast + X-ray
  - No **MRI**
  - No intraoperative arthrogram
  - No confirmation of true reduction
  - No examination at 3 months → ligaments, capsule still lax?
  - No pre-op traction for irreducible hip
  - → ↑ problems, complications
- Result → no spared operations; repeated **anaesthesia**
  - → 66–85% need further surgery → near future, not distant
- Not in favour of closed reduction → no respectable **traction method**
  - Sequence of events
  - → more cost, longer casting
- Worldwide trend → **open reduction**
  - Conferences → "did you reduce the anaesthesia rate?" → no

Pavlik Follow-Up Ask AI

- Follow-up → **examination and ultrasound**
- Worn until **alpha angle** 60° (normal)
- **Weaning** previously advised → recently no difference vs stopping at once
- Stopped at once when
  - Alpha angle 60°
  - Clinically stable → no subluxation or dislocation
  - Ultrasound → **more than half of head** inside acetabulum

Neonatal Diagnosis and Screening Ask AI

- Q (Dr. Osama Abdel-Rashid) → diagnosed at birth → Pavlik or wait?
- DDH → **spectrum** (e.g. subluxatable hip)
  - No neonatal ultrasonography → 80% may show acetabular dysplasia
- Clinically stable → nothing
- Positive Barlow, Ortolani, or both
  - → **Pavlik immediately**
- High-risk, normal exam → family history (sister with DDH), **breech** → ultrasound at 6 weeks
- **Selective screening** in Egypt, not universal → 6 weeks
- Clinical signs of instability → treat immediately
- Think in terms of **instability**, not "DDH" → wide spectrum

Bilateral Cases: Timing Ask AI

Bilateral

- Bilateral open reduction + capsulorrhaphy + Dega → **one side**, other side after 2 weeks
  - Cast removed → second side → less total cast time
- Sequential with full casts → ~4 months; this protocol → ~2.5 months *[C4 · transcript: "two 2-month casts = about three months"]*
  - Both sides completed
- Not simultaneous
  - **Bilateral simultaneous pelvic osteotomy** → very demanding
  - Heavy bleeding → CBC before second side → anaemia → transfusion
  - Unilateral → transfusion basically never needed
- **Hypotensive anaesthesia** → ↓ bleeding
- Iron supplements → improve CBC before second operation
- All ages → department policy

Upper Age Limit for Open Reduction Ask AI

- **Controversial**
  - Unilateral → almost any age → up to ~9–10 years (missed / neglected)
  - Bilateral → beyond 8–9 years
    - Surgery not preferred
    - Sometimes beyond 7 years → very high dislocation, walking well
- **High bilateral symmetric DDH**, walking well → leave alone
  - Very mobile, no disability
  - High DDH → no high incidence of later osteoarthritis
  - Unlike dysplastic subluxated hip → leave; later **total hip** when osteoarthritic, painful
- Late intervention → **stiffness** → worse gait, pain → worse than before
- Bilateral symmetric late (> 8–9 years) → **better left alone**
- **No cut-off point**; no complete agreement

	Unilateral	Bilateral
Upper age for open reduction	~9–10 years	8–9 years (sometimes 7 if very high, walking well)
Agreement	Controversial	More agreement

Capsule Suturing Ask AI

Medial

- Medial approach → capsule **not sutured** → medial tight capsule left open
- Anterolateral → inferior part opened
  - Left open; **superior part** sutured

Consensus: Open Reduction + Pelvic Osteotomy ± Femoral Shortening Ask AI

- Moderator → all cases → **open reduction + pelvic osteotomy ± femoral shortening** → yes
- Femoral osteotomy → reduction without **tension**
  - Primary aim → shortening; not anteversion reduction
- No **varus osteotomy**
- Pelvic osteotomy → **consensus** → all cases with open reduction → primary pathology in acetabulum
- Open reduction + femoral osteotomy without pelvic osteotomy → illogical, unacceptable

K-Wire Fixation After Closed Reduction Ask AI

- Many cases seen → **K-wire + closed reduction**
- True reduction confirmed? → no
- Needs wire → not smoothly reducible, unstable → **redislocates** after wire removal
- No reputable textbook → closed reduction with wire fixation
- No **publication** → don't do it
  - "Works for us"
  - → where published? reputable journal → world judges
  - Unpublished experience → **least level of evidence**
  - Convince them or be convinced
  - Own complications → seen by others, not you

Waiting for the Ossific Nucleus Ask AI

- Q → wait for **ossific nucleus** before reduction?
- Theory → ↓ avascular necrosis → **no consensus;** unproven hypothesis
  - Main proponent → Nicholas Clarke, England
  - No proof → no need to wait for open reduction
- Applies to **open reduction**, not closed
  - Pavlik = form of closed reduction → not applicable
- Moderator → topic needs detailed sessions
- **Polio** cases → large area → detail needed to understand osteotomy choice

Post-Cast Stiffness Ask AI

- Q → stiffness after cast → how, and
  - Approach?
- After **cast removal** → hip abducted, flexed → no extension, no adduction
  - Do not try → child restores movement alone
  - Tell parents → leave child
  - Sit, walk, crawl freely
  - Sitting without walking 1–4 months → do not interfere
  - Walks with legs apart → gradually brings them together
- Only requirement → **prevent falls** (not on bed, not with other children)
  - Fall → fracture → recast → delayed motor recovery
- Abduction stiffness → DDH leg appears longer → **apparent lengthening**, not true
  - Range restores itself
- **Stone-stiff** hip → caused injury, infection, or severe cartilage injury
  - Seen in post-septic hip dislocation, not DDH (developmental dysplasia of the hip)

Inverted Labrum Ask AI

- Q → **inverted labrum** preventing reduction?
- Cause → **trial of closed reduction** → labrum pushed inside by head → iatrogenic
  - Never encountered in whole career; reported though
- Never multiple **radial incisions;** never resect labrum
  - At most → single radial incision → evert with hook → head reduces
- Cause → bad closed-reduction technique + **femoral head fixation**
  - When not truly reduced → inverted labrum
- **No excision**
- Moderator → closing thanks

Perthes Disease: Pathology

(Aetiology, stages, classification and prognosis)

Prof Dr Hisham Abdel-Ghani

Watch Original Lecture

Lecture transcript processed with AI assistance.

Contents

↑ Book contents

Introduction and Waldenström Stages . . . 143
Overview . . . 143
Waldenström Classification . . . 143
Normal Capital Femoral Epiphysis . . . 144
Sources of Knowledge . . . 144
Structure of the Epiphysis . . . 144
Blood Supply and Nutrition . . . 145
Primary Ischaemic Attack and Revascularisation . . . 145
Ischaemia and Growth Arrest . . . 145
The "Smaller" Ossific Nucleus . . . 146
Revascularisation . . . 146
Creeping Substitution . . . 146
Increased Density Explained . . . 147
Subchondral Fracture and Second Ischaemic Attack . . . 147
Subchondral Fracture and Symptoms . . . 147
Head-Inside-Head Appearance . . . 148
Imaging the Subchondral Fracture . . . 148
Second (Mechanical) Ischaemic Attack . . . 149
Pathology of the Second Attack . . . 149
Fragmentation, Biological Plasticity and Healing . . . 150
Fragmentation Explained . . . 150
Biological Plasticity . . . 151
Subchondral Ossification on X-ray . . . 151
Deformities and Subluxation . . . 152
Head Deformities . . . 152
Subluxation and Shenton's Line . . . 152
Age and Cartilage Thickness . . . 153
Metaphyseal Changes . . . 153
Radiographic Examples . . . 153
Not Acetabular Dysplasia . . . 154
Repeated Mechanical Ischaemia . . . 154
Principles of Classification . . . 154
Purpose of Classification . . . 155
Active Versus Healed Stage . . . 156
Catterall Classification . . . 156

Types . . . 156
Why the Anterior Head First . . . 157
Head-at-Risk Signs . . . 157
Evaluation . . . 158
Salter-Thompson Classification . . . 158
Lateral Pillar (Herring) Classification . . . 158
Origin . . . 158
Identifying the Lateral Pillar . . . 159
Grading . . . 159
B/C Border Group . . . 159
Evaluation . . . 160
Choosing an Active-Stage Classification . . . 160
Stulberg Classification . . . 161
Types . . . 161
Radiographic Examples . . . 162
Congruency and Age . . . 162
Timing of Assessment . . . 162
Prognostic Value . . . 163
Linking Back to the Active Stage . . . 163
Prognostic Factors . . . 163
Prognostic Factors Study . . . 163
Age: The Egg Analogy . . . 164
Factor Significance . . . 164
Prognosis by Lateral Pillar and Age . . . 165
Stulberg 1–2 by Group . . . 165
Which Age Counts . . . 165
Limits of Prediction . . . 166
Stulberg 4–5 by Group . . . 166
Treatment Implications . . . 167
Treatment by Group and Age (Preview) . . . 167
Limitations and Future Directions . . . 167
Summary . . . 168
Questions and Answers . . . 168
Gender and Normal Variants . . . 168
Sequelae Beyond the Epiphysis . . . 168
Perthes as a Distinct Entity . . . 169
Skeletal Age . . . 169
Subchondral Fracture vs Subchondral Ossification . . . 169

Osteoid Osteoma vs Perthes 170
Onset, Duration and Other Questions 170

AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 158) “type B matches Catterall 2 and 3” → “type B matches Catterall 3 and 4”

C2 (p. 162) “aspherical incongruent” → “aspherical congruent”

C3 (p. 165) “surgery for Herring A acceptable” → “surgery for Herring A unacceptable”

C4 (p. 169) “subchondral ossification occurs in dense stage” → “subchondral fracture occurs in dense stage”

Queried - kept as spoken, but looks wrong

Q1 (p. 168) as spoken; same lecture gives Stulberg 3 ~47% joint space narrowing, so near-normal longevity is doubtful

Introduction and Waldenström Stages Ask AI

Overview Ask AI

Perthes

- **Perthes disease** → a major pillar of paediatric orthopaedics
- Still not fully understood
- Full name → **Legg-Calvé-Perthes disease**, after those who first described it
- Understanding requires **pathogenesis** — itself still poorly understood
- **Natural history** → differs for every Perthes patient
- No black and white in medicine; the unknown exceeds the known

Waldenström Classification Ask AI

Waldenström

- Poorly understood disease → poorly understood behaviour
- Often called pathological → actually a **radiological classification**

Stage	X-ray appearance
Condensation	Increased density
Fragmentation	Fragmented epiphysis; radiolucent and radiodense areas
Healing	Radio-opaque and radiolucent areas; close to fragmentation
Healed (end stage)	Subchondral ossification

- Fragmentation on X-ray ≠ fragmented head
  - X-ray shows **ossific nucleus** only — not cartilage, not whole head
  - Opened hip, any stage → smooth cartilage, mostly round head
  - May be irregular; never resembles the X-ray
- At surgery → head never found fragmented or broken
- **Healing stage** → little different from fragmentation on X-ray
- Fragmentation vs healing → distinguished by **follow-up**
  - Single first X-ray → cannot tell
  - Symptomatic `~4 months` → fragmentation
  - Symptomatic `2 years` → healing started
  - Serial X-rays: same fragmentation + increased density → healing started
- Healing → continues to **end (healed) stage**
- Healed stage marker → **subchondral ossification**
  - Ossification in subchondral bone beneath the cartilage
  - Example: density + fragmentation → later head density increases → healing

Exam Pearl

- Waldenström → radiological, not pathological
- Fragmentation vs healing → only by follow-up
- Healed stage → subchondral ossification

Normal Capital Femoral Epiphysis Ask AI

Sources of Knowledge Ask AI

Perthes

- Radiological stages → **explained by pathological changes**
- Knowledge sources:
  - Very few **human specimens**
    - Femoral head examined after death of a child with Perthes
  - **Animal studies** → induced ischaemia in living hips → observed changes
- Plan: normal epiphysis → pathological changes
  - → radiological appearance explained by pathology
  - → pathogenesis of **deformity**

Structure of the Epiphysis Ask AI

- **Capital femoral epiphysis** → ossific nucleus; metaphysis = neck
- Ossific nucleus → surrounded by **epiphyseal cartilage**
  - Covered externally by ordinary hyaline articular cartilage
- Cartilage layers of the growing head:
  - **Superficial cartilage zone** → ordinary articular cartilage, as in adult
    - Adult → cartilage lies directly on bone
    - **Immature hip** → cartilage lies on more (epiphyseal) cartilage
  - Centre of epiphyseal cartilage → ossific nucleus
    - Covered by **germinal cell layer**
- **Germinal cell layer** → forms bone through usual physeal stages
  - Proliferation → hypertrophy → apoptosis → ossification → new bone formation
- Ossific nucleus enlarges → occupies more epiphyseal cartilage until maturity
- At **maturity** → all cartilage → bone; only articular cartilage remains
- **Physeal plate** → between epiphysis and metaphysis; the normal growth plate
- Second growth plate → **around the ossific nucleus**
  - Enlarges head → encroaches on epiphyseal cartilage
  - → all epiphyseal cartilage → bone, as in all bone ends
- Layers: superficial articular cartilage → epiphyseal cartilage → **growth layer** of the epiphysis

Blood Supply and Nutrition Ask AI

- Cartilage → **no blood supply**
- Cartilage becoming vascularised → stage before ossification

Structure	Nutrition	Ischaemia
Superficial (articular) cartilage	Synovial fluid	Continues to function
Epiphyseal cartilage	Synovial fluid	Continues to function
Germinal cell layer of ossific nucleus	Epiphyseal vessels (arterial)	Cells die; growth stops

- **Synovial fluid** → nourishes articular and epiphyseal cartilage, as elsewhere
- No vessel crosses the **physeal plate** → epiphyseal supply mandatory
- Bone ends: all cartilage → ossific nucleus appears
  - → **enchondral ossification** via germinal layer
  - → epiphyseal cartilage disappears → only hyaline articular cartilage
- **Metaphyseal supply** → cannot cross the physis
- **Growing child** → direct vessels to ossific nucleus required

Primary Ischaemic Attack and Revascularisation Ask AI

Ischaemia and Growth Arrest Ask AI

- Ossific nucleus → **cannot use synovial fluid**; needs arterial supply
- Nucleus takes whole epiphysis → physis disappears → supply reaches new bone directly
  - Only articular hyaline cartilage remains
- Perthes → **acute ischaemia**
- **Epiphyseal vessels** occluded — cause not discussed
  - Many theories: coagulation defects, others
- Supply to ossific nucleus cut → ischaemia
- First to die → **cells** (they need the blood supply)
- **Germinal cell layer** affected → proliferation stops
- → nucleus stops growing; stays at its size
- Normal serial X-rays → nucleus larger each time
- Nucleus **does not shrink** → stops growing
- Surrounding cartilage → fluid-nourished → keeps growing normally
- → head (cartilage) grows; ossific nucleus static

The "Smaller" Ossific Nucleus Ask AI

- "Smaller" = **relatively smaller ossific nucleus**
  - At ischaemia → equal to contralateral nucleus
  - Affected side → cartilage grows, nucleus static
  - Normal side → cartilage and nucleus both grow
  - → affected nucleus appears smaller
- Never shrank → **ceased growth**
- Same in animals with artificial ischaemia: cartilage grows, nucleus stops
- Thick cartilage → not abnormal; normal growth continuing
- → **smaller-size epiphysis** on X-ray
- Example X-ray → incidental, at very onset of complaint
- Wider **articular space** than other side
  - No effusion → cartilage enlarged, nucleus static

> **Clinical Pearl**
> - Small ossific nucleus → growth arrest, not shrinkage
> - Wider joint space → growing cartilage, not effusion

Revascularisation Ask AI

- Ischaemia in any living structure → body responds → **revascularization**
- **Bone architecture** intact → spaces, cavities, vascular channels preserved
- Building analogy: gas kills residents (cells)
  - Building intact → stairs, lift, rooms, flats
  - Living part → dead
- Dead tissue → **ingrowth of new vessels**

Creeping Substitution Ask AI

- Vessels re-enter → remove **dead bone** → replace it
- Architecture intact → vessels enter via existing channels → **new bone**
- New bone = **woven bone**
  - Haphazardly arranged; not yet lamellar
  - Collagen + calcium crystals only
- Dead tissue → **bone resorption** with many osteoclasts → new bone
- **Bone formation** always preceded by bone resorption
  - Bone metabolic cycle: osteoclasts → blood vessels → new bone
- New bone laid before **complete resorption** of old bone → bone on bone
- Cut section (not for memorising):
  - AVN → vessels enter as **granulation tissue** onto dead bone
  - → resorption → **creeping substitution**: old bone out, new bone in
- New bone laid > bone resorbed → new bone on **dead lamellae**

Increased Density Explained Ask AI

- **Increased density** = revascularization: vessels in, new bone on old bone
- Analogy: dead skeleton still visible on X-ray (calcium)
  - Here → new bone added on top of it
- → density increases
- Wider articular space → cartilage normal and growing → **normal head size**
  - Ossified proportion reduced → arrest of growth
- No complaint up to this stage → **clinically silent**

High Yield

- Primary attack → cells die; architecture intact
- Revascularization → creeping substitution with woven bone
- Increased density → new bone laid on dead bone
- Up to this point → clinically silent

Subchondral Fracture and Second Ischaemic Attack Ask AI

Subchondral Fracture and Symptoms Ask AI

- Trigger → trauma (jumping, falling) or spontaneous, at a critical bone state
- **Subchondral bone** weak, woven > lamellar → fracture
- → **subchondral fracture** → intra-articular inflammatory products
- → inflammation, effusion → symptoms: pain and **limp**
- Parents notice: morning inability to walk on the leg, tiredness, limp
- Limp resolves → new fall → new subchondral fracture
  - → **recurrence of the same manifestation**
- No subchondral fracture → process completes → **epiphysis reforms** as if nothing happened
- Incidental X-ray → "had Perthes and it healed"
- Subchondral fracture → manifestations → **seeks medical advice**

Head-Inside-Head Appearance Ask AI

- No subchondral fracture → **silent** course
- Full revascularization → sclerosis → growth resumes
  - Insult leaves a scar → **head-inside-head appearance**
- Analogy: fractured epiphysis in a young child → growth resumes
  - → **Harris line** appears

Harris line	Meaning
Parallel to epiphysis	Growth restored normally
Not parallel	Partial growth arrest
Absent	Complete growth arrest

- Harris line → routinely followed in **long bones**
- Sign of **restoration of normal growth** of the epiphysis
- Apparent subchondral thickening → not thickening
  - Evidence of past ischaemia + renewed growth → dense line
- **Very rare** on X-ray
- Meaning → child had ischaemia; passed, over
- Usually **incidental** on X-ray taken for another reason

Imaging the Subchondral Fracture Ask AI

- Subchondral fracture → seen on **frog lateral**, not on AP
- Never image a hip complaint with AP only
- Required → **AP and frog lateral**
- Frog lateral → lateral view of **femoral head**, not acetabulum
- Subchondral fracture = **onset of clinical manifestation**
  - Inflammatory material → articular space → inflammation, effusion
  - → pain and limp

Pitfall

- Subchondral fracture → missed on AP view
- Hip complaint → always AP + frog lateral

Second (Mechanical) Ischaemic Attack Ask AI

- Subchondral fracture = **second attack of ischaemia**
- Occurs on a **revascularized epiphysis** → weaker
- **Primary ischaemic attack** → any (unknown) cause
- Second attack → **mechanical**
- Tissue present at the time:
  - **Dead bone** → cells dead
  - New vascular tissue
- New bone → **haphazardly arranged**; bone, but not lamellar
- Not lamellar → cannot resist **strains**
- Analogy: heap of steel bars + cement ≠ **reinforced concrete**
  - Reinforced concrete → steel shaped and spaced → resists stresses in set directions
  - Random steel + cement → may not resist applied stresses

Pathology of the Second Attack Ask AI

- Findings after second attack:
  - **Dead lamellar bone** — as usual
  - New woven bone
  - **Dead woven bone**
    - → creeping substitution spoiled by trauma
  - **Dead granulation tissue** → part of new vascularity died
- These = **hallmarks** of a second ischaemic attack
- Articular cartilage → continues; no **enchondral ossification** in its deep part
- Primary ischaemia → cartilage intact
- Subchondral fracture → involves cartilage too
  - Partly → **necrosis**
  - Partly → replaced by fibrous tissue
- Bulldozer analogy → second attack demolishes the building
  - Before: channels present → easy re-occupation
  - Demolished building → hard to re-occupy
- **Revascularization** → much harder; channels and routes damaged
- Dead trabeculae present
- → new beginning; new woven bone again
- Traumatic second attack → **loss of structural stability**
- → head **liable to deformation**
  - = liable to change of shape
- **Obliteration of the vascular channels** → slow revascularization
- Restarts from the beginning; not simple vessel re-entry
- During revascularization after second attack:
  - **Dead trabeculae**
  - Woven bone
  - Dead woven bone
  - Vascular connective tissue
  - **Fibrous tissue** → from repeated injury
- → **collapse of the epiphysis** may begin

	Primary attack	Second attack
Cause	Any (unknown)	Mechanical — subchondral fracture
Substrate	Normal epiphysis	Revascularized, weaker epiphysis
Analogy	Gas — building intact	Bulldozer — building demolished
Architecture	Intact channels	Loss of structural stability
Cartilage	Intact	Partial necrosis, fibrous replacement
Revascularization	Easy, via existing channels	Slow; channels obliterated
Clinical	Silent	Pain, limp
Consequence	Reforms if no fracture	Liable to deformation, collapse

Fragmentation, Biological Plasticity and Healing Ask AI

Fragmentation Explained Ask AI

- Patch after patch of changes → **fragmented appearance**
- Some parts revascularized; others not yet reached
- Access very difficult → building demolished

Biological Plasticity Ask AI

- Key term → **biological plasticity**
- Head in fragmentation stage → **moldable** (deformable)
  - Round mould → round
  - Square mould → square
  - Irregular space → irregular
- Goal → head **well contained** inside the acetabulum
- Acetabulum → **mold for the head** → keeps it round
- Subluxed head → acetabular edge bites → **deformed**
- Aim → replace all dead bone with new bone
  - → then **remodeling** → round shape
- Subchondral ossification → head **healed** → shape fixed
- Head-preserving surgery → only while moldable
  - Active stage = fragmentation stage
- Healed stage → shape will not change
- Any operation → no change in head shape
- Every X-ray feature has meaning → still active, or over?
- **Containment** in a healed head → useless
  - No longer moldable, plastic, or plastically deformable

Must-Know

- Fragmentation stage → head moldable (biological plasticity)
- Acetabulum → mould for the head
- Subchondral ossification → healed; shape fixed
- Containment after healing → useless

Subchondral Ossification on X-ray Ask AI

- Healing recognised by **subchondral ossification**
- Very fine white line in **subchondral bone**, between epiphysis and acetabulum
- Stippled ossification → very fine bone in subchondral region
- Line seen → head healed
- Same case → line visible
- Line very fine → later denser
- Not fragmentation → **healing stage**; shape will not change
- Progression: dots → thin line → fully formed
- Line seen → disease over, even with **fragmentation** elsewhere
  - Need not involve the whole head
- Always search subchondral bone for this line
- From first appearance of the line → **shape fixed**
- Example → dense subchondral bone
- Pathology explains: increased density, fragmentation, healing
- → **radiological appearance** of Perthes disease

Deformities and Subluxation Ask AI

Head Deformities Ask AI

- **Coxa plana** → flattening of the head
  - Subchondral fracture → depressed, flat head
- **Coxa magna** → healed but large head
  - Explanation unclear; said to be **reactive hyperaemia**
  - Increased blood supply → stimulated nucleus growth → larger head
- Old belief → Perthes from **tuberculous arthritis** or bone TB
  - Old rule: acetabulum affected → TB; head only → Perthes
  - **Not correct**
  - Acetabulum moulds head; deformed head also deforms acetabulum
- Most famous head deformity → **mushroom**

Subluxation and Shenton's Line Ask AI

- **Subluxation** = interruption of Shenton's line
- Cartilage maintains head size; only nucleus relatively small
  - → Shenton's line intact
- Shenton's line disrupted → **true collapse of the head**
  - Whole head collapsed; cartilage size also reduced
  - Not merely nucleus collapse under intact cartilage
- Subluxation → **very alarming**, very significant sign
- Assessed on X-ray
  - No **MRI** needed
- Head moved up inside acetabulum → only possible with true collapse
- Most patients → small ossific nucleus
  - Shenton intact → head size maintained by **normal growth of the cartilage**
  - Large cartilage → nourished from synovium
- True collapse → typically **older patient**
  - Articular cartilage only; little epiphyseal cartilage

Age and Cartilage Thickness Ask AI

- True collapse of epiphysis = true collapse of head
- Older patient: nucleus collapse → **femoral head collapse** → Shenton interrupted
- Subluxation → proves true collapse occurred
- Young patient: nucleus may collapse to a line
  - Shenton intact → cartilage maintains **head shape and height**
  - → no subluxation

	Young child	Older child
Epiphyseal cartilage	Abundant	Little; articular only
Nucleus collapse	Head height maintained	True head collapse
Shenton's line	Intact	Interrupted
Subluxation	Absent	Present

High Yield

- Subluxation in Perthes = interruption of Shenton's line
- Shenton interrupted → true collapse of the femoral head
- Diagnosed on X-ray; no MRI needed
- Typical in older child with little cartilage

Metaphyseal Changes Ask AI

- Epiphyseal supply interrupted → metaphyseal supply may also be affected
- → metaphyseal changes
- = **metaphyseal cysts**
  - True explanation unknown
  - Most likely bone attempting to grow

Radiographic Examples Ask AI

- Nucleus fully collapsed, Shenton intact → **thicker cartilage**
- Shenton disrupted → whole head collapsed
  - Not enough cartilage for head size
- **Growth arrest** → partial or complete
  - Complete arrest → small whole head, small metaphysis
  - Part still working → **coxa magna**
  - Larger part affected → **coxa vara**
- Head moulds acetabulum, as acetabulum moulds head
- Head shape changes → acetabulum changes
  - Example: very large head

Not Acetabular Dysplasia Ask AI

- Acetabulum shaped to the head → **not acetabular dysplasia**
- No acetabular procedure to cover the head
  - Large head **contained** inside acetabulum; acetabulum follows its shape
- Example → marked **flattening**; acetabulum follows head shape

Pitfall

- Acetabulum moulded to a large head → not acetabular dysplasia
- Acetabular involvement does not separate TB from Perthes

Repeated Mechanical Ischaemia Ask AI

- Perthes ≠ mere **avascular necrosis**
- = complication of AVN secondary to subchondral fracture
  - → further ischaemic attack on a revascularized capital femoral epiphysis
- **Secondary ischaemic insult** → traumatic or mechanical
  - Subchondral fracture on revascularized epiphysis
- Subchondral fractures → may recur once, twice, three times
- Protect child from jumping and blows → avoids **repeated attacks** of ischaemia

Must-Know

- Perthes = AVN complicated by subchondral fracture
- Fracture → mechanical ischaemia on revascularized epiphysis
- Repeated attacks → keep the process going

Principles of Classification Ask AI

- Explanation → **current understanding** worldwide; mysteries remain
- Complex topic

Purpose of Classification Ask AI

- Previous scheme → **radiological stages**, explained by pathology
- Aims of any classification:
  - **Enhance communication** → "grade X of classification Y" conveys the patient
  - Gives needed data at a glance
    - Like recognising a familiar face
    - Twenty types → unremembered, unused
  - **Reliability**
    - Same X-rays shown again, graded differently → not reliable
    - Same grades every time, even twenty times →
      - **Intra-observer reliability**
    - Other observers give same grades → **inter-observer reliability**
    - Higher reliability → more credible, dependable, usable
- Communication requirements:
  - **Simple**
  - Easy to remember
  - High inter- and intra-observer reliability
  - Covers full spectrum of the pathology
- Usefulness → **insight into prognosis**
  - Grade `1` → grade `2` worse → grade `3` worse still
- Ideal → **guide treatment**: type `1` → X, type `2` → Y, type `3` → Z
- Three tests: communication, prognosis, treatment

Active Versus Healed Stage Ask AI

- Aim → best classification for prognosis and treatment
- All Perthes classifications → **based exclusively on radiology**
- Asked to classify → first ask which stage

	Active stage	Healed stage
Head	In fragmentation; will keep changing	Shape fixed
Purpose	Prognosis of active stage → what it ends as	Prognosis of later osteoarthritis
Classifications	Catterall, Salter-Thompson, lateral pillar (Herring)	Stulberg

- Healed patient → never "Catterall X" or "**lateral pillar** Y"
- **Stulberg** → sphericity of head + congruity with acetabulum
  - → insight into prognosis of osteoarthritis
- Active-stage aims: communication, full spectrum, **prognosis**, treatment guidance
- Three active-stage classifications: Catterall, Salter-Thompson, lateral pillar (Herring)
- Classify in **fragmentation** → patches or fragmentation seen
- Healing with subchondral ossification → active-stage classification invalid
  - → assess sphericity → **Stulberg**
  - → rate of osteoarthritis
- Fragmentation without subchondral ossification → **active stage**

Exam Pearl

- All Perthes classifications → radiological only
- Catterall, Salter-Thompson, Herring → active stage only
- Stulberg → healed stage; predicts osteoarthritis
- Subchondral ossification present → use Stulberg

Catterall Classification Ask AI

Types Ask AI

Catterall

- 1971, **Catterall** → four types
- Type 1 → **anterolateral quadrant** of femoral head

Why the Anterior Head First Ask AI

- Heel strike, start of **stance phase** → full body weight descends
- Hip at that moment → **flexed**
- → anterior head receives most impact during walking
- → anterior part **first involved** by subchondral fracture
  - → fragmentation, mechanical insult of capital femoral epiphysis

Type	Involvement of femoral head
1	Anterolateral quadrant (quarter)
2	50% (half)
3	50% to full (three-quarters)
4	Complete

Head-at-Risk Signs Ask AI

Catterall

- Added later as qualifiers → present → **poorer prognosis**
  - **Lateral calcification** → lateral to acetabular edge
    - Edge bites head → extrusion or ossification of lateral part
  - **Subluxation**
    - Most important head-at-risk sign
    - True collapse; no cartilage to grow and compensate
    - More frequent → poorer prognosis
  - **Gage sign** → triangular radiolucency on frog lateral or AP
    - = very osteoporotic epiphysis
  - **Transverse (horizontal) physis** → greater compression loads
    - In fact → little significance

Evaluation Ask AI

Catterall

- **Descriptive** of morphology
- Intra-observer reliability → **poor**
  - One reader type 1, another type 2
  - Others → 3 or 4
  - Same reader, 3–4 viewings → 2 once, 3 another
  - → lacks high inter- and intra-observer reliability
- Second defect → classification changes over time
  - Classified in June → may change after 6 months
  - **Spread of involvement** increases
  - → no fixed prognosis; changes with time

High Yield

- Catterall → 25, 50, 75% and complete involvement
- Anterior head involved first → flexed hip at heel strike
- Head-at-risk: lateral calcification, Gage sign, transverse physis, subluxation
- Subluxation → most important head-at-risk sign
- Defects → poor reliability; changes over time

Salter-Thompson Classification Ask AI

- 1984, **Salter and Thompson**
- Based only on **subchondral fracture**

Type	Subchondral fracture extent	Catterall equivalent
A	Less than 50% of head	Catterall 1 and 2
B	More than 50% of head	Catterall 3 and 4 *[C1 · transcript: "type B matches Catterall 2 and 3"]*

- No **morphology** → one pathological feature only
- Subchondral fracture → seen in only ~30% (very early presenters)
- → 70% unclassifiable → **not very useful**

Lateral Pillar (Herring) Classification Ask AI

Origin Ask AI

Herring

- **Anthony (Tony) Herring** → lateral pillar classification, 1992
- Dallas, Texas → all Perthes X-rays with **long-term follow-up**
- → very strong correlation: lateral pillar ↔ **prognosis**
- **Lateral pillar** → lateral portion of capital femoral epiphysis
  - AP view
  - Early fragmentation stage
- Active-stage classification → patient must be in fragmentation
- Lateral part collapsed or not → **Herring grade**

Identifying the Lateral Pillar Ask AI

Herring

- **Radiolucent line** → separates lateral portion from rest of epiphysis
- Always at 5–30% of epiphyseal width (1/20 → 1/3)
- Lateral to the radiolucent line → **lateral pillar**
- Examples: very small piece vs a whole third
- No radiolucent line → **lateral-most 25%** of epiphyseal width
  - = a quarter of the epiphysis → measured for height

Grading Ask AI

Herring

- Loss of height vs **contralateral hip**
  - Bilateral → vs middle pillar

Group	Lateral pillar
A	Not affected; height maintained
B	Depressed; more than 50% height maintained
B/C	B-level height + very thin, poorly ossified, or exactly 50%
C	Less than 50% height remaining

- Images → **Tony Herring's** original paper
- Type A example → normal height; radiolucent line visible
- Lateral pillar normally slightly lower than **middle pillar**
  - Middle pillar → forms convexity of the head
- No height loss → **Herring A**
- Depression, >50% height remaining → **Herring B**
- Less than 50% height → **Herring C**

B/C Border Group Ask AI

Herring

- 2004 (12 years later) → **B/C** group added
  - Prognosis → not as good as B, not as bad as C
- Essentially B (≥50% height) + one of:
  - **Very narrow** lateral pillar → 2–3 mm wide
    - Not the full 20–30% or 25%
  - **Very poorly ossified** → barely visible; very weak bone
  - **Exactly 50%** depressed vs middle pillar
- → prognosis poorer than B
- Final groups → A, B, **B/C**, C

Evaluation Ask AI

Herring

- **Easy** and simple
- Very good inter- and intra-observer reliability
  - Repeat viewings → same grade; other observers → same grade
  - Contrast **Catterall** → very poor reliability
- Defect → no **overall extent** of the pathology
  - = how the whole head is doing
- Shared defect with Catterall → early grade may change in 3–6 months
- → classify today; **reclassify** after 6 months of follow-up

Exam Pearl

- Herring → lateral pillar height vs contralateral hip
- Bilateral → compare with middle pillar
- No radiolucent line → lateral-most 25%
- B/C → thin (2–3 mm), poorly ossified, or exactly 50%
- Reclassify after 6 months

Choosing an Active-Stage Classification Ask AI

- Best → enhances communication, gives prognosis, **guides treatment**
- Criteria → full spectrum, simple, easy, high reliability
  - Best fit → approximately **lateral pillar**
- Prognosis = will the healed head be deformed?
  - → feature correlating with **final appearance** of capital femoral epiphysis
- Requires knowing the **healed-stage classification** first

	Catterall	Salter-Thompson	Lateral pillar
Year	1971	1984	1992 (B/C 2004)
Basis	Extent of head involvement	Extent of subchondral fracture	Lateral pillar height
Morphology	Descriptive	None	No overall extent
Reliability	Very poor	—	Very good
Main defect	Changes over time	Fracture seen in only 30%	Changes over time

Stulberg Classification Ask AI

- **End-stage classification** → related to active-stage grades

Types Ask AI

Stulberg

Type	Head	Acetabulum	Category
1	Round, normal size	Round	Spherical
2	Round, larger; same AP and lateral diameter	Spherical	Spherical
3	Oval; AP ≠ frog lateral diameter	—	Oval
4	Flat or irregular	Takes head shape	Aspherical congruence
5	Aspherical	Spherical (or vice versa)	Aspherical incongruence

- **Stulberg** → very easy; nothing to memorise
- Type 2 → same diameter on **AP and lateral views** → no problem
- Types 1 and 2 → **spherical hip** → best outcome
  - Only difference → size
- Type 3 → like an **egg**: different from each angle
- Types 4–5 → neither round nor oval; flattening
  - Or **irregularity**
- Avoid "incongruent incongruity" / "congruent incongruity" → confusion
  - Use sphericity terms: **aspherical congruence** = type 4
- Congruence = **coaptation** → shapes match, not spherical

Radiographic Examples Ask AI

Stulberg

- Healed Perthes, same size as other side → **type 1**
- AP diameter = lateral diameter, larger than normal side → **type 2**
  - Spherical hip = normal hip
- Normal and spherical but **larger** than other side → type 2
- Round everywhere, no flat part, AP ≠ frog diameter → **type 3** (oval)
- Oval, other view → spherical contour, no **flattened area**
- **Mushroom** head → not spherical
  - Acetabulum → biconcave; two concavities accommodate the head
  - → aspherical, good coaptation → **aspherical congruent**
- Use **aspherical congruent** / aspherical incongruent → avoids confusion
- Acetabulum takes head shape → **congruent** → type 4
- Marked **flattening** → not oval
  - Trace the head contour → round all over? Not here
  - Notch in head reproduced in acetabulum → **type 4**
- **Mushroom-shaped head** + same-shaped acetabulum → aspherical congruent *[C2 · transcript: "aspherical incongruent"]*
- **Flat head**, rounded acetabulum → aspherical incongruent
  - → type 5; worse than type 4

Congruency and Age Ask AI

Stulberg

- Type 4 → worse head shape, but congruent
- What matters → **congruency**; deformed but congruent is better
- **Younger child** → more mutual moulding, head ↔ acetabulum
- Older child → head may deform
  - Acetabulum lacks **growth** to be moulded → stays round
  - → aspherical incongruity
- Worst → severe deformity; acetabulum bears **no relation** to head
- Deformed head + completely **normal acetabulum**

Timing of Assessment Ask AI

Stulberg

- Old view → apply only at skeletal maturity → **not currently correct**
- Subchondral ossification begun → **final shape**, even at young age
- → Stulberg applicable immediately

Prognostic Value Ask AI

Stulberg

Type	Osteoarthritis outcome
1	No osteoarthritis at all
2	No radiologic joint space narrowing at 40 years follow-up
3	Joint space narrowing in ~47%
4	Similar to type 3 (~47%)
5	61% → joint space narrowing and osteoarthritis

- Types 1–2 → **excellent**
- Types 3–4 → intermediate
- Type 5 → poor prognosis for **osteoarthritis**

High Yield
- Stulberg 1–2 → spherical; no osteoarthritis
- Stulberg 3 → oval; ~47% narrowing
- Stulberg 4 → aspherical congruent
- Stulberg 5 → aspherical incongruent; 61%
- Congruency matters more than head shape

Linking Back to the Active Stage Ask AI

- Question → which active-stage classification to use?
- Answer → the one predicting the **final Stulberg class**
- → gives insight into prognosis → useful
- Predicted Stulberg class → **incidence of osteoarthritis**

Prognostic Factors Ask AI

Prognostic Factors Study Ask AI

- Study → all Perthes parameters vs **prognostic significance** (final Stulberg class)
- Greatest significance → **age** and lateral pillar

Age: The Egg Analogy Ask AI

- Young children → abundant cartilage, small **ossific nucleus**
- Egg dropped 5 cm onto hard surface → cracks
- Egg inside a **silicone ball** → larger ball
  - → egg may not break
- Less silicone → more breakage and **deformity**
- Thicker cartilage → maintains shape of ossific nucleus (the egg)
  - Even if broken → **sphericity** maintained
- Older child → less epiphyseal cartilage
  - → nucleus more liable to **deformation**
  - → deformation reflected in femoral head shape

Factor Significance Ask AI

Factor	Prognostic significance
Age	+++ — most important
Lateral pillar (Herring)	+++ — very great correlation with outcome
Subluxation	++
Mobility (range of motion)	++
Gender	++
Degree / extent of necrosis	No value
Metaphyseal involvement, metaphyseal cysts	No value
Gage sign	No value
Transverse physeal plate	No value

- **Range of motion** → keeps the acetabular mould working
  - Full ROM of head inside acetabulum required
  - ROM lost → head stuck under acetabular edge → edge bites → mushroom
- Extent of necrosis meaningless → undermines **Catterall**
  - Catterall type 3 → much poorer prognosis than type 4
  - Less necrosis may be worse than complete necrosis
- Age + **shape of the lateral pillar** → correlated with Stulberg outcome
- → main tools for prognosis during the active stage
- **Salter-Thompson** and Catterall → no prognostic help
  - **Extent of avascular necrosis** not related to final outcome

Must-Know

- Two most important prognostic factors → age and lateral pillar
- Extent of necrosis → no prognostic value
- Catterall 3 → worse than Catterall 4
- Loss of ROM → acetabular edge bites → mushroom head

Prognosis by Lateral Pillar and Age Ask AI

Stulberg 1–2 by Group Ask AI

Herring

- **Herring A**, no lateral pillar collapse →
  - At all ages → Stulberg 1 and 2
  - Spherical, normal hip → **normal longevity**
- Herring A at any age → surgery **unacceptable** *[C3 · transcript: "surgery for Herring A acceptable"]*

Group	Onset before 8 years	Onset after 8 years
A	Stulberg 1–2, all ages	Stulberg 1–2, all ages
B	70–90% → Stulberg 1–2	50–70% → Stulberg 1–2
B/C	~50% (0–50%) → Stulberg 1–2	Almost none → 1–2
C	Less than 20% → Stulberg 1–2	Almost none → 1–2

- **Herring B**, onset before 8 → 70–90% normal hips → keep hands off
- B/C separated → findings **significantly affect** prognosis

Which Age Counts Ask AI

- Perthes children → **delayed skeletal maturity**
- **Chronological age** → age from birth
- 8 years chronological ≈ 6 years **skeletal**
- Cut-off → 8 years chronological = 6 years skeletal
- Example: age 9, complaining 3 years → onset at 6
- Complaining 2 years → onset at 7, not 9
- Use **age of onset**, not current age
- B, onset before 8 chronological → 70–90% Stulberg 1–2
- After 8, **B/C or C** → almost none Stulberg 1–2

Exam Pearl

- Prognosis uses age at onset, not current age
- Cut-off → 8 years chronological = 6 years skeletal

Limits of Prediction Ask AI

- Data → **percentages**, not individual prediction
- Child B, aged 4–6 → 70–90% chance of Stulberg 1–2
  - Remaining 10–20% → may be 3 or 4
  - Individual outcome → unpredictable
- Problem of Perthes → deal with percentages, not patient **per se**
- Patient placed in a category, not given an exact grade
- Poorer prognosis after 8 → does not itself justify surgery
- Scientific approach → research comparing surgery vs no surgery
  - **Not deduction**
- Poor prognosis ≠ need for surgery
  - Must show surgery changes the **natural history**
- Model study:
  - 100 patients, onset after 8, **Herring B or C** → operate
  - 100 matched patients → no treatment
  - Significant difference, natural history changed → patient benefits
  - No change → no need to intervene

Stulberg 4–5 by Group Ask AI

- Onset before 8, B, B/C or C → almost never **Stulberg 4–5**
  - → Stulberg 1–2, or at most 3 → **very good** prognosis
  - → generally no intervention before 8
- After 8 → Stulberg 4–5 incidence **increases** in B
  - Markedly after ~10 years
- **B/C and C** after 8 → very high incidence of 4–5
- Herring C, onset ~11 years → 100% Stulberg 4–5 → **very poor prognosis**
- Age + lateral pillar → predicted Stulberg class → **osteoarthritis rate**

Treatment Implications Ask AI

Treatment by Group and Age (Preview) Ask AI

- **Lateral pillar A**, any age of onset → no treatment
  - Almost all → Stulberg 1–2
- **Lateral pillar C** → poor prognosis, especially onset after 8
  - Below 8 → acceptable prognosis
  - **Intervention** → no difference to prognosis
- B and B/C, onset under 8 → no treatment needed; treatment will not help
- B or B/C, onset after 8 → **benefits from treatment** (proven by studies)
  - Significant improvement with surgical intervention
  - Surgery type → next lecture
- C → **doubtful value** of surgery
  - Onset 8 or more → no proof surgery changes natural history

Group	Onset under 8 years	Onset at or after 8 years
A	No treatment	No treatment
B	No treatment	Treat — surgical benefit proven
B/C	Surgical outcome doubtful	Treat — surgical benefit proven
C	Surgical outcome doubtful	No proven surgical benefit

- Surgery indicated → only B and B/C, onset ≥8 chronological (6 skeletal)
  - Only category with **evidence** of surgical benefit

Must-Know

- Operate only on B and B/C with onset ≥8 years
- = 6 years skeletal age
- Herring A → never needs treatment
- Herring C → no proven benefit from surgery

Limitations and Future Directions Ask AI

- Drawback → age + lateral pillar do not give the **exact Stulberg grade**
  - Only a category or percentage → 90% good, 10% may not be
- Research focus → diagnose **early, before fragmentation** → early prognosis
- New tools exist
  - Not yet standard; need time
- **Perfusion MRI** → predicts lateral pillar involvement in early cases
  - Dedicated software (hip vascularity) → measures perfusion of capital femoral epiphysis
  - → extent of head involvement → **prognosis**
  - Still needs **long-term study**; must withstand test of time

Summary Ask AI

Perthes

- Active stage → **lateral pillar** combined with age of onset
- Look for **subluxation** → true collapse of the femoral head
- **Range of motion** → very important
  - Acetabulum → mould for femoral head → maintain good ROM
- **Treatment** planned according to expected outcome
- Treatment → next lecture

Questions and Answers Ask AI

Gender and Normal Variants Ask AI

- **Gender** → slight male predominance; little significance
  - Gender → 2 pluses
  - Age and lateral pillar → 3 pluses
- No normal variant of femoral ossific nucleus → no normal **fragmented** proximal epiphysis
- **Bilateral symmetrical** abnormal proximal femoral epiphyses → not Perthes
  - Most likely skeletal dysplasia
  - Possibly spondyloepiphyseal dysplasia, or multiple epiphyseal dysplasia

Clinical Pearl

- Bilateral symmetrical epiphyseal changes → think skeletal dysplasia, not Perthes

Sequelae Beyond the Epiphysis Ask AI

- Q: AVN of neck, metaphysis or trochanter during/after healing? Pain when healed?
- AVN in Perthes → **capital femoral epiphysis** only
- Neck AVN (e.g. after neck fracture) → unrelated to Perthes
- Deformities (covered in first lecture):
  - Coxa plana, breva, vara, valga, magna
  - All → **sequelae** of capital femoral epiphysis AVN
  - Not direct AVN of neck or greater trochanter
- **Overgrowth of the greater trochanter** → relative, not true
  - Trochanter grows normally; head stopped
  - → trochanter relatively larger than femoral head
  - Trochanter itself → unaffected
- Pain after healing → **Stulberg-dependent**
  - Stulberg 4–5 → osteoarthritis → pain
  - Stulberg 1, 2, 3 → pain very unlikely; longevity almost normal *[Q1 · as spoken; same lecture gives Stulberg 3 ~47% joint space narrowing, so near-normal longevity is doubtful]*

Perthes as a Distinct Entity Ask AI

- Association with **septic hip** → none
- Any identifiable AVN cause → not Perthes
- Perthes = **idiopathic avascular necrosis**
- Not Perthes → AVN secondary to:
  - Trauma
  - Infection
  - Tumour
  - Surgical intervention
  - **Slipped capital femoral epiphysis**
- Reason → specific entity
  - With a specific prognosis
- Presentation **after 12 years** → not Perthes → juvenile AVN

Exam Pearl

- Perthes = idiopathic AVN of the capital femoral epiphysis
- Identifiable cause → not Perthes
- Onset after 12 years → juvenile AVN

Skeletal Age Ask AI

- Children → small for their age (8 looks younger than 8)
- Skeletal age → lags ~2 years behind **chronological age**
- Determined from **wrist** or elbow signs
- Skeletal cut-off 6 years = chronological 8 years
  - Chronological = **birth certificate** age
  - Certificate 8 → skeletal 6; certificate 10 → skeletal 8
  - Subtract 2 years → skeletal age

Subchondral Fracture vs Subchondral Ossification Ask AI

- **Subchondral fracture** → start of mechanical AVN cascade
- Seen in ~30% of cases; not seen in 70%
- Re-ossification of an empty area → not necessarily
  - Fracture often never seen
- Fracture line → appears then disappears → **fragmentation** begins
- Occurs in **dense stage** (increased density), not fragmentation *[C4 · transcript: "subchondral ossification occurs in dense stage"]*
- Sequence: subchondral fracture → fragmentation → **subchondral ossification** in healing

Osteoid Osteoma vs Perthes Ask AI

- Misdiagnosis as **osteoid osteoma** → great difference between them
- Hip osteoid osteoma → near **lesser trochanter**
- May present with coxa magna
  - Whole proximal femur **osteoporotic** vs other side

	Osteoid osteoma	Perthes
Presentation	Persistent pain	Limp
Pain pattern	Night pain; not during activity	Intermittent; activity-related
Analgesics	Relieved by NSAIDs	—
Range of motion	Little limitation	—
Density site	Near lesser trochanter or femoral neck	Epiphysis
Bone scan	Hyperactivity away from epiphysis	—

- Density near lesser trochanter or **femoral neck** → think osteoid osteoma

Onset, Duration and Other Questions Ask AI

- Lateral pillar reference → **other hip**
  - Both affected → middle pillar
- Onset → based on **symptoms**
  - No symptoms, incidental X-ray → onset unknown
  - Symptoms require a **subchondral fracture**
  - → ask about earlier complaints
  - Onset = month complaint began, not **X-ray date** (e.g. two months ago)
  - Treatment decision → based on age at onset
- Stage sometimes undeterminable on one film
  - X-ray today → repeat at 3 and 6 months → **fragmentation or healing**
- Disease duration:
  - Old teaching → healing in 1, 1.5 or 2 years
  - In fact → depends on several factors
  - Careful child, no jumping or falls → shorter course
  - Repeated **new subchondral fractures**, falls, inflammation → longer cycle
    - → repeated mechanical ischaemia
- Acetabulum → **no role in prognosis**
  - Role → Stulberg only: takes head shape → 4; does not → 5
- Irritable hip:
  - Not shown to cause Perthes
  - **Irritable hip resistant to treatment** → MRI → possible early Perthes
  - Rapid recovery → no problem
- Treatment questions → next week's lecture

Perthes Disease: Treatment

(Containment principles and treatment options)

Prof Dr Hisham Abdel-Ghani

Watch Original Lecture

Lecture transcript processed with AI assistance.

Contents

↑ Book contents

Treatment Concept: Active versus Healed Disease . . . 175
Illustrative Case . . . 175
Healing Stage and Treatment Aims . . . 176
Non-operative Management of Active Disease . . . 176
Range of Motion . . . 177
Avoiding Trauma . . . 177
Containment: Principle and Methods . . . 177
Non-operative versus Operative Containment . . . 178
Containability and Hinge Abduction . . . 178
Mechanism . . . 178
Prerequisites for Containment . . . 179
True versus False Hinge Abduction . . . 179
Containable versus Uncontainable Head . . . 179
Femoral and Pelvic Osteotomy . . . 180
Femoral Varus Osteotomy . . . 180
Pelvic Reorientation Osteotomy . . . 180
Shelf Acetabuloplasty . . . 181
Concept: Old versus Current . . . 181
Catterall Technique . . . 182
Spitzy Technique . . . 183
Personal Protocol . . . 183
Surgical Steps . . . 184
Results . . . 184
Management of the Uncontainable Head . . . 184
Valgus Osteotomy . . . 185
Chiari Osteotomy . . . 185
Summary of Active Disease Treatment . . . 185
Adjunct Procedures: Drilling and Distraction . . . 186
Epiphyseal Drilling . . . 186
Hip Distraction (Arthrodiastasis) . . . 186
Residual Deformities . . . 187
Case: Treat the Patient, Not the X-ray . . . 187
Indications for Correction . . . 188
Femoral Head Reduction Osteotomy . . . 188
Evidence . . . 188

Overall Summary and Treatment Algorithm 189
Questions and Answers 190
Shelf Position 190
Aftercare 191
Shelf Technique Details 191
Cheilectomy 191
Combining Valgus and Shelf 191
Indications and Adjuncts 192
Residual-Deformity Case Revisited 192
Distraction and Ilizarov: Evidence and Practice 193
Technique and Indication Queries 194
Closing Comments 195

AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 185) “no abduction deformity” → “prevents adduction deformity (likely)”

C2 (p. 189) “asymptomatic hip” → “symptomatic hip”

Queried - kept as spoken, but looks wrong

Q1 (p. 189) As spoken; same 2018 series quoted as 6 hips earlier, then five here.

Treatment Concept: Active versus Healed Disease Ask AI

- Natural history and final outcome → graded by **Stulberg classification**
- Asked about classification → first clarify: **active stage** or end stage
- Asked about treatment → clarify: active disease or **residual deformities**
  - Concept and idea behind treatment differ completely
- **Active disease**: not yet healed
  - Head plastic, deformable
  - Aim → **keep sphericity** of femoral head → best end-stage result
- **Healed disease**: subchondral ossification started
  - Head no longer moldable, no longer plastic → no further deformation
- Healed hip → "containment to preserve the head" is wrong
  - Containment will not change femoral head shape
- Active disease → head deformable; **biological plasticity**
- After subchondral ossification → final shape attained
  - Any intervention → will not change its shape

Illustrative Case Ask AI

Case

- 9-year-old with Perthes, presented today
- Onset at 5 years (reportedly 4) → hip presented fully healed
- Family told: operation to preserve sphericity → treated as active disease
- **Perthes on X-ray ≠ active disease**

Healing Stage and Treatment Aims Ask AI

- Hallmark of **healing stage** → subchondral ossification
  - After it appears → head not moldable → final shape
- Active stage goal → reduce deforming stresses on femoral head
- Healed stage → no aim to change final result; head shape unchangeable
- Active stage treatment → **head containment**
  - Head held inside acetabulum → acetabulum acts as mold
- Mold → keeps sphericity of head
- Healed stage treatment → **correction of residual deformities**
  - Only **clinically significant** residual deformities treated
  - Treat a patient with a problem, not an X-ray
  - Radiographic correction → should correct the patient's problem
  - Good-looking X-ray, patient gained nothing → failure

	Active disease	Healed disease
Marker	Not yet healed	**Subchondral ossification**
Head	Plastic, deformable, moldable	Final shape; not moldable
Aim	Keep sphericity; reduce deforming stresses	Not changing head shape
Treatment	**Head containment**	**Correction of residual deformities**

Pitfall

- Perthes on X-ray ≠ active disease
- Subchondral ossification → healed → head shape final
- Containment in a healed hip → will not change head shape
- Residual deformity → treat only when clinically significant

Non-operative Management of Active Disease Ask AI

- First part → **treatment of active disease**
- Options → non-operative and operative
  - Operative indications → covered in previous lecture
- Non-operative measures → every patient, operated or conservative
- Indication recap:
  - Herring **B**, onset < 8 years → conservative
  - **B** and **B/C**, onset > 8 years (chronological age) → operated; benefit
  - **C**, > 8 years → no evidence surgery helps

Range of Motion Ask AI

ROM

- All patients, operated or not → **maintain range of motion**
- Head must stay mobile inside acetabulum
  - Movement within acetabulum → acetabulum molds the head
- Most important of all → **full motion**
- Parents taught to perform exercises at home
  - Not physiotherapy modalities: infrared, vibrations, pulses → useless
  - Requirement → range of motion
- **Abduction exercises** → full abduction of both hips
  - One side only → pelvis rotates → no true abduction
  - Result → contralateral normal side abducted instead
- Schedule → 2–3 sessions per day
  - About a quarter of an hour each
- Child "spreads his legs" himself → through **pelvic rotation**
  - Diseased hip → not in full abduction
  - Parents → full hip abduction by hand
- No range, young child, surgery not considered → **adductor tenotomy / adductor release**
  - Needed very, very rarely
- **Casting** 2–3 weeks → advocated by some
- Long-term bracing → almost disappeared worldwide
- Whether operating or not → maintain full range of motion

Avoiding Trauma Ask AI

Activity

- Parents instructed → avoid **trauma to the head**
- First infarction → biological
- Second, third, fourth, further infarctions → traumatic
  - Result from trauma
- Minimize trauma → no ball games, no activities with falls
- **Swimming** → allowed
- Cycling → may be allowed
- Avoid any trauma or impact to the hip
- **Bisphosphonates** → given by some
  - No clear regimen or role in Perthes

Clinical Pearl

- Every patient, operated or not → maintain full range of motion
- Abduct both hips → unilateral abduction → pelvic rotation
- Exercises by parents at home; physiotherapy modalities useless
- Repeat infarctions traumatic → no ball games or falls
- Swimming allowed; cycling may be allowed

Containment: Principle and Methods Ask AI

- True benefit from surgery → patients with **active disease** (previous lecture)
- **Containment** → head kept inside acetabulum

  - Acetabulum as tool (mold) → keeps sphericity
  - Of the femoral head
- **Lateralization of femoral head** + lost range → acetabulum shapes head → irregularities
- Two methods → non-operative and operative containment

Non-operative versus Operative Containment Ask AI

- Many casts and **abduction hip orthoses** described
  - Example → **Petrie cast**
- Use decreased greatly → not effective
- May be harmful → very severe **osteopenia**
  - Osteopenia → predisposes to further collapse
- No longer used in large proportion of centers worldwide
- Containment in use → **surgical containment**
- **Operative containment** types:
  - Acetabulum brought over head → pelvic osteotomy
  - Head seated inside acetabulum → femoral osteotomy
  - Both, or other procedures
- Aim → protect weak head from deforming forces until it can reform
  - Until healing occurs
- Non-operative containment illogical → disease runs ~2 years
  - Child in cast that long → unreasonable
- Head inside → femoral head extrusion and deformation prevented

Containability and Hinge Abduction Ask AI

- Every head entering acetabulum → every head containable
- Some heads do not enter → cannot be contained
- Key term → **hinge abduction**
- Head during active disease:
  - Rounded or even oval → no hinge abduction
  - Or deforming → hinge abduction developing

Mechanism Ask AI

Hinge

- Head moving inside acetabulum → **center of rotation** = center of head
- Part extruded, hinging on lateral acetabular edge → center of rotation shifts to hinge point
  - Axis of rotation → shifted laterally to acetabular edge
  - Point of impingement of femoral head
- On lateral edge of acetabulum
- Abduction → head not brought inside → impinges on lateral edge
- Abduction → head comes away from acetabular floor
  - → **Medial joint space (medial gap) increases with abduction**
  - Hinging point → lateral acetabular edge, not femoral head center
- Head not entering acetabulum → **uncontainable head**

Prerequisites for Containment Ask AI

- **Active disease**
  - Healed patient → containment meaningless; aim is sphericity
- **Indicated patient** → per classification lecture
- **Containable** femoral head → no hinge abduction

True versus False Hinge Abduction Ask AI

Hinge

- Hinge abduction diagnosis → not simple; needs precision
- **True hinge abduction** → implies false hinge abduction exists
- True hinge abduction → lateral edge or extrusion **not reducible**
- Deformed head may appear hinged
  - X-ray → seems certain hinge abduction
  - **Abduction under anesthesia** → prominent part enters acetabulum → reducible
  - → Not true hinge abduction → head containable
- Diagnosis → **examination under anesthesia (EUA)**
- **Arthrography** → may be performed with abduction
- Hinging not reducing + medial joint space widens → true hinge abduction
- Reference → **"Determining hinge abduction in Legg-Calvé-Perthes disease"**
  - Journal of Pediatric Orthopaedics, 2017

	True hinge abduction	False hinge abduction
Extrusion	**Not reducible**	Reducible
Abduction under anesthesia	Prominent part stays out; medial joint space widens	Prominent part enters acetabulum
Head	Uncontainable	Containable (if active stage)

Containable versus Uncontainable Head Ask AI

- **Irreducible hinge abduction** → prominent part not entering with abduction under anesthesia
  - → Uncontainable head
- Head enters → shape irrelevant → **containable head**, if active stage
- **Aspherical congruity** → another contraindication
  - Usually toward healed stage, not active stage
- Active stage → irreducible hinge abduction = contraindication to containment procedures
  - In active stage of Perthes disease
- **Containable head** = active disease + no true hinge abduction + no aspherical congruity
- Aspherical congruity → femur and acetabulum congruent despite deformed shapes
  - Changing femur–acetabulum relationship → creates incongruity

Must-Know

- Containable head = active disease + no true hinge abduction + no aspherical congruity
- Hinge abduction → center of rotation shifts to lateral acetabular edge
- Abduction → medial joint space widens
- True hinge abduction → irreducible on EUA / arthrography with abduction

Femoral and Pelvic Osteotomy Ask AI

- Each procedure performed by many surgeons; each has merits
- No one can claim one is better
  - Many centers → **femoral osteotomy**
  - Many centers → **pelvic osteotomy**
- Abu El Reesh → pelvic osteotomy almost never performed
- Papers support each → neither superior; each has pros and cons
- Currently favored → something different (the shelf)
- Most important prerequisite before any containment → **very good range of motion**
  - Range lacking, especially under anesthesia → head not containable
  - Femur–acetabulum congruity must be preserved

Femoral Varus Osteotomy Ask AI

Varus

- **Femoral varus osteotomy** → performed by many
- Lateral part of femoral head → brought inside acetabulum
- Prevents extrusion and deformation of **lateral pillar** → what matters most
- Disadvantages:
  - **Shortens lower limb**
  - **Abductor insufficiency** → abductor muscles relaxed
  - Remodeling back to valgus → variable with age
  - May need **another osteotomy after healing**
- Aim of second osteotomy → restore **neck-shaft angle**
  - Corrects mechanical deficiency produced by varus

Pelvic Reorientation Osteotomy Ask AI

Salter

- Second option → **pelvic reorientation osteotomy**
- Most famous in Perthes → **Salter osteotomy**
- Salter indications → decreased greatly worldwide
- Pre-COVID pelvic osteotomy course, Switzerland → only current indication for **innominate osteotomy:**
  - Essentially one condition → reported as bladder exstrophy
  - Otherwise not indicated for orthopedic or hip problems
  - Alternatives → more effective, easier, better results, better fixation
- Alternative → **triple pelvic osteotomy**
- Aim of these osteotomies → rotate pelvis
- To cover the femoral head
- Case: Perthes patient → underwent
  - **Salter osteotomy**
  - Very good healing
- Disadvantages of Salter:
  - **Lengthening of lower limb** → not significant
  - Rotation → **increases intra-articular pressure**
  - Raised pressure → theoretically risky to head; may increase deformity
- Lengthening → may help slightly
  - Existing collapse and mild shortening → corrected by pelvic osteotomy
- Never with **limited range of motion** or aspherical congruity

  - → Severe **stiffness**
- Only in hips with very good range of motion

	Femoral varus osteotomy	Salter osteotomy
Mechanism	Lateral head brought inside acetabulum	Pelvis rotated to cover head
Limb length	**Shortening**	Lengthening (not significant; may help)
Other drawbacks	Abductor insufficiency; variable valgus remodeling → second osteotomy	↑ Intra-articular pressure → possible ↑ deformity
Contraindication	—	Limited range of motion; aspherical congruity → stiffness

Pitfall
- Salter in limited range of motion or aspherical congruity → severe stiffness
- Varus osteotomy → shortening + abductor insufficiency
- Very good range of motion → prerequisite before any containment

Shelf Acetabuloplasty Ask AI

- Third option beyond pelvic reorientation and femoral varus osteotomy
  - → **Shelf acetabuloplasty**
- Not a reorientation pelvic osteotomy → **shelf osteotomy**

Concept: Old versus Current Ask AI

Shelf

- Old definition → bone placed above hip joint capsule
  - Theory → **cartilaginous metaplasia of capsule**
  - Capsule supported by bone → capsule turns into cartilage
  - → Acetabulum enlarged; cartilage acts as part of joint
- Theory persisted very long → concept of shelf osteotomy changed
- Current name → **labral support osteotomy**
- Perthes → head goes outside → **shearing force** on edge, on labrum
- **Labrum** → center of acetabular growth and depth increase
- Shearing force → inhibits acetabular growth → **acetabular dysplasia**
  - → Head extrudes further
- Shelf of bone → prevents femoral head shear on labrum
- Acts as labral support
- No cartilaginous metaplasia → **increased rate of growth of acetabulum**
  - → Shelf gradually disappears
- Images: shelf size → incorporated over time
  - Compare initial **center-edge angle** with its later increase
- ↓ Labral shear stresses → acetabular growth stimulated
  - → ↑ Depth, ↑ center-edge angle

  - → Shelf **resorbs** later
- Aim of shelf → not cartilaginous metaplasia under hip capsule
- Literature → currently preferred operation
  - Avoids congruity / incongruity issue
  - Avoids shortening of femoral osteotomy
  - Avoids lengthening of pelvic osteotomy
- About 50 techniques of **shelf acetabuloplasty**
  - Two used at Abu El Reesh

	Old concept	Current concept
Name	Shelf above capsule	**Labral support osteotomy**
Mechanism	Cartilaginous metaplasia of capsule → enlarged acetabulum	↓ Labral shear → ↑ acetabular growth
Fate of shelf	Part of joint	Incorporates, resorbs

Catterall Technique Ask AI

Catterall

- First technique used → **Catterall shelf technique**
- Hip opened → **outer table of iliac bone** exposed
- **U-shaped cut** in outer table → exactly above acetabulum
- Inferior part separated → becomes a **tent**
- Tent opened → packed underneath with **bone graft**
- U lifted up; graft packed under it
- Breaks → **K-wire** inserted for fixation
- Graft underneath → very good result, no particular problems
- Bone graft under tent → provides support
- **Long shelf** not wanted
  - Aim → not bone over hip joint; just labral support
  - 3 cm shelf → **impingement**
- **Elastic recoil** of U (outer table) → prevents graft extrusion
- Fractured case → K-wire fixation, seen on follow-up
- Shelf always **anterolateral**
- Later heals (incorporates)

Spitzy Technique Ask AI

Spitzy

- More recently used → **Spitzy shelf**
  - Easier, very convenient technique
- Originally not described for Perthes → used more in some acetabular dysplasia
- **Tricortical graft** harvested
  - Immature patient → actually bicortical
  - Opened **iliac apophysis** → no cortex there
- Graft → **trapezoid** appearance
- Graft → impacted above acetabulum → also a type of shelf

	Catterall	Spitzy
Bone	U-cut outer iliac table lifted as tent; bone graft beneath	Tricortical (bicortical if immature) trapezoid graft
Placement	Exactly above acetabulum	Impacted above acetabulum
Fixation	**Elastic recoil**; K-wire if fractured	Impaction; no additional fixation
Status	Used many years; excellent results	Current; easier

Personal Protocol Ask AI

Protocol

- Herring **B** or **B/C**, onset > 8 years → indicated for surgery
- Containable, good-looking head → surgery done
- Everything feasible done when operating
- Current choice → **shelf acetabuloplasty**
  - Neither Salter nor varus osteotomy
- Added → **soft tissue release**, essential
  - **Adductor longus**
  - Tendinous part of **iliopsoas**
  - Aim → ↑ abduction range
  - Aim → prevent **flexion deformity**
- Optional addition → **percutaneous drilling of epiphysis**
  - Similar to **core decompression**
  - **Physis (growth plate)** → must not be injured
  - 3 mm K-wire → three drill holes in epiphysis
  - Theoretical aim → ↑ vascularity
  - Only alongside surgery → no indication for drilling alone

Surgical Steps Ask AI

Protocol

- Step 1 → drilling with K-wire
  - Three drill holes, 3 mm K-wire
  - **Smooth wire**, not drill bit → avoids physeal injury
- Step 2 → adductor region opened → **adductor longus** released
- Step 3 → **bikini incision**
  - Two procedures → iliopsoas release + shelf acetabuloplasty
- Abdominal muscle (reportedly external oblique) elevated
  - **Apophysis** found → opened
- Plane developed → **tensor fasciae latae** / sartorius
  - **Lateral cutaneous nerve of thigh** lies in this plane
- **Iliopsoas tendon** delivered → released
- Tendon shown
- Shelf placed under **image intensifier**
  - Exact localization of acetabular edge → shelf directly above it
- **Saw** → trapezoidal piece harvested for shelf
- **Drill holes** exactly above acetabulum
  - Depth increased gradually with burr → about thinnest graft thickness
- Graft placed → impacted with **impactor and hammer**
  - → Very stable in place
- No additional fixation needed
- Long shelf not needed → labral support only

Results Ask AI

Shelf

- Largely maintains head sphericity
- Indicated patients after shelf → 84% **Stulberg III or less**
  - **Full painless range of motion**
  - **Improved acetabular morphology**
- Past Salter cases → some ended with severe **stiffness**
  - Cause → unrecognized **aspherical congruity**
- Shelf advantage → even with some incongruity, acetabulum–head relationship unaltered
- Much safer → fewest complications
- Lowest incidence of **post-operative stiffness**

High Yield

- Shelf = labral support osteotomy, not capsular cartilaginous metaplasia
- ↓ Labral shear → ↑ acetabular growth, ↑ center-edge angle → shelf resorbs
- Avoids femoral shortening, pelvic lengthening, congruity issue
- 84% → Stulberg III or less; lowest post-operative stiffness
- Shelf always anterolateral; long shelf → impingement

Management of the Uncontainable Head Ask AI

- Active disease + hinge abduction → **uncontainable head**
- EUA with abduction, or arthrography → hinge abduction found

  - **Irreducible** → true hinge abduction, not false
  - → Containment not possible
- Options:
  - **Shelf acetabuloplasty**
  - Useful; improves head sphericity → lower success than in containable head
  - **Subtrochanteric valgus osteotomy**
  - Valgus → ↑ distance impinging edge–acetabulum → ↑ abduction range
  - **Chiari osteotomy** → another type of shelf

Valgus Osteotomy Ask AI

Valgus

- Case: hinge abduction → elevation, step, notched head
- No abduction even under anesthesia → **irreducible hinge abduction**
- Treatment → **valgus osteotomy**
- Valgus → ↑ distance prominent lateral edge–acetabulum
  - → No **impingement**
- Spherical part kept inside acetabulum →
  - ↑ Mobility and outcome
  - Abduction range provided
  - Prevents a deformity (likely adduction) *[C1 · transcript: "no abduction deformity"]*
  - ↓ Extrusion
- Final outcome of patient shown
- Good result after valgus osteotomy
- **Varus osteotomy** → contains femoral head
- **Valgus osteotomy** in Perthes → irreducible hinge abduction
  - → Not a containable hip

Chiari Osteotomy Ask AI

Chiari

- **Chiari osteotomy** → type of shelf
- Somewhat larger operation
- **Supra-acetabular** cut → medialization of distal fragment
- vs ordinary shelf (Spitzy, Catterall, Staheli) → one difference only: **medialization**
- Medialization → ↓ stresses
  - ↓ Lever arm of body weight on hip
- Chiari → still a type of shelf

Summary of Active Disease Treatment Ask AI

- Indications for surgery → end of classification lecture
- Today → operation types and aims
- All of the above → **active disease**, not healed stage
- Aim → **maintain sphericity**
- Healed head → not treated as active disease
- Operation → **containment** of head inside acetabulum:
  - **Femoral varus osteotomy**
  - **Salter (pelvic) osteotomy**
  - Salter → reorients acetabulum to cover head

  - **Shelf** → induces acetabular growth; prevents femoral head shear on labrum
- Other operations exist
- Personal standard treatment → **shelf acetabuloplasty with soft tissue release**

Adjunct Procedures: Drilling and Distraction Ask AI

Epiphyseal Drilling Ask AI

Drilling

- **Drilling** added → nothing lost
  - Less than 5 minutes
  - Patient already anesthetized
- No indication to operate solely for core decompression / epiphyseal drilling
- Literature → no clear indication
- Drilling → **improves pain only, not outcome**
  - Femoral head decompressed → ↓ pain
- Anesthesia just for drilling → no evidence of effectiveness
  - Contrary claims → incorrect

Hip Distraction (Arthrodiastasis) Ask AI

Distraction

- External fixator for distraction → much discussed
- Claim → distraction creates negative pressure drawing head in
  - → Entirely illogical
- Syringe analogy: closed nozzle + pull → negative pressure
  - Maintaining it → syringe walls must be **rigid**
- Nylon-bag wall → collapses on suction → space not maintained
- Hip joint → **living structure**
  - Negative pressure → fluid imbibition into acetabulum → pressure neutralized
- Capsule ≠ rigid, impermeable, non-living wall
  - Pressure falls → head does not expand → cannot work
- Real aim of **distraction** → remove stresses from head → prevent collapse
  - Must be done at particular age and time, before any collapse
- Only literature with results → **all height gained lost after fixator removal**
  - Distraction + prolonged immobilization → severe **osteopenia**
  - Osteopenia → bone liable to collapse
- Possible indication → **very painful hip**, no other option
- Practice: Perthes children aged 4, 5, 6 years
  - Told they will undergo **distraction arthroplasty**
  - Rejected → as is any operation at this age
  - No strong, real evidence of surgical indication at this age
- Already healed cases → told containment + distraction arthroplasty
  - Results reported excellent
  - Child left untouched → result even better
- Prognosis very good without intervention
- **Good outcome cannot be referred to the intervention**
- Intervention traumatized the child; without it, outcome possibly better

Pitfall

- Drilling → improves pain only, not outcome; never done alone
- Distraction → all height gained lost after fixator removal
- Distraction + immobilization → osteopenia → collapse
- Healed or young (4–6 years) cases → good outcome expected anyway

Residual Deformities Ask AI

- Active disease concluded → next: **residual deformity**

Case: Treat the Patient, Not the X-ray Ask AI

Case

- Perthes patient:
  - **Short neck**
  - **Overgrown greater trochanter**
  - Shortening of 2 cm
- Common plan → valgus osteotomy + greater trochanter transfer
- Same patient walking → shoe raise on right (short side) only
- Stable **single-limb stance**
- No positive **Trendelenburg test**
- X-ray → radiological problem only
- Before measuring angles and planning → ask the patient's problem
- No previous surgery → what would surgery gain?
- Opening hip, osteotomies → no better gait or stability
- Hip → **mechanically sound** and functionally sound
- Intervention → only harms this joint

Indications for Correction Ask AI

Residual

- **Residual deformity treated when clinically significant**
  - Patient complaint → not X-ray treatment
- Treating X-rays → common, very sad confusion
- **Coxa vara** + related symptoms → valgus osteotomy:
  - Positive Trendelenburg test
  - Abductor insufficiency
  - Hip pain from abductor fatigue
- Valgus osteotomy → always with **lateralization of distal femoral shaft**
- Coxa vara planning → full separate lecture
- Options:
  - **Transfer of greater trochanter**
  - Valgus alone
  - Transfer alone
  - Combination

Must-Know

- Residual deformity → operate only when clinically significant
- Symptomatic coxa vara → valgus osteotomy
- Valgus osteotomy → always with lateralization of distal femoral shaft
- Mechanically sound, painless hip → no intervention despite X-ray

Femoral Head Reduction Osteotomy Ask AI

- Names → **head reduction osteotomy** / femoral head reshaping osteotomy
- Adopted locally → "Egyptian touch"
- Changing femoral head shape → very difficult, very major operation
- Aim → remove **necrotic central part** of femoral head
  - Close again → spherical femoral head
- **Lateral retinacular vessels** preserved → vascularity
- Difficult operation
- Central part → **avascular**
- Lateral part holding on → **vascular**
- After resection → fragment brought back, closed in place

Evidence Ask AI

HRO

- Developed and performed by **Ganz** center, Bern, Switzerland
  - And by **Dror Paley**, America
- Paley's paper → very well written; honest self-critique
- **Siebenrock** (with Ganz, Bern) → head reduction osteotomy + additional containment surgery
- Ganz center → very large **referral center** for hip preservation surgery
- 10 years → only 11 patients suitable
  - vs claims of 15–20 cases a year
- **10 of 11 needed pelvic osteotomy** → logical
  - Smaller head → smaller acetabulum, or changed acetabular congruity / shape

  - Pelvic osteotomy → required in almost all cases
- Conclusion → **radiological appearance improved**
  - **Function of patient not improved**
  - Scoring systems and grading systems → **no significant pre- vs post-operative difference**
- Second paper → **Bone & Joint Journal**, 2018, US center
  - Strict selection criteria for **femoral head reshaping osteotomy**
  - 5 years → 6 hips suitable of 847 hip preservation surgeries (not total hip)
- Indication very rare; selection criteria very strict
- Only in **mature** patients, not immature
- Take-home → **rare but distinct indication** for severe femoral head deformity
- Head reduction osteotomy + **periacetabular osteotomy** → salvage, young (< 20 years):
  - Symptomatic hip *[C2 · transcript: "asymptomatic hip"]*
  - Aspherical oversized femoral head
  - Advanced central hip degeneration
- **Salvage procedure, not reconstructive** → buys time until total hip
  - < 20 years → very young for **total hip replacement**
- Not to be taken lightly
- ~847–850 operations in 5 years → only five suitable (as quoted) *[Q1 · As spoken; same 2018 series quoted as 6 hips earlier, then five here.]*
- Claims of ten cases a year, or for a thesis → unacceptable

	Siebenrock (Ganz center, Bern)	Bone & Joint Journal 2018 (US)
Period	10 years	5 years
Suitable	11 patients	6 of 847 hips
Finding	10/11 needed **pelvic osteotomy**; X-ray improved, function not	Rare, distinct indication; salvage with PAO

Exam Pearl

- Head reduction osteotomy → removes necrotic central part; lateral retinacular vessels preserved
- Almost always needs pelvic osteotomy
- Radiological improvement; no significant functional improvement
- Salvage, not reconstructive → young (< 20 years), mature patients

Overall Summary and Treatment Algorithm Ask AI

- Such claims → misleading
- Deciding who needs surgery → a million times more important than the operation
- **Most Perthes patients do not need surgery**
  - Minority need surgery
- Active stage surgery → Herring B and B/C, onset > 8 years
- Residual deformity surgery → only when **clinically significant**
- Current consensus in pediatric orthopedics:

Herring group	Onset < 8 years	Onset > 8 years
A	No treatment	No treatment
B	No surgery	Treatment: pelvic or femoral osteotomy; preferred shelf + soft tissue release + drilling
B/C	Treated almost like B	Treated almost like B
C	No evidence surgery changes outcome	No evidence surgery changes outcome

- Personal preference → **shelf osteotomy with soft tissue release**
  - + Drilling → patient already open; nothing lost
- Surgical options in active stage:

	No hinge abduction	Hinge abduction
Options	Salter, varus or shelf + drilling	**Valgus osteotomy**, shelf or Chiari osteotomy

- Drilling alone → never done
- **Arthrodiastasis** → very, very questionable option
- Every case, operated or not → maintain range of motion

Questions and Answers Ask AI

Shelf Position Ask AI

Q&A

- Q: shelf anterolateral, not posterior → why?
- A: femoral head almost **over-covered posteriorly**
  - Maximum head–acetabulum support → posterior
- Posterior hip → **bony stability**; well covered
- Anterior hip → deficient
  - Cause → **anteversion of acetabulum** + anteversion of femoral head
  - Needed for flexion → anterior coverage like posterior → no flexion
- Bony stability → provided where needed
- Despite strong posterior stability → commonest **hip dislocation is posterior**
  - Most stress, especially axial loading → directed posteriorly
- → **Extrusion always anterolateral** → part not covered by bone

Aftercare Ask AI

Q&A

- Q: aftercare after surgery?
- A: **short spica cast,** 3 weeks only
  - Aim → prevent pain-related flexion deformity
- After 3 weeks → cast removed → mobility + described exercises
- Parent performs exercises
- **Weight bearing restored at 6 weeks** after surgery
  - First 3 weeks → cast
  - Second 3 weeks → no cast

Shelf Technique Details Ask AI

Q&A

- Q: Catterall U technique, K-wire for broken graft → **wire migration** concern?
- A: does not occur
- Shifted from Catterall → **Spitzy** → easier
- Some → **Staheli**; some → plate; a hundred shelf techniques
- What matters → notion behind the shelf, why it is done
- Catterall → used many years, excellent results
- Tent fractures on lifting → single K-wire very stable
  - Second wire → not needed

Cheilectomy Ask AI

Cheilectomy

- Q: **cheilectomy**?
- A: not preferred
- Lateral portion → actually healthy part; retains vascularity
- **Necrotic part → central**
  - Lateral part → separated from necrotic central part
- Lateral bump removed → vascular part lost
  - → Weak, avascular central part exposed to weight bearing
- Lateral part → preserved
- Alternatives:
  - Active stage → **shelf**; lateral bump may **remodel** (literature)
  - Before healing → head shape improves with shelf osteotomy
  - Or → **valgus osteotomy**
- **Excision of lateral part → not a preferred option now**

Combining Valgus and Shelf Ask AI

Q&A

- Q: hinge abduction → **valgus femoral osteotomy + shelf** at same time?
- A: meaningless → part will not be put inside anyway
  - Achieves nothing
- Hinge abduction → usually near healing stage, not height of active stage
- Shelf → from start, to change shape
- Valgus → existing shape accepted as final outcome
- Moderator → classification questions covered in previous lecture
  - Both lectures uploaded together soon

Indications and Adjuncts Ask AI

Q&A

- Q: 7-year-old, Herring B, no subluxation
  - Containment surgery → very, very extremely rare
- Extrusion → not measured as percentage in Perthes
  - Unlike lateral migration / **Reimers index** in cerebral palsy
- Very good range of motion → maintain it
- Abduction limited, not responding:
  - **Anti-inflammatories** → ↓ inflammation and pain
  - Still no range → **adductor release**
  - Or **Botox** into adductors → maintain range
- Full abduction maintained → no surgery needed
  - Prognosis → **Stulberg I or II** in 70–90%
  - Final part of previous lecture
- Q: routine **soft tissue release**?
  - A: yes, necessarily → ↓ stresses on femoral head
- Q: K-wire removed? A: almost never left; always removed (as transcribed)
- Q: **bisphosphonates** dose, duration? A: no standardization
  - No proof they truly improve or remodel head; dose unknown
- 8-year threshold → relates back to indication
- Q: bone marrow injection as stem cell source? A: **no evidence**
  - Not even in ordinary avascular necrosis in adults
- **Inter-observer reliability** → discussed by Dr. Hisham, previous lecture
- Q: Herring reliability given observer variability?
  - **Herring classification** → high inter- and intra-observer reliability
  - **Catterall classification** → poor inter- and intra-observer reliability

Residual-Deformity Case Revisited Ask AI

Residual

- Q: short neck, overgrown greater trochanter, short limb → valgus osteotomy to stop progression?
- Claim 1 → **Trendelenburg** gait
  - False → no Trendelenburg gait, no limp
- Claim 2 → surgery **prevents progression** → progression of what?
  - Hip **healed**
  - Disliked X-ray shape → normal function
  - **Painless full range of motion**; mechanically sound
  - No **abductor insufficiency**
- Interference → may itself create a problem
- Even uncomplicated → risks such as **infection**
  - Patient not needing surgery pays the price, even if only financially
- Cost → 6–8 weeks in bed, physiotherapy to walk, pain → no benefit
- **Leg length discrepancy** → contralateral epiphysiodesis
  - Two years of growth left; discrepancy 2 cm
  - Walks from next day, no problem
  - 2 cm corrected over those two years

Distraction and Ilizarov: Evidence and Practice Ask AI

Distraction

- Q (regular attendee): why distraction not supported at a higher level?
- A: bring a lecture, strong paper, sound study
  - Showing **distraction arthrodiastasis** changes anything
- Active disease aim → change **natural history**
  - Maintain sphericity → change outcome
- Proof needed → change in **final outcome**
- Many papers → **very weak**
  - Much published in non-genuine journals
  - Strong editorial office claiming very good results
- Series → children aged 4–5 years, **Herring B**
  - Herring B, C, even C < 8 at age 2–4 → poor outcome not expected
  - "Good results in 70%" → meaningless
- Expected anyway → **Stulberg I or II** in 70–90%
- What was offered to the patient?
- **Ilizarov** frame around child's hip:
  - Child cannot sit or sleep
  - Child, parents, whole family suffer
  - Winter → cannot wear clothes, move, or leave home
- Burden of operations and **external fixator** → not underestimated
- Ilizarov indication → **superior results** over any other intervention
  - Same result by other intervention → other one preferred
  - Reason → more comfortable, easier for patient
- Knowing Ilizarov, not knowing the operation → no justification for Ilizarov
- Choosing what one knows for lack of the appropriate technique → unacceptable
- → **Malpractice**
- Cannot perform operation → refer, or learn it
- World uses osteotomies → Perthes by Ilizarov alone → no supporting paper

Technique and Indication Queries Ask AI

Q&A

- Q: reflected head of rectus femoris re-sutured at closure? A: no
- Spitzy, Catterall → no need to cut **reflected head of rectus femoris**
- **Staheli** → reflected head released → graft placed → reflected head fixed back over it
- Not cut at all in Perthes
  - Exception → existing **flexion deformity** needing release (rare)
- Surgical indication in Perthes → very rare
- Q: **adductor contracture** → how to test hinge abduction?
  - No true adductor contracture → abduction limited secondary to pain
  - Confirming true hinge abduction → **examination under anesthesia**
- Abroad → **arthrography** always with abduction
  - Medial dye pooling + true hinging at acetabular edge → confirmed
- Anesthetized + full abduction obtained → no contracture
  - No real shortening of capsule or adductors
- Q: **lateral pillar C** → observe during active stage
  - Until final deformities occur?
- A: Herring C < 8 years → exercises to maintain range of motion
- Maintaining range of motion → possibly the most important task
- Good outcome in many → **maintenance of range of motion**, not merely surgical containment
- **Herring C, onset > 8 years** → some operate
  - Honesty → tell parents: no evidence of true benefit
- B or B/C → evidence: patient benefits; outcome improved
- C → improvement not guaranteed; no true or strong evidence
- Questioner → Jordanian, not Iraqi
- Q: **femoral shortening with varus osteotomy** for containment? A: no
  - Varus already shortens → added shortening → disaster

Closing Comments Ask AI

Q&A

- Moderator → topic questions concluded
- Dr. Hisham's lectures → about fifty viewers next day
- All previous lectures uploaded; series continues next week with Dr. Hisham
- Comment → **"Surgeon's ego is disastrous"**
  - Response → no ego; point is surgeon choosing operation he performs well
  - Choosing only what one knows → patient denied better chance → **malpractice**
- **Coxa magna** → later lecture
  - Coxa magna + acetabulum shaped to it → **aspherical congruity**
  - Acetabulum cannot be reduced; reduced → impingement
- **Activity limitation:**
  - No contact sports, no jumping
  - No ball games → falls and impacts
  - **Swimming** and cycling allowed
  - Total movement ban → child driven mad → redirect to other activities
  - Forbidden → jumping, running, falls → ball games
  - Girls also play football now
- Instructions apply to girls and boys
- Closing remarks

Exam Pearl

- Extrusion anterolateral → shelf anterolateral
- Aftercare → short spica 3 weeks; weight bearing at 6 weeks
- Herring → high observer reliability; Catterall → poor
- Cheilectomy removes vascular lateral part → not preferred
- Herring C → no evidence surgery changes outcome

Lower Limb Deformity Planning - Part I

(Axes, joint orientation lines and malalignment analysis)

Prof Dr Hisham Abdel-Ghani

Watch Original Lecture

Lecture transcript processed with AI assistance.

Contents

↑ Book contents

Introduction to Deformity Planning . . . 199
Scope . . . 199
Course Outline . . . 199
Why Planning Matters . . . 199
Historical Background . . . 199
This Lecture . . . 200
Joint Orientation Lines — Frontal Plane . . . 200
Principle . . . 200
Ankle . . . 200
Knee . . . 200
Hip . . . 201
Hip Line 2 and Functional Coxa Vara . . . 201
Joint Orientation Lines — Sagittal Plane . . . 201
Ankle . . . 201
Knee . . . 202
Summary of Joint Orientation Lines . . . 202
Bone Axes — Anatomical and Mechanical . . . 203
Terminology . . . 203
Mechanical Axis . . . 203
Anatomical Axis . . . 203
Femoral Anatomical Axis and the IM Nail . . . 204
Tibial Axes and Femorotibial Relationship . . . 205
Malalignment Test . . . 205
Measure, Do Not Eyeball . . . 205
The Test . . . 205
Aligned Does Not Mean Normal . . . 206
Frontal Plane Orientation Angles . . . 206
Naming Convention . . . 206
LPFA and LDFA . . . 206
JLCA . . . 207
MPTA . . . 207
LDTA and the Contralateral Limb . . . 207
Mechanical versus Anatomical Femoral Planning . . . 207
Lower Limb Obliquity and Knee Joint Line Design . . . 208
Obliquity to the Ground . . . 208

Normal Knee — Valgus or Varus? . . . 209
Proximal Tibial Varus and the Posterior Condyles . . . 209
Sagittal Plane Axes and Angles . . . 210
Femur . . . 210
Tibia . . . 210
Exercises — Locating the Source of Deformity . . . 211
Exercise 1 . . . 211
Exercise 2 . . . 211
Exercise 3 . . . 211
Exercise 4 . . . 212
Wrap-up and Questions . . . 212
Next Lecture . . . 212
Imaging for Measurement . . . 213
Young Children . . . 213
Long-Leg Film Availability in Egypt . . . 213
Bone Ninja . . . 214
The Workbook . . . 214

Introduction to Deformity Planning Ask AI

Scope Ask AI

- Lecture series → **deformity planning:** planning and correcting deformity
  - Number of lectures not yet fixed
- Not limited to paediatric orthopaedics → essential for every orthopaedic surgeon

Course Outline Ask AI

- Today → **malalignment (MAL)** in the frontal plane
  - Tibia today if time allows; otherwise next time with MAL
- Following sessions
  - **Femoral uniapical** and femoral multiapical deformity
  - **Osteotomy concepts**
  - **Sagittal plane deformity**
  - **Oblique plane deformity**
  - Useful concepts every surgeon should know
- Unfamiliarity with these concepts unacceptable, regardless of subspecialty

Why Planning Matters Ask AI

- Required to perform properly
  - **Arthroplasty**
  - **Osteotomy**
  - **Intramedullary nail** → correct entry point and exit (distal) point
- Small fractures → bone handled as **lines** → correct alignment
- No post-fixation drift into **varus** or valgus
  - Lines and angles known in advance
- Not restricted to paediatric orthopaedics; paediatrics highlighted it

Historical Background Ask AI

- Residency era → many competing planning techniques
  - Each author → a different method
- Foundation → **Dror Paley** and John Herzenberg
  - Sinai Hospital, Baltimore, USA
- Annual **Deformity Planning Course** → still held today
- Lecture material → that course's material
  - Old version, not recent; concepts identical, working material differs
  - Old version → **drawings**
  - Recent version → actual **X-ray images**
- Priority → understand **concepts**; no hand drawing → easier still
- Practice → Paley–Herzenberg **workbook** drawings, designed for drawing and practising
- Later → **digital planning** with recent techniques
  - Apps on Apple Store or Google Play

This Lecture Ask AI

- First chapter of the deformity-planning index
  - **Malalignment and malorientation** of lower limb, frontal plane
- Bone = **lines and angles** only; no "bone shaped like this"

Joint Orientation Lines — Frontal Plane Ask AI

Principle Ask AI

Frontal

- Frontal X-ray (frontal view of lower limb) → bone as lines
  - Plus **joint orientation lines**
- First step → learn to draw joint orientation lines
- Workbook distributed → draw the lines yourself

Ankle Ask AI

Frontal

- Reference → **tibial plafond** (distal articular surface of tibia)
- Two points
  - Lateral part of plafond
  - Medial: **junction of medial malleolus** with articular surface
- Plafond absent or unclear → **dome of the talus**
- Result → **ankle joint orientation line**

Knee Ask AI

Frontal

- **Two joint orientation lines**: femoral and tibial
- Drawing any line → two ways
  - **Two points**
  - **Start point + angle** relative to an already-drawn line
- Principle → bone handled as lines
- Knee joint orientation line → two lines: tibia and femur
- **Tibial line** → deepest points of medial and lateral plateau
  - = proximal tibial (lower) knee joint orientation line
- Femoral line
  - **Most distal points** of medial and lateral femoral condyles
  - = femoral knee joint orientation line
- Two lines → **parallel;** undisputed
  - Not parallel → **abnormality**

Hip Ask AI

Frontal

- Ankle straight → line intuitive, drawn correctly even without training
- Hip **round**, no flat surface
  - Circular → no obvious line across it
- Two lines relate to the hip
  - **Line 1 — neck-shaft angle**
    - Tubular bone centre → point medial + point lateral
    - Bisect → **axis of the neck**
    - Head centre → 2–3 lines across **widest diameter** → intersection
    - **Bone Ninja** and similar apps → centre found easily; shown later
    - Neck-shaft angle → known to all
    - Its value must be known
  - Coxa vara lecture → second relationship affecting hip function: **abductors**

Hip Line 2 and Functional Coxa Vara Ask AI

Hip

- Normal neck-shaft angle + **greater trochanter overgrowth** → lax abductors → not working
- Behaves like coxa vara → **functional coxa vara**
  - Neck-shaft angle intact; **abductor insufficiency**
- Detection → relationship of femoral head to greater trochanter
- **Line 2 — hip joint orientation line**: femoral head centre → greater trochanter tip
  - Detects trochanteric overgrowth even with intact neck-shaft angle

Joint Orientation Lines — Sagittal Plane Ask AI

Ankle Ask AI

Sagittal

- Sagittal lines also drawn distal → proximal
- Line → **anterior and posterior lips** of tibial plafond
- Fallback → **dome of the talus**, anterior-most and posterior-most ends
- Talus → easy on AP view, very difficult on lateral view
  - Sagittal → always use **plafond lips**

Knee Ask AI

Sagittal

- Knee → two lines again
- Knee problem → **ligamentous constraint,** not bony constraint
- Normal distal femur + normal proximal tibia, yet possible
  - **Intra-articular source** of deformity
  - **Ligamentous laxity**
  - **Ligamentous injury**
- These → **source of deformity**
  - Both articular surfaces drawn and related to each other
- Tibia → line **parallel** to proximal tibial articular surface
  - Very easy
- Femur on lateral view → **circular distal femur,** like the hip
  - Drawing ambiguous → different people draw differently
- Agreed **convention** needed
  - Lines not created by nature → defined for planning
  - Analogy → gait **stance / swing phase:** agreed terms, common language
- **Paley and Herzenberg** → lines defined as parameters and angles for sagittal planning
- Femoral convention
  - Physis present → anterior-most and posterior-most tips of **physeal plate**
  - Adult, or physis not drawable
    - **Articular surface–shaft meeting point,** anterior and posterior → draw line

Summary of Joint Orientation Lines Ask AI

Joint	Frontal plane	Sagittal plane
Ankle	Plafond: lateral point + medial malleolus junction; talar dome fallback	Anterior + posterior lips of plafond
Knee — tibia	Deepest points of medial + lateral plateau	Parallel to proximal tibial articular surface
Knee — femur	Most distal points of both condyles	Physeal plate tips; adult → articular surface–shaft junction
Hip	Neck-shaft angle; hip JOL (head centre → greater trochanter tip)	Not drawn

- Hip line in sagittal plane → **no significance,** not used
  - Unused lines not burdened

Bone Axes — Anatomical and Mechanical Ask AI

Terminology Ask AI

- **Axis of the bone** → two types: anatomical and mechanical
- "Mechanical axis" here = **bone axis**, not body axis between the two limbs
- Always complete the phrase → **segment named**
  - Lower limb, femur, tibia, proximal femur, distal tibia
- **Any segment** → both anatomical and mechanical axes drawable
- Whole bone available → both axes drawable
  - Used frequently

Mechanical Axis Ask AI

- Definition → **centre of joint above** to centre of joint below
- Femur → **hip centre** → knee centre
- Partial femur → proximal / distal femoral mechanical axis from **angles** (shown later)
- See bone as lines; aim → **lines aligned**, not bone looking aligned
- Tibia → **proximal articular surface centre** → ankle centre
- Normal limb → femoral and tibial mechanical axes **collinear**
- **Mechanical axis of lower limb** → hip centre → ankle centre
  - Passes through knee centre → **mechanically aligned**
- Normal limb → femoral mechanical axis **collinear** with tibial

Anatomical Axis Ask AI

- Correction on anatomical axis → **anatomical correction**
- Drawable on even a **small bone segment**
  - Mechanical axis hard when only a small segment visible
  - → anatomical axis easy
- Anatomical axis difficult → bone **widened and dilated**, no medulla, not tubular
- **Two methods** → same final result
- Definition → **mid-diaphyseal line**
  - Bisect medulla of any diaphysis → draw line
- Femoral shaft → two points, bisect
  - → **anatomical axis of the femur**

Femoral Anatomical Axis and the IM Nail Ask AI

Femur

- Femoral anatomical axis = **intramedullary nail** axis
  - Nail in normal femur → centre of medulla
- Proximal exit → approximately **piriform fossa**
- Distal exit → **just medial** to knee centre, not at centre
  - Adult → slightly medial to centre of femoral notch
- Distal femoral fracture with IM nail
  - Nail into lateral condyle → **varus**
  - Nail into medial condyle → **valgus**
- Wide medulla → adjust **before locking screws,** even without whole bone
- Target → nail exits **medial to trochlear centre**
  - Or femoral notch, or tibial spine
  - → good **anatomical alignment** without measuring whole femur
- Proximal entry → **piriform fossa** → direct medullary passage
  - Entry too lateral → misses medulla
- **Anatomical axes** parallel intramedullary fixation devices

Clinical Pearl

- Femoral nail entry → piriform fossa
- Distal nail exit → just medial to notch / trochlear centre
- Nail into lateral condyle → varus
- Nail into medial condyle → valgus
- Correct before locking screws

Tibial Axes and Femorotibial Relationship Ask AI

- Tibia → **mid-diaphyseal lines** → anatomical axis; plus mechanical axis
- Exit points at centre of bone / articular surface
  - → frontal plane tibia: **anatomical = mechanical**
  - Tibial angles → no anatomical / mechanical difference
- Femur → axes **differ markedly**
- Femoral and tibial anatomical axes **not collinear**
  - **Valgus angle** ~6° between them
  - Femur → **curved bone** with neck-shaft angle
  - Purpose (gait) → **narrow base of support**

Feature	Mechanical axis	Anatomical axis
Definition	Joint centre above → joint centre below	Mid-diaphyseal line
Easy when	Whole bone visible	Small segment visible
Difficult when	Only small segment visible	Widened, dilated, non-tubular bone
Femur	Hip centre → knee centre	Nail axis; piriform fossa → just medial to knee centre
Tibia (frontal)	= anatomical	= mechanical

Malalignment Test Ask AI

Measure, Do Not Eyeball Ask AI

- X-ray → **never rely on eyes**; findings missed
  - Pen, ruler, protractor → measure the axes
- Now → photograph on iPad → **Bone Ninja**
  - No hand measurement, protractor, ruler, pen or calculation
- First step on any X-ray → **malalignment test**

The Test Ask AI

- Draw **mechanical axis of lower limb**: hip centre → ankle centre
- Through knee centre?
  - Through centre → **no malalignment**
    - No malalignment ≠ normal lower limb
  - Medial to centre → **medial mechanical axis deviation** → genu varum
- **Genu varum** = hip–ankle line passes medial to knee centre
  - Lateral to centre → **lateral mechanical axis deviation** → genu valgum
- Next step (coming lectures) → **source of deformity**
  - Tibial or femoral → exact site within that bone
  - Or joint / articular surface → exact site

Aligned Does Not Mean Normal Ask AI

- Normal mechanical axis, collinear segments → normal limb?
  - Hip–ankle line through knee centre, yet **not normal**
- Compensation patterns
  - **Femoral varus** compensated by tibial valgus
  - **Femoral valgus** compensated by tibial varus
- → **aligned lower limb**
  - Mechanical axis corrected, not normal → **acceptable compensatory deformity**
- Initial alignment drawing → **insufficient**
- Normal limb = **normal axis** + normal joint orientation lines
- Axes and JOLs never judged alone → **related to each other**
  - Normal relationship → aligned limb with normal JOLs
- Axis through knee centre + **abnormal JOL** → aligned, not normal

Pitfall

- Aligned lower limb ≠ normal lower limb
- Femoral varus + tibial valgus → aligned, compensatory
- Normal = normal axis + normal joint orientation lines

Frontal Plane Orientation Angles Ask AI

- Frontal plane: **joint orientation lines** + mechanical axis → angles

Naming Convention Ask AI

Paley

- Memorise angles; else apps (**Multiplier**, Bone Ninja) include reference image
- Two crossing lines → **two angles**
  - Unless 90° → one <90°, one >90°
- Convention → always the angle **expected <90°**
- Abbreviation order
  - **L** lateral / M medial
  - **P** proximal / D distal
  - **F** femoral / T tibial
  - **a** anatomical / m mechanical
  - **A** → angle
- Example: L + P + F + A
  - → **LPFA** (lateral proximal femoral angle), ≈90°
- Almost all angles → 87°–90° → **easy to memorise**

LPFA and LDFA Ask AI

Femur

- Distal angle → specify **mechanical or anatomical** beforehand → LDFA
- **LPFA** → always mechanical
  - Hip JOL vs mechanical axis
  - Anatomical lateral proximal angle >90° → **not measured**
  - Always report angle <90° → unified language, no confusion
- Distal → prefix small **m** or small a
- **mLDFA** = 88° or 87°; up to 90°

JLCA Ask AI

Knee

- Angle between two knee JOLs → **joint line convergence angle**
- Normal 0°; maximum 2°
- >2° → **abnormal** joint line convergence
  - → **intra-articular source** of deformity

MPTA Ask AI

Tibia

- Draw tibial mechanical axis (= anatomical in tibia)
- Parallel articular surfaces → **opposing angles**
  - **mLDFA = MPTA:** 87° or 88° → 87° or 88°
- **mLDFA** → among most important angles; equals MPTA
- No m / a prefix → tibial mechanical = **anatomical**
  - Medial, proximal, tibial, angle

LDTA and the Contralateral Limb Ask AI

Tibia

- Distal tibial articular surface → smaller angle **lateral** → LDTA
  - Slight **valgus;** ≈87°, 88°, 89°, 90°
- Patient's own range → **contralateral normal limb**
  - Contralateral 90° → correct to 90°; 87° → 87°
  - **Both sides abnormal** → reference values
- Tibia → **MPTA and LDTA;** no anatomical / mechanical distinction

Mechanical versus Anatomical Femoral Planning Ask AI

- Femur → **mLDFA,** relative to mechanical axis
- Anatomical planning → proximal smaller angle **medial**
  - **MPFA** → anatomical planning
  - **LPFA** → mechanical planning
- LDFA → m / a must be specified; **both lateral,** both <90°
- **aLDFA** < mLDFA by ~6°, whichever planning used
  - mLDFA 88° → subtract 6 or 7 → 81°
- → femoral **anatomical axis inclined** to mechanical by 6–7°
- Illustration: hip JOL + femoral mechanical axis + **LDFA** (~90°)
  - Anatomical axis → 6–7° less than mechanical angle
- Arithmetic: 6–7° difference → 90° becomes 84°
  - Same 6° separating **mLDFA and aLDFA**
- Tilting red femoral line to anatomical → **angle reduced**
  - 96° → 84°
- Anatomical planning → use **medial angle** (<90°)
- Proximal femur → **no m / a prefix** needed
  - Mechanical → lateral angle; anatomical → medial angle
  - MPFA → **anatomical planning;** LPFA → mechanical planning
- Tibia → same angle either axis; **anatomical = mechanical**

Angle	Site	Lines	Normal
LPFA	Proximal femur	Hip JOL + femoral mechanical axis	≈90°

Angle	Site	Lines	Normal
MPFA	Proximal femur	Hip JOL + femoral anatomical axis	Anatomical planning only
mLDFA	Distal femur	Femoral knee JOL + mechanical axis	87°–88°, up to 90°
aLDFA	Distal femur	Femoral knee JOL + anatomical axis	≈81° (mLDFA – 6–7°)
JLCA	Knee	Femoral vs tibial knee JOL	0°, max 2°
MPTA	Proximal tibia	Tibial knee JOL + tibial axis	87°–88° (= mLDFA)
LDTA	Distal tibia	Ankle JOL + tibial axis	87°–90°

High Yield

- mLDFA 87°–88°, maximum 90°
- MPTA = mLDFA when knee JOLs parallel
- JLCA 0°, maximum 2°; >2° → intra-articular source
- aLDFA = mLDFA – 6–7°
- Femoral–tibial anatomical axes → ~6° valgus

Exam Pearl

- LPFA → mechanical planning
- MPFA → anatomical planning
- LDFA → always needs m or a prefix
- Tibial angles → no prefix; axes coincide

Lower Limb Obliquity and Knee Joint Line Design Ask AI

Obliquity to the Ground Ask AI

- Femoral anatomical axis → oblique to femoral and limb mechanical axis by ~6°
- Limb mechanical axis → **not perpendicular** to ground
  - Inclined 3° to vertical
- Cause → **wide pelvis** + neck-shaft angle + feet together → oblique limbs
- Tibia: shaft–articular surface angle → 87°, not 90°
  - → these are the 3°

Normal Knee — Valgus or Varus? Ask AI

- Recurrent question → normal knee in **valgus or varus**?
- Distal femur anatomical axis vs articular surface → **valgus**
- Proximal tibial angle <90° → **proximal tibial varus**
- Ligament-dependent articular surface → tilted 3° valgus
- Limb axis perpendicular to ground → knee surface **not parallel** to ground → impossible
- Solution → limb **oblique to ground**
  - 3° between vertical line and limb mechanical axis
- Reason → **pelvis wider** than base of support (feet side by side)
  - Walking → limb axis oblique; normal gait along a line
- Obliquity → **knee surface parallel** to ground → best case
- Vertical line vs mechanical axis → 3°
- No **oblique knee articular surface** in nature → parallel to ground
- 3° varus between tibial axis and proximal tibia + limb axis 3° oblique
  - → oblique surface becomes **parallel to ground**

Proximal Tibial Varus and the Posterior Condyles Ask AI

- Second benefit of **proximal tibial varus**
- **Medial femoral condyle** → more prominent posteriorly than lateral
- Posterior condylar line (end-on view) → inclined 3°
- **Total knee arthroplasty** → cuts rotated
  - More removed from **posterior medial condyle** than lateral
  - Purpose → parallel cuts; prosthesis cannot follow natural tilt
- Medial condyle more prominent
  - Lateral less prominent → posterior condylar line **oblique**
- Flexion with tibial surface perpendicular to shaft → leg deviates to **valgus**
  - Femur more prominent posteromedially
- 3° **proximal tibial varus** → compensates 3° posterior condylar difference
  - → leg comes down straight on flexion
- On flexion → 3° **compensation** immediate

Sagittal Plane Axes and Angles Ask AI

- Frontal plane complete; sagittal → **proximal femur** and neck-shaft angle ignored

Femur Ask AI

Sagittal

- Sagittal → always **anatomical axis** (mid-diaphyseal line)
- Normal femur → **anterior bow**
  - Separate proximal and distal anatomical axes → angle between them
- **Hip JOL** → not used in sagittal plane
- Proximal and distal mid-diaphyseal lines → check distal **exit point**
  - Intersects distal femur → **anterior / middle third junction**
- **DCS** or distal femoral plate → positioned along this axis
  - Correct position → plate projection on shaft, neither anterior nor posterior
  - → mid-diaphyseal line passes through this point
- Distal femoral sagittal joint line → intersects at same junction → **PDFA**
  - Angle <90° → **posterior**; ≈83°

Tibia Ask AI

Sagittal

- Tibial mid-diaphyseal line → exits at **anterior fifth of plateau**
- **Plate** → along shaft; too posterior → projects outside
- **Nail entry point** → this point → nail straight in bone
- Proximal tibia → **posterior slope** ~10°
  - → **PPTA** 81° or 80° (or 77°)
  - Exact difference immaterial → measure against normal side
- Still know **normal / acceptable parameters**
- Distal tibia → **posterior lip** extends more distally than anterior
  - Angle <90° → anterior → **ADTA**; ~10° slope (≈80°)
- Tibia between **opposite slopes**
  - Proximal → posterior 10°
  - Distal → anterior 10°
- Axial load → proximal slides **anterior**, distal slides posterior
  - → anterior support proximally, posterior support distally
- Few simple angles → **easy to memorise**

Angle	Site	Normal
PDFA	Distal femur	≈83°
PPTA	Proximal tibia	81° or 80° (or 77°); posterior slope ~10°
ADTA	Distal tibia	≈80°; anterior slope ~10°

Exercises — Locating the Source of Deformity Ask AI

Exercise 1 Ask AI

Ex 1

- Draw lines yourself on the sheet
- Mechanical axis hip → ankle → through **knee centre**?
  - **No** → MAD (mechanical axis deviation)
  - Medial → **genu varum**
- Second step → **source:** femur or tibia
  - Femoral and tibial mechanical axes
  - Femoral and tibial joint orientation lines
- **mLDFA** 94°; normal 87°
  - Never forget 87°–88°; maximum 90°
  - → 7° **distal femoral varus**
- **JLCA** 0° → normal
- **MPTA** 87° → normal
- → abnormality entirely **femoral**
- Exact site within bone → **CORA** (centre of rotation of angulation), later lecture

Pitfall

- Myth: valgus → femoral origin; varus → tibial origin
- No such rule → draw lines, find source
- Attack the source of deformity

Exercise 2 Ask AI

Ex 2

- Varus or valgus? **Do not eyeball** → draw limb mechanical axis
- **Medial MAD** → genu varum
- Source → femoral + tibial mechanical axes; knee, ankle, hip **JOLs**
- **mLDFA** 87°
  - → femur normal
- **JLCA** normal
- Tibial angle 82°; should be 87°, like femur
  - → source **tibial**, not femoral

Exercise 3 Ask AI

Ex 3

- By eye → varum; lines → **severe varum**
- Large MAD → axis almost entirely **outside knee**
- Medial MAD → **genu varum**
- mLDFA 94° vs 87° → **femoral varum**
- Tibia 82° → ~5° **tibial varum**
- → genu varum from **both** femoral and tibial deformity
- Missing either → **undercorrection**
  - Tibia-only correction → residual femoral varum
  - Not solvable by **eyeballing** in theatre

Exercise 4 Ask AI

Ex 4

- Apparent varum → **severe varum** on close look
- mLDFA → normal
- **JLCA** 7°; normal 0–2°
- **Tibial varus** present
- MAD source → tibial varus + **joint-line source** (ligamentous laxity)
  - 7° above normal → increases MAD

Exercise	MAD	mLDFA	JLCA	Tibia	Source
1	Medial → genu varum	94°	0°	MPTA 87°	Femur (7° distal femoral varus)
2	Medial → genu varum	87°	Normal	82°	Tibia
3	Medial, severe	94°	—	82°	Femur + tibia (~5° tibial varum)
4	Severe varum	Normal	7°	Tibial varus	Tibia + ligamentous laxity

Wrap-up and Questions Ask AI

Next Lecture Ask AI

- First lecture complete → repeated next time in a different way
  - For absentees; same content, entirely different slides from those distributed
  - Attendees → **consolidation**
- Next → builds on this; **uniapical deformity,** likely tibial

Imaging for Measurement Ask AI

- Q: standing films, ordinary films or scanogram?
- **Standing films** required
  - JLCA from ligamentous laxity → appears only standing
- Film **exactly frontal** → patella facing anteriorly
  - Malorientation → **false angles** (genu varum / valgum lecture)
- Request → **digital radiograph of lower limb, AP standing**, patella anterior
- Patella not central in **trochlear notch** → do not measure
- **CT scanogram** → length measurement only; not for planning
  - Not **weight bearing**
  - Knee ligamentous laxity → invisible lying in CT
  - Large part of deformity → disappears lying, appears standing
  - **Patella** cannot be fixed exactly anterior in CT
  - Use → **CT scanogram of lower limb** for bone lengths

Feature	Standing digital AP radiograph	CT scanogram
Weight bearing	Yes	No
Use	Deformity planning	Bone length measurement
Ligamentous laxity / JLCA	Visible	Not visible
Patella	Exactly anterior required	Cannot be fixed anterior

Must-Know

- Planning film → digital AP standing, patella facing anterior
- Patella off-centre in trochlear notch → do not measure
- CT scanogram → lengths only, not planning

Young Children Ask AI

- Method → patients **older than 7–8 years**
- Younger → measurement inaccurate; articular surface largely **cartilage**, not bone

Long-Leg Film Availability in Egypt Ask AI

- Q: few Egyptian centres perform standing long-leg films, especially for adults?
- More than 50% of centres → **device available**
- Problem → **technician** positioning
  - Rotational problems: **external tibial torsion**, increased femoral anteversion
  - Untrained → feet together only
  - → wrong
- Instruction → **patellae exactly forward**, not inward or outward
  - Rotate leg until patellae face anterior → then image
- Most centres → **digital radiograph of both lower limbs**, AP standing
  - Routine
  - Difference → skilled vs untrained technicians checking **patellae centred**

Bone Ninja Ask AI

- Measuring program → **Bone Ninja**; not a phone app
  - Apple Store, iPad
- Not on **Google Play;** not downloadable on iPhone
- Separate lecture worthwhile → **simplifies line drawing**

The Workbook Ask AI

- Book → Paley–Herzenberg **workbook,** drawn on in this lecture
- This workbook old; **new workbook** completely different
- New one unavailable; digital version → old workbook → used here

Lower Limb Deformity Planning - Part 2

(Planning corrections and osteotomy geometry)

Prof Dr Hisham Abdel-Ghani

Watch Original Lecture

Lecture transcript processed with AI assistance.

Contents

↑ Book contents

The MAP Approach to Limb Alignment . . . 219
Why Deformity Planning Matters . . . 219
Origin of the Methodology . . . 219
MAP: Measure, Analyze, Pick . . . 220
Bones as Lines: Frontal Plane Bone Axes . . . 220
Two Ways to Draw a Line . . . 220
Mechanical Axis of the Bone . . . 220
Anatomical Axis . . . 221
Clinical Relevance: Femoral Nailing . . . 221
Segmental Axes . . . 221
Mechanical Axis of the Lower Limb . . . 222
Definition and Alignment . . . 222
Measuring the MAD . . . 222
Aligned Is Not Normal . . . 222
Joint Orientation Lines . . . 223
Step Two: Analyze the Angles . . . 223
Ankle . . . 223
Knee and JLCA . . . 223
Hip: Center of Femoral Head and Neck Shaft Angle . . . 223
Greater Trochanter . . . 223
Frontal Plane Angles . . . 224
Convention of Angle Selection . . . 224
Naming the Angles . . . 224
Lateral Distal Femoral Angle (LDFA) . . . 224
Proximal Femoral Angles . . . 225
Tibial Angles . . . 225
Weight-Bearing Axis and Proximal Tibial Varus . . . 226
Valgus or Varus Knee? . . . 226
Why the Proximal Tibia Has 3 Degrees of Varus . . . 226
Full-Length Radiography . . . 226
Standing AP View . . . 226
Measurement in Children . . . 226
Tibiofemoral Angle . . . 227
Case: Effect of Rotation . . . 227
Leg Length Discrepancy . . . 228

Pelvic Obliquity and Foot Height . . . 228
Standard Reference Point and Shoe Lift . . . 228
Flexion Deformity and Apparent Shortening . . . 229
Frontal Plane Summary . . . 229
Sagittal Plane: Hinge Point and Limb Axis . . . 229
Sagittal vs Frontal Plane . . . 229
Knee Hinge Point . . . 230
Mechanical Axis of the Lower Limb . . . 230
Recurvatum and Ligamentous Laxity . . . 230
Sagittal Plane: Bone Axes and Angles . . . 231
Same Steps as the Frontal Plane . . . 231
Modified Mechanical Axis of the Femur . . . 231
Posterior Distal Femoral Angle (PDFA) . . . 231
Tibia . . . 231
Normal Sagittal Values . . . 231
Anatomical Axis . . . 232
Sagittal Plane Steps . . . 232
Anterior Cortical Lines and Sagittal Rules . . . 232
Anterior Cortical Lines . . . 232
Sagittal Plane Rules . . . 233
Worked Examples: Analyze the Angles and Pick the Bone . . . 233
Two Ways to Draw the Tibial Axis . . . 233
No Proximal Diaphysis . . . 233
Pick the Bone: Examples . . . 234
Acceptable vs Unacceptable Correction . . . 235
Next Lecture and Questions . . . 235
MAP and the Coming ABC Lecture . . . 235
Q: Positioning for the Lateral X-ray . . . 235

AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 234) “the femur is at 93” → “MPTA (tibia) is 93”

C2 (p. 234) “medial proximal tibial angle 85” → “mLDFA 85”

Queried - kept as spoken, but looks wrong

Q1 (p. 225) As spoken; LDTA (about 89) matches MPTA (87), not MPFA (about 84); likely slip for MPTA.

The MAP Approach to Limb Alignment Ask AI

Why Deformity Planning Matters Ask AI

- **Limb alignment** and deformity planning → distinguish one surgeon from another
  - How the limb is seen
  - Ability to plan a deformity correction
- Based on measurements, angles and lines — not on feeling
- Previous lecture (frontal plane) covered:
  - **Anatomical axis**
  - Mechanical axis
  - Joint orientation lines
- This lecture → **normal nomenclature** and standard measurements
  - Revision of previous lecture, in more depth
  - Needs simple mathematics and geometry → major difference in practice

Origin of the Methodology Ask AI

- Standards and methodology set by **Dror Paley** and John Herzenberg
  - Herzenberg now holds Paley's place → Baltimore, Maryland, USA
  - Yearly course and **planning workshop** there
- Lecturer attended once, in 2017
- Phone planning apps:
  - **Multiplier** → Android and Apple Store; possible future lecture
  - **Bone Ninja** → Apple Store only, iPad only
    - Not on Google Play; does not work on Android
- Electronic drawing replaces ruler drawing → angles given immediately

MAP: Measure, Analyze, Pick Ask AI

- Two new terms beyond the basic previous lecture
- Every lower extremity x-ray → **MAP**
  - **M** → measure mechanical axis deviation (MAD)
    - Plus check for **leg length discrepancy (LLD)**
  - **A** → analyze the angles
  - **P** → pick the bone
- Sequence:
  - MAD present → **global deformity**
  - Lines of each bone → angles measured → number of deformities
  - Pick the bone to correct
    - Not every deformed bone needs correction
    - One bone may be overcorrected at the expense of another

High Yield

- MAP = Measure MAD (+ LLD) → Analyze angles → Pick the bone
- MAD present → global deformity
- Not every deformed bone must be corrected

Bones as Lines: Frontal Plane Bone Axes Ask AI

Two Ways to Draw a Line Ask AI

- Bone on x-ray → treated as **lines,** not bone shape
- Two methods:
  - **Two points** → line between them
  - **Starting point on a line + angle**
    - Example: middle of red line (intercondylar notch) → 87 degrees laterally
- Line = two points, or starting point on a line + angle

Mechanical Axis of the Bone Ask AI

Frontal

- More advanced; repeated for those absent
- **Mechanical axis of a bone** → straight line joining centers of proximal and distal joints
  - Femur → center of hip → center of knee (**intercondylar notch**)
  - Tibia → **tibial spines** → center of tibial plafond / dome of talus

Anatomical Axis Ask AI

Frontal

- **Anatomical axis** → mid-diaphyseal line
  - Center points of diaphysis at two levels → line
- Tibia:
  - Mechanical axis = **anatomical axis**
  - Difference very small → negligible in calculations
- Femur:
  - Proximal → **piriformis fossa**
  - Distal → about 1 cm **medial** to center of trochlear notch (adults)

	Femur	Tibia
Mechanical axis	Center of hip → intercondylar notch	Tibial spines → plafond / talar dome
Anatomical axis, proximal	Piriformis fossa	Mid-diaphyseal line
Anatomical axis, distal	About 1 cm medial to trochlear notch center	Mid-diaphyseal line
Mechanical vs anatomical	Differ by 6 degrees	Same

Clinical Relevance: Femoral Nailing Ask AI

- **Straight nail** entry point → piriformis fossa
- Supracondylar femoral fracture + **intramedullary nail:**
  - Nail pointing medial to center → correct; angle restored
  - Otherwise → **malalignment**

Segmental Axes Ask AI

- Bone breaks → anatomical and mechanical axes break too
- Any bone segment → own **anatomical and mechanical axes**
  - Mechanical axis sometimes difficult; anatomical axis mostly possible
- Example: distal femur not visible
  - Mid-diaphyseal line of proximal part + of distal part
  - Not collinear → **deformity**
- Mechanical axis of a segment:
  - Distal femur → **joint orientation line** → up at 87 degrees
  - Proximal femur → drawable without seeing distal femur (exercises, coming lectures)
- Bones always seen as relations between lines → anatomical or mechanical axes

> **Clinical Pearl**
> - Straight femoral nail entry → piriformis fossa
> - Distal femoral anatomical axis → about 1 cm medial to trochlear notch center
> - Supracondylar IM nail pointing medial to center → alignment restored

Mechanical Axis of the Lower Limb Ask AI

Definition and Alignment Ask AI

- Axes so far → **frontal plane** only; sagittal plane differs
- **Mechanical axis of the lower limb** → center of hip → center of ankle joint
  - Normally → through center of knee
  - Femoral and tibial mechanical axes → **collinear**
- **Aligned limb** → femoral mechanical axis collinear with tibial
  - Center of hip → center of knee → center of ankle
- Axis exactly through joint centers → "my limb is aligned"; memorize the term

Measuring the MAD Ask AI

- MAP placed at top of every slide → memorized
- First step on any x-ray → line from center of hip to center of ankle
  - Through center of knee or not?
- Shifted medially → **medial MAD**
  - Medial MAD = **genu varum** (definition)
- Ruler technique:
  - Ruler on center of femoral head and center of ankle → **short line** only
  - Purpose → film not cluttered
  - Bone Ninja or similar → not needed; line moved and changed freely
- Limb axis → global picture: **malalignment** / MAD present or not
  - Femoral mechanical axis alone → no problem
- Shifted laterally → **genu valgum** = lateral MAD
  - Again ruler → short line

Aligned Is Not Normal Ask AI

- Normal mechanical axis → normal lower limb? → no
  - Figure: deformed bone, yet hip–ankle line through knee center → **aligned limb, not normal**
- Previous slides → "aligned limb" only
  - No MAD, no **malalignment** → still not called normal
- Normal mechanical axis alone → does not make the limb normal
- **Normal lower limb** = normal mechanical axis + normal joint orientations
  - Joint relations to mechanical / anatomical axis → within normal
  - Figure: joint relations to axes → not normal
- Requirement → normal relation of **joint orientation lines** to mechanical axis
- **Joint orientation angle** → joint orientation line + mechanical or anatomical axis

Exam Pearl

- Genu varum = medial mechanical axis deviation
- Genu valgum = lateral mechanical axis deviation
- Normal lower limb = normal mechanical axis + normal joint orientations
- Normal mechanical axis alone → aligned limb, not normal limb

Joint Orientation Lines Ask AI

Step Two: Analyze the Angles Ask AI

- No malalignment → case not closed
  - Hip–ankle line through knee center → not enough
  - Joint orientation line vs anatomical or mechanical axis → must be measured
- Second step after MAD → **analyze the angles**

Ankle Ask AI

- Line along **dome of the talus** / tibial plafond
- Single joint orientation line

Knee and JLCA Ask AI

- Knee → **two joint orientation lines**
  - Proximal tibia; distal femur
- Purpose → detect **knee joint source of deformity**
- Measured by **joint line convergence angle (JLCA)**
  - Tibial vs femoral joint orientation lines → not more than `2` degrees (parallel)
  - More → **joint source of deformity**

Hip: Center of Femoral Head and Neck Shaft Angle Ask AI

- First step → **center of the hip**
- Methods:
  - **Circle** template
  - Mid of neck + widest point of head → perpendicular / joining centers → center of femoral head
  - **Bone Ninja** → click on line → circle
    - Circle placed on head → center of femoral head given (shown later)
- **Neck shaft angle** → center of neck + center of head → femoral neck axis vs femur
  - Less than normal → coxa vara
  - More than normal → coxa valga

Greater Trochanter Ask AI

- **Overgrowth of the greater trochanter** → abductor insufficiency
  - Possible with neither coxa vara nor coxa valga
- Hip analysis → needs a line related to greater trochanter
  - As in coxa vara lecture: normal neck shaft angle + trochanteric overgrowth
  - → Trochanter addressed (e.g. advanced) → improved **abductor function**
- **Hip joint orientation line** → as described: tip of greater trochanter → center of femoral head

Frontal Plane Angles Ask AI

Convention of Angle Selection Ask AI

- Joint orientation line with normal relation to axis → **normal limb**
  - Aligned limb + abnormal joint → within acceptable range?
- Convention → angles **less than 90** (or 90) chosen → fewer possibilities
- Each angle explained below

Naming the Angles Ask AI

- **Letter code:**

Letter	Meaning
m (small)	Mechanical axis
a (small)	Anatomical axis
L	Lateral
M	Medial
P	Proximal
D	Distal
F	Femur
T	Tibia

- Small m / a at start → refers to mechanical / anatomical axis
- Example → **mLDFA** (mechanical lateral distal femoral angle)
  - Femoral mechanical axis vs femoral joint orientation line

Lateral Distal Femoral Angle (LDFA) Ask AI

- Lateral chosen → less than 90
- **mLDFA** → 87 degrees (range 87 to 90); memorize
- **aLDFA** → about 6 degrees less → 81; lateral again
- Femoral mechanical vs **anatomical axis** → 6 degrees
  - On mechanical → 87; on anatomical → 81
  - Mechanical axis start → center of knee
  - Anatomical axis start → **medial to center of knee**
- Repetition and use → never forgotten

Proximal Femoral Angles Ask AI

- Most important femoral angle → **LDFA**: anatomical 81, mechanical 87
- Hip joint orientation line vs mechanical axis → angle less than 90 mostly lateral
  - → **LPFA** (lateral proximal femoral angle) = mechanical
- Hip joint orientation line vs anatomical axis → angle less than 90 is medial
  - → **MPFA** (medial proximal femoral angle)
  - MPFA → automatically = **anatomical axis**
  - LPFA → automatically = mechanical axis
- Proximal femur → no need to state anatomical or mechanical
  - Lateral → mechanical; medial → anatomical; no mixing
- LDFA → must state **anatomical or mechanical**; both lateral

Tibial Angles Ask AI

- Tibia → anatomical = mechanical → no problem
  - Parallel joint lines + femoral mechanical axis = tibial axis → 87 continues as 87
- Most important tibial angle → **MPTA** (medial proximal tibial angle)
  - Same mechanical or anatomical → no small m or a
- Distal → **LDTA** (lateral distal tibial angle) → 86 to 90; 90 acceptable
- Most important angles in **malalignment test** / angle analysis:
  - LDFA → anatomical or mechanical
  - MPTA → same on either axis
- Distal tibial angle → approximately equal to MPFA, but named LDTA *[Q1 · As spoken; LDTA (about 89) matches MPTA (87), not MPFA (about 84); likely slip for MPTA.]*
- Femur → mechanical and anatomical axes differ by 6 degrees; tibia → **equal**

Angle	Axis	Normal (degrees)
mLDFA	Mechanical	87 (87–90)
aLDFA	Anatomical	81
LPFA	Mechanical	—
MPFA	Anatomical	—
MPTA	Mechanical = anatomical	87
LDTA	Mechanical = anatomical	86–90
JLCA	Femoral vs tibial joint lines	Not more than 2

Must-Know

- mLDFA 87 (87–90); aLDFA 81
- Femur: mechanical vs anatomical axis → 6 degrees; tibia → identical
- LDFA and MPTA → most important frontal angles
- JLCA more than 2 degrees → joint source of deformity
- LPFA → mechanical; MPFA → anatomical

Weight-Bearing Axis and Proximal Tibial Varus Ask AI

Valgus or Varus Knee? Ask AI

- Apparent dilemma:
  - Relative to femur → knee in valgus
  - Relative to tibia → knee in varus
  - Resolution → simple
- Femoral anatomical axis and limb mechanical axis ≠ **mechanical axis of body weight**
  - Body-weight axis vs limb mechanical axis → about 3 degrees
- Cause → wide pelvis + narrow base of support during walking
  - Real limb mechanical axis → **not vertical** to ground
- Lower limb → **oblique to ground** by about 3 degrees
  - → Inclined joint orientation line becomes horizontal (parallel to ground)

Why the Proximal Tibia Has 3 Degrees of Varus Ask AI

- Further advantage → **3 degrees of varus** in tibia (also lecture 3)
  - Back of medial condyle → posteriorly prominent vs lateral condyle
  - → Tibia stays under knee in flexion
- Flexion → medial condyle obliquity = 3-degree obliquity of proximal tibia
  - → Leg not thrown into **valgus**
- **Total knee** → more cut from medial condyle than lateral
  - Tibia cut perpendicular (90-90) → natural configuration not reproduced

Full-Length Radiography Ask AI

Standing AP View Ask AI

- Request → **digital radiograph of both lower limbs, AP, standing**
- Anterior position of limb = anterior position of patella (knee)
- **Patella anterior** → whatever the foot position
  - External tibial torsion → severe out-toeing
  - Internal tibial torsion → marked in-toeing
- First look on any x-ray → patella position
  - Not anterior → x-ray not measurable

Measurement in Children Ask AI

- Only ossified part visible → **articular surfaces** of distal femur and tibia not seen
- **Mechanical axis test** → easy
  - Hip center → ankle center → MAD present?
  - → Genu valgum (lateral MAD) or medial MAD
- Angles:
  - Do not represent true articular surface
  - Not valid before 7–8 years → until **femoral epiphysis** almost takes articular shape
- **Greater trochanter** fully cartilaginous → hip joint orientation line not measurable
  - Coxa vara and similar → other angles (coxa vara lecture)

Tibiofemoral Angle Ask AI

- Solution → femoral axis measured vs tibia
- **Anatomic tibiofemoral angle** or mechanical tibiofemoral angle
- Mechanical → tibia **collinear** with femoral mechanical axis
- Anatomical → 6 degrees valgus accepted
  - Known 6 degrees valgus between femur and tibia
- Normally aligned limb → femoral mechanical axis collinear with tibial

Tibiofemoral angle	Normal
Mechanical	Femoral and tibial axes collinear
Anatomical	6 degrees valgus

Case: Effect of Rotation Ask AI

Case

- Boy with **proximal tibial valgus** → eight-plates → corrected clinically
- First x-ray (outside centre):
  - Mechanical axis drawn
  - **Medial MAD** → apparent overcorrection
  - Patella not visible centrally; boy has external tibial torsion
  - Technician → feet parallel (common habit)
- Rule → **patella facing anterior** = anterior projection of limb, not foot direction
- Repeat x-ray elsewhere:
  - Patella exactly central
  - No genu varum; no **overcorrection**
- Decision on first x-ray → plate moved to other side for non-existent genu varum
- Correct x-ray → plate removed; **correction achieved**

	First x-ray	Repeat x-ray
Positioning	Feet parallel	Patella anterior
Patella	Not central	Center of condyles
Finding	Medial MAD (apparent overcorrection)	No genu varum
Decision	Plate moved to opposite side (wrong)	Plate removed

- Attention to **rotation** → taking, calculating and measuring the x-ray
- **Exact AP view** → patella at center of distal femoral condyles
- Severe malrotation in femur, tibia, or both in opposite directions:
  - Each bone imaged in its own AP position → analyzed separately

Pitfall

- Feet parallel ≠ AP view; patella must face anterior
- Patella not central → x-ray not measurable
- Malrotated x-ray → false MAD → wrong surgical decision
- Children → angles not valid before 7–8 years

Leg Length Discrepancy Ask AI

Pelvic Obliquity and Foot Height Ask AI

- LLD measurement → part of the M of MAP
- Normal bilateral standing → no pelvic **obliquity**
- Normal bone lengths + oblique pelvis → one of:
  - **Flexion deformity**
  - Foot height depression on one side
    - Fibular hemimelia
    - Very severe unilateral flat foot
- **Foot height depression** → adds about 1–2 cm LLD; may be significant
- Whole limb height measured, not bone lengths only
  - Equal to bone lengths → same
  - More than femur + tibia lengths → extra discrepancy **reduced foot height**

Standard Reference Point and Shoe Lift Ask AI

- Measured from **standard reference point**
  - End of film, or a grid line if x-ray on grid lines
  - Object of exactly known measure placed (explained later)
- Example: **shoe lift** of 5 cm → no pelvic obliquity
  - Distance difference zero → shoe lift added → discrepancy = 5 cm
- **LLD equation** → hip-to-reference heights: A – B + any shoe lift

Flexion Deformity and Apparent Shortening Ask AI

- Common error in bone length measurement → **CP hemiplegia**
- Computer measurement → apparent leg shortening; no true shortening
  - Cause → **flexion deformity of the knee**
- Single-plane image of flexed knee → height reduced; femur length not reduced
  - Bone tilted, not perpendicular → height appears less
- Solution → **lateral view** of femur + lateral view of tibia → added → true length
- AP view → false LLD
  - Rays **oblique to the bone**, not perpendicular
  - Measures height seen, not true bone length
- No centre in Egypt measures the lateral view → lecturer measures it himself

Pitfall

- Knee flexion deformity (e.g. CP hemiplegia) → false LLD on AP
- True length → femur and tibia measured on lateral views
- Foot height depression → adds 1–2 cm LLD
- LLD = A – B + shoe lift

Frontal Plane Summary Ask AI

- Frontal plane completed → MAD measurement + angle analysis
- Angles → **joint orientation lines** vs mechanical and anatomical axes
- Take-aways:
  - Correct positioning; correct AP view
  - Correct mechanical and anatomical axes
  - Joint orientation line drawn correctly → normal relation to axes
  - Axes correct alone → **aligned limb**, not normal limb
  - Angles analyzed even when **compensating each other** → acceptable or not

Sagittal Plane: Hinge Point and Limb Axis Ask AI

Sagittal vs Frontal Plane Ask AI

- Sagittal plane → quite different from frontal
  - Frontal → used much more in planning
  - Sagittal → explains many things; useful tricks
- Film → **standing**, full knee extension

Knee Hinge Point Ask AI

- First line drawn → **knee hinge point**; paediatric ≠ adult

	Line	Line
Paediatrics	Physis / physeal scar of distal femur	Posterior cortex of distal femur
Adults	Blumensaat's line (roof of intercondylar notch)	Posterior cortex of distal femur

- **True lateral view** → intersection of the two lines = hinge point
- Adults → no physis, no physeal scar → **Blumensaat's line**

Mechanical Axis of the Lower Limb Ask AI

Sagittal

- Lateral view → **limb mechanical axis** → center of hip → center of ankle
  - Frontal plane → through center of knee
  - Sagittal plane → **anterior to hinge point**
- Anterior → good, no limits; posterior → bad
- Posterior to hinge point → **flexion deformity of the knee**
  - → Sagittal plane deformity directly
- Anterior → sagittal deformity may or may not exist; other factors
- Posterior → something wrong, 100%
- Anterior → may be good or bad
  - → **Further analysis**; axis should be anterior to knee

Recurvatum and Ligamentous Laxity Ask AI

- **Hyperlax** individual → stands in recurvatum; walks without it → normal
- Significant recurvatum → during **stance phase** of walking → problem
- Recurvatum only for **knee locking** while standing → no problem
- Variable standing recurvatum individual **ligamentous laxity**
- Sequence → knee hinge point → limb mechanical axis
  - Bad → clear-cut; good → no limits

High Yield

- Sagittal limb mechanical axis → anterior to knee hinge point
- Posterior → knee flexion deformity; 100% wrong
- Anterior → no limit; deformity still possible
- Hinge point: paediatrics → physis; adults → Blumensaat's line

Sagittal Plane: Bone Axes and Angles Ask AI

Same Steps as the Frontal Plane Ask AI

- Frontal → limb axis → MAD → each bone analyzed
  - Mechanical and anatomical axes of each bone → planning
- Sagittal → same steps; difference → **knee hinge point** defined
- Frontal definition (joint center above → joint center below) → fails in sagittal
  - Center of distal femur in lateral view → undefined
- → **Modified mechanical axis;** definition of mechanical axis unchanged

Modified Mechanical Axis of the Femur Ask AI

- Knee joint orientation line → along physis (child) / **physeal scar** (mature)
  - Absent → two points at **junction of cortex with condyle,** anterior and posterior
- Axis → as described, from center of femoral head
  - → Junction of **anterior third** with posterior two-thirds of this line

Posterior Distal Femoral Angle (PDFA) Ask AI

- Angle less than 90 → posterior → **PDFA**
- About 83–84
  - Less than 90 by about 6 degrees

Tibia Ask AI

Sagittal

- Tibial anatomical / mechanical / modified mechanical axis → simple
- Knee joint orientation line → anterior and posterior ends of **plateau**
- Intersection point → **anterior fifth**
- **Posterior slope** of proximal tibia about 10 degrees → angle 80–81
- Modified mechanical axis use → **detect deformity**
- Exact deformity location and axis → **anatomical axis**

Normal Sagittal Values Ask AI

- Lateral view → **normal values:**

Angle	Normal (degrees)
PDFA (posterior distal femoral)	About 83 (83–84)
PPTA (posterior proximal tibial)	About 80–81
ADTA (anterior distal tibial)	About 80

- Tibia → proximally posterior 80; distally anterior 80
- No **hip joint orientation line** in the sagittal plane

Anatomical Axis Ask AI

Sagittal

- Same as frontal → **mid-diaphyseal line**
- Femur → normal **anterior bow**
  - → Proximal and distal diaphyseal anatomical axes differ
- Femoral mid-diaphyseal line meets knee joint orientation line → junction of anterior third
- AO manual: **DCS** / fixed-angle distal femoral plate
  - Entry → junction of anterior third and middle third
  - Reason → femoral diaphysis axis; further back → exits bone posteriorly
- **Tibial mid-diaphyseal line** → meets joint orientation line at anterior fifth
  - Proximal tibial plate (lateral or medial) → axis at anterior fifth / posterior four-fifths of plateau
  - → Plate projection along tibial shaft
- Nomenclature → PDFA, ADTA; straightforward

	Femur	Tibia
Axis meets knee joint line	Anterior third / posterior two-thirds	Anterior fifth / posterior four-fifths
Implant application	DCS / fixed-angle plate entry point	Proximal tibial plate axis
Normal angle	PDFA 83–84	PPTA 80–81; ADTA 80

Sagittal Plane Steps Ask AI

- **Sequence:**
  - Knee hinge point
  - Limb mechanical axis → anterior to hinge point; posterior → problem
  - Mechanical axis of each bone → deformity present?
  - Deformity → anatomical axis → exact location → planning and osteotomies

Anterior Cortical Lines and Sagittal Rules Ask AI

Anterior Cortical Lines Ask AI

- New concept, absent in frontal plane → **anterior cortical lines**
  - Anterior cortex of distal femur + anterior cortex of tibia → collinear
- Frontal analogue → angle between femoral and tibial joint orientation lines
  - → Knee / **joint source of deformity**
  - Measured by joint line convergence angle
- Sagittal → femoral condyles round → no joint line
  - → Anterior cortical line used instead
- Collinear, or normal **recurvatum** → angulated up to 5 degrees only
- Intersection → always at **joint line**
- Intersection distal or proximal to joint line → something wrong
  - Anterior cortical lines not continuous →
    1. **Anterior subluxation of the tibia** — not just recurvatum
    2. Posterior sloping, procurvatum or recurvatum of the bone itself
- Joint source only → intersection any direction, but **at knee level**

Sagittal Plane Rules Ask AI

- Rules:
  - Limb mechanical axis → **anterior to knee hinge point**
  - Anterior cortical lines → collinear or recurvatum up to 5 degrees
    - Intersection at level of knee joint
  - Problem → mechanical axis vs femoral and tibial proximal and distal
    - Joint orientation lines
  - Deformity → planning; location by **anatomical axis**
- Tibial anatomical axis ≈ **modified mechanical axis** of tibia
- "Modified" → not joint center above → joint center below
  - Femur → junction of anterior third with posterior two-thirds
  - Tibia → anterior fifth

Exam Pearl

- Anterior cortical lines → collinear, or recurvatum up to 5 degrees
- Intersection → always at knee joint line
- Intersection off joint line → subluxation or bony deformity
- Modified mechanical axis → femur anterior third; tibia anterior fifth

Worked Examples: Analyze the Angles and Pick the Bone Ask AI

Two Ways to Draw the Tibial Axis Ask AI

- Goal → identify the deformed bone
- Proximal tibial **mid-diaphyseal line** + distal tibial mid-diaphyseal line
  - Should be a single line
  - Angle between them → deformity
- Alternative → **mechanical axis:**
  - Knee joint orientation line → down at 87 degrees
  - Ankle joint orientation line → up at 84 degrees
  - Same intersection tibial mechanical = anatomical

No Proximal Diaphysis Ask AI

- Proximal tibial segment too short → **no mid-diaphyseal line**
  - Joint orientation line → 87 degrees → mechanical axis of proximal tibia
  - Second line → intersection point
- Proof → **both drawing methods** needed
  - Anatomical-only approach → fails; no proximal segment for mid-diaphyseal line
  - Reverse → joint orientation line → line at 87 degrees medially
  - → **Mechanical axis of proximal tibia,** even if only articular surface remains
- Biggest proof → bone dealt with as **lines,** not bone shape
- Distal line → drawn normally; proximal line → second method
  - No bone segment available proximally
- MAP so far → **measurements** + analyze angles; next → pick the bone

Pick the Bone: Examples Ask AI

- Which bone to correct → key message, **frontal plane**
- Example 1 → **genu varum**
  - mLDFA 94 vs normal 87
  - → 7 degrees varus
  - MPTA 85 on both sides → within normal
  - Deformity → **femoral only**
- Myth → "varum = tibial deformity, valgum = femoral deformity" → erase
  - Rule → look where the **deformity** is
  - Femur corrected; tibia untouched → easy decision
- Example 2 → **mLDFA** 107 vs 87 → 20 degrees femoral varus
  - MPTA 72 → 15 degrees tibial varus
  - Large femoral + large tibial deformity → **both corrected**
- Example 3 → one key lesson
  - MPTA 91 and 93 → valgus; mLDFA 85 *[C1 · transcript: "the femur is at 93"] [C2 · transcript: "medial proximal tibial angle 85"]*
  - Very marked valgus; red mechanical axis → severe shift
  - 85 → only 2 degrees
  - → About 5 degrees **tibial valgus** + 2 degrees femoral valgus
- Options → all in tibia, all in femur, or each bone separately (2 and 5 degrees)

Example	mLDFA	MPTA	Deformity	Decision
1	94 → 7 varus	85 bilaterally (normal)	Femoral only	Correct femur; tibia untouched
2	107 → 20 varus	72 → 15 varus	Large femoral + tibial	Correct both
3	85 → 2 valgus	91 / 93 → about 5 valgus	Femoral + tibial valgus	See acceptable options

Acceptable vs Unacceptable Correction Ask AI

- Completely unacceptable → any **residual tibial varus**
  - Correcting 6 degrees tibial valgus + 3 or 2 femoral degrees in tibia → tibia into varus
- Proximal tibial varus → body weight axis **medial to knee**
  - → Force pushes femur to slide medially if **medial slope** in proximal tibia
  - → Not acceptable
- Never end with **valgus femur + varus tibia**
- Allowed → corrected or **overcorrected tibia** → valgus tibia + varus femur
  - Small tibial + femoral genu valgum → correct from femur → femur into some varus
  - LDFA 95 instead of 87 → compensates **tibial valgus**
- Limits → about 7 degrees, not more
- Pick the bone → overcorrect one bone → **one osteotomy** instead of two

Pitfall

- Any residual tibial varus → unacceptable
- Never end with valgus femur + varus tibia
- Valgus tibia + varus femur → acceptable; limit about 7 degrees
- "Varum = tibial, valgum = femoral" → false; locate the deformity

Next Lecture and Questions Ask AI

MAP and the Coming ABC Lecture Ask AI

- Today → **MAP** only: Measure, Analyze, Pick
- Next lecture → **ABC:**
  - Apex of the deformity
  - Bone cut
  - Correction
- Content → **osteotomy rules**; angulation and angulation-translation osteotomy
- "**Failure to plan** is planning to fail"

Q: Positioning for the Lateral X-ray Ask AI

- Required appearance → **medial femoral condyle posterior** to lateral condyle
  - Condyles superimposed as one → patient in external rotation
- Fibula → as described, over posterior half of tibia
- Patient **standing**, legs apart
  - Imaged leg forward; other leg back; weight borne on it

Lower Limb Deformity Planning - Part 3

(Applying deformity planning to cases and fixation choices)

Prof Dr Hisham Abdel-Ghani

Watch Original Lecture

Lecture transcript processed with AI assistance.

Contents

↑ Book contents

Deformity Planning Framework: MAP Recap . . . 240
MAP Steps . . . 240
Approach to the X-ray . . . 240
ABC Steps: Apex and Bone Cut . . . 240
ABC Sequence . . . 240
Finding the Apex . . . 241
Choosing the Bone Cut Site . . . 241
Correction Level: Transverse Bisector Line . . . 241
Definition . . . 241
Origin of the Concept . . . 242
Axis of Rotation, Not Point . . . 242
Osteotomy Types and the Axis of Correction . . . 242
Common Osteotomies . . . 242
Axis of Correction of Angulation . . . 243
Angulation Versus Angulation Plus Translation . . . 243
Open Wedge: ACA on Convex Cortex . . . 243
Closing Wedge: ACA on Concave Border . . . 243
Osteotomy Away from the Apex . . . 243
Distance from Apex and Amount of Translation . . . 244
Rules . . . 244
Example: Mid-Shaft Tibial Deformity . . . 244
Dome Osteotomy . . . 245
Proximal Tibial Deformity at the Physis . . . 245
Centre of the Dome . . . 245
External Fixator Hinges and the ACA . . . 246
Hinge Placement . . . 246
Planning Sequence Summary . . . 246
Uniapical Tibial Deformity . . . 247
Example 1: Mid-Shaft Deformity . . . 247
Example 2: Proximal Tibial Varus Near the Knee . . . 248
Example 3: Distal Tibial Deformity . . . 248
Example 4: Femoral Varus + Distal Tibial Valgus . . . 249
Key Rule: Single-Level Deformity . . . 249
Multiapical Tibial Deformity . . . 250
Example 1: Proximal Varus + Mid-Shaft Valgus . . . 250

Example 2: Intersection at Far Cortex 250
Resolution CORA Versus True CORAs 251
Why It Matters 251
Resolution (Summation) CORA 251
Markers of Multiapical Deformity 251
Multiapical Tibial Deformity: Further Examples 253
Example 3: Mid-Shaft Valgus + Distal Varus 253
Osteotomy for Each CORA 253
Example 4: Proximal Varus + Mid-Shaft Valgus 253
Summary: Uniapical Versus Multiapical 254
Recommended Reading and Practice 254
Questions and Answers 255
Open Versus Closing Wedge 255
Degree of Correction 255

AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

- **C1** (p. 248) “30° made more MAD than 13°” → “13° made more MAD than 30°”
- **C2** (p. 248) “82° or 90° distal angle” → “89° or 90° distal angle”
- **C3** (p. 248) “12° in previous knee example” → “13° in previous knee example”

Deformity Planning Framework: MAP Recap Ask AI

MAP Steps Ask AI

MAP

- Continuation of previous lecture on **deformity planning**
- **MAP** — written at top of every slide
  - M → measure **mechanical axis deviation (MAD)**
  - A → **analyze the angles**
  - P → **pick the point** to work upon
- Slide design → **John Herzenberg**
  - X-rays and drawings from Herzenberg's book
  - And from his yearly course in **Baltimore**

Approach to the X-ray Ask AI

MAP

- Step 1 → measure **mechanical axis deviation**
  - Any abnormality → MAD present
  - No MAD ≠ normal limb
- Step 2 → **mechanical axis of each bone** + its joint orientation line
  - Abnormal relation → that bone carries the deformity
- Step 3 → analyze the angles
  - Mechanical and anatomical axes of **proximal fragment**
  - Mechanical and anatomical axes of **distal fragment**
- Result → exact site of deformity
  - **Single bone** → correct at deformity site
  - More than one bone / site → **over-correct** one bone to compensate the other
- Acceptable scenario → no residual **proximal tibial varus**

High Yield

- Never accept **proximal tibial varus**
- Acceptable → distal femoral varus + **proximal tibial valgus**
- Reverse combination → completely unacceptable

- Sequence → bone with problem → site within bone → **bone cut** site

ABC Steps: Apex and Bone Cut Ask AI

ABC Sequence Ask AI

ABC

- A → **determination of the apex**
- B → **bone cut**
  - Not always at apex of deformity
- C → **correction**

Finding the Apex Ask AI

Apex

- Example: visible **mid-shaft tibial deformity**
  - Draw **joint orientation line** of proximal and distal tibia
  - Draw mechanical or anatomical axis of tibia
  - Tibia → **mechanical axis = anatomical axis,** proximal and distal segments
  - **MPTA** = 87° → normal
  - **LDTA** = 90° → normal
  - Only abnormality → **visible deformity** = apex
- Apex = **CORA (center of rotation of angulation)**
  - Point around which deformity has rotated
- Every bone → **proximal axis** + distal axis
  - Middle segment → may have its own axis
- Proximal + distal axes intersect → angle
  - Angle carries the **transverse bisector line** → basis of correction level

Choosing the Bone Cut Site Ask AI

Bone cut

- Deformity centre **in the physis** of growing child → no cut through physis
- Cut site governed by
  - **Biological factor**
  - **Anatomical factor**
  - **Fixation factor**
  - Others → e.g., desired site unsuitable
- **Mechanical** limits → osteotomy not feasible at some sites
  - E.g., skin of region unsuitable for opening and cutting
  - → forced to cut elsewhere
- Cutting at apex → not always possible
- Rule → osteotomy **as close as possible to the apex** whenever feasible

Correction Level: Transverse Bisector Line Ask AI

Definition Ask AI

TBL

- C → draw the **correction level**
- Correction level = **transverse bisector line**
  - Line bisecting the **obtuse angle** at the apex
- Every apex → **longitudinal bisector line** + transverse bisector line
  - Work on the transverse one

Origin of the Concept Ask AI

TBL

- Lines drawn on bones → **human conventions**
  - Surgeon 1 → **centre (mid-diaphyseal) line** as axis
  - Surgeon 2 → **medial border** of tibia
  - Surgeon 3 → **lateral border** of tibia
- Each osteotomizes at own apex → all three correct the tibia
  - Line choice irrelevant → question is what unites them
- Line drawn entirely **outside the tibia** → same deformity angle, same CORA degree
  - Cutting + rotating around it → still corrects
- Apex of every line (blue, red, green) → lies on **transverse bisector line**
  - → transverse bisector line = **multiple CORAs**
- Correction → rotation around axis **perpendicular to any point of the CORA** line
- Transverse bisector line = **correction line**
  - Rotation around any point on it → bone corrected
- Axis of correction → **need not lie inside the bone**

Axis of Rotation, Not Point Ask AI

ACA

- Correction → rotation around an **axis** (a line), not a point
- Door analogy → rotation around **axis line** through all hinges
  - Door axis → line perpendicular to a point
- Bone correction (close, angulate, manipulate) → needs a **hinge**
  - Hinge perpendicular to transverse bisector line
- Axis → **perpendicular to any point** of transverse bisector line

Osteotomy Types and the Axis of Correction Ask AI

Common Osteotomies Ask AI

Osteotomy

- **Open wedge**
- **Closing wedge**
- **Dome osteotomy**

Osteotomy	Axis of rotation	Movement
Open wedge	Apex of triangle	Angulation only, or angulation + translation
Closing wedge	Apex of triangle	Angulation only, or angulation + translation
Dome	Centre of curved cut	**Obligatory** angulation + translation

- Open / close alone → **angulation only**
- Open / close + translate → **angulation + translation**
- Dome → cuts a curve, rotates along it
  - Never angulation alone → **obligatory translation** component

Axis of Correction of Angulation Ask AI

ACA

- Any osteotomy → rotates around **axis of correction of angulation (ACA)**
- Axis of osteotomy = ACA → must be known when cutting
- Open or closing wedge, angulation only → ACA at **wedge apex**
- Translation added → ACA no longer at wedge apex
- Apex → centre of rotation of the bone
- ACA → **perpendicular to any point** of transverse bisector line
  - Any CORA on it → represents apex of deformity
- ACA on transverse bisector line → rotation corrects deformity
- Key question → **angulation only** vs angulation + translation

Angulation Versus Angulation Plus Translation Ask AI

Open Wedge: ACA on Convex Cortex Ask AI

Open wedge

- Bone with proximal + distal parts, visible apex
  1) Draw proximal and distal segment axes
  2) Draw transverse bisector line at apex
  3) Osteotomy → **red point = ACA**, the rotation centre
- ACA → on transverse bisector line, at apex, at **far cortex** (convex side)
- Rotation → **open wedge** → perfect correction
- Open wedge ACA → **apex of wedge** (apex of triangle)
- Translation → not needed

Closing Wedge: ACA on Concave Border Ask AI

Closing

- ACA on **concave border** → **closing wedge**
  - Rotation axis → apex of removed wedge, on concave side
- Apex on transverse bisector line → wedge removed → alignment corrected
- Key → ACA on a point of **transverse bisector line**

Osteotomy Away from the Apex Ask AI

Translation

- Intuition → grasp bent bone at apex to straighten
- Apex not usable when
  - **Physis**
  - **Scarring**
  - Space needed for a particular fixator
- → cut **below the deformity**, e.g., open wedge
- Rotation around **wedge apex** → not on transverse bisector line
  - Angulation only → axis not perpendicular to any line point
- → unacceptable; ACA wrongly positioned
- Fix → angulation **+ translation**
- After translation → ACA on CORA point, no longer wedge apex
- Translation → like a **compass** swing around circle centred on CORA

Distance from Apex and Amount of Translation Ask AI

Translation

- Osteotomy further from CORA → **more translation**
- Closer to apex → less translation
- Direction of translation → watch bone movement → draw circle → check direction
- Stay close to apex → less translation
- Too far → translation may exceed **bone diameter**
  - → bone ends **off-ended** → insufficient contact for union
- With translation → ACA on transverse bisector line (correction line)

Rules Ask AI

Translation

	Angulation only	Angulation + translation
Osteotomy	Passes through a point of transverse bisector line	Passes through no point of transverse bisector line
Wedge apex (= ACA without translation)	On transverse bisector line	Off transverse bisector line
Correction	Rotation perpendicular to CORA → corrected	Translation added → rotation around line point

Must-Know

- Osteotomy through any line point → **angulation only**
- Any point on transverse bisector line will do
- Osteotomy off the line → angulation **+ translation**

Example: Mid-Shaft Tibial Deformity Ask AI

Tibia

- **Mid-shaft tibial deformity** ≈ 30°
  - Draw transverse bisector line → rotate around any of its points
  - Cut + ACA at **convex border** → open wedge → sufficient
  - Osteotomy along the line → **angulation only**
  - Rotation around triangle apex = ACA at deformity apex
  - ACA on **concave border** → closing wedge, not open wedge

Dome Osteotomy Ask AI

Proximal Tibial Deformity at the Physis Ask AI

Dome

- **Very proximal tibial deformity** → exactly at physis level
  - CORA inside physis → closing wedge there impossible
  - Cut there → **physeal damage**
  - Osteotomy placed elsewhere → correction possible
- Lecturer's early dome osteotomy experience
  - Leg corrected, looked very good
  - Patient → persistent **medial joint pain** after surgery
  - Fault → surgical, not patient exaggeration
- **Angry face** vs smiling face dome → depends on bone site + deformity site
- Old teaching → "angry face" dome in proximal tibia

Centre of the Dome Ask AI

Dome

- Dome rotates around **red point**
  - = circle centre → draw circle before "angry face" cut
- Angry face here → red point (ACA) not on transverse bisector line
  - → rotation **opposite** to required direction
- Result → **severe mechanical axis deviation**
- Intraoperative axis check → cable held as mechanical axis (diathermy cable)
  - Aimed from target point → forced **over-valgus**
  - → **valgus stress on ankle joint**
  - Over-correction needed → hip centre → knee centre → ankle centre
- Correcting only required 37° → axis misses ankle centre
  - Line femoral head centre → ankle centre → residual **medial MAD**
  - Angulation fully corrected → **shift** remains
  - → medial joint pain despite full angular correction
- Correct approach → circle drawn **around the CORA** → then site osteotomy
  - Proximal tibia, this case → **smiling face**
  - Same configuration in distal femur → angry face
- Angry vs smiling irrelevant → circle of rotation is what matters
- Circle centre = **CORA centre** → smiley face here
- → mechanical axis properly restored
- Alternative → open / closing wedge + translation, same principle
- Dome advantage → **obligatory angulation + translation**
  - Correct circle → correct osteotomy shape
  - → correct rotation + translation direction

Pitfall

- Dome circle not centred on **CORA** → rotation in wrong direction
- → residual medial MAD + medial joint pain
- Forced over-correction → over-valgus + **ankle valgus stress**

External Fixator Hinges and the ACA Ask AI

Hinge Placement Ask AI

Ex-fix

- Proximal construct → **perpendicular to proximal tibial axis**
- Distal construct → perpendicular to distal tibial axis
- Usually **two hinges** → together = ACA
  - Line between hinges → perpendicular to any point of transverse bisector line

Hinge (ACA) position	Result
Far cortex	Regular **open wedge**
Far from convex cortex, outside bone, on transverse bisector line	Angulation corrected + **lengthening**

- Hinges at far cortex → regular open wedge
  - Same as without fixator
- Hinges **far from convex cortex** → outside bone
  - Parallel lines → same angle, all linked via transverse bisector line
  - ACA perpendicular to line point, away from convex cortex
  - → angulation corrected + **lengthening**
  - Wedge apex no longer at convex cortex border
  - Gap → **part of a trapezoid**, not a triangle

Planning Sequence Summary Ask AI

- Previous lecture → measure MAD, analyze angles, pick the point
- This lecture → apex, bone cut, axis of correction, correction mode
- Correction mode → angulation only vs **angulation + translation**
- Sequence
  1) **Determine the apex**
  2) Determine **bone cut level**
  3) Determine **correction level** → CORA / transverse bisector line / any point on it
  4) **Osteotomy** as close as possible to CORA
- Rotation → around CORA or any point of transverse bisector line
- Rotation around point off the line → **error**

Uniapical Tibial Deformity Ask AI

Example 1: Mid-Shaft Deformity Ask AI

Uniapical

- Examples → lessons for analysing any deformity
- **Uniapical** → single deformity only; not multiapical
- Whole-limb **mechanical axis** → medial MAD → varus deformity
- **Visible deformity** in tibia
- Femur → femoral joint orientation line
  - **mLDFA** = 87° → normal; number to memorise
- Next → mechanical axis of proximal tibia
  - From joint orientation line at **MPTA** 87°
  - = extension of same red line
  - Preceded by tibial mechanical axis
- Tibial mechanical axis → **MPTA** = 72°
- Distal tibial angle → also deviated
- → deformity in tibia, nowhere else
- Mechanical axis + joint orientation line first → spot **deformed bone**
- Then mechanical axes of **proximal tibia** and distal tibia
  - 90° distally, 87° proximally
- Distinguishing features
  1) CORA at **visible deformity**
  2) Lines lie
  - in **center of bone**
- Intersection in center of bone + at visible deformity → **single CORA**
  - = single-level, uniapical tibial deformity
- Alternative → proximal + distal **mid-diaphyseal lines** → same angle
  - MPTA normal, LDTA normal
  - Only abnormality → angle in center of bone, at visible deformity

Example 2: Proximal Tibial Varus Near the Knee Ask AI

Uniapical

- Mechanical axis → **very severe MAD**
- Femoral and tibial mechanical axes → both appear abnormal
- Femoral axis → **not extended** as proximal tibial axis
  - Instead → 88° or 87° from joint line centre
- Femur → **normal** on second step
- Tibia → clearly abnormal
- Tibial mechanical axis → through bone centre; shaft normal
- **MPTA** = 74° instead of 87° → 13° varus
- 13° varus → large MAD
  - More MAD than 30° of previous image *[C1 · transcript: "30° made more MAD than 13°"]*
- **Deformity near joint** → greater MAD
- Proximal tibial mechanical axis → from **joint orientation line** centre at 87°
  - ≈ extension of normal femoral mechanical axis
- CORA → **very proximal**
  - At joint line
  - Or in proximal tibial physis in a child
- Alternative → **mid-diaphyseal line** + angle drawn here
  - + second line at 87° → CORA at intersection

Example 3: Distal Tibial Deformity Ask AI

Uniapical

- Very obvious **visible deformity** in distal tibia
- → **no MAD**
  - Same degree slightly more proximal → severe MAD
- Distance from knee → determines MAD
  - Further from knee → **less MAD**
  - Closer to knee → more MAD
- Tibial mechanical axis abnormal → problem in tibia
- M done → A: **analysis of the angles**
  - Draw proximal tibial mechanical axis
  - Draw **distal tibial joint orientation line**
  - Distal tibial mechanical axis → at 89° or 90° from it *[C2 · transcript: "82° or 90° distal angle"]*
- Deformity here ≈ 40° (figure as heard)
  - → **no MAD**
  - vs 13° near knee (example 2) → considerable MAD *[C3 · transcript: "12° in previous knee example"]*
- Deformity far from knee (near ankle) → no severe MAD
- All previous examples → lines intersect **in center of bone**
  - At level of visible deformity
  - Bone not rounded
  - → **single-level deformity**

Example 4: Femoral Varus + Distal Tibial Valgus Ask AI

Uniapical

- **Medial MAD**
- Distal tibia → **valgus** 16°
- Femur → **varus** 6°
- 6° outweighed 16° → **medial MAD**
  - 16° far from knee → no MAD
  - 6° close to knee → produced the MAD
- Medial MAD ≠ **single deformity** responsible
  - Another deformity may coexist

Key Rule: Single-Level Deformity Ask AI

Uniapical

Example	Deformity	Site	MAD
Previous image	30°	Mid-shaft tibia	Less than 13° near knee
2	13° varus (MPTA 74°)	Proximal tibia, near knee	Large
3	≈40°	Distal tibia	None
4	6° varus vs 16° valgus	Femur vs distal tibia	Medial, from the 6°

High Yield

- CORA at visible deformity + center of bone → **single-level deformity**
- Deformity near knee → **more MAD**
- Deformity far from knee → little or no MAD

- Next → **tibial multiapical deformity**
  - Tibia chosen → simple, useful lessons

Multiapical Tibial Deformity Ask AI

Example 1: Proximal Varus + Mid-Shaft Valgus Ask AI

Multiapical

- Mechanical axis → **mild lateral MAD**
- Obvious **valgus of mid-shaft tibia** → why no severe MAD?
- Femoral mechanical axis → normal (87°)
- Tibial mechanical axis → abnormal
- Normal femoral axis → **extended** into tibia → 87°
- Distally → **joint orientation line** + distal tibial mechanical axis
- Proximal + distal tibial axes intersect → **not at visible deformity**
- Uniapical criterion → CORA at visible deformity, in center of bone
  - Here → in center of bone
  - Not at visible deformity → **more than one CORA** + middle segment
- Middle segment → **anatomical axis**
- Proximal segment (metaphysis) → no mid-diaphyseal line → **mechanical axis**
  - Joint orientation line → 87° medial angle → metaphyseal mechanical axis
  - Mechanical = anatomical axis in this segment → **green line**
- → **two CORAs**
- Varus higher, nearer knee → valgus effect small despite larger magnitude
- → **varus in proximal tibia**
- Mid-shaft → 14° **valgus**
- Proximal → 75° vs normal 87° → ≈12° **varus**
- Take the radiograph → draw → measure
  - Correct varus only → **genu valgum**
  - Correct valgus only → **genu varum**
- Intersection not at visible deformity → **more than one CORA**
- **Rounded bone** → more than one CORA immediately

Example 2: Intersection at Far Cortex Ask AI

Multiapical

- Whole-limb mechanical axis → **medial MAD**
- Spot deformed bone(s) → femoral, tibial
- Method → **mechanical axis of each bone** vs joint orientation line
- Tibia → joint orientation lines proximal + distal
  - 87° proximally, ≈90° distally
  - = normal values
- Single CORA → lines intersect **in center of bone**
- Here → intersection at **far cortex**
  - May intersect outside the bone
  - → more than a single CORA

Resolution CORA Versus True CORAs Ask AI

Why It Matters Ask AI

IM nail

- **Single osteotomy** → mechanical correction possible
  - Roundness persists → no **intramedullary nail**
- IM nail → bone restored **anatomically**
- No IM nail after osteotomy with **translation**
- IM nail after anatomical correction → **every CORA** corrected → straight bone
- Applies to **weak bone** / osteopenic bone

Resolution (Summation) CORA Ask AI

CORA

- These cases → **anatomical** not mechanical correction
  - Several CORAs → **several osteotomies** → intramedullary fixation
- Otherwise → correction from **summation CORA** (combined) → sufficient
- Combined CORA = **resolution CORA** = summation CORA
  - Usable for correction
- Finding CORA count → connect point on proximal **mid-diaphyseal line** with point on distal
- Correction from resolution CORA → **curved portion** remains
  - → intramedullary device not inserted well
  - Unless **very thin** intramedullary device
- Alternative → mid-diaphyseal line of any **straight cortex** portion
  - + mechanical axis above and below
  - → shows the **true CORAs** for correction
- Two CORAs → relevant for **intramedullary fixation**
  - Otherwise → one CORA, single osteotomy
  - Anatomical correction → **two osteotomies**

Markers of Multiapical Deformity Ask AI

Multiapical

- Uniapical → lines intersect **in center of bone**
- Markers of **multiapical deformity**
  1) Intersection not at visible deformity
  2) Intersection not in center of bone
- That intersection = **resolution CORA**
- True CORAs → point on blue mid-diaphyseal line + point on red → **green line**
  - → **two CORAs**
  - Blue, red, green → all mid-diaphyseal
- Both CORAs corrected → lines straight → **anatomical** restoration

	Resolution CORA	True CORAs
Osteotomies	Single	One at each CORA
Restores	Normal mechanical axis; normal joint orientation lines	Anatomical, straight bone
Alignment	Ankle centre → knee centre line; normal knee–ankle joint line angle	Same, plus straight shaft

	Resolution CORA	True CORAs
Residual	Curved portion	None
Intramedullary nail	Not possible (except very thin device)	Possible

- Multiapical + intramedullary device → **osteotomy at each CORA**
- Indications for multiple osteotomies + intramedullary device
  1) **Osteogenesis imperfecta**
  2) Osteopenic bone
  3) **Vitamin D–resistant rickets**
  4) Geriatric, very osteoporotic bone
- Intramedullary device → find true CORAs → cut at each → **anatomically normal** bone
- Incompatible with intramedullary device
  - Osteotomy with **translation**
  - Curved portion left + cut below it for mechanical correction

Pitfall

- Correction from **resolution CORA** → curve persists → nail cannot pass
- Translation osteotomy → intramedullary device cannot pass

- **Anatomical planning** → tibia ideal
  - Anatomical + mechanical axes both work
- Curved segment
  - Middle segment → **mid-diaphyseal line** of curved bone
  - Proximal and distal parts → mechanical axes
  - → two CORAs appear immediately

Multiapical Tibial Deformity: Further Examples Ask AI

Example 3: Mid-Shaft Valgus + Distal Varus Ask AI

Multiapical

- **Mechanical axis deviation** + visible mid-shaft tibial valgus
- Femur → **normal**
  - Mechanical axis at 87° to joint orientation line
- Tibial mechanical axis → **abnormal**
- Analysis → femoral mechanical axis extended as tibial axis → 87°
  - Along **mid-diaphyseal line**; angle normal
- Distally → joint orientation line + line at normal 90°
- Lines far apart → **not intersecting**
- Distal mid-diaphyseal line vs joint orientation line → 122° instead of 90°
  - → 32° **varus in distal tibia**
- Red mechanical axis + blue **mechanical axis of distal tibial articular surface** → not meeting
- Proximal + distal tibial axes not meeting → **multiple CORAs** → multiapical
- Markers → any one → **multiapical deformity**
  1) Intersection not at visible deformity
  2) Intersection not in center of bone
  3) **No intersection** at level of bone
- CORAs → 34° and 32° → roughly equal
- Yet **MAD** present → 32° very distant → almost no effect on mechanical axis
- Alternative → proximal + distal **mid-diaphyseal lines** → meeting point
  - Distal articular surface axis → line at 90° → same result

Osteotomy for Each CORA Ask AI

Multiapical

- First CORA → cut at it
  - Any type → **open wedge,** closing wedge
- Second CORA → at joint line → cut **away from deformity**
  - Must include **translation** → rotation around CORA
  - Osteotomy apex cannot lie at **articular surface** level

Example 4: Proximal Varus + Mid-Shaft Valgus Ask AI

Multiapical

- **Lateral MAD** → disproportionate to visible valgus
- Tibial mechanical axis → **green line**
- → whole deformity in tibia
- **Varus** in proximal tibia
- **Valgus** in mid-shaft tibia

Summary: Uniapical Versus Multiapical Ask AI

Multiapical

Feature	Uniapical	Multiapical
Intersection level	At visible deformity	Not at visible deformity
Intersection site	Center of bone	Outside center of bone
Intersection	Present	May be absent
Bone shape	Not rounded	Rounded

Exam Pearl

- Axes not intersecting at **visible deformity** → multiapical
- Intersection outside **center of bone** → multiapical
- **No intersection** → multiapical
- **Rounded bone** → multiapical immediately

- End of lecture

Recommended Reading and Practice Ask AI

- Moderator → thanks to **Dr. Hisham**
- Recommended book → **Dror Paley** (title as heard: *Principles of Deformity Correction*)
  - Best book; essential reading for this topic
- ≈3 years ago → courses at **Kasr Al-Ainy**
  - Workshop **CDs** distributed → home practice
  - Material widely available → practise, draw, exercise
- Moderator → topic has **rules**, not guesswork
  - Smiling vs angry face → decided by rules
  - Advice → practise drawing on examples
- Slides → from **workbook**, available in libraries
  - Prepared by **Abdel-Hamid Rashed**; printed at Kasr Al-Ainy courses
  - CD widely held; distributed at conferences
- Practice → draw and analyze **every radiograph**

Questions and Answers Ask AI

Open Versus Closing Wedge Ask AI

Wedge

	Closing wedge	Open wedge
Indication	Excess length / length unimportant	Length matters
Metaphysis	—	May heal itself; **cancellous bone** fills
Diaphysis	—	Bone graft, or gradual Ilizarov correction

- **Open wedge in metaphysis** → self-filling, cancellous bone
- Open wedge in diaphysis → **bone graft**
  - Or gradual correction with **Ilizarov** frame → regenerate bone, no graft

Degree of Correction Ask AI

Wedge

- Calculated from the drawing
- **Degree of correction** → known before entering operating room
  - Determines open wedge size
  - E.g., **Puddu plate** → plate (wedge) size per angle
- Moderator → essential lecture
- Lecturer → topic **very, very basic**
- Moderator → Dror Paley's book difficult; this explanation easier

Frontal Plane Malalignment of the Knee

(Frontal plane analysis, mechanical axis and joint-line orientation around the knee)

Prof Dr Hisham Abdel-Ghani

Watch Original Lecture

Lecture transcript processed with AI assistance.

Contents

↑ Book contents

Frontal Plane Malalignment: Definitions . . . 261
Introduction . . . 261
Clinical Definitions . . . 261
Mechanical Axis Definition . . . 261
Imaging Pitfall: Rotation . . . 262
Apparent Bilateral Genu Valgum . . . 262
Apparent Genu Varum From External Tibial Torsion . . . 262
Obesity and Apparent Genu Valgum . . . 263
Soft Tissue Effect . . . 263
Recap of the Two Rules . . . 263
Obese Normal Default . . . 264
Obese Patient With Normal Clinical Alignment . . . 264
Imaging Standard . . . 264
Vitamin D Deficiency and Rickets . . . 265
Commonest Cause . . . 265
Active Rickets . . . 265
Deformities Produced . . . 265
Blount Disease: Classification and Features . . . 265
Classification . . . 265
Blount Versus Rachitic Genu Varum . . . 266
Obesity and Vitamin D Link . . . 266
Lateral Thrusting . . . 266
Infantile Blount . . . 266
Diagnosis of Blount Disease . . . 267
Peaking Is Not Diagnostic . . . 267
Radiographic Angles . . . 267
Progression Despite Medical Treatment . . . 267
Late-Onset Types . . . 269
Treatment Options and Physiological Alignment . . . 269
Overview . . . 269
Physiological Alignment . . . 270
Response to Medical Treatment . . . 270
Age Limits for Waiting . . . 271
Vitamin D Supplementation . . . 272
Sources and Requirements . . . 272

Dosing Protocol . . . 272
Inactive Versus Active Forms . . . 273
Problems With the Active Form . . . 274
Preparations . . . 274
Duration of Treatment . . . 275
Guided Growth (Hemiepiphysiodesis) . . . 275
Principle . . . 275
Indications and Cases . . . 276
8-Plate Complications and Design . . . 276
Rebound Growth After Removal . . . 277
Osteotomy . . . 277
Acute Versus Gradual Correction . . . 277
Indications: Salvage Procedure . . . 277
Complications of Osteotomy . . . 277
Acute Correction Without Fibular Osteotomy . . . 278
Historic Blount Cases . . . 279
Late-Onset Varus: Metaphyseal Varus Versus True Blount . . . 279
Collinear Plateaus . . . 279
The Multiplier App . . . 279
Why a Simple Valgus Osteotomy Fails . . . 280
Medial Plateau Elevation . . . 281
Acute Versus Gradual Elevation . . . 281
Gradual Elevation Technique . . . 281
Elevation Cases . . . 282
Shortening With Deformity . . . 282
Lecture Summary . . . 282
Questions and Answers . . . 283
Progressive Infantile Blount . . . 283
Genu Varum With Torsion . . . 283
Hemiepiphysiodesis in a Multiplanar Deformity . . . 283
Acute Versus Gradual Correction . . . 283
Lateral Thrust in Rickets . . . 284
Double Elevating Osteotomy . . . 284
Flatfoot and Genu Valgum . . . 284
MRI in Blount . . . 284
Combining Active and Inactive Vitamin D . . . 285
Windswept Deformity Before 6–7 Years . . . 286
Duration of Vitamin D . . . 286

Timing of 8-Plate Removal 287
Osteoclasis Technique 287
Fasciotomy in Acute Correction 287

AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 273) “Decal (and P-Decal)” → “Devarol (and P-Devarol)”

C2 (p. 273) “alfacalcidol is 25-hydroxy only” → “only 1-hydroxylated”

C3 (p. 273) “converted to 1,25 in liver” → “converted to 25-hydroxy”

Queried - kept as spoken, but looks wrong

Q1 (p. 270) as spoken; physiological genu varum normally resolves by about 2 years, valgum peaks at 3–4

Frontal Plane Malalignment: Definitions Ask AI

Introduction Ask AI

- Topic relevant to all orthopaedic surgeons, not only paediatric
  - Met as patients, relatives, or personally
  - Opening greetings ahead of Eid
- Thanks to Dr. Mohamed → organising the lectures during COVID
  - Topic: **frontal plane malalignment around the knee**
  - Frontal plane → patient seen while standing

Clinical Definitions Ask AI

- Frontal plane = **coronal plane** → genu varum, genu valgum
- Normal stance → knees touching, ankles (medial malleoli) touching
  - **Genu valgum** (knock knees) → ankles separated
    - Mild or severe → gap between medial malleoli
- Gap = **intermalleolar distance** → increased in genu valgum
  - Severity in fingers → e.g. 12 fingers vs 4 fingers → compare patients
  - Rough, inaccurate → usable as follow-up measurement
- **Genu varum** → ankles touching, knees separated
  - Measured: **intercondylar distance** → between medial femoral condyles
  - Can be very severe
- Very severe or mild genu varum
  - Mild clinical varum ≠ absence of severe deformity
- Example: mild clinical varum, intercondylar distance ≈ two fingers
  - X-ray → **depression of the medial plateau**
  - Inspection and clinical measurement → not enough
- Clinical assessment → possibly **falsely reassuring**
  - Severe knee affection despite mild appearance → scientific definition needed

Mechanical Axis Definition Ask AI

- **Mechanical axis of the lower limb** → center of hip → center of ankle
  - Should pass through center of knee
  - Shifted laterally → **lateral mechanical axis deviation** → genu valgum
  - Shifted medially → **medial mechanical axis deviation** → genu varum
- Correct definition → clinical judgement alone insufficient

	Genu valgum	Genu varum
Clinical picture	Knees touching, ankles apart	Ankles touching, knees apart
Distance measured	Intermalleolar	Intercondylar (medial femoral condyles)
Mechanical axis	Lateral deviation	Medial deviation

Imaging Pitfall: Rotation Ask AI

Apparent Bilateral Genu Valgum Ask AI

- Clinically → severe bilateral genu valgum
- First X-ray → right genu varum, left genu valgum
  - Repeated with **rotation** corrected
- Corrected film → right limb straight, left severe genu valgum
  - Only change → rotation of lower limb during imaging
  - Rule: **standing anteroposterior view**, patellae facing anterior
- Patellae facing inward → **poor measurements** → incorrect diagnosis
  - Same patient, two films → different mechanical axis position
  - Entire difference → limb rotation

Apparent Genu Varum From External Tibial Torsion Ask AI

Torsion

- Patient told to stand feet together → appears genu varum
  - Same patient, two images → which to believe
  - **Patellae facing inwards**
- Incorrect imaging or examination → patellae must face anterior
  - Patellae anterior → neither genu varum nor genu valgum
  - Actual finding: **external tibial torsion** → tibia rotated externally beyond normal
  - Feet placed side by side →
- → internal rotation proximally → false interpretation of genu varum
  - Normal stance + feet turned inward → **apparent genu varum**, not true genu varum
- Correct stance, patellae anterior → no genu varum
  - Check patellar position → clinical examination and imaging

Must-Know

- Rule number one: take care of rotation
- Standing AP view, patellae facing anterior
- Patellae facing inward → apparent genu varum, false measurements

Obesity and Apparent Genu Valgum Ask AI

Soft Tissue Effect Ask AI

- Patellae anterior, correct stance → technique not the problem
  - **Lateral mechanical axis deviation** → very mild
  - Yet very marked clinical genu valgum
- Wide ankle gap ≠ actual mechanical axis deviation
  - **Soft tissue shadow** → knees touching in soft tissue, bones far apart
  - Cause → **obesity**
- Thick thighs → knees touch early → ankles cannot meet → apparent genu valgum
  - True valgum absent or very mild
  - Soft tissue prevents knees meeting → marked clinical appearance
  - → **rule number two**

Recap of the Two Rules Ask AI

Recap

- Definition → by mechanical axis deviation
- Lateral → **genu valgum;** medial → genu varum
  - First patient: severe left genu valgum only on incorrect film
  - Actual deviation → very mild
- Rotation corrected, patellae anterior → right within normal, left lateral deviation
  - → take care of rotation
  - Second patient: feet together → apparent genu varum
- Correct stance → no genu varum
  - Clinical exam or X-ray → **patellae facing anterior** → rule number one
  - Rule number two: soft tissue → no intercondylar distance, large ankle gap
- Mechanical axis → no severe lateral deviation
  - Knees kept apart by soft tissue **secondary to obesity** → ankles far apart
  - → **false impression of genu valgum**
  - Another patient → **intercondylar distance** present
- → rather than intermalleolar → clinically no genu valgum, knees not touching
  - Obese patient → interpret against expected apparent genu valgum
  - Obesity masking → **increased intercondylar distance** with ankles touching

Obese Normal Default Ask AI

Obesity

- Obese patient → normal mechanical axis + increased intermalleolar distance
  - Obese default → **apparent genu valgum,** not true genu valgum
  - **Excessive soft tissue** → knees apart → ankles apart
  - Expected → **increased intermalleolar distance** → image before diagnosing
- X-ray → possibly **normal alignment,** no genu valgum
  - Obese + **no clinical malalignment** (knees and ankles touching) → genu varum
  - **Thin patient** → knees and ankles touch normally
  - **Obese patient** → ankles apart → clinical apparent genu valgum

Clinical stance	Thin patient	Obese patient
Knees and ankles touching	Normal	Genu varum
Knees touching, ankles apart	Genu valgum	Normal (apparent genu valgum)

Obese Patient With Normal Clinical Alignment Ask AI

Obesity

- Obese + knees and ankles touching → **genu varum**
  - Not genu valgum, not straight, not correct alignment
  - Diagnostic errors → take care of **rotation** and **obesity**
- Example: knees and ankles touching → looks good, but obese
  - Soft tissue keeps knees apart → ankles touching only if varus
  - → severe **bilateral** genu varum, left better than right
  - After correction → should become apparent genu valgum
- Obese → knees apart → ankles apart → normal = **apparent genu valgum**
  - Obese + ankles touching → genu varum
  - Correcting to knees and ankles touching →
- → **undercorrection**
  - Target → normal physiological (apparent) genu valgum, no malalignment on X-ray
  - Diagnose with **patellae facing forward** → rule number 1
- **Measurement of knee and medial malleolar distances unreliable** → varies with patient size
  - Obese → genu valgum expected → within limits or not → X-ray and axis measurement
  - Obese + ankles close together → genu varum certain → X-rays

Pitfall

- Obese default → apparent genu valgum, increased intermalleolar distance
- Obese with knees and ankles touching → genu varum
- Correcting obese varum to knees and ankles touching → undercorrection
- Clinical distances vary with patient size → unreliable

Imaging Standard Ask AI

- **Anteroposterior view of the lower limb, standing**, patellae facing anterior
  - Check → **shadow of the patella** over center of femoral condyles

Vitamin D Deficiency and Rickets Ask AI

Commonest Cause Ask AI

- Scope → common causes; not rarities, skeletal dysplasias, major problems
  - **Vitamin D deficiency** → most common cause of frontal plane malalignment, by far
  - Can produce genu varum, genu valgum, or one of each
- One varum + one valgum → **windswept deformity**
  - Always confused with rickets → **Blount disease**
  - Formerly considered unrelated to vitamin D deficiency
- Actually **related to vitamin D deficiency** → possible cause of Blount
  - Most important cause of vitamin D deficiency → **nutritional rickets**

Active Rickets Ask AI

Rickets

- **Active rickets** features:
  - Rarefaction of the metaphysis
  - Widening of the physes
  - Cupping
- Active rickets → **all the physes**, never one only
  - Femur, proximal tibia, distal tibia
  - Not only in babies aged one, two or three years
- Example: girl, 13 years old, **active rickets**
  - Marked physeal widening and rarefaction → even **capital femoral epiphysis**
  - Any growing age with physis and epiphysis → severe deficiency → **hypophosphataemia** → rickets
  - Separate lecture on rickets
- Active rickets seen at 11, 12, 13 and 14 years
- Catastrophic in a country with so much sun

Deformities Produced Ask AI

Rickets

- **Nutritional rickets** → genu varum (medial deviation) or genu valgum (lateral deviation)
- → **windswept deformity** → valgus one side, varus other side
  - → association with Blount disease

Blount Disease: Classification and Features Ask AI

Classification Ask AI

Blount

- **Blount disease** → associated with vitamin D deficiency
  - Classic: infantile, juvenile, adolescent
  - Current: **infantile (early onset)** vs late onset
    - Late onset = juvenile + adolescent

Blount Versus Rachitic Genu Varum Ask AI

Blount

- **Rachitic genu varum** → proximal tibial varus
- + **distal femoral varus** → whole limb bowing
  - Blount → mostly **proximal tibial metaphysis**

	Rachitic genu varum	Blount disease
Site	Proximal tibial varus + distal femoral varus	Mostly proximal tibial metaphysis
Extent	Whole limb bowing	Proximal tibia

Obesity and Vitamin D Link Ask AI

Blount

- Mostly **obese**
- Not always → thin child does not exclude Blount
  - Very thin girl shown at start → Blount disease
  - Obesity → **vitamin D deficiency increases**
  - Vitamin D **deposited and stored in the fatty tissue** → low blood level
- → **relative vitamin D deficiency**
  - Blount vs active rickets with rachitic varum → why → not understood

Lateral Thrusting Ask AI

Blount

- **Lateral thrusting** → varus increases in **stance phase**
  - Cause → depression of **posteromedial corner of the proximal tibia**
- Weight-bearing → varus → called "laxity" → actually **instability of the knee**
  - Femoral condyle falls into proximal tibial depression
  - Not lateral laxity → depression of **posteromedial proximal tibial condyle**
- Femur **settles in this depression** → **rocking movement** → lateral thrusting
  - Current naming → infantile and late onset
- Late onset = juvenile + adolescent → no strong differentiation between them

Infantile Blount Ask AI

Infantile

- Clear **metaphyseal changes** → no doubt Blount disease
  - **Subluxation;** always image patient standing
- Plateau depression present but hidden → incomplete **ossification**
  - Ossified medial proximal tibial epiphysis → occupies **intercondylar notch of the femur**
  - **Subluxation** → part of lateral tibial plateau
- → outside boundary of **distal femoral epiphysis** → Blount certain

Diagnosis of Blount Disease Ask AI

Peaking Is Not Diagnostic Ask AI

Blount

- Audience vote (≈888 attendees, ≈72 responses) → 50% / 50%
- No **subluxation** → no lateral tibial epiphysis uncovered by distal femoral condyle
  - Diagnosis on **peaking** and irregularity → wrong
  - Never diagnose Blount by **peaking of the proximal tibial metaphysis**
- → **the most common mistake**
- Several **signs of Blount**

Radiographic Angles Ask AI

Blount

- **Metaphyseal-diaphyseal angle:**
  - Line along **lateral border of the tibia**
  - Line joining lateral and medial metaphyseal peaks → perpendicular to first line
  - More than 16° → one criterion favouring Blount, not diagnostic per se
- Never relied upon alone
  - **Femoral metaphyseal-diaphyseal angle**
- + **tibial metaphyseal-diaphyseal angle** → tibial > femoral → favours progression and Blount
  - "More in favour" → not certain
  - No gross metaphyseal changes → Blount only if **progression** proven

Progression Despite Medical Treatment Ask AI

Blount

- Disagreement → **labs** → exclude vitamin D deficiency → treat → follow six months
  - Genu varum increases → Blount
  - Stable → wait more
  - Improving → not Blount
  - Some Blount → **spontaneous correction without surgical treatment**
- Sign number one → **genu varum progressive in spite of medical treatment**
  - Number two → **subluxation** → absent here → not Blount
- Features encouraging diagnosis:
  - **Bilateral and asymmetric** → also common post-rachitic
  - **Lateral thrusting** → also post-rachitic without Blount
  - **Positive family history**
  - **Obesity**
  - **Internal tibial torsion**
- **Peaking** → least significant sign → all rachitic patients have peaking
  - Rachitic X-ray with peaking shown → not Blount disease
  - Never base diagnosis on peaking
- No real value in diagnosis → most important → **progression in spite of medical treatment**

Weight	Sign	Note
Number one	Progression despite medical treatment	Most important sign
Number two	Subluxation	Lateral tibial epiphysis uncovered

Weight	Sign	Note
Supportive	Bilateral asymmetric, lateral thrust, family history, obesity, internal tibial torsion	Many also post-rachitic
Criterion only	Metaphyseal-diaphyseal angle > 16°; tibial > femoral angle	Not diagnostic alone
Least	Peaking	All rachitic patients

High Yield

- Most important Blount sign → progressive varum despite medical treatment
- Peaking → least significant sign → never diagnostic
- Metaphyseal-diaphyseal angle > 16° → criterion only
- Peaking-based diagnosis → the most common mistake

Late-Onset Types Ask AI

Late onset

- Applies to both types, mainly **infantile**
  - Proximal tibial epiphysis largely not visible → still **radiolucent**
- Late onset → genu varum, somewhat obese boy or girl
  - Onset 7–8 years, 10 years, 12–13 years
- Formerly juvenile + adolescent → two types to differentiate
  - **Type 1** → no plateau depression
    - Medial and lateral tibial plateaus **collinear**
    - Deformity **metaphyseal**; CORA in metaphysis or at physis
- Rickets → possible cause in older child while growing
    - Rickets not confined to age one or two, or start of walking
  - **Type 2** → compare lateral and medial plateau
  - Literature mixes both → adolescent / late-onset / juvenile Blount
- No depression of **posteromedial part of the medial tibial plateau** → no Blount disease
  - → ordinary **metaphyseal deformity**
    - Any **bone-softening disease**, post-trauma, post-tumour, other causes
  - Collinear plateaus, no **intra-articular deformity** → not Blount
- **Depression of the medial plateau** → definitely Blount disease
  - Easier at adolescent and juvenile ages → epiphysis no longer all cartilage
  - Infants → many parts not visible

	Type	Type
Medial plateau	Collinear with lateral, no depression	Posteromedial depression
Deformity	Metaphyseal; CORA metaphysis/physis	Intra-articular
Diagnosis	Not Blount → metaphyseal deformity	Blount disease
Causes	Rickets, bone softening, trauma, tumour	Blount

Treatment Options and Physiological Alignment Ask AI

Overview Ask AI

- **Medical treatment** → relies on **remodeling power**
  - Healthy bone → remodeling → **spontaneous correction**
- **Surgical treatment:**
  - **Osteotomy** → acute or gradual correction
    - Gradual → **external fixator**
    - Acute → external fixator, any internal fixation, or casts
  - **Hemiepiphysiodesis**
- **No physiotherapy** for genu varum or genu valgum
  - **Braces** → no evidence of efficacy
    - "Blount" corrected with brace → would have corrected with medical treatment anyway

Physiological Alignment Ask AI

- Question → who gets medical treatment; varum vs valgum by age
  - Start of walking → about 3–4 years → **physiological genu varum** *[Q1 · as spoken; physiological genu varum normally resolves by about 2 years, valgum peaks at 3–4]*
- → corrects alone → may turn into **genu valgum** → corrects by 6–7 years

Response to Medical Treatment Ask AI

- Given in **active stages of rickets** → aim healed rickets → continue until correction
  - Example: **asymmetric** deformity, very clear peaking
- + **metaphyseal deformity** → looks like Blount → treat six months
- → corrected → not Blount
  - Rushed surgery → wrong
  - Risk → **overcorrection** → opposite deformity → further operations
- **Active rickets** → treatment → full correction
  - No surgery, no braces, no physiotherapy referral

Age Limits for Waiting Ask AI

- Medical treatment alone, how long:
  - **Genu varum** → until age 4–5 years
  - **Genu valgum** → may improve until 6–7 years
  - No surgery for genu varum at 2 or 3 years → unless proven Blount
  - No surgery for genu valgum at early age (likely 3–4 years) → corrects by 6–7 years
  - Follow-up every six months
- Improving → be patient
  - Improvement **plateaus** → correct the remainder
  - Complete correction → nothing needed
  - Lecturer's teaching → 4 or 5 years, not less
- Colleague operating at 3 years → not in literature → "everyone does it"
  - → completely wrong → self-justification
- "I will do it better" → no justification
  - Do the right thing; not responsible for others; not a saviour
- 800–900 attendees doing right → no early unnecessary surgery
  - Spread right practice; never follow others
- Improving → never rush into surgery
  - Next → vitamin D sources, dosing, deficiency

	Physiological until	Wait with medical treatment until
Genu varum	3–4 years	4–5 years
Genu valgum	6–7 years	6–7 years

Must-Know

- Genu varum → wait until 4–5 years
- Genu valgum → wait until 6–7 years
- Improving → be patient; plateau → correct remainder
- No physiotherapy, no braces for genu varum or valgum

Vitamin D Supplementation Ask AI

Sources and Requirements Ask AI

Vitamin D

- Long lecture on YouTube
  - **Vitamin D essentially not present in food**
  - Daily requirement from eggs → about 25 eggs per day
  - Active disease or deformity → **not rely on the sun**
- Sun, healthy life, milk, cheese, yoghurt, eggs → advised
  - Rely on **medical treatment** → optimise until healed
  - **Calcium** → if no milk, cheese, yoghurt, eggs
  - After about 10 months of age → 800–1000 IU/day → **recommended daily allowance**
- Achievable → 20 minutes to half an hour of **direct sun**, any time of day
  - Early morning / before sunset advice (low ultraviolet) → foreigners, sun-sensitive skin
  - Local skin adapted to local sun → unless very fair
- Mid-winter → 2 or 3 PM sun pleasant → need not be 10 AM
  - **Morning-only belief** → sun avoidance
  - Morning 10–12 °C → no half-sleeves; 2–3 PM at 16–20 °C → fine

Dosing Protocol Ask AI

Vitamin D

- **Maintenance** → 1000 IU/day
  - **Healed rickets**, vitamin D not very low → 1000 IU/day
  - **Active rickets**, very low vitamin D, high PTH (parathormone) → 3 × RDA
    - About 2000–3000 IU/day for 2–3 months → until labs normal
- Labs corrected → continue on **daily recommended allowance**
  - Start 2000–3000 IU for 3 months → maintenance 1000 IU/day → until full correction

Situation	Dose	Duration
Healed rickets, vitamin D not very low	1000 IU/day	Maintenance
Active rickets, very low vitamin D, high PTH	2000–3000 IU/day (3 × RDA)	2–3 months → labs normal
After labs normalise	1000 IU/day	Until complete correction

Inactive Versus Active Forms Ask AI

Vitamin D

- Two types on the market:
- **Inactive forms** → Vidrop, Devarol, local products, imported 1000/2000/5000 IU *[C1 · transcript: "Decal (and P-Decal)"]*
  - **Active forms** → One-Alpha, Alpha D3 (name unclear in recording), alfacalcidol
- **Alfacalcidol** → behaves as active form
  - Claim: only 1-hydroxylated, not 1,25-dihydroxy → "semi-activated" → misleading *[C2 · transcript: "alfacalcidol is 25-hydroxy only"]*
  - **Cannot be stored**; body **cannot control** it
  - Consumed immediately in activation cycle of **1,25-dihydroxycholecalciferol**
  - Rule → give the **inactive form**
- Regular form → **never give the active form**
  - Inactive form → converted to 25-hydroxy → **stored in the liver** *[C3 · transcript: "converted to 1,25 in liver"]*
  - Misconception (doctors, paediatricians) → inactive form needs **activation by the sun**
  - Sun → **forms** vitamin D (inactive form); does not activate it
- Liver → **25-hydroxylated** → stored → released and activated per body needs
- Inactive form → **physiological** → body controls, releases, benefits

Problems With the Active Form Ask AI

Active D

- Body forced to consume it → active
- **No recommended daily allowance** → dose too low or too high
- Leaflet → not usable as **nutritional supplement** or for nutritional rickets
- **Kidney stones** → caused by active form, not inactive form
- **Half-life** short (about 3–4 hours, as heard) → one month of use
- → stopping two days → **state of deficiency**
- Trap: on One-Alpha / Alpha D3 → measuring **25-hydroxycholecalciferol** → no rise
- Dose increased → still no rise
  - Giving **1,25-dihydroxy** → cannot raise 25-hydroxy
  - Raising 25-hydroxy → inactive form → 25-hydroxylated
  - Routine test → **25-hydroxy** level, not 1,25-dihydroxycholecalciferol
- Active form → **expensive**
  - Drug representatives' claims ("active, faster, better") → no excuse → read and know

	Inactive form	Active form
Examples	Vidrop, Devarol, imported 1000/2000/5000 IU	One-Alpha, Alpha D3, alfacalcidol
Storage	Stored in liver as 25-hydroxy	Cannot be stored
Body control	Released and activated per needs	Forced consumption, no control
RDA	800–1000 IU/day	No recommended daily allowance
Nutritional rickets	Indicated	Not a nutritional supplement
Kidney stones	No	Yes
Raises 25-hydroxy level	Yes	No
Cost	Much cheaper	Expensive

Preparations Ask AI

Doses

- Replacement → **non-activated (ordinary) form;** know concentrations
  - Calculate volume for 1000 IU and 3000 IU
  - Inactive forms → much cheaper than active forms

Preparation	Concentration
Devarol	1000 IU per 5 mL
P-Devarol	400 IU per 5 mL
Vidrop	100 IU per drop

Duration of Treatment Ask AI

Vitamin D

- Treated → improved → stopped → same lifestyle → rickets recurs → genu varum again
  - No milk, no cheese → no calcium
  - No sun exposure → no vitamin D
- Recurrence at older age → treated under four years → presents at five or six → surgery
  - Genu varum at this age → no correction with medical treatment
  - Genu valgum after seven years → likewise
  - Lifestyle unchanged → continue medical treatment
- Calcium or vitamin D not obtained → **supplementation** → else recurrence
  - Analogy: anaemia from poor eating → eat well three months → stop → recurs
  - Kidney stones claim → too much of anything harmful
- Food in huge quantities → obesity, diabetes
- Excess or misuse ≠ avoid entirely → deficiency also harmful
  - Inadequate sun exposure → **vitamin D supplementation**

Guided Growth (Hemiepiphysiodesis) Ask AI

Principle Ask AI

Guided

- Most patients → correct with medical treatment
  - Surgery → **hemiepiphysiodesis** → most important in paediatric age group
- **Angular deformity with CORA near the physis** → osteotomy or guided growth
  - Conditions:
    1. CORA near physis
    2. Still growing → **growth capacity**
    3. Time remaining until **maturity**
- **Maturity** → girls 14 years, boys 16 years
  - Growth remaining → guided growth instead of osteotomy
- Example: mild genu valgum → tether growth on **convex side** (medial)
  - **Planning** → tibial or femoral origin → fix the bone carrying the deformity
- → **increased growth on the lateral aspect**, decreased medial → corrected
  - No bone cutting, no casts, no bed rest
  - 2 cm incision, very quick → **bear weight on the same day**
  - No large plates, no big wound
- No osteotomy complications → corrected → remove **8-plates**

Indications and Cases Ask AI

8-plate

- **8-plates** → routine genu varum and valgum + Blount disease
  - Blount corrected by hemiepiphysiodesis without osteotomy
  - Strong trend → hemiepiphysiodesis over extensive osteotomies
- Girl with genu varum → **late-onset Blount disease**
  - Very clear depressed medial plateau + **subluxation** of lateral tibial condyle
  - Contrast → previous **infantile Blount** case
  - 8-plate screws initially **parallel to each other**
- Correction → increased medial growth vs lateral → **divergence of the screws**
  - Plate **unlocked** → screws move in plate holes
  - Complete correction → even **medial plateau depression corrected**
- Very severe genu valgum, deformity mostly **femur** → 8-plate → corrected
  - No osteotomy, no **common peroneal nerve** worry
  - No bed rest for a month and a half to two months until union
  - Moves next day, walks; 2 cm incision
- No cast → plates removed later

8-Plate Complications and Design Ask AI

8-plate

- Commonest failure → **metaphyseal screw** breakage
  - Literature → Blount, obese, older age
  - **Plate failure** → rare (seen)
- Personal observation (unpublished) → **failure and fracture of the screws** in non-compliant patients
  - Vitamin D and calcium deficiency → treatment stopped → **bone softening** → relapse
  - → stresses on the screws
- → **fatigue failure** → screw fracture
  - Lecturer's speculation → may be wrong or right
- Previously **staples** → strong **compression of the physis** + extrusion with growth
- Plate → mobile screws → no strong physeal compression
- Mobile screws → no **backing out**
- Cheaper alternatives (Dr. Mohamed Hegazy) → cut **one-third tubular plate**, 3.5 reconstruction plate
  - Cheap 8-plates now available
- Ordinary screws, small plate, **two-hole plate** → same results reported
- 8-plate advantage → **thin profile** → not palpable
  - Always **subcutaneous**
  - High-profile implant → palpable → bothers child

	Staples	-plate
Physeal compression	Strong	None strong (mobile screws)
Displacement	Extrusion with growth	No backing out
Profile	—	Thin, not palpable

Rebound Growth After Removal Ask AI

8-plate

- After removal → **rebound growth** with recurrence → possible
  - Especially younger age
  - **Femur more than tibia**
- Also with **staples**
  - Very young child → **overcorrect** a little
  - Amount → no agreement → **a matter of experience**
- Most former osteotomy cases → now 8-plates
  - Very small group → still needs osteotomy

Osteotomy Ask AI

Acute Versus Gradual Correction Ask AI

Osteotomy

- **Acute correction** → cut and correct in same operation
- Fixation → **cast**, wires, plate, or external fixator
- **Gradual correction** → external fixator only, not plates or wires

Indications: Salvage Procedure Ask AI

Osteotomy

- **Osteotomy currently a salvage procedure** → when guided growth impossible:
  - Additional severe deformity → **multiaxial deformity**
  - Severe **rotation**
  - **Intra-articular source** of deformity
    - → now **relative contraindication** to guided growth → 8-plates used in Blount
  - **Shortening plus deformity** → gradual correction and lengthening → external fixator
  - **Skeletal dysplasias**, late presentation, slow **rate of growth** (e.g. dwarfism)
    - → guided growth ineffective → **osteotomy**

Complications of Osteotomy Ask AI

Osteotomy

- Early genu varum + **proximal tibial osteotomy** → Cozen phenomenon → genu valgum
- → severe **valgus deformity** of the tibia
  - Example: crude osteotomy + **fibular osteotomy** → corrected, but unneeded
  - **Femoral varus with tibial deformity** → tibia only corrected → femur remodels → genu valgum
- → allow **spontaneous correction and remodeling power**
  - Medical treatment first → avoid unneeded operations, wounds, complications
  - **Compartment syndrome** → osteotomy complication; no concern with 8-plates
- **Fibular osteotomy** → dangerous
  - Proximal → **common peroneal nerve** injury
  - Middle segment → **peroneal vessels** injury
  - Complications may exceed tibial osteotomy

Acute Correction Without Fibular Osteotomy Ask AI

Technique

- Acute correction fixed with
- **wires** or plate
  - Goal → tibial cut only, fibula intact
  - Avoids paralysis of **extensor hallucis longus**, common peroneal nerve, peroneal vessels injury
  - CORA + **transverse bisector line** →
- → **proximal to the fibula**
  - Osteotomy **below the tibial tuberosity** → spares proximal tibial physis
  - Exit just distal to physis, above (or at) fibular head
  - → no **fibular osteotomy** needed
  - Under **image intensifier**
  - Alternative fixation → **Puddu plate**
- **Puddu plate** → open wedge osteotomy plate
  - **Spacer** → prevents collapse of the wedge
  - **Mature** patient → screws spare physis
  - Osteotomy opened above fibula, no fibular osteotomy
  - Open **physeal growth plate** → screws distal to physis
- Requires somewhat **older age**
  - Example: boy, 15 years old, one year of growth remaining → insufficient for 8-plate
  - Presenting at 13 or 14 → 8-plate; no acute correction, no Puddu plate, no osteotomy
  - **Whenever there is a chance for an 8-plate, do nothing else**

Historic Blount Cases Ask AI

Blount

- Girl with Blount disease → today 8-plate straight away
  - 8-plate → replaced many interventions
  - Previously → **transphyseal osteotomy**
- Fixed with wires, fibula intact → corrected, walked well
  - Risk → **physeal growth arrest**
  - Treated before 8-plate era
- Doctor's daughter → **metaphyseal deformity** + physeal deformity
- **Severe depression of the medial plateau** + severe knee subluxation
- Surgery → complete correction:
  - **Transphyseal osteotomy** → elevate medial plateau
  - Proximal tibial osteotomy → rotation and varus
  - Fibular osteotomy
- Very good follow-up → today 8-plate, not **double osteotomy**
- Another girl → Blount + **lateral thrusting**
- Clear **depression of the medial plateau** → not merely peaking; visible knee subluxation
- Osteotomy then → today 8-plate

Clinical Pearl

- Whenever there is a chance for an 8-plate → nothing else
- Osteotomy → salvage procedure when guided growth impossible
- Fibular osteotomy → common peroneal nerve (proximal), peroneal vessels (middle)
- Early varum + proximal tibial osteotomy → Cozen phenomenon → valgus

Late-Onset Varus: Metaphyseal Varus Versus True Blount Ask AI

Collinear Plateaus Ask AI

Late onset

- **Collinearity of the medial and lateral plateaus** → not Blount
  - → osteotomy case → **proximal tibial varus**
  - **Enough growth remaining** → 8-plate

The Multiplier App Ask AI

- **Multiplier** app ("Multiplier HD")
  - Input → relevant angles, physeal width, age
  - Output → correction time with 8-plate
  - Example: 15-year-old boy → two years needed → no 8-plate
- 6 months needed → 8-plate possible
- Also → **leg length discrepancy**, many other options
  - Time **enough for the 8-plate** → 8-plate; not enough → osteotomy

Why a Simple Valgus Osteotomy Fails Ask AI

Blount

- True **Blount disease** → depression of the medial plateau → different story
- True Blount treated as **proximal tibial metaphyseal varus** → grave error
  - Example: **proximal tibial valgus osteotomy** in Blount
  - **Source of the deformity intra-articular** → rocking and instability from depressed
- **medial plateau** → correcting from below → creates another deformity
  - Not corrected → still thrusting
  - One deformity (proximal tibial varus) → varus + valgus
  - **Intra-articular instability** / rocking uncorrected
- Differentiate → metaphyseal varus vs true proximal tibial varus (**Blount disease**)
  - After ordinary valgus osteotomy → **the knee is still subluxed**
  - → plateau must be **elevated**
- **Valgus osteotomy** of proximal tibia alone → not enough → elevation required

	Proximal tibial metaphyseal varus	True Blount disease
Plateaus	Collinear	Medial plateau depressed
Source	Metaphyseal	Intra-articular
Knee	Stable	Rocking, instability, subluxation
Treatment	8-plate if growth remains; else osteotomy	Medial plateau elevation (8-plate now first)
Valgus osteotomy alone	Adequate	Fails → knee still subluxed

Medial Plateau Elevation Ask AI

Acute Versus Gradual Elevation Ask AI

Elevation

- **Elevation of the plateau** → acute
- or gradual → neither proven better → use your skill
  - Acute drawbacks:
    - Very stiff deformity → **graft**
    - Intraoperative correction thought adequate → actually not enough
  - Acute fixation advice → screws or **Schanz pins**
- → from **lateral aspect** → elevate medial plateau → complete screws, wires or pins
  - Medial entry → fracture during correction → undercorrection, graft
  - → very high **tension on the skin** → necrosis of the skin
- Lecturer prefers **gradual elevation** of the medial plateau
  - Very simple frame → ring or **half-ring Ilizarov**
  - Very easy to build; very small incision

	Acute elevation	Gradual elevation
Evidence	Neither proven better	Neither proven better
Stiff deformity	Graft may be needed	Gradual
Correction	Risk of undercorrection	Step by step
Skin	High tension → necrosis risk (medial entry)	No skin tension
Device	Screws/Schanz pins from lateral	Ring or half-ring Ilizarov

Gradual Elevation Technique Ask AI

Elevation

- Step `1` → **K-wire** defines center of the knee
- Step `2` → two **Schanz screws** in epiphysis, under image
  - One anterior, one posterior
  - Posterior depression > anterior → anterior more proximal
- → posterior more distal
- Never **parallel** → posterior medial plateau depressed vs anterior
  - Both in epiphysis
  - Step `3` → osteotomy reaches **center of the physis** (K-wire)
- Step `4` → **osteoclasis** of the epiphysis only → open → let close
  - Nothing more acutely → gradual correction with Ilizarov
- Upper **half-ring** → holds two Schanz screws in epiphysis
  - **Hinge** → behind them, at center of the knee
  - Ring made parallel to tibia → elevates epiphysis via Schanz pins
- Posterior > anterior → acute correction of **posterior slope** (posterior depression)
  - Then wedge opened gradually → rings parallel
- → correction achieved

Elevation Cases Ask AI

Elevation

- Example → first complete ring, later **half-ring**
- Depression corrected → apparent **overcorrection** (genu valgum)
  - Obese → normal = increased intermalleolar distance on standing
  - Blount → overcorrect a little
- → protects from **recurrence**
- Another patient → **bilateral Blount**
- Very small incision; osteotomy under image
- **Hinge** behind and in front; two Schanz pins **oblique** before mounting
  - Frame aligned → posterior part straightened → posterior slope corrected
  - → **posteromedial depression** of Blount corrected
- Gradual correction → **wedge of bone** forming
- Elevated until **medial and lateral plateaus collinear**
  - = **gradual open wedge osteotomy** → limited to medial plateau, not whole proximal tibia
- All patients → **full range of knee motion**; last follow-up > 5 years
- Patient about 16–17 years → lecturer almost never operates on adults
- **Physeal scar**, mature → medial plateau elevation
- Same frame right and left
- **Final appearance** → rocking, subluxation, instability corrected → correct operation
  - Medial plateau depression + **proximal tibial osteotomy** only → alignment restored clinically
  - Mechanically → rocking and instability persist → nothing done for patient

Shortening With Deformity Ask AI

Ex-fix

- **Shortening** + deformity → gradual correction with external fixator → lengthening
- No option other than **external fixator**

Lecture Summary Ask AI

- Most important cause → **vitamin D deficiency** → varus, valgus, windswept, Blount
- Obese → normally **apparent genu valgum**
  - Obese + normal clinical alignment → **genu varum**
- Imaging → **patellae facing anterior**, standing
- **Planning** (another lecture) → CORA, center of rotation
- Guided growth possible → **guided growth**, not osteotomy
  - Osteotomy → only when guided growth will not work

High Yield

- Vitamin D deficiency → most common cause of frontal plane malalignment
- Obese + normal clinical alignment → genu varum
- Standing AP, patellae facing anterior
- Guided growth whenever possible; osteotomy → salvage

Questions and Answers Ask AI

- Moderator → thanks Dr. Hisham ("game changer") → limited questions within topic

Progressive Infantile Blount Ask AI Q&A

- Q: Progressive **infantile Blount** on follow-up → still wait until four years?
  - A: No → waiting only for uncertain diagnosis; certain Blount → intervene
  - Straight away → infantile type → **8-plate**, not osteotomy

Genu Varum With Torsion Ask AI Q&A

- Q: **Genu varum and internal tibial torsion** → which first, or both together?
  - A: Correcting varus → torsion often corrects with it
  - Very severe **internal femoral torsion** + tibial varum → femoral osteotomy needed
    - → osteotomy of femur and tibia in single operation
    - Cast and bed six weeks anyway
    - Avoid guided growth then second operation later
  - Generally → no combining
  - → torsion usually corrects with varus correction

Hemiepiphysiodesis in a Multiplanar Deformity Ask AI Q&A

- Q: Blount not a pure coronal plane deformity → how does hemiepiphysiodesis work?
  - Valid point → **posteromedial aspect** depressed
    - Genu varum + **procurvatum** (as if knee flexion deformity) + some internal rotation
  - Some advise **anterolateral** 8-plate → not proven different → lateral as usual
  - Interpretation → **medial physis** "breathes" → catches up with faster lateral part
  - Not fully understood → but works
  - → deficiency in **understanding of Blount**, not of plate or technique

Acute Versus Gradual Correction Ask AI Q&A

- Q: Angle threshold → acute vs **gradual correction**?
  - A: None → acute possible for almost all genu varum, even depressed medial plateau
  - **Acute correction** not wrong → improper acute correction is
  - Gradual preferred → **versatility**, step by step, no skin tension or skin problems
- Genu valgum osteotomy → **common peroneal nerve** issue
  - Femoral varus osteotomy → no **stretch** of nerve
  - Tibial varus osteotomy → stretches nerve → major problem → gradual preferred
  - Acute required → **exploration of the common peroneal nerve** before osteotomy
  - + **release** → no strong nerve tension → avoid common peroneal nerve palsy

Genu valgum osteotomy	Common peroneal nerve	Preference
Femoral varus osteotomy	Not stretched	No major problem
Tibial varus osteotomy	Stretched	Gradual; if acute → explore and release nerve

- Rickets questions → deferred to Dr. Hisham's rickets lecture

Lateral Thrust in Rickets Ask AI

Q&A

- Q: **Lateral thrust** in nutritional rickets? Still wait until four years?
  - A: **Lateral thrust is not pathognomonic**
  - Rickets + lateral thrust → improves with medical treatment
  - No progression with treatment → wait → corrects without surgery
  - **Surgical intervention** → progression on treatment
  - → with **proper dose** and normal metabolic profile

Double Elevating Osteotomy Ask AI

Q&A

- Q: **Double elevating osteotomy**?
  - A: Depressed medial plateau + **subcondylar varus**
    - → elevation osteotomy + proximal tibial osteotomy → **V-osteotomy**
  - Girl shown earlier → **double osteotomy**
  - First osteotomy → **depression of the medial plateau**
  - Second osteotomy → **subcondylar varus** + rotation
  - Today → **8-plate**, not double osteotomy
  - Cutting near physis → osteotome injury → **complete physeal growth arrest**
    - Very big problem → girl aged two and a half years

Flatfoot and Genu Valgum Ask AI

Q&A

- Q: Role of **flatfoot** in genu valgum?
  - A: No
  - Overweight, good alignment → walks with **genu valgum**
  - Obese + knees and feet touching → genu varum
  - Genu valgum gait, legs apart → **varus of the heel**
  - → **compensation** for valgus, not flatfoot
  - Flatfoot → more with genu varum → compensation for varus
  - Flatfoot → not a cause of either → **unrelated**
  - **Foot compensates for tibial alignment above it:**
    - Tibia into valgus → foot into varus
    - Tibia into varus → foot slight valgus → **plantigrade**

MRI in Blount Ask AI

Q&A

- Q: Role of **MRI** in Blount diagnosis?
  - A: Not a diagnostic tool per se
  - Late onset → good **ossification** → degree of depression visible
  - Infantile → sometimes MRI → depression severity, need for **elevation**
  - No clear role → used in past osteotomy planning → rare with 8-plates

Combining Active and Inactive Vitamin D Ask AI

Active D

- Q: **Combination of active and inactive forms**?
  - A: No → active rickets + vitamin D → **increased activation** above physiological → plateau
  - Only the body reproduces this curve
    - **Very complicated mechanism for activation and deactivation** of enzymes and metabolism
    - Any active dose → less (delays patient) or more (wrong)
  - Starved patient + **inactive form** → activated same day → works
    - Active levels possibly higher than with active form
    - Active form → interferes with **physiological pathway of activation**
  - Never give active form
  - Not for orthopaedic prescription → **endocrinology and nephrology**
  - Indications → **vitamin D-resistant rickets**, failure of activation
  - → **deficiency of the enzymes** activating vitamin D
  - Not a **nutritional supplement** → only for absent activation pathway
  - Orthopaedic surgeon alone → no → needs **nephrologist**
  - or **endocrinologist** → acid-base balance, kidney function
  - Harm risk → unnecessary **renal stones**
  - Different monitored parameters in activation problems
  - Active form needed → other specialty → **guide through proper treatment pathway**

Exam Pearl

- Nutritional rickets → inactive form only
- Active form → no RDA, no storage, kidney stones
- Active form → does not raise 25-hydroxy level
- Active form indications → vitamin D-resistant rickets, activation failure
- Sun → forms vitamin D; does not activate it

Windswept Deformity Before 6–7 Years Ask AI

Windswept

- Q: **Windswept deformity** presenting before 6–7 years?
  - A: 3 years old → **medical treatment** only + follow-up
    - Genu varum part → corrects until 4–5 years
    - Genu valgum part → until 6–7 years
  - At 6–7 years → medical correction very difficult
  - Varum correction age passed; valgum about to pass
  - Correct **metabolic error** → 8-plates with planning
    - Tibial or femoral? Exact plate sites?
    - One side tibial only, other femoral + tibial
    - Both sides femoral + tibial, opposite to each other
    - → **proper measurements**
  - Separate lecture → **planning of frontal plane**
  - **Malalignment of the lower limb** → where to correct, CORA
- Never operate while metabolically deficient → treat until bone strong
  - → fixation holds, good **healing**
  - **Very poor** / **osteopenic** bone:
    - Screws → very loose
    - Osteotomy → **collapse** from bone weakness
    - Healing rate → low and delayed
  - Always improve patient **metabolically**
  - → before surgery

Duration of Vitamin D Ask AI

Vitamin D

- Q: Continue vitamin D until **closure of the physes**?
  - A: No such rule → no natural vitamin D → artificial supplementation
  - → for life
  - Example: adult never seeing sun (office, operating theatre)
    - → **bone pains**, muscle weakness, depressed immunity, osteomalacia
  - **Mature** → no deformity; immature → deformity
  - Immature → adequate vitamin D avoids deformity
  - Later → avoids **osteomalacia**, osteopenia
  - Also → immunity, immune diseases, bone disease, muscle weakness, pains, fracture rate

Timing of 8-Plate Removal Ask AI

8-plate

- Q: When to remove the **8-plate**?
  - A: As soon as corrected → never judged clinically → patient may be **obese**
  - "Knees and tibiae touching" in obese → still genu varum
  - Going into genu valgum → which degree suits his obesity?
  - **Long-film radiograph**, standing, patellae anterior → measure **mechanical axis**
  - Axis corrected and normal → remove; not corrected → keep
  - **Blount disease** → overcorrect a little (amount unknown)
    - Not removed exactly at correction → lower incidence of recurrence
  - Very young + **femoral correction by 8-plate** → overcorrect → rebound overgrowth
  - Tibia → no worry about rebound overgrowth
- App link → Dr. Hisham → posted on **Telegram group**

> **Clinical Pearl**
> - Plate removal → long-film standing radiograph, mechanical axis normal
> - Blount → overcorrect a little before removal
> - Very young femur → overcorrect → rebound overgrowth
> - Tibia → no rebound overgrowth worry

Osteoclasis Technique Ask AI

Elevation

- Q: Elasticity of osteotomy technique → **osteoclasis of the epiphysis**
  - Osteotome stops at **physis** → tibia lifted and moved → osteoclasis
  - Osteotome not driven into epiphysis
  - X-ray → destruction on **medial side**
  - Blount → **posterior and medial depression**
  - Posterior part → elevated by **Schanz pins** once inserted
  - Pins not **perpendicular** → raised when frame perpendicular to tibia
  - Medial part elevated → corrects **medial slope of the epiphysis**

Fasciotomy in Acute Correction Ask AI

Q&A

- Q: **Fasciotomy** needed in acute correction?
  - A: Many recommend → proximal tibial osteotomy → main risk **compartment syndrome**
  - Particularly **anterior compartment**
  - Fibula cut → anterior compartment fasciotomy
- Moderator → remaining questions → planning lecture; thanks Dr. Hisham
- Next lecture → more questions

Management of Rickets

(Aetiology, biochemical and radiological features, and medical and surgical treatment)

Prof Dr Hisham Abdel-Ghani

Watch Original Lecture

Lecture transcript processed with AI assistance.

Contents

↑ Book contents

Definition of Rickets . . . 292
Lecture Plan . . . 292
Definition . . . 292
Causes of Inadequate Mineralization . . . 293
Physeal Mineralization and Alkaline Phosphatase . . . 293
Radiological Hallmark: Widened Physis . . . 293
Physeal Zones . . . 293
Mineralization Triggers Apoptosis . . . 294
Pyrophosphate and Alkaline Phosphatase . . . 294
Compensatory Rise of Alkaline Phosphatase . . . 295
Clinical and Radiological Diagnosis . . . 295
Diagnostic Hallmarks . . . 295
Clinical Picture . . . 296
Radiological Picture . . . 296
Causes and Classification of Rickets . . . 297
Nutritional vs Genetic . . . 297
Genetic Types . . . 297
Why Calcium Deficiency Acts Through Phosphorus . . . 297
Practical Classification: Calciopenic vs Phosphopenic . . . 298
Low-Phosphorus Environment at Any Age . . . 298
Laboratory Investigations . . . 298
Basic Tests . . . 298
Hypophosphatasia . . . 299
Rachitic Picture with Normal Labs . . . 299
Vitamin D Metabolism . . . 300
Synthesis and Activation . . . 300
Levels at Which Metabolism Fails . . . 301
Stages of Vitamin D Deficiency . . . 301
Priority: Blood Calcium . . . 301
Stage 1: Hypocalcemic Stage . . . 302
Later Stages: Active Rickets . . . 302
Nutritional Calciopenic Rickets . . . 303
Mutual Unmasking . . . 303
Differentiation . . . 303
Inactive vs Active Vitamin D . . . 304

Radiological Response and Deformity Correction . . . 305
Summary of the First Two Causes . . . 305
Genetic Calciopenic Rickets . . . 306
25-Hydroxylase Deficiency . . . 306
1-Alpha-Hydroxylase Deficiency . . . 306
End-Organ (Receptor) Unresponsiveness . . . 307
Phosphopenic (FGF23-Dependent) Rickets . . . 307
PTH-Dependent vs FGF23-Dependent . . . 307
Phosphate Homeostasis . . . 308
FGF23 . . . 308
PTH and Vitamin D Respond to Low Calcium . . . 309
Physiological Effects of Elevated FGF23 . . . 309
Pathological FGF23 Excess . . . 309
PTH Level as the Key Discriminator . . . 310
Random Active Vitamin D and Nephrocalcinosis . . . 311
Take-Home Messages . . . 311
Questions and Answers . . . 312
Renal Failure and Activation . . . 312
"Renal Rickets" . . . 313
Ionized Calcium . . . 313
Why Tetany Is Not a Feature of Rickets . . . 313
Low 25-Hydroxyvitamin D Despite Storage . . . 313
Practical Management: Vitamin D Dosing and Referral . . . 314
Vitamin D Dose in Active Rickets . . . 314
Vitamin D Preparations . . . 314
FGF23 Source and Urine Analysis . . . 314
When to Refer . . . 315
Vitamin D Frequency, Injections and Calcium . . . 315
Oral Frequency . . . 315
Injections . . . 316
Vitamin D Toxicity . . . 316
Calcium Dosing . . . 316
Moderator's Summary . . . 316

AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 297) “remaining 15% (genetic)” → “13% of cases”

Queried - kept as spoken, but looks wrong

Q1 (p. 299) as spoken; hypophosphatasia typically has normal/high phosphorus and calcium with low-normal PTH; also contradicts next lines

Q2 (p. 300) as spoken; chronic renal failure characteristically reduces 1-alpha-hydroxylation, hence active vitamin D use in CKD

Q3 (p. 300) as spoken; osteoblasts and chondrocytes do express vitamin D receptors

Q4 (p. 308) as spoken; FGF23 was first identified and published in 2000

Q5 (p. 316) as spoken; children's daily calcium requirement is about 500-1300 mg, not 250-400 mg

Definition of Rickets Ask AI

Lecture Plan Ask AI

- Difficult topic → requires concentration
- Lecture on YouTube → understand, don't take notes
- Topic → **rickets** from the paediatric orthopaedic perspective
- Much confusion in reaching the diagnosis
- Studied in 5th-year paediatrics, master's and doctorate → still not an ordinary topic
- Orthopaedic surgeon → must know his **limits**
  - Diagnosis
  - Recognising types
  - Treatment and management
- Lecture gives the **core** → base for further reading
- Plan
  - Definition
  - Clinical diagnosis
  - Types
  - Diagnosing the type → frame for treatment

Definition Ask AI

- **Metabolic bone disease**
- Inadequate mineralization of the **growing bone**
  - Poor mineralization of bone
- Cause → disrupted calcium, phosphorus and/or vitamin D metabolism
- Essential difference → the **physis**

	Rickets	Osteomalacia
Skeleton	Growing	Mature (adult)
Physis	Present	Absent
Defect	Poor mineralization	Poor mineralization

- Metabolic process → reflected entirely in **physeal changes**
  - Radiological diagnosis
  - Pathological diagnosis

Causes of Inadequate Mineralization Ask AI

- **Calcium deficiency** → building without bricks
- **Phosphorus**
  - Abundant in all foods → absorbed automatically from intestine
  - **Nutritional phosphorus deficiency** → does not exist
  - Contrast → nutritional calcium deficiency exists
    1. **Milk allergy**
    2. No milk or yoghurt
    3. Famine; poor populations
    4. No calcium-containing or **calcium-fortified** products
    5. Legume-heavy diet → **phytates** → block calcium absorption
  - Calcium → deficient by low intake; phosphorus → never
- **Vitamin D metabolism** → nutritional or non-nutritional causes
  - No "vitamin D deficiency" label → until other **metabolic errors** excluded
- Bulk of pathology → in the **physis**
- One, two, or all three → rickets

Physeal Mineralization and Alkaline Phosphatase Ask AI

Radiological Hallmark: Widened Physis Ask AI

- X-ray → look at the physis first
- Normal physis → about 2 mm
- Rickets → about 5 mm → **widened physis**
- Large part remains **cartilage** → no mineralization of newly formed cartilage
- Normal → each added portion → mineralized → bone
- Rickets → remains cartilage → widening from poor mineralization

Physeal Zones Ask AI

- Epiphysis on one side, metaphysis on the other
- **Germinal cell layer** → epiphyseal side
  - Only multiplying cells in the physis
  - Injury → **growth arrest**
    1. Partial or complete
- Sequence
  - Germinal cells multiply → **columns of cells**
  - Cells enlarge → **hypertrophic zone** → store material
  - **Apoptosis** → calcification of cartilage (not ossification) → **zone of enchondral ossification**
- Order → **proliferative zone** → hypertrophic zone
- Hypertrophic zone → **weakest point** of the cartilage
  - Fractures → always here
- Storage + enlargement → cell death → **vascular invasion** → bone formation

Mineralization Triggers Apoptosis Ask AI

- Trigger of **apoptosis** → mineralization around and within the cells
- Mineralization → forming **hydroxyapatite**
- Requires calcium + phosphorus → exceed **equilibrium** → crystal formation
- One crystal → **propagates** → many crystals
- Crystals → cell death
- **Failure of apoptosis**
  - Primarily → **hypophosphatemia**
- Calcium always providable; insufficient phosphorus → no crystals
  - **Solubility product** of calcium and phosphorus never exceeded → no crystallization
- Low phosphorus → crystallization never occurs
- No nutritional phosphorus deficiency, yet physis may have a **low phosphorus environment**
  - Prevents apoptosis
  - → no calcification → no ossification
- Normal people → calcium and phosphorus in blood → why no tissue calcification?

Pyrophosphate and Alkaline Phosphatase Ask AI

- **Pyrophosphate** → natural inhibitor of hydroxyapatite formation
  - Prevents calcium binding phosphorus
- **Alkaline phosphatase** → antagonist of pyrophosphate
  - → calcium joins phosphorus → hydroxyapatite crystal
- Growing child → crystallization + physeal bone formation → alkaline phosphatase present
- Many types; relevant one → **bone alkaline phosphatase (BALP)**
- BALP inhibits pyrophosphate → **inorganic phosphorus** deposited → calcification
- No inorganic phosphorus → alkaline phosphatase keeps rising
  - Attempt at calcification → fails
- Alkaline phosphatase → makes available and increases phosphorus
- Pyrophosphate binds inorganic phosphorus → blocks its **availability**
  - → no ossification, no hydroxyapatite
- Rising alkaline phosphatase → stops pyrophosphate
  - → ↑ **local inorganic phosphorus**
  - → phosphorus + calcium join → crystal
- Crystal inside cell → cell death → propagation → calcification
  - In **hypertrophic zone** and calcification zone of physeal plate
- Without alkaline phosphatase
  - Pyrophosphate inhibition unopposed
  - Inorganic phosphorus can't exceed the solubility product
  - → no hydroxyapatite deposition

Compensatory Rise of Alkaline Phosphatase Ask AI

- No phosphorus → **no mineralization**
  - No alkaline phosphatase, or
  - Alkaline phosphatase present but phosphorus low
- Phosphorus low → its provider rises
  - **Biofeedback mechanism** → like insulin with blood sugar
- Phosphorus unavailable → **alkaline phosphatase rises**
  - → inhibits pyrophosphate → ↑ inorganic phosphorus → ossification
- Absent or very low phosphorus → rickets
- **Elevated alkaline phosphatase** → known feature of rickets
- Cause → ↓ inorganic phosphorus → no mineralization
- Basic function → **provide phosphorus**
  - Frees phosphorus from pyrophosphate → free inorganic → hydroxyapatite
- **Compensatory elevation** → covers the phosphorus deficit
  - Provides maximum inorganic phosphorus

Clinical and Radiological Diagnosis Ask AI

Diagnostic Hallmarks Ask AI

- Radiological → **wide physeal plate**
- Laboratory → **low phosphorus**

High Yield

- Rickets hallmark → **wide physeal plate** + low phosphorus
- Active rickets → never without **low phosphorus**
- Rising alkaline phosphatase → compensation for phosphorus deficit

Clinical Picture Ask AI

- **Genu varum** or genu valgum
- **Windswept deformity** → previous lecture (frontal plane malalignment)
- Every physis **swollen** → no ossification
- Thin physis → now large cartilage component
- Cartilage → **viscoelastic,** deformable
  - Axial loading → width increases
  - Even **muscle tone** alone, not only body weight
  - Width increases relative to height
- → **Broad metaphysis** → every bone end broad
- **Osteochondral junction** of ribs → broad
  - → **rachitic rosary** (rosary chest)
- **Pigeon chest** → diaphragmatic traction from inside
- **Flaring** of distal rib cage + indentation above → diaphragm
- These → signs of **severe rickets**
- **Pain**
  - Vitamin D deficiency
  - Bone weakness
  - **Micro-fractures**
  - Generalized non-specific musculoskeletal pain
- Vitamin D → **immunity**
  - Running nose
  - Influenza-like manifestations
- **Miserable baby**
  - Weak, can't run, tires quickly
  - **Growth failure**

Radiological Picture Ask AI

- Physis → **broad;** normal → not exceeding about 2 mm
- Metaphysis → also affected → appearance of **active rickets**
- Same physeal picture + normal neighbouring bone → not rickets
- Normal bone, abnormal physis only → **skeletal dysplasia**
  - **Chondrometaphyseal dysplasia** → three types
- Rickets → adjacent bone **osteopenic,** abnormal
  - Black dots → **moth-eaten appearance**

Causes and Classification of Rickets Ask AI

Nutritional vs Genetic Ask AI

- Nutritional → calcium or vitamin D deficiency
  - 87% of cases
- Genetic
  - 13% of cases *[C1 · transcript: "remaining 15% (genetic)"]*
- No nutritional phosphorus deficiency → phosphorus in all foods, not only fish
- Main vitamin D source → **sun**
  - Deficiency → **lack of sun exposure**, not dietary lack
- Calcium deficiency → no **dairy products** (main calcium source)
- Genetic → **enzyme** or absorption problems

Genetic Types Ask AI

- **Hypophosphatemic rickets** → primary loss of phosphorus
  - Intestinal absorption normal, independent of anything else
  - Control of phosphorus level
    - Through renal **excretion**
    - ↓ excretion → phosphorus rises
    - ↑ excretion → phosphorus falls immediately
  - Primary pathology → excessive loss → impaired **renal tubular reabsorption**
- **Vitamin D–dependent rickets**
  - Vitamin D must be activated
  - Then acts on receptors → defect in **activation** or receptors
  - → calcium deficiency → rickets

Why Calcium Deficiency Acts Through Phosphorus Ask AI

- Rickets → basically phosphorus deficiency, not calcium deficiency
- **Active rickets** → never unless phosphorus is low
- Deficient vitamin D activation → **low calcium environment**
- → ↑ **parathormone**
  - ↑ renal calcium reabsorption
  - ↑ phosphorus excretion
- → **secondary loss of phosphorus** → hypophosphatemic state → active rickets
- Active rickets → requires **phosphorus deficiency**
  - Primary → ↓ renal tubular reabsorption → **hypophosphatemic** rickets
  - Secondary → ↑ parathormone → **vitamin D–dependent** rickets
- Both → genetic types → 13% of all rickets
- Summary
  - Nutritional → calcium, vitamin D
  - Genetic (13%)
    - Vitamin D activity → acting through calcium
    - Primary phosphorus loss → **excessive renal loss** of phosphorus

Practical Classification: Calciopenic vs Phosphopenic Ask AI

- All rickets → **calciopenic** or phosphopenic
- Calciopenic → calcium deficiency → ↑ parathormone → ↓ phosphorus
  - All nutritional causes (vitamin D, calcium)
  - **Vitamin D–dependent** genetic causes
- Phosphopenic → **primary renal pathology** → phosphorus loss
  - Mediated by **fibroblast growth factor 23 (FGF23)**
  - FGF23 → still not taught among calcium/phosphorus homeostasis mechanisms

	Calciopenic	Phosphopenic
Primary defect	Calcium deficiency	Renal phosphorus loss
Cause of low phosphorus	↑ Parathormone	FGF23
Includes	Nutritional (vitamin D, calcium); vitamin D–dependent genetic	Primary renal phosphorus loss

Low-Phosphorus Environment at Any Age Ask AI

- Low phosphorus environment
  - **Failure of apoptosis** of hypertrophic chondrocytes
  - Failure of vascularization and bone formation
- Open **growth plate** + low phosphorus → active rickets
  - Any age → 13 years or 2 years
- 13-year-old girl → active rickets, **widened physes**

Laboratory Investigations Ask AI

Basic Tests Ask AI

- Calcium, phosphorus, alkaline phosphatase, **parathormone**
- ± Vitamin D → **25-hydroxycholecalciferol**
  - ± **1,25-dihydroxycholecalciferol**

Test	Expected finding in rickets
Calcium	Low or normal → never elevated
Phosphorus	Low or low-normal; high or normal → doesn't fit active rickets
Alkaline phosphatase	High → detaches phosphorus from pyrophosphate for calcification
Parathormone	Normal, high or very high → differentiates types
Vitamin D	Not discussed here

- Basic tests → roughly indicate the **type** of rickets

Hypophosphatasia Ask AI

- Broad physes, **broad metaphyses**, widened physes on X-ray
- Looks rachitic; adjacent bone similarly affected
- Phosphorus low, parathormone high → fits rickets *[Q1 · as spoken; hypophosphatasia typically has normal/high phosphorus and calcium with low-normal PTH; also contradicts next lines]*
- Exception → alkaline phosphatase **low**, very low
- → **Hypophosphatasia**
  - No alkaline phosphatase secreted
  - **Severe form** → sometimes lethal
  - Calcium, vitamin D, phosphorus present → can't ossify
    1. Can't detach phosphorus from **pyrophosphate**

Rachitic Picture with Normal Labs Ask AI

- X-ray, radiological and clinical picture → rickets (broad metaphyses)
- Labs good; alkaline phosphatase almost normal
- → **no active rickets**, despite the X-ray
- Look for **chondrometaphyseal dysplasia** or similar

Condition	X-ray	Alkaline phosphatase	Other labs
Active rickets	Wide physis; osteopenic adjacent bone	High	Low phosphorus
Hypophosphatasia	Rachitic	Low, very low	Phosphorus low, parathormone high
Chondrometaphyseal dysplasia	Abnormal physis	Almost normal	Good

Vitamin D Metabolism Ask AI

Synthesis and Activation Ask AI

- Formation → **sun exposure**
- **Misconception** → oral vitamin D (V-drop, Devarol) activated in skin
  - → useless without sun → no physiological basis
- No activation of vitamin D in the skin
- Skin → site of **formation**
  - Sun induces formation, doesn't activate
- Skin synthesis → storage
- Intestinal absorption → all to **liver**
- Liver → **25-hydroxycholecalciferol** → stored
- Blood 25-hydroxycholecalciferol → size of store → deficient or not
  - Reflects liver store → main store drawn upon
- Need → **1-alpha-hydroxylase** in the kidney → 1,25-dihydroxycholecalciferol
- Renal activation → **endocrine function,** unrelated to excretory function
  - Renal failure → not necessarily lacking the enzyme *[Q2 · as spoken; chronic renal failure characteristically reduces 1-alpha-hydroxylation, hence active vitamin D use in CKD]*
- Active form acts on → **intestine** and immune cells
  - → vitamin D important for immune function
- **T-cells** → activate vitamin D themselves
  - No renal activation needed
- **Bone** → no receptors for vitamin D or its active form *[Q3 · as spoken; osteoblasts and chondrocytes do express vitamin D receptors]*
- Action on bone → via ↑ calcium availability, not bone receptors
- Calcium provided → through the intestine

Levels at Which Metabolism Fails Ask AI

- Nutritional → no sun → **decreased synthesis**
- Liver → no 25-hydroxylation
- Kidney → **1-alpha-hydroxylase deficiency**
- All working → defect in **intestinal receptors**
  - No calcium absorption → no ↑ calcium availability
  - → **hypocalcemia** → ↑ parathormone
  - → ↑ calcium absorption + phosphorus excretion → **hypophosphatemic state** → rickets
- **Calcium deficiency** → equivalent of vitamin D deficiency
  - Vitamin D works on intestine, but no calcium to absorb or make
- Calcium → same line as vitamin D
  - Very low calcium → as if vitamin D can't work
- Any failing arrow → rickets
  - Deficient **intake**
  - Deficient activation in liver
  - Deficient activation in kidney
  - Non-response of target organ → intestine
  - No calcium → nothing for vitamin D to act upon

Stages of Vitamin D Deficiency Ask AI

Priority: Blood Calcium Ask AI

- Stages → rarely studied
- Vitamin D deficiency → no calcium absorption → calcium falls
- Calcium below a level
  - **Tetany**
  - Cardiac muscle stops
  - → death
- Body priority → maintain **blood calcium**
- Normal blood calcium → "you are fine" → wrong
  - Parathormone raised maximally to compensate
  - Absorbs any available calcium
- Not enough → parathormone acts on **bone** → stores mobilized to blood
  - ↑ renal reabsorption
- Blood calcium → **mandatory for life**
- Bone calcium → a store
- Patient may be sick with calcium at **lower limit of normal** or normal

Pitfall

- Normal blood calcium → doesn't mean the patient is fine
- Calcium maintained by **parathormone** at the cost of everything else
- Calcium deficiency → assess by dietary history, not calcium test

Stage 1: Hypocalcemic Stage Ask AI

- Vitamin D falls → calcium falls first (blue line)
- Parathormone must increase
- **Polypeptide hormone** → sudden rise → delayed response
- Parathormone rises but can't raise calcium → calcium falls
- Early stage → **tetany**, spasms
- Stage 1
  - Calcium → low
  - Phosphorus → elevated, not low
    - Parathormone high in blood but **not yet effective**
    - → no excessive phosphorus excretion
  - Vitamin D → low

Later Stages: Active Rickets Ask AI

- Parathormone effective, without **resistance**
  - ↑ calcium absorption → calcium rises
  - ↑ phosphorus excretion → phosphorus falls
  - ↑ calcium release from bone
- Calcium rises slightly → parathormone falls again
  - Each stage → responds to the previous
- Then parathormone keeps rising → compensating vitamin D deficit
- → **marked fall in phosphorus** → active stage of rickets
- Vitamin D–**PTH** interplay → controls calcium and phosphorus
  - Phosphorus → takes the hit
- Hypocalcemic (first) stage → **convulsions**, tetany, no rickets
  - Phosphorus not low
- Phosphorus low → active rickets

	Stage 1 (hypocalcemic)	Active rickets stage
Calcium	Low	Raised by parathormone
Phosphorus	Elevated	Markedly low
Parathormone	High, not yet effective	Effective, keeps rising
Vitamin D	Low	Low
Clinical	Tetany, convulsions; no rickets	Active rickets

Nutritional Calciopenic Rickets Ask AI

Mutual Unmasking Ask AI

- **Calciopenic rickets** → parathormone-dependent
- First type → vitamin D deficiency; second → calcium deficiency
- Mild vitamin D deficiency + plenty of dairy
  - → may not develop rickets
- Trivial vitamin D deficiency + no dairy, no calcium → **active rickets**
  - Parathormone sky-high, phosphorus low
- Minor deficiency of calcium or vitamin D
  - → **unmasked** by deficiency of the other
- Slightly greater calcium deficiency → unmasks vitamin D deficiency, and vice versa
- Complementary → ↑ calcium **intake** → ↑ absorption even without vitamin D
- Vitamin D deficiency + deficient intake → each unmasks the other

Differentiation Ask AI

Nutritional

- Both conditions
  - Calcium → low or normal
  - Phosphorus → low or low-normal → required for active rickets
  - Alkaline phosphatase → **high**
  - Parathormone → very high
    - Calcium deficiency → greatest **stimulus** for parathormone secretion
    - Parathyroid → detects any fall → emergency release
- Discriminator → **25-hydroxyvitamin D**
  - Vitamin D deficiency → low
  - Calcium deficiency → normal
- Normal vitamin D → needs **calcium**
- Low vitamin D → needs vitamin D
- Combined calcium deficiency?
  - → **history taking**, not calcium test
    - Calcium maintained by parathormone
  - Milk, yoghurt, dairy intake
- **Milk allergy** → milk and dairy stopped, no calcium supplement → common paediatric error
- 25-hydroxyvitamin D → **basic test**, available in all laboratories
- 1,25 → less routine
- Not needed here → calcium given anyway
- Precise differentiation → **active form** (1,25)
  - Vitamin D deficiency → low
  - Calcium deficiency → increased
    - Body activates more vitamin D to raise calcium
- Usually → stop at **25-hydroxyvitamin D** level

	Vitamin D deficiency	Calcium deficiency
Calcium	Low or normal	Low or normal

	Vitamin D deficiency	Calcium deficiency
Phosphorus	Low or low-normal	Low or low-normal
Alkaline phosphatase	High	High
Parathormone	Very high	Very high
25-hydroxyvitamin D	Low	Normal
1,25-dihydroxyvitamin D	Low	Increased
Basic need	Vitamin D	Calcium

Inactive vs Active Vitamin D Ask AI

Nutritional

- Treatment → **vitamin D and calcium**
- Inactive form given; active 1,25 measured
  - → **super-rise** in vitamin D activation
  - → **supraphysiological level**
    - Physiological range → from 20 to about 60 (upper value unclear in recording)
  - → gradual decline → normalized
- Active form → can't reproduce the natural physiological curve
  - Unknown level reached, dose, when and where to reduce
- Best → **inactive form**
  - No word "alpha" in its name
- Body → stores, releases and activates it **physiologically**
  - Complex control, activation and inactivation
- Active form → below or above need; no taper guidance
- Natural curve → **normal physiological mechanisms** of calcium/phosphorus homeostasis

	Inactive form	Active form
Storage	Stored	Not stored; short half-life
Missed doses	Store buffers	3 days off → deficient again
Cost	Cheap	Very expensive
Role	True medical supplement	Can't imitate natural curve
Calcium homeostasis	Follows normal mechanism	Blocks normal regulation

Radiological Response and Deformity Correction Ask AI

- Before treatment
  - Very wide, **flared** physis
  - Moth-eaten bone
  - Marked osteopenia
- After treatment
  - Ossification of non-calcified cartilaginous physis → normal width
  - Adjacent bone quality improves
- Residual line → **Harris line**
  - Marks the time of rickets
- Deformities → may correct
  - **Genu varum** → with treatment up to 4 years
  - Genu valgum → up to 6 years
- Never operate before allowing **physiological remodeling**

Summary of the First Two Causes Ask AI

- Still **calciopenic** → ↑ parathormone → hypophosphatemic state → rickets
  - ↑ calcium absorption + phosphorus excretion

Genetic Calciopenic Rickets Ask AI

25-Hydroxylase Deficiency Ask AI

25-OHase

- Third cause → adequate sun, inactive vitamin D, calcium (milk, yoghurt) → still rickets
- No activation to **25-hydroxycholecalciferol**
- Labs → classic rickets
  - Phosphorus → low or low-normal
  - Calcium → low or normal
  - Alkaline phosphatase → high → **active rickets**
  - Parathormone → sky-high
  - 25-hydroxyvitamin D → low
- Low level mimics vitamin D deficiency → give vitamin D + calcium
  - Repeat labs after three months with sure compliance
  - Unchanged; vitamin D still low → **no activation**
- Key → **low 25-hydroxyvitamin D despite good intake**
- 1,25 → also low
- Danger → starting with the active form → patient improves
  - Inactive form first → **test treatment**
  - Skipped → diagnosis lost
- **Lifelong** active form → compensates defective activation
  - vs deficiency rickets → stop after treatment, better lifestyle, sun
- Treatment
  - Some → **high doses** of ordinary vitamin D → ↑ liver activation
  - Others → active form → 1-alpha form (heard as alfacalcidol) or calcitriol (as heard)
- Dose control → **labs**
  - Parathormone normalizes
  - Alkaline phosphatase normalizes
  - Phosphorus rises
- Strong parathormone **inhibition** begins
  - → reduce dose slightly; keep following
- Duration → **for life**
- No active form before investigation → can't distinguish from **deficiency rickets**

1-Alpha-Hydroxylase Deficiency Ask AI

1α-OHase

- Good liver activation; no renal activation to **1,25-dihydroxycholecalciferol**
- Labs → **classic calciopenic rickets**
- 25-hydroxyvitamin D → normal or high
- High parathormone, high alkaline phosphatase, low phosphorus + normal/high 25 → paradox
- Almost the only patient needing **1,25** measurement
  - Very expensive; sent abroad; not standard
- 1,25 low + 25 normal/high → **renal activation** missing
- Treatment → **active form**

End-Organ (Receptor) Unresponsiveness Ask AI

End-organ

- Sun, vitamin D, calcium, liver and kidney activation → all good → still rickets
  - Low phosphorus; parathormone sky-high
- Cause → **no intestinal receptors**
- Labs → classic calciopenic
  - Calcium → normal or mildly low
  - Phosphorus → low
  - Alkaline phosphatase → high
  - Parathormone → high
  - 25-hydroxyvitamin D → normal or increased
  - 1,25 → **increased** → body keeps activating, can't act
- → **Lack of responsiveness**
- Many, not all → **alopecia**
  - No hair, no eyebrows + signs of active rickets (broad metaphyses)
- No calcium absorption → regardless of calcium given
- Active vitamin D → **useless**
  - No intestinal receptors to act on
- **Oral calcium** supplementation → not absorbed
- Treatment → **calcium infusion** (intravenous)
  - Not at home
  - Hospital, **ICU**
  - Managed by nephrology or endocrinology
- Improves somewhat **at maturity**
  - **Sex hormones** → non-vitamin D–dependent intestinal calcium absorption
  - Absent in the immature
- **Repeated calcium infusion** → until maturity

Clinical Pearl

- Alopecia + active rickets → **end-organ unresponsiveness**
- 25 normal/increased, 1,25 increased
- Oral calcium and active vitamin D → both useless
- IV calcium infusion in ICU → until maturity

Phosphopenic (FGF23-Dependent) Rickets Ask AI

PTH-Dependent vs FGF23-Dependent Ask AI

- Rickets → calciopenic or phosphopenic
- Calciopenic → **PTH-dependent**
  - Calcium deficiency → PTH → phosphorus deficiency → active rickets
  - Any defect in vitamin D **metabolic pathway** or calcium absorption
- Phosphopenic
  - Phosphorus → regulated through **excretion**, not absorption
  - → **FGF23-dependent**
  - FGF23 → causes the low phosphorus

Phosphate Homeostasis Ask AI

- Controlled basically by **renal excretion;** not mainly by intestine
- Vitamin D
  - ↑ phosphorus absorption; ↓ phosphorus excretion
  - ↑ **renal reabsorption**
- Parathormone → ↑ renal phosphorus excretion
- FGF23 → acts on **phosphorus only**

	Calcium (kidney)	Phosphorus (kidney)
Vitamin D	↑ Reabsorption	↑ Reabsorption
Parathormone	↑ Reabsorption	↑ Excretion
FGF23	—	↑ Excretion (only action)

FGF23 Ask AI

FGF23

- Source → **bone**
- Discovered 1999 → still rarely discussed outside internal medicine, nephrology *[Q4 · as spoken; FGF23 was first identified and published in 2000]*
- Target → **kidney**
- Function → ↑ renal phosphorus excretion → **hypophosphatemic states**
- Main stimulus → **hyperphosphatemia**
- Inhibitor?
  - Phosphorus falls → FGF23 stops
  - → vitamin D ↑ **renal phosphorus reabsorption**
- Simplified → lowers phosphorus → stimulated by ↑ phosphorus
- Inhibited by → **phosphorus deficiency** → ↓ renal excretion
- Logic → large phosphorus load (e.g. **fish**) → risk of ectopic calcification
  - → ↑ renal **excretion**
  - → ↓ vitamin D activation → stops 1-alpha-hydroxylase
    - → no ↑ phosphorus absorption
- Real topic → more complex; simplified here

PTH and Vitamin D Respond to Low Calcium Ask AI

- Parathyroid and vitamin D → main stimulus **low calcium**
- Parathormone
  - Releases calcium from bone
  - ↑ renal **calcium reabsorption**
  - ↑ renal phosphorus excretion
  - **Activates vitamin D**
- Vitamin D
  - ↑ calcium and phosphorus absorption
  - **Corrects** parathormone's renal phosphorus loss → more phosphorus in blood
  - Prevents **marked phosphorus drop**
  - Normalizes calcium
- Normal blood calcium → result of compensation; all else disturbed
  - **Hypocalcemic state** → not tolerated

Physiological Effects of Elevated FGF23 Ask AI

- FGF23 rises → high blood phosphorus
  - Vitamin D adds to it
- Actions
  - ↑ renal **phosphorus excretion**
  - Stops vitamin D → ↓ renal reabsorption
- → ↓ hyperphosphatemic state → ↓ blood phosphorus
- ↓ vitamin D activation → ↓ intestinal calcium [and phosphorus] absorption
- Net → **phosphorus wasting**
- Key part of **phosphorus homeostasis**

Pathological FGF23 Excess Ask AI

- Basis for lab differentiation → **calciopenic** vs phosphopenic
- Marked FGF23 excess → main target **kidney**
  - Excessive phosphorus excretion
- Effects on parathormone
  - **Hypophosphatemia** → stops parathormone
    - Main trigger → calcium; very low phosphorus still stops it
    - Inhibition → weaker than low-calcium stimulation
  - **Direct inhibition** of parathormone secretion by FGF23
  - ↓ vitamin D activation → ↓ calcium → **stimulates** parathormone
    - Only drive keeping parathyroid secreting
    - Slight **hypocalcemia** → keeps bone going
    - Opposes **negative biofeedback** of hypophosphatemia + direct FGF23 effect
- Labs → almost ordinary rickets
  - Only difference → parathormone **not sky-high**, unlike calciopenic

PTH Level as the Key Discriminator Ask AI

- Phosphopenic → two parathormone **inhibitors,** one stimulator
- Calciopenic → everything stimulates parathormone
  - Low calcium
  - Low vitamin D → no calcium
- Low or not very high parathormone + very low phosphorus → **primary phosphopenic**
  - Hypophosphatemia → not parathormone-driven → **excessive FGF23**
- Labs
  - Calcium → normal
  - Phosphorus → low, very low
  - Alkaline phosphatase → elevated → **active rickets**
  - Parathormone → normal or mildly elevated
  - 25-hydroxyvitamin D → normal
- → **phosphopenic**, FGF23-dependent, not PTH-dependent
- 1,25-dihydroxyvitamin D
  - Calciopenic → high, keeps rising
    - Compensatory
  - Phosphopenic → **low** → FGF23 inhibits activation to the active form

Must-Know

- Phosphopenic vs calciopenic → **parathormone level**
- Calciopenic → parathormone very high, sky-high
- Phosphopenic → normal or mildly elevated
- 1,25 → high in calciopenic, low in phosphopenic

Random Active Vitamin D and Nephrocalcinosis Ask AI

- Scenario → **random treatment**
- Only parathormone drive → slight hypocalcemia from vitamin D shortfall
- Active vitamin D given → ↑ calcium absorption → calcium rises
  - → **negative feedback** → parathormone stops completely
  - → almost **silent bone**, very poor activity
    - Parathormone → most important activator of the **metabolic unit of bone**
  - → ↑ vitamin D function
- Parathormone → **calcium reabsorption** from kidney + phosphorus excretion
  - Phosphorus excretion → already FGF23-driven
- No parathormone → ↑ renal calcium excretion + plenty of phosphorus
  - → **nephrocalcinosis**
  - → renal stones
- Previously prevented → parathormone reabsorbing calcium, keeping it out of urine
- Random active vitamin D → removes the only parathormone stimulus
- → Origin of belief "vitamin D causes **nephrocalcinosis**"
  - True only when misused in a patient not needing it
- Treated by **nephrology**; these types very numerous
- Target parathormone → **slightly high**
  - Normal or strongly inhibited parathormone
    - → nephrocalcinosis at once

Pitfall

- Active vitamin D in phosphopenic rickets → parathormone stops
- → ↑ urinary calcium + phosphorus → **nephrocalcinosis**, renal stones
- Normalized parathormone → no reason for satisfaction
- Keep parathormone slightly high

Take-Home Messages Ask AI

- Rickets → **clinical and radiological diagnosis**
  - Labs → differentiate types, not confirm diagnosis
- Low alkaline phosphatase → exclude **hypophosphatasia**
- Everything fine + alkaline phosphatase not elevated → **skeletal dysplasia**
- Main discriminator → **parathormone level**
  - Hypophosphatemic → not very high → two inhibitors
  - Calciopenic → many stimulators
- Not every patient needs **25 and 1,25**
  - Mostly → 25 only
  - 1,25 → very few cases
    - Confirm **end-organ failure** or other problem
- **Non-active form** → gold standard of treatment
  - Empirical active form → spoils the picture, delays diagnosis
  - Treats conditions truly needing active form unknowingly
- **Correct treatment** → starts with correct diagnosis

- Active form treating 25- or 1,25-inactivation unknowingly
  - These → **metabolic errors**, genetic not nutritional
  - → treatment stopped → relapse
- Genetic types → **lifelong treatment**
- Active form → mostly prescribed by nephrology or endocrinology

Type	Calcium	Phosphorus	ALP	PTH	25(OH)D	1,25	Treatment
Vitamin D deficiency	Low or normal	Low or low-normal	High	Very high	Low	Low	Inactive vitamin D + calcium
Calcium deficiency	Low or normal	Low or low-normal	High	Very high	Normal	Increased	Calcium
25-hydroxylase deficiency	Low or normal	Low or low-normal	High	Sky-high	Low despite good intake	Low	High-dose vitamin D or active form, lifelong
1-alpha-hydroxylase deficiency	Classic calciopenic	Low	High	High	Normal or high	Low	Active form
End-organ unresponsiveness	Normal or mildly low	Low	High	High	Normal or increased	Increased	IV calcium infusion until maturity
Phosphopenic (FGF23)	Normal	Very low	Elevated	Normal or mildly elevated	Normal	Low	Nephrology

Questions and Answers Ask AI

Renal Failure and Activation Ask AI

- Q → active form mandatory in **renal failure** with hypophosphatemic rickets?
- A → no
  - Renal activation → **endocrine function**, not excretory
  - Renal failure → problem in excretion
  - Activation may still be good → **not mandatory**

"Renal Rickets" Ask AI

- Q → meaning of **renal rickets**?
- First classification → genetic vs nutritional
  - Genetic 13%, nutritional 87%
- Replaced by → **calciopenic** vs phosphopenic
- "Renal" covers both
  - 1,25-hydroxycholecalciferol inactivation → calciopenic
  - All phosphopenic conditions
- Rare in books → confuses physiology and diagnosis
  - Covers both **PTH-dependent** and FGF23-dependent → drop the term

Ionized Calcium Ask AI

- Q → benefit of **ionized calcium** measurement?
- A → no benefit for orthopaedic surgeons
  - Paediatricians → tetany → hypocalcemic **convulsion** or not
  - Must measure at the time of the attack
  - After the **spasm** → may find nothing
- Occurs in **first stage** of vitamin D deficiency

Why Tetany Is Not a Feature of Rickets Ask AI

- Tetany → not a manifestation of **rickets**
  - Nutritional, vitamin D deficiency, calciopenic rickets
- Reason → **PTH high** → calcium maintained
- Tetany → when PTH not working
  - First stage of vitamin D deficiency → **resistance** to parathyroid supercharging

Exam Pearl

- Tetany → not a feature of **rickets**
- Tetany → first (hypocalcemic) stage of vitamin D deficiency
- First stage → phosphorus not low → no active rickets

Low 25-Hydroxyvitamin D Despite Storage Ask AI

- Q → inactive form stored → why low 25-hydroxyvitamin D at diagnosis?
- Ordinary vitamin D → 25 form → **stored in liver**
  - Drawn according to needs
- Main source → **sun**
- Winter → little sun → store in summer for winter
- Stored level ↔ blood level → **equilibrium**
- Blood level → reflects **sufficiency** and store
- 25-hydroxycholecalciferol → enough vitamin D or not
- Active form → unsuitable
  - Short life
  - Activated from 25 to 1,25 **according to need**
  - End product of vitamin D

Practical Management: Vitamin D Dosing and Referral Ask AI

Vitamin D Dose in Active Rickets Ask AI

Vit D

- Vitamin D or calcium deficiency
  - Alkaline phosphatase high; calcium low or normal
  - Phosphorus low; parathormone sky-high
  - → **active rickets** → 3 times the daily requirement
- Daily requirement → 1000 units/day after 9–10 months
  - **Constant** throughout life → adult = child
  - Not by age or weight
- Active rickets → 3000 units
  - Per day → average

Age	Daily dose in active rickets
2 years	2000 units only
2–4 years	2000 units
5–6 years	3000 units
Over 9 years	4000–5000 units

- Duration → three months → **repeat labs**
- Labs improved, vitamin D normal, no sun or lifestyle change
  - → reduce to **daily requirement** → 1000 units/day

Vitamin D Preparations Ask AI

Preparation	Content	Suitability
V-drop	100 units/drop → 10 drops = 1000 units	Suitable
Decal B12 (as heard)	1000 units per 5 cc spoon	Suitable
Bidecal (as heard)	400 units per 5 cc	Not suitable
Calcid (as heard) and similar	200 units	Not suitable

- Not any vitamin D preparation fits → choose by **content**

FGF23 Source and Urine Analysis Ask AI

- **FGF23** → secreted from bone
- Urine analysis in rickets → not orthopaedic role
  - Needs **24-hour phosphorus excretion** → nephrology
- Phosphopenic, PTH not very high, suspected FGF23 excess
  - → **paediatric nephrologist**, not ordinary nephrologist
  - → or paediatric endocrinologist

When to Refer Ask AI

- Engage others vs continue → key **management** decision
  - Proven **nutritional** → treat
  - 25-hydroxyvitamin D deficiency
    - → continue
  - 1,25-hydroxyvitamin D deficiency → **active form**, continue
    - Confirm calciopenic → PTH sky-high (800, 700, 1000)
  - End-organ failure + **alopecia** → calcium infusion
    - Not the surgeon's job → dosing unknown
    - **Endocrinologist** + paediatric ICU → protocols, repeated admissions
    - Calcium all at once or concentrated → kills
  - Manifestations of rickets + **stunted growth**
    - Physes not classic rickets; labs severe
    - PTH 100, 150, 200 pg → doesn't fit
    - → probably **phosphopenic**, not calciopenic
    - → immediate transfer to paediatric nephrologist
    - Surgeon → **deformities** only
    - Not metabolic errors

Vitamin D Frequency, Injections and Calcium Ask AI

Oral Frequency Ask AI

- Q → calcium dose; V-drop once weekly or daily?
- **Oral** route
  - Daily > weekly > monthly → limited **capacity of absorption**
- 1000 units/day → monthly 30,000 at once (30 × 1000 or 6 × 5000)
  - Absorbing 30,000 at once → unproven
- Distribute across days → **absorption capacity** and variation
- Daily > weekly > monthly > every 6 months > **single dose**
- Weekly option → 10 drops × 7 → 70 drops in water once a week
- Three bottles at month's end → no data on absorption
- Best → every day, at most **every week**

Injections Ask AI

- **Injection** → certain delivery, independent of intestinal absorption
  - Absorption varies → day to day, prior meal
  - Fat → ↑ absorption; no fat → ↓
- **Devarol** → 200,000 units/ampoule → enough for 200 days
- One dose covering 200 days → **no evidence**
- Preferred → half ampoule every 3 months
- Evidence of limited storage
  - Optimized level + 3–4 months without treatment or sun → **deficiency**
  - Body storage capacity → insufficient
- Months protected by an optimized level → unknown to date
- Whole ampoule for six months → excess not stored
  - → converted to **24,25 form** (inactive), not 1,25 → discarded

Vitamin D Toxicity Ask AI

- Three weekly injections → 600,000 units in a one-year-old
  - 600,000 units → **two-year dose**
  - → **vitamin D toxicity**

Calcium Dosing Ask AI

- Two cups of milk (250 mL each) per day → **enough**
- Or one cup of milk + two cups of yoghurt
- **Calcium supplement** → only if these not taken
- Many calcium preparations
- Requirement → 250–400 mg *[Q5 · as spoken; children's daily calcium requirement is about 500-1300 mg, not 250-400 mg]*
- Higher requirement
  - Pregnancy
  - **Lactation** → even more
  - Small child
  - Adolescent → growth spurt
- Calcium requirements → **variable**
- Medicinal calcium → a little calcium dissolved in water
- Milk and yoghurt → many growth **factors** beyond calcium
- **Calcium injection** → doesn't exist
- Milk allergy + refusal of daily medicine
  - Vitamin D → injection possible
  - Calcium → no injection, no weekly dose
  - → must be taken **every day**

Moderator's Summary Ask AI

- **Oral** → better than injection
- Injection → half ampoule every three months → 100,000 units
- **French ampoules** → 600,000 units → not advised
- Egyptian **Devarol** → 200,000 units
  - Half ampoule → 100,000 units → 100 days → about 3 months

Supracondylar and Lateral Condyle Fractures of the Humerus

(Classification, assessment, reduction and fixation in children)

Prof Dr Hisham Abdel-Ghani

Watch Original Lecture

Lecture transcript processed with AI assistance.

Contents

↑ Book contents

Supracondylar Humerus Fracture: Overview . . . 322
Epidemiology . . . 322
Mechanism of Injury . . . 322
Radiological Assessment . . . 322
Fat Pad Sign . . . 322
Anterior Humeral Line . . . 323
Baumann's Angle . . . 323
Gartland Classification . . . 323
Associated Injuries . . . 324
Bony Injury . . . 324
Nerve Injuries . . . 324
Vascular Injury . . . 325
Treatment Indications . . . 325
Non-operative Treatment . . . 325
Rationale of Type II Conditions . . . 325
Operative Treatment . . . 326
Reduction and Fixation Technique . . . 326
Closed Reduction: Principles . . . 326
Reduction Sequence . . . 327
Forearm Rotation and Distal Fragment . . . 327
Positioning for Imaging . . . 327
Pin Insertion . . . 328
Open Reduction: Indication . . . 328
Open Reduction: Approaches . . . 328
Timing and Urgent Fixation . . . 329
Timing of Reduction . . . 329
Urgent Fixation Indications . . . 329
Compartment Syndrome in Children . . . 330
Type II Management and Pin Fixation . . . 330
Type II Controversy . . . 330
Lateral vs Crossed Pins . . . 330
Lateral Pin Configuration . . . 331
Pin Technique and Errors . . . 331
Crossed Pins . . . 332
Anterior Approach and Vascular Exploration . . . 332

Approach for Open Reduction . . . 332
Absent Pulse: Decision Making . . . 333
Indications for Vascular Exploration . . . 333
Supracondylar Fracture: Summary Pearls . . . 333
Supracondylar Fracture: Questions and Answers . . . 334
Vascular Surgeon Sequence . . . 334
Intramedullary Wire . . . 334
Post-operative Care . . . 334
Compartment Syndrome in Children (Revisited) . . . 335
Fasciotomy . . . 335
Slab vs Cast . . . 335
Post-operative Nerve Injury . . . 335
Vascular Exploration Tips . . . 335
Closed Reduction in the ER . . . 336
Lateral Condyle Fracture: Classification and Imaging . . . 336
Characteristics . . . 336
Milch Classification . . . 336
Jakob Classification . . . 337
Treatment Logic . . . 337
Type I vs Type II on X-ray . . . 338
Song Classification . . . 338
Identifying Song 3 and Hidden Displacement . . . 339
Lateral Condyle Fracture: Management . . . 339
Management Principles . . . 339
Treatment by Stage . . . 340
Open Reduction Rationale . . . 340
Surgical Approach and Technique . . . 341
Fixation . . . 341
Open Reduction Key Points . . . 342
Why the Lateral Condyle Matters . . . 342
Final Questions and Answers . . . 343
Late Presentation . . . 343
Tourniquet . . . 343
CT and MRI . . . 343
Internal Oblique View . . . 343
Acceptable Rotation and Angulation . . . 343
Unfused Epiphysis vs Fracture . . . 344
Moderator Remarks . . . 344

Myositis Ossificans 344
Redo for Poorly Fixed Supracondylar Fracture 344
Counselling About Myositis Ossificans 345
Associated Ligament Injury 345
Pulse Loss After Reduction 345

AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

- **C1** (p. 322) “fat appears somewhat radio-opaque” → “fat appears radiolucent”
- **C2** (p. 329) “middle and ring fingers” → “index and middle fingers”
- **C3** (p. 338) “lateral gap less than medial gap” → “lateral gap greater than medial gap”
- **C4** (p. 339) “rotated fragment became type 4” → “rotated fragment became type 5”
- **C5** (p. 340) “ORIF especially if undisplaced” → “ORIF especially if displaced”
- **C6** (p. 343) “cubitus varus will remodel” → “cubitus varus will not remodel”

Queried - kept as spoken, but looks wrong

- **Q1** (p. 339) As spoken; intact-hinge pattern given 27% and 20-25% elsewhere; 47% is the equal-gap figure
- **Q2** (p. 340) As spoken; same equal-gap pattern was given 42% displacement earlier in the lecture

Supracondylar Humerus Fracture: Overview Ask AI

Epidemiology Ask AI SCH

- **Supracondylar fracture of the humerus** in children → seen by every grade, resident to professor
- Lecturer → Dr. Hisham Abdel-Ghani; YouTube channel under his name, further lectures
- **Most common fracture around the elbow** in children → ~59% (≈60%)
  - Lateral condyle fracture → ~11%
- **Peak incidence** → 5–7 years
- More common in **non-dominant hand**
  - Dominant hand busy with activity → fall onto non-dominant hand

Mechanism of Injury Ask AI SCH

- **Extension type** → vast majority
  - Fall on outstretched hand → elbow hyperextension
  - Supracondylar area → thin bone; contains thin **olecranon fossa**
  - **Olecranon process of ulna** acts as lever → breaks supracondylar region
- More common with **ligamentous hyperlaxity** (hypermobile, lax joints)
  - Less common with stiff ligaments
- **Flexion type** → 2%
  - Direct fall on elbow → backward force from olecranon
  - → supracondylar fracture with **flexed distal fragment**
- Split → extension 98% / flexion 2%

High Yield

- Most common pediatric elbow fracture → supracondylar, ~59%
- Peak age → 5–7 years; non-dominant hand
- Extension type → 98%; flexion type → 2%
- Extension mechanism → olecranon levers through thin olecranon fossa

Radiological Assessment Ask AI

Fat Pad Sign Ask AI SCH

- Very minimal fracture → **positive fat pad sign** may be only finding
  - Fat around every joint → radiolucent on X-ray *[C1 · transcript: "fat appears somewhat radio-opaque"]*
  - Any fracture → **hemarthrosis** (supracondylar, lateral condyle, other)
  - → fat pad elevated → radiolucent zone
  - Anterior and posterior pads; **posterior fat pad** → most significant
- Fracture line often invisible → becomes visible with **callus** at 10–12 days
  - Callus confirms fracture, even without visible displacement
- Positive fat pad sign → **something wrong in this elbow**

Anterior Humeral Line Ask AI

- **Anterior humeral line** → assesses displacement and adequacy of reduction
  - Along anterior cortex of humerus → distally through center of capitulum
  - Assessed only on **true (dead) lateral view**
- Never diagnose or call undisplaced from **oblique view**; do not accept it
  - Usual referral → AP + oblique; obtain **true lateral view**
  - True displacement → assessable only on true lateral
  - True lateral → **hourglass appearance** of condylar region and olecranon fossa

Age	Anterior humeral line should pass through
Younger than 4 years	Anterior third of capitulum (acceptable)
Older than 4 years	Center of capitulum, always

- Uses
  - Posterior displacement in **minimally displaced** or impacted fracture
  - Acceptability of **reduction** during manipulation
- Capitulum posterior to line → **posterior displacement** (extended distal fragment)
- Capitulum far anterior → **flexion type**, distal fragment excessively flexed

Baumann's Angle Ask AI

- **Baumann's angle** → radiological angle for alignment
  - Rationale → many elbow bones unossified, mostly cartilage → poorly seen
  - Used instead of articular line, articular surface or radiographic bony ends of distal humerus
  - Value → about 20 degrees valgus

Gartland Classification Ask AI

- Classification → **extension type** only; flexion type has none
- Two big categories → extension (98%) / flexion
- Extension sub-classification → **Gartland**; very easy

Type	Features
I	Undisplaced; fat pad sign; later callus; fracture line in place
II (IIA)	Intact posterior cortex; anterior cortex fractured; distal fragment hyperextended → anterior periosteum torn; some subtypes rotated; no loss of posterior cortical contact
III	Completely off-ended; distal humerus fully displaced from proximal fragment
IV	Unstable in flexion and extension; distal fragment highly mobile anteriorly and posteriorly; mostly iatrogenic

- **Type IV** mechanism → repeated manipulation (once, twice, three times) → posterior periosteum torn
  - Extension type + torn posterior periosteum → unstable in flexion and extension

- Posterior **Steinmann pin** or K-wire to lever fragment → tears posterior periosteum
  - → unstable in flexion and extension; key point in reduction
- Type IV reduction → K-wires in both fragments as **joysticks**

Pitfall

- Aggressive or repeated manipulation → torn posterior periosteum → Gartland IV
- Posterior pin used as lever → same result

Associated Injuries Ask AI

Bony Injury Ask AI SCH

- **Distal radius** → most common associated bony injury
- Never accept elbow X-ray without **forearm X-ray** → possible distal radial fracture
  - Changes manipulation → avoid heavy loading
- No excessive rotation or **force over the fracture**

Nerve Injuries Ask AI SCH

- **Anterior interosseous nerve (AIN)** → most common associated nerve injury
- Rule → injured nerve or vessel lies on **side of the torn periosteum**

Type	Torn periosteum	Nerve at risk
Extension	Anterior → bony spike penetrates anteriorly	AIN (anteromedial periosteum); radial nerve (anterolateral periosteum)
Flexion	Posterior → nerve exposed to bony end, unprotected	Ulnar nerve, most common

- **Radial nerve** → lateral aspect, anterior to elbow
- Possible exam question; revisited later
- Radial, ulnar and AIN injuries → mostly **self-recovering**
  - AIN or ulnar injury → reassure parents → recovery within ~3 months
  - **Preoperative diagnosis** → no one blames reduction for the injury
- **Post-treatment nerve injury** → alarming; a real problem
  - Causes → iatrogenic, **K-wire induced**, excessive manipulation
  - Hence preoperative documentation essential

Exam Pearl

- Nerve injury → side of torn periosteum
- Extension → AIN most common; radial with anterolateral periosteum
- Flexion → ulnar nerve
- Preoperative nerve injury → self-recovering, ~3 months; post-treatment → alarming

Vascular Injury Ask AI

SCH

- **Vascular injury** → uncommon, ~1%
  - Otherwise frequently seen, given high fracture incidence

Presentation	Capillary refill	Radial pulse	Significance
Pink, warm hand, intact pulse	Very good	Present	Excellent
Pink pulseless hand	Present	Absent	Observation; no problem
Pale pulseless hand	Absent or sluggish	Absent	Main concern

- **Pale pulseless hand** → the scenario that worries us

Treatment Indications Ask AI

Non-operative Treatment Ask AI

SCH

- ED presentation → residents and consultants alike must know management
- Options → non-operative / operative, like any fracture
- Non-operative → **posterior molded cast**
  - Above-elbow cast; posterior mold on olecranon → supports extended distal fragment
- Candidates → extension type only; **flexion type** → no non-operative option
- **Type I** (undisplaced) → posterior molded cast, very good results
- Cast position → **90 degrees of flexion** or slightly less
  - Excessive flexion → once used to hold bone → **compartment syndrome**
  - Some surgeons flex gradually → stop at **radial pulse attenuation**, short of 90
- **Type II** → possible if good reduction achieved
  - Good reduction → anterior humeral line through center of capitulum
  - Plus minimal swelling + no medial comminution → offer parents non-operative option

Rationale of Type II Conditions Ask AI

- More swelling → marked **soft tissue injury** → unstable fracture (any fracture)
  - Even after good reduction → swelling subsides → **redisplacement**
  - Excessive swelling → uncontrollable, very unstable, despite innocent first look
- **Medial comminution** → fragment sinks into varus
  - → **cubitus varus**, usual complication after supracondylar fracture

Operative Treatment Ask AI

SCH

- **Type III** (complete displacement) → all operative
  - No point in non-operative hassle
- Type II + excessive swelling, medial comminution or **unstable reduction** → operate directly
- **Flexion type** → always operative, directly
- Decision easy → type III and flexion type; difficult → **type II**

Category	Non-operative	Operative
Type I	Posterior molded cast	—
Type II	Good reduction, minimal swelling, no medial comminution	Excessive swelling, medial comminution, unstable reduction
Type III	—	All
Flexion type	Never	Always

Must-Know

- Type II operative triggers → excessive swelling, medial comminution, unstable reduction
- Cast → 90 degrees or less; excessive flexion → compartment syndrome
- Type III and flexion type → always operative

Reduction and Fixation Technique Ask AI

Closed Reduction: Principles Ask AI

SCH

- **Closed reduction and percutaneous fixation** → most commonly used operative option
  - → vast majority of cases
- First step → **gentle traction**
  - Lightest orthopaedic surgeon ~60–70 kg ; some heavier
  - Body-weight pull → **severe soft tissue injury**
- No body-weight leverage on a two-, three- or five-year-old child
  - → **compartment syndrome**
  - Anaesthetized muscles → **no resistance to traction** → soft tissues torn
- Always **gentle** with children → technique, not force
- Traction purpose → **disimpact** fracture or restore overlap; nothing more
  - Irreducible by gentle hand traction → irreducible by body weight; only tears tissues

Reduction Sequence Ask AI SCH

- Steps
  - **Gentle traction** → disimpaction
  - Supination and pronation → correct **rotation**
  - **Mediolateral pressure** → medial displacement: push medial epicondyle laterally
  - Lateral displacement → push medially; displacement mostly **medial**
  - Direct pressure on **posterior olecranon** → corrects posterior displacement
  - Elbow flexion → locks posteromedial fragment
- **Intact posterior periosteal hinge** → medial
  - Use it as **tension band**
- Olecranon pressure gentle; **aggressive manipulation** tears posterior periosteum
  - → very unstable → **Gartland type IV**

Forearm Rotation and Distal Fragment Ask AI SCH

- **Posteromedial fracture** → intact periosteum medial
- Elbow flexed → distal fragment direction controlled by **supination and pronation**
  - Distal fragment rotates with forearm in either direction
- Medial hinge intact → **excessive pronation** → closes lateral gap → prevents cubitus varus
- **Posterolateral fracture** → posterolateral periosteum as hinge (tension band)
  - → supination → closes medial gap

Displacement	Intact hinge	Forearm position	Gap closed
Posteromedial	Medial periosteum	Pronation	Lateral
Posterolateral	Posterolateral periosteum	Supination	Medial

- **No fixed forearm position** for every patient → depends on distal fragment
- Video → gentle traction by surgeon's own hands; no one else pulling

Positioning for Imaging Ask AI SCH

- Correct **medial displacement** first → posteromedial most common
- Traction + olecranon pressure → corrects **posterior displacement** or angulation
- Goal → hold fragment + good **lateral view** for K-wires
  - Sequence → traction → mediolateral correction → **pronation with flexion**
  - Internal rotation for imaging (posteromedial) → distal fragment moves
  - Solution → **external rotation,** arm beside patient
  - → fragment locked → **true lateral view** under image intensifier

Pin Insertion Ask AI SCH

- Preference → **lateral wires first**, on true lateral, after satisfactory reduction
- Good lateral configuration → nothing further needed
- Still unstable → **medial wire**
  - Never in **full flexion** → ulnar nerve stretched over medial condyle → injury
  - Reduce flexion slightly → palpate **ulnar nerve** → roll it posteriorly
  - Entry → **tip of medial epicondyle**
  - Never posterior aspect of medial epicondyle → aiming at ulnar nerve
  - Reduce flexion → palpate medial condyle → insert
- Large **hematoma** → severe soft tissue injury + unstable fracture
  - Example → 2 lateral pins → still unstable → **medial pin** added
- **Closed reduction** → successful in most cases

Open Reduction: Indication Ask AI SCH

- ORIF only when **fracture unreducible** by closed manoeuvres
  - "Cannot" → by a **professional orthopaedic surgeon**, not a junior resident
  - Concept → fracture not reducible by **closed manoeuvres**

Open Reduction: Approaches Ask AI SCH

- Options → posterior, lateral, combined lateral + medial, **anterior**
- **Posterior approach** → not preferred, abandoned long ago
  - Triceps adhesions → **stiffness**
  - Enters intact posterior periosteum → the **hinge** relied on for reduction
  - Obstacle anterior in extension type → posterior entry illogical
- Later practice → medial or lateral, or lateral only
- **Current trend** → **anterior approach**
  - Causes of failed closed reduction → **anterior problems**
  - Neurovascular entrapment at fracture site; muscle interposed between fragments
  - Removal + **neurovascular bundle exploration** → need anterior approach
- Obstacle to reduction → always anterior; mostly **neurovascular structures**
- Supracondylar fracture with **vascular injury needing exploration** → explore anteriorly
- One approach → **anterior** for both reduction and exploration

Approach	Verdict
Posterior	Not preferred → triceps adhesions, stiffness, violates intact posterior hinge
Lateral / combined lateral + medial	Older alternative
Anterior	Current trend → removes anterior obstacle + allows neurovascular exploration

Timing and Urgent Fixation Ask AI

Timing of Reduction Ask AI

SCH

- Supracondylar fracture → many unresolved **controversial issues** ; normal in medicine
- Scenario → 2 a.m. ED arrival: whoever present, or wait for experienced surgeon?
  - **Safe to delay surgery** 12–18 hours
  - Not urgent → closed reduction day 1 vs day 7 → same results (papers)
  - Junior → wait until morning → operate under **senior supervision**
  - → best results, fewer complications

Urgent Fixation Indications Ask AI

SCH

- Classic rules for urgent fixation of any fracture
  - **Open supracondylar fracture**
  - **Pale avascular limb** → no vascularity, no pulse; alarming dysvascular limb
  - = pale limb + no palpable radial pulse
  - **Floating elbow** → supracondylar + forearm fracture
  - **Skin puckering** → fragment about to penetrate skin
  - → reduce as soon as possible
- Median nerve injury + vascular injury
  - Avascular limb → opened directly
  - **Pink pulseless hand + median nerve injury** → worrying
  - → risk of **painless compartment syndrome**
  - **Median nerve** → most sensation of forearm and hand
  - Median injury → very liable to painless compartment syndrome → urgent
- Diagnosis not easy → must actively examine **median nerve** and AIN
- Median nerve test → absent flexion of **index and middle** fingers *[C2 · transcript: "middle and ring fingers"]*
  - Poor flexion → **median nerve injury** → urgent treatment
- Urgent treatment → closed reduction and fixation; not necessarily **open reduction**
- Also urgent → **evolving compartment syndrome**
- Also urgent → **cognitive disability**, uncooperative child, parents unable to manage

Compartment Syndrome in Children Ask AI

SCH

- **Diagnosed differently** from adults

Population	Diagnostic criteria
Adults	5 P's
Children	3 A's → agitation, irresponsiveness to analgesia, anxiety

- Child crying, collapsed, agitated; analgesics fail
  - Fracture + this picture → **compartment syndrome** developing or present
- Do not wait for pulse loss, **capillary loss** or other adult findings

> **Clinical Pearl**
> - Children → 3 A's: agitation, anxiety, irresponsiveness to analgesia
> - Pink pulseless hand + median nerve injury → painless compartment syndrome → urgent
> - Median nerve test → index and middle finger flexion

Type II Management and Pin Fixation Ask AI

Type II Controversy Ask AI

SCH

- Some closed, some open → what, and when?
  - Excessive swelling, medial comminution, **reduction not maintained** → operative directly
  - All absent → parents choose: surgery, or cast
  - Cast → **weekly X-ray**
  - Fragment moves or reduction lost within 1–2 weeks → **fixation**
  - Decision → left to parents
- Case → type II meeting **operative indications**
  - Mother physician; other child with **cognitive disorder** , non-verbal
  - Injured child → her only talking child; terrified of anaesthesia; accepting deformity over surgery
  - Non-operative trial not wrong → **posterior molded cast**, good reduction
  - Weekly X-ray follow-up, patience → outcome reported favourable
- No **sharp cut-off point** in type II → room to manoeuvre
- **Medial comminution** → cast in pronation → locks lateral side
  - Prevents distal fragment collapse → cubitus varus
  - **Pronation** locks lateral fragment → corrects distal fragment varus
  - Lateral comminution → opposite → **supination**
- X-ray every 5–7 days; displaces → **closed reduction and internal fixation**

Lateral vs Crossed Pins Ask AI

SCH

- Third controversy → **lateral pins only** vs crossed pins
  - Lateral only → good mechanical stability
  - Innumerable variables → same configuration stable in one, unstable in another
  - **Crossed pins** → mechanically superior; excellent stability
- **Medial pin** drawback → ulnar nerve injury ~3%
- Avoid medial pins where possible → **lateral pins only**

Lateral Pin Configuration Ask AI

SCH

- Two to three pins, **divergent at the fracture site**
  - Not crossing at fracture → divergent, or even **parallel**
  - Key → **far apart** at fracture level
- Stability → **test under image intensifier** after fixation
  - Internal + external rotation → stable → configuration adequate, whatever shape
  - Unstable → configuration insufficient → add **medial pin**
  - Stable on image intensifier and real-time exam → no **medial pin**
- Three good pins but unstable → add **medial pin**
- Avoid medial pin if possible; unstable at end of surgery → must insert

Pin Technique and Errors Ask AI

SCH

- **Thick pins** → thin pins fail to penetrate far cortex → travel in medulla
- Avoid excessive penetration and trials → **physis** injury → growth arrest
  - Fracture far from physis → growth arrest is **iatrogenic**
- Second pin difficult if first left long
  - Tip → after first pin **cut and bend** → correct orientation for second
  - Repeat → cut and bend second → insert third
  - Long wires block orientation; cut and bent → easy textbook insertion
- **Technical errors** with lateral pins
  - Thin pins
  - One pin intramedullary, engages far cortex; other **does not cross fracture**
  - One good; other thin → **slips inside medulla**
  - Good entry points , far cortex engaged, but **crossing at fracture level**
  - Crossing at fracture → no stability → unacceptable
- Guide → **stability test** under image intensifier after fixation
  - Unstable → immediately add **medial wire**

Crossed Pins Ask AI

SCH

- **Crossed pins** → best, ideal fixation
  - Must cross **above** fracture, not at its level → otherwise unstable
- Medial pin drawback → **ulnar nerve injury** 3–8%
- Current medial pin indication → **lateral pins failed** to achieve stability

	Lateral pins only	Crossed pins
Stability	Good mechanical stability; variable between patients	Mechanically superior; ideal
Configuration	2–3 pins, divergent or parallel, far apart at fracture	Cross above fracture, not at its level
Drawback	Fail if configuration inadequate	Medial pin → ulnar nerve injury 3–8%
Use	First choice	When lateral pins fail stability test

High Yield

- Lateral pins → divergent, far apart at fracture; never crossing at fracture
- Test stability under image intensifier → unstable → add medial pin
- Thick pins; avoid repeated passes → physeal growth arrest
- Medial pin → ulnar nerve injury 3–8%

Anterior Approach and Vascular Exploration Ask AI

Approach for Open Reduction Ask AI

SCH

- **Controversial issue;** possible exam question
- Best → **anterior approach**
  - Open reduction indication now → mostly **failure to reduce**
  - Failure → **anterior obstacle** to remove
- Usually also meets **vascular exploration** indication → same anterior approach for reduction
- No additional approach needed beyond anterior exploration of neurovascular structures

Absent Pulse: Decision Making Ask AI

SCH

- Absent pulse → the problem; **capillary refill** matters more than pulse

Situation	Action
Pink hand, capillary refill, no pulse	No exploration → closed reduction + fixation → observe at least 24 hours
Any capillary refill change during observation	Exploration
Pulseless at presentation → urgent closed reduction → refill and pulse return	Observe at least 24 hours → possible intimal injury → later thrombosis
Pulseless → manipulation → refill returns, pulse does not	No exploration, provided good refill and pink hand
Good pulse before → absent after closed reduction	Open reduction → free vessels
Aggressive manipulation → vascular injury	Exploration

- Summary → **capillary refill** decides
  - Problem at presentation or after manipulation → **observe**
  - Refill not returning or affected after reduction → **exploration with open reduction**

Indications for Vascular Exploration Ask AI

SCH

- **Persistent ischemia** after reduction
- **Median nerve injury + pulseless hand**
  - Brachial artery usually entangled in fracture with median nerve
- **Ischemia or nerve injury** developing after reduction
- **Emerging ischemia** in pink pulseless hand during 24-hour observation
  - After reduction → capillary refill decreases → exploration

Must-Know

- Capillary refill, not pulse, drives decision
- Pink pulseless hand → reduce, fix, observe at least 24 hours
- Restored pulse → still observe 24 hours → intimal injury, thrombosis
- Pulse lost after reduction → open reduction + exploration

Supracondylar Fracture: Summary Pearls Ask AI

- **Never pin without acceptable reduction**
  - AP view → no displacement; lateral view → **anterior humeral line**
  - Most helpful check during closed reduction
- **No small K-wires**

Age	K-wire size
2 or 2.5 years	1.8 mm
5–6 years	2 mm or 2.5 mm

Age	K-wire size
10 years	2.5 mm

- **Do not start high** in distal fragment → far from fracture
  - Very high insertion → proper lateral configuration impossible
- **Medial pin** → never in full flexion → ulnar nerve injury
- Wires → **divergent at the fracture site**
- After fixation → **test stability;** still unstable → fixation not enough
- After fixation → forearm in **mid position**
  - No excessive supination or pronation → avoids compartment syndrome
- Bending wire may pull it out → **hold steady** while bending
- After closed reduction and internal fixation → check **neurovascular status**
  - Sluggish capillary circulation or lost pulse after intervention → **exploration**

Supracondylar Fracture: Questions and Answers Ask AI

Vascular Surgeon Sequence Ask AI

- Q → frequent **vascular injury** cases, weekly
  - With vascular surgeon → who opens anteriorly first?
- A → **vascular surgeon** opens anteriorly, frees vessels and nerves
  - → distal fragment then easy to reduce and fix
  - Real issue → **definition of vascular injury**
  - **Absent pulse alone** → not an indication for exploration
  - Avoid overtreatment → no unnecessary surgery

Intramedullary Wire Ask AI

- Q → wire into medulla acceptable?
- A → **intramedullary K-wire** accepted as third wire
  - Condition → other 2 wires engage far (medial) cortex
- Past technique for juniors → **trans-olecranon intramedullary wire** + lateral pin
  - Not standard in closed reduction and internal fixation

Post-operative Care Ask AI SCH

- Cast → 3–4 weeks, until **good union** on X-ray
- After cast and wire removal → **no physiotherapy**
  - If any → very gentle, painless
  - Most children → **full range of movement** regained spontaneously
  - Time frame → two weeks, a month or a month and a half, own pace
- **No aggressive physiotherapy** or manipulation
  - → more injury, more stiffness , more complications
  - Range regained without **physiotherapy intervention**

Compartment Syndrome in Children (Revisited) Ask AI

- Cast change in children → invasive ordeal; avoid unless needed
- **Not diagnosed like adults**
  - Do not wait for pulse diminution, **ischemia** or pale hand
  - 3 A's → anxiety, agitation, **non-responsiveness to analgesia**
  - **Irritable**, fearful of anyone approaching
  - Crying, screaming, **tremendous pain**; staff unable to manage
  - Analgesia → no response
- **Passive stretch** test → not applicable; children less cooperative than adults
- Such a child → immediately think **compartment syndrome**; do not wait

Fasciotomy Ask AI SCH

- **Impending compartment syndrome** → fasciotomy; safest, life- or hand-saving
  - Release **all compartments**, including carpal tunnel
  - Prolonged **vascular injury** → vascular surgeon may release prophylactically

Slab vs Cast Ask AI SCH

- Q → after good closed reduction: single posterior slab, double slab or full cast?
- A → no slabs → unsuccessful
  - **Plaster of Paris slab** → loose within two days → moves → pressure point
  - Child moves broken slab; compression bandage slips → **failure**
  - Ordinary **cast**; slab effect → bivalve, or keep posterior half + compression bandage
- Preference → cast in **fibreglass**; plaster of Paris acceptable
  - Cast dries → **bivalve**
  - Doubt about decompression → discard half, keep **posterior part**
  - Posterior part → rigid, stays 3–4 weeks without weakening

Post-operative Nerve Injury Ask AI SCH

- Q → **post-operative nerve injury** (e.g., medial pin) → remove K-wire or wait?
- A → no definitive answer
  - Wire through nerve → removal won't help nerve; fracture reduction lost
  - Nerve often **compressed by wire** → remove wire
- Safer overall → **remove and replace** the wire

Vascular Exploration Tips Ask AI

- Q → tips from experience beyond literature?
- A → none special → free entangled structures → return to pins
  - **Vascular surgeon** completes exploration if needed
  - Vascularity may return well; mostly **intimal tear** present anyway
  - Artery looks good → later **occlusion** from intimal tear
  - Most prefer **resection and grafting**

Closed Reduction in the ER Ask AI

SCH

- Q → closed reduction in ER?
- A → planned **non-operative** treatment → yes, in ER
  - Type III or type II planned for surgery, no vascular injury → **pointless**
  - → excessive manipulation + soft tissue damage; child awake, pulling against you
  - Cast → **redisplaces** on entering theatre
- Only ER reduction without anaesthesia → **mild type II**
  - Flexion only → cast; non-operative agreed with parents
  - Repeat X-ray after a week → stability check
- Surgery planned → leave **first manipulation** to operating room
  - No ER manipulation followed by second theatre manipulation
  - Exception → **vascular injury** → reduce in ER to restore vascularity until theatre

Lateral Condyle Fracture: Classification and Imaging Ask AI

Characteristics Ask AI

LC

- Supracondylar → 59% (≈60%); **lateral condyle** → 12% to 15%
- **Average age** → about 6 years
- **Intra-articular fracture** → key difference from supracondylar
  - Supracondylar → distal humeral fracture; no articular incongruity issue
  - Lateral condyle → **intra-articular elbow fracture**
- **Transphyseal** → physeal (Salter–Harris) injury
  - → **physeal arrest** or partial growth arrest

Milch Classification Ask AI

- Most asked question → **Milch classification**

Milch	Mechanism	Fracture line	Salter–Harris
I	Impaction / shear → direct trauma to hand pushes lateral condyle upward	Crosses physis through ossific nucleus of capitulum	IV
II	Avulsion → varus stress	Transphyseal, between capitulum and trochlea ossific nuclei	II

- **Mechanistic classification** → tells mechanism only (avulsion vs impaction)
  - No relation to **planning of treatment** or management
  - No relation to **long-term prognosis** or displacement rate
  - Standard and memorised, yet **not functional** for decision making
  - No meaning in **planning of management**

Jakob Classification Ask AI

- Not new → earlier classification (reportedly 1964); **Jakob** → 1975
- Three types → must know; guides treatment well
- **Song** → alternative, covered later

Jakob	Features	Stability
I	Metaphysis → across physis → stops before articular surface; intact cartilaginous hinge; gap < 2 mm	Theoretically stable
II	Same gap, but fracture reaches articular surface	Liable to much greater displacement than I
III	Displaced and rotated	Unstable

- Stage I → partial metaphyseal break entering epiphysis; **hinge of cartilage** intact
  - Articular cartilage intact → displacement cannot be marked
  - Gap → never exceeds 2 mm
- Stage II → articular surface fractured, intra-articular extension
  - Even < 2 mm → **liable to displace** → must be distinguished from I
- Not included by Jakob
  - **Displaced but not rotated**
  - In place but displacement > 4 mm

Treatment Logic Ask AI

Jakob

- Type I (not reaching articular surface) → **conservative**
- Type II → high liability to displacement
  - Conservative + follow-up → fix if displaces; or **fix from start**
  - Other factors also affect decision
- **Rotated** → reduce and fix; articular incongruity unacceptable

Type I vs Type II on X-ray Ask AI

Jakob

- Detecting **intra-articular extension** → very big, very difficult problem
  - X-ray may suggest no extension when present
- Jakob memorised → **minimally displaced** vs displaced (III)
  - Minimally displaced → not reaching (I) or reaching (II) articular surface

Gap pattern	Interpretation	Displacement incidence
Lateral gap > medial gap	Intact intra-articular (cartilage) hinge → opens laterally only	27% *[C3 · transcript: "lateral gap less than medial gap"]*
Lateral gap = medial gap	Fracture reaching articular surface, with separation	42%

- **Not definite rules** → radiology can mislead; not an anatomical specimen
- Minimally displaced
  - Intact articular hinge → **stable fracture**
  - Complete → **potentially unstable** → liable to displacement
- Lateral > medial still displaces in 27%
  - → fracture may still reach articular surface; non-articular fracture would not displace
- Equal gaps → 42% displacement
- Gap appearance → indicates **displacement risk** → guides management decision

Song Classification Ask AI

- 2008 → **Song** expanded Jakob → adds omitted types
- Additions
  - Stage 4 → **displaced without rotation** (absent from Jakob)
  - Displaced + rotated → Jakob 3 becomes Song 5 → most severe
  - Stage 1 → **metaphyseal fracture** only; line limited to metaphysis, not reaching physis; very rare
- Imaging → AP + lateral insufficient → add **oblique views**
  - **Internal oblique view** → most important
  - Determines **true displacement**; AP + lateral may wrongly show undisplaced

Song	Features	Jakob
1	Fracture line in metaphysis only	—
2	Not reaching articular surface; lateral gap > medial gap	I
3	Reaches articular surface; undisplaced, gap < 2 mm; lateral gap = medial gap	II
4	Displaced > 2 mm; no rotation; completely unstable	—
5	Displaced > 2 mm with rotation	III

- **Song type 3** → potentially unstable

Identifying Song 3 and Hidden Displacement Ask AI

Song

- **MRI** → shows intra-articular extension easily
- **Internal oblique view** → never call undisplaced on AP alone
- Equal gaps, ≤ 2 mm, reaching articular surface → **potentially unstable**
  - Whole problem → **type 3**, liable to future displacement
  - Types 4 and 5 → reduce and fix
  - Types 1 and 2 → no problem
- Cases
  - AP → completely undisplaced; **internal oblique** → > 0.5 cm displacement
  - → basically type 4 → **ORIF**; not conservative
  - AP → in place; **oblique view** → displacement visible
  - AP and lateral → very innocent
  - Internal oblique → displaced **metaphyseal flake** → type 4, not 1 or 2
  - Very young child → fragment completely rotated → **type 5** *[C4 · transcript: "rotated fragment became type 4"]*

Pitfall

- Lateral condyle → never trust AP + lateral alone
- Internal oblique view → reveals true displacement
- "Undisplaced" on AP may be > 0.5 cm displaced → type 4
- Song recommends 4 views → AP, lateral, internal and external oblique

Lateral Condyle Fracture: Management Ask AI

Management Principles Ask AI

LC

- **Intra-articular + transphyseal** fracture → implications
  - **Anatomical reduction** → no acceptable displacement
  - Physis opposite physis, articular surface anatomical → restore or maintain elbow ROM
  - **Intra-articular derangement** → beyond a physeal injury
- "< 2 mm" → on **AP, lateral and oblique** views
  - Oblique may show > 4–5 mm, even a centimetre
  - → a different entity
- < 2 mm + metaphyseal only or **intact hinge** (lateral > medial) → conservative
  - Still serial X-rays → 47% displacement incidence *[Q1 · As spoken; intact-hinge pattern given 27% and 20-25% elsewhere; 47% is the equal-gap figure]*
- **MRI** for all → not feasible
  - Expensive
  - Needs **anaesthesia** in a child
  - Exception → fixation indicated but parents refuse surgery
  - MRI as agreed **point of decision** → intra-articular → fix; not → may not fix

Treatment by Stage Ask AI

LC

Pattern	Song	Displacement risk	Treatment
< 2 mm, lateral gap > medial gap (articular hinge)	1, 2	20–25%	Cast 6 weeks; close follow-up for displacement
< 2 mm, equal gaps (intra-articular extension)	3	~50%	Surgery → percutaneous pinning; open reduction if it moves
> 2 mm or rotation	4, 5	—	Closed reduction + percutaneous pinning, or ORIF → mostly ORIF

- Types 4 and 5 → mostly **ORIF**
  - Intra-articular → excellent articular reduction + restored articular congruity
- Equal gaps → **potentially unstable**
  - 50-50 → of ten patients, five displace during follow-up
  - Not displaced yet → **percutaneous pinning** sufficient; no open reduction
- **Doubtful pattern** (type 2 vs 3 → extension vs intact hinge)
  - **Ultrasonography** → very hard to make reliable
  - **MRI** → shows intra-articular extension
- **Surgery indications**
  - Displacement > 2 mm → no doubt
  - < 2 mm + **stability concern** → medial gap = lateral gap
  - → 50% (exactly 47%) displacement incidence → surgery preferred *[Q2 · As spoken; same equal-gap pattern was given 42% displacement earlier in the lecture]*
- Clinical sign of instability → **excessive lateral soft tissue swelling**
  - → excessive trauma → torn **soft tissue stabilizers** (as in supracondylar)

Open Reduction Rationale Ask AI

LC

- Displaced, needing reduction → **open reduction preferred**
  - Radiological findings unreliable; articular surface must be restored
  - **Lateral condyle fragment** → three-quarters cartilage in children
  - No reliable landmarks → see articular surface directly
  - → preserves **articular congruity** and elbow ROM
- **Closed reduction** → not preferred
- **Percutaneous fixation** without opening → only if undisplaced on all views
  - All views → AP, lateral and obliques ; never two views alone
- **ORIF** → **gold standard** for lateral condyle fracture
  - Especially displaced, rotated, or stability concern (medial gap = lateral gap) *[C5 · transcript: "ORIF especially if undisplaced"]*

Surgical Approach and Technique Ask AI LC

- **Lateral approach** → very easy
  - Under anaesthesia; arm by side; no side table
  - Along distal humerus → between **triceps** (posterior) and extensor carpi radialis (anterior)
  - Incise **periosteum** → find fracture → reflect periosteum anteriorly
- **Keep dissection anterior**
  - Fragment blood supply comes from **posterior**
  - Posterior dissection → **avascular necrosis** of lateral condyle fragment
- Open **anterior capsule** → evacuate hematoma → see articular surface
- Irreducible → **K-wire joystick** on articular fragment
- **Biggest mistake** → judging reduction by metaphyseal congruity
  - Rely on **articular surface congruity**; never on lateral humeral border
  - Fragment → excess cartilage → always **prominent** vs metaphysis
  - → false impression of poor reduction despite excellent reduction

Fixation Ask AI LC

- Two or three wires after reduction
- **Percutaneous wires** → removed at 3–4 weeks , rather 4 weeks, after union
- Alternatives → **buried bent wires** or screws; results similar, none proven better
- **Screw** → must be removed → transphyseal → growth arrest otherwise
- **Smooth wires** → may cause nothing; buried → still better removed
- Buried wires → must not cross **olecranon fossa** → would block joint motion
- Wires outside skin → removed anyway; drawback → **higher infection** incidence
- **Configuration**
  - One wire → far cortex of metaphysis
  - One wire → anterior intra-articular part of fracture

Open Reduction Key Points Ask AI

LC

- **Anatomical reduction** of articular surface → must see it
  - Assistant → **proper anterior retraction**
  - **Anterior capsule** opened
  - Intra-articular **hematoma** evacuated → lateral condyle and articular reduction visible
- Internal fixation tips
  - **Sterile tourniquet**, placed high, out of the way
  - Never use **lateral metaphysis** to assess reduction
  - **Stay anterior** → protects posterior blood supply of lateral condyle
  - Open **anterior capsule**
  - Assess reduction by **articular surface**
- Fixation options → screw / buried pins / **pins outside skin** (preferred); none superior

Must-Know

- ORIF → gold standard; lateral approach
- Stay anterior → blood supply from posterior → avascular necrosis risk
- Judge reduction by articular surface, never metaphysis
- Screw → transphyseal → must be removed
- Wires → never through olecranon fossa

Why the Lateral Condyle Matters Ask AI

LC

- **Neglected** or displaced late case → very difficult surgery, poor results
  - Sometimes → accept **elbow deformities**, many problems; avoidable case
- Never AP + lateral only → **internal oblique view** for displacement
- **Open reduction** → gold standard when reduction needed
- **Percutaneous fixation** → when no reduction needed (fracture in place)
- Undisplaced **Jakob 2 / Song 3** → surgical, especially if
  - Non-compliant with follow-up X-rays
  - Severe soft tissue injury or **massive swelling** → high displacement potential
  - Regardless → 50% displacement probability
- Goal → avoid **missed fracture** or displaced late intra-articular fracture → worst case
- Not a "junior operation" → among most difficult fixations in orthopaedics
  - Femur fixation → much easier than **lateral condyle**
  - **Junior staff** → supervised by experienced senior

Clinical Pearl

- Surgery → displacement > 2 mm, rotation, or equal medial and lateral gaps
- Equal gaps → ~50% (exactly 47%) displacement
- Excessive lateral swelling → torn soft tissue stabilizers → instability
- Undisplaced in all views → percutaneous pinning; displaced → open reduction

Final Questions and Answers Ask AI

Late Presentation Ask AI LC

- Q → presentation after 1.5–2 months with **full range of movement**
- A → do nothing → surgery may injure **physis,** creating a new physeal injury
  - Follow up → treat resulting deformity
- Prevention → **correct initial diagnosis**
- Fixing → safe side; never wrong
  - vs neglected displacement → **stiffness,** limited ROM
  - Late intervention → very difficult, poor results

Tourniquet Ask AI

- **Tourniquet** → only for open reduction

CT and MRI Ask AI LC

- **CT** → no role at young age
  - Fragments cartilaginous → only ossific nucleus seen; articular surface invisible
  - Older child (10, 12, 13 years) → may help
  - Cartilage visible only on **MRI,** ultrasonography (difficult in elbow), arthrography
  - Young patients → **no CT;** no benefit to treatment

Internal Oblique View Ask AI LC

- **Internal oblique** → taken as if internally rotating the shoulder
- Request oblique views → internal and external
- **Song recommends 4 views**

Acceptable Rotation and Angulation Ask AI SCH

- **Avoid rotation**
  - Never pure → **rotation with angulation;** fracture not transverse
  - Transverse → pure rotation; any obliquity → rotation means angulation
  - **Internal rotation** → cubitus varus → unacceptable
- Slight extension or flexion → acceptable → **remodeling**
  - Plane of **elbow range of movement** → corrects with growth
- **Mediolateral angulation** (cubitus varus or valgus) → no correction with growth
  - Rare pediatric fracture → no remodeling; **cubitus varus** persists *[C6 · transcript: "cubitus varus will remodel"]*
  - **Valgus** → also not corrected

Deformity	Remodels	Accept
Hyperextension / flexion	Yes	Some
Rotation (always with angulation)	No	No
Varus / valgus	No	No

Unfused Epiphysis vs Fracture Ask AI

- Q → non-fused lateral condyle epiphysis vs fracture?
- A → know **physis orientation** and appearance
  - **Lateral condyle fracture** → always starts from metaphysis ; must involve it
  - Difficulty → very young age; distal humerus almost entirely cartilaginous
  - **Clinical examination** → swelling and tenderness only laterally → lateral condyle fracture
- Age ≤ 1 year or slightly older → **slipped distal humeral epiphysis**
  - Misdiagnosed as lateral condyle or supracondylar fracture
  - Whole **ulna with radius** displaced relative to humerus
  - Doubt → **arthrogram** → cartilage outline

Moderator Remarks Ask AI

- Textbook questions skipped → focus on **lecturer's experience**
- Two-hour session; 750 attendees
- Closing thanks between moderator and lecturer

Myositis Ossificans Ask AI

- **Heterotopic ossification** → more accurately myositis ossificans
  - More in **prone** individuals
  - Causes → excessive, aggressive or **repeated manipulation** / reduction trials
  - Body-weight traction on child → severe **soft tissue damage**
  - → muscle damage, significant hematoma → myositis ossificans
  - **Aggressive physiotherapy** manipulation → myositis ossificans
  - Failed fixation → reoperation, more manipulation and swelling → higher risk
- Management → wait for **complete maturation** of new bone
  - Monitors → **alkaline phosphatase** + radiological corticalization of new bone
  - Mature → **excise**
  - Immature excision → 100% recurrence

Pitfall

- Myositis ossificans → aggressive, repeated manipulation; aggressive physiotherapy
- Excise only after maturation → alkaline phosphatase + corticalization
- Early excision → 100% recurrence

Redo for Poorly Fixed Supracondylar Fracture Ask AI SCH

- Q → operated elsewhere for supracondylar fracture, slipped → redo criteria?
- A → **papers:** closed reduction or surgery safe up to about seven days from trauma
- **Unacceptable reduction** (completely displaced, varus, excessive rotation)
  - → correct within a week to ten days
- After two weeks (from third week) → healing, **new bone**, callus
  - → heroic surgery; another surgery needed anyway
  - Better → heal in malunited position → **osteotomy** → much safer
- Only flexion or extension of distal fragment → leave; **remodels**
  - Even beyond a week, especially young
- Redo limit → a week to ten days at most
  - Preferred only if **completely off-ended**, untreated, no callus → now rare

Counselling About Myositis Ossificans Ask AI

- Q → explain to parents **high chance of myositis ossificans**?
- Scenario → 3–4 manipulations, then **percutaneous wires**, unacceptable reduction at two weeks
  - Plan → open **osteotomy** to correct deformity

Timing of osteotomy	Myositis ossificans risk
Within ~two weeks	High
After full healing and maximum remodeling, callus gone (6 months–1 year)	Low; same correction achieved

- → be patient; **delayed osteotomy**

Associated Ligament Injury Ask AI

- Q → associated **medial collateral ligament** injury?
- A → very rare or absent in children
  - Children → **ligaments stronger than bone**
  - Bone fails first → ligaments cannot then fail
  - Lateral condyle or **intercondylar fractures** with ligament injury → adults, adolescents
  - Late adolescents → treated like adults ; not pediatric supracondylar fracture

Pulse Loss After Reduction Ask AI SCH

- Q → good pulse before; after reduction and fixation → **weak or absent**, hand pink, good refill
- A → a problem → immediately **undo fixation** → restore pulse
- Usually → **open reduction via anterior approach** + neurovascular exploration
- Pulse diminution after reduction → reduction rarely 100% acceptable
  - → open, free **compressed or entrapped** vessels

Traumatic Radial Head Dislocation

(Mechanism, Bado-type recognition and reduction of the dislocated radial head)

Prof Dr Hisham Abdel-Ghani

Watch Original Lecture

Lecture transcript processed with AI assistance.

Contents

↑ Book contents

Traumatic Radial Head Dislocation . . . 351
Why the Diagnosis Matters . . . 351
Radiocapitellar Line . . . 351
Cause: Monteggia Lesion . . . 351
Malunited Ulna Blocks Reduction . . . 352
Congenital versus Traumatic Dislocation . . . 352
Features of the Congenital Type . . . 353
Missed Monteggia Lesion . . . 353
Late Presentation . . . 353
Ulnar Bow Sign . . . 354
Natural History . . . 354
Clinical Presentation . . . 354
Treatment of the Missed Lesion . . . 355
Principle . . . 355
Prerequisites for Open Reduction . . . 355
Case: Ten-Year Painful Missed Dislocation . . . 355
Role of Ulnar Osteotomy . . . 356
Open Reduction and Ulnar Osteotomy: Technique . . . 356
Case: 12-Year-Old, Missed Monteggia . . . 356
Positioning and Incision . . . 356
Exposure . . . 356
Abnormal Ulnohumeral–Radial Head Relationship . . . 357
Ulnar Osteotomy . . . 357
Temporary K-Wire and Stabilisation . . . 358
Postoperative Results and Illustrative Cases . . . 358
Results . . . 358
Second Case: Three Months Missed . . . 359
Confirming Stability . . . 359
Plate Position and Screws . . . 359
K-Wire Role . . . 359
Remodeling of Overcorrection . . . 359
Case: Three-Year-Old, Very Proximal Angulation . . . 360
Take-Home Message . . . 360
Discussion: Postoperative Care and Nerve Injury . . . 360
Postoperative Regimen . . . 360

Radial Osteotomy . . . 361
Combined Forearm Fractures . . . 361
Posterior Interosseous Nerve . . . 361
Tardy Ulnar Nerve Palsy . . . 361
Discussion: Technique, Imaging and Congenital Cases . . . 361
K-Wire Entry Point . . . 361
Congenital: Don't Touch . . . 362
Oblique versus Transverse Osteotomy . . . 362
Role of MRI . . . 362
Preventing Recurrence: Key Tip . . . 362
Double Approach . . . 362
Discussion: Indications, Timing and Assessment . . . 363
Painful Flattened Head Before Maturity . . . 363
Timing versus Fresh Trauma . . . 363
Preoperative Assessment of the Head . . . 363
Radiographic Check After Closed Reduction . . . 363
Nail-Patella Syndrome . . . 363
Annular Ligament Reconstruction . . . 364
Discussion: Fixation Adjuncts, Healing and Complications . . . 364
Transcapitellar Wire . . . 364
Splints and Casts . . . 365
Healing of the Osteotomy Gap . . . 365
Plate Screw Crossing the Physis . . . 365
Osteotome versus Saw . . . 365
Complications . . . 366
Longitudinal Instability and Recurrence . . . 366
TRASH Lesions . . . 366
Discussion: Acute Reduction and Fixation Choices . . . 367
Contouring the Plate . . . 367
Closed Reduction in the Acute Case . . . 367
Centre of the Capitulum in Young Children . . . 367
Fixation Method of the Osteotomy . . . 367
Forearm Position in the Cast . . . 368
Discussion: Revision, Physes and Late Indications . . . 368
Why Not Reduce the Congenital Case . . . 368
Contouring Locked Plates . . . 368
Ilizarov . . . 368
Management of Recurrence . . . 368

Arresting the Physes 369
Radial Shortening 369
Age Limit 369
Pulled Elbow and Acute Monteggia Management 370
Pulled Elbow versus Radial Head Dislocation 370
Acute Bado Type I Monteggia 370
Mid-Prone versus Mid-Position 370
Plastic Deformation: Closed or Open 371
Irregular Radial Head Preoperatively 371

AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 353) “pseudo-cubitus varus” → “cubitus deformity, probably valgus”

C2 (p. 364) “sigmoid notch had become concave” → “sigmoid notch had become convex”

Traumatic Radial Head Dislocation Ask AI

Why the Diagnosis Matters Ask AI

- Frequently **missed** diagnosis
  - The eye sees what it knows; misses what it does not know
  - No prior idea → abnormality overlooked on the X-ray
- Early treatment → easy; managed as **acute trauma**
- Left missed → major problem → major operation
  - Results inferior to ordinary fracture treatment

Radiocapitellar Line Ask AI

- Line drawn → **central axis of the radial neck**
  - Neck of proximal radius, not radial shaft
- Normal → long axis of radial neck **crosses the capitulum** in all views
  - AP view → transects capitulum
  - Lateral view → transects capitulum
  - Line in place → no dislocation
- AP view, abnormal → line **not transecting** capitulum
  - Dislocation diagnosable from AP alone; no need to wait for lateral
  - Long axis not through capitulum centre
- Lateral view → long axis should cross **capitulum centre**
  - Dislocated → axis not bisecting capitulum
  - Example shown → **anterior radial head dislocation**

High Yield

- Long axis of radial neck → through capitulum centre
- Applies in both AP and lateral views
- Off-centre in any view → radial head dislocation

Cause: Monteggia Lesion Ask AI

Monteggia

- **Isolated radial head dislocation** from trauma → very questionable
  - Never proven so far
- Certain cause → **Monteggia** or Monteggia-variant fracture-dislocation
- Components → proximal ulnar fracture + **radial head dislocation**
- Ulnar fracture seen → draw **radial neck axis** → off-centre = dislocation
  - Then reduce ulnar fracture + reduce radial head
- Real problem → ulna treated alone**, dislocation unnoticed**
  - Detected only by relating radial neck axis to capitulum centre
- Original injury most often → **Bado type I**
  - Anterior Monteggia / anterior radial head dislocation
  - Accompanied by **anterior bowing** of the ulna

Malunited Ulna Blocks Reduction Ask AI

- Ulna unites; radial head dislocation persists
  - Ulna heals **anteriorly bowed**
- Any **closed manipulation** of radial head → fails
- Bado I anterior angulation → ulna unites normally but **bowed anteriorly**
  - Anterior bow present → closed reduction never restores radial head
- Sometimes no complete fracture → **plastic deformation** only
  - May look healed, with nothing wrong

Pitfall

- Isolated traumatic radial head dislocation → unproven; look at the ulna
- Ulna fixed, radial head ignored → missed lesion
- Anterior ulnar bow → closed reduction fails

Congenital versus Traumatic Dislocation Ask AI

- Differentiation → major difference in **treatment**
  - Congenital → no reduction, no open reduction, no attempt
  - Traumatic → chance to treat
  - Congenital type → **not very uncommon**

Features of the Congenital Type Ask AI

Congenital

- Radial neck axis passes **posterior** to capitulum
  - Posterior dislocation → most likely congenital, not traumatic
- Often **bilateral**
- Mimics cubitus deformity (probably valgus) → termed **pseudo-cubitus** *[C1 · transcript: "pseudo-cubitus varus"]*
- Congenital anomaly
- Look at the **proximal ulna**
  - **Trochlear notch** → very shallow
  - Very abnormal proximal ulnar shape
- Radial head **convex** rather than concave
- Deformed proximal ulna + convex head → suspect **congenital**
- Distal humerus → also abnormal
- Congenital **maldevelopment** + posterior radial head dislocation
- No open or **closed reduction**
  - Radial head convex → exact opposite of normal concavity
  - Proximal ulna → very deformed, malshaped

Feature	Traumatic	Congenital
Direction	Anterior (Bado I)	Posterior
Laterality	—	Often bilateral
Radial head	Shape preserved early	Convex, no concavity
Proximal ulna	Fracture, anterior bow, plastic deformation	Shallow trochlear notch, malformed
Distal humerus / capitulum	—	Malformed
Treatment	Reduce	Don't touch

Missed Monteggia Lesion Ask AI

Late Presentation Ask AI

Missed

- Trauma → **cast** → radial head left unreduced
- Presents after cast removal → after **ulnar union**
  - About 6 weeks or 2 months
  - Or 3 months, 1 year, 2 years, 5 years
- Question → how does it present; **natural history**?
- Comparison X-ray → left normal, right abnormal
  - Right → radial head axis dislocated

Ulnar Bow Sign Ask AI

- Lateral forearm X-ray → line along **posterior ulnar border**
  - Posterior tip of olecranon → posterior border of distal ulna
- Normal ulna → almost completely straight
- Dislocated side → mid-ulna **elevated** off the line
- Elevation = **anterior ulnar bowing**
- Distance > 2 mm → **ulnar bow sign**
- Implies prior ulnar fracture
  - → **malunited** in anterior angulation
  - Anterior angulation → blocks radial head reduction
- Fresh fall + this appearance → **plastic deformation** (paediatric fracture type)
- Late → **malunion**

Exam Pearl

- Ulnar bow sign → line from olecranon tip to distal ulna posterior border
- Bow > 2 mm → anterior ulnar bowing
- Bow + dislocated head → malunited or plastically deformed ulna

Natural History Ask AI

- Left 1–3 years → ulna **remodels** straight
  - Ulnar bow sign disappears
  - Deceptive → looks like isolated radial head dislocation
  - Reality → ulnar fracture, malunited in anterior angulation, ulnar bow
- Ulna regains shape; radial head **never reduces spontaneously**
- Long-standing dislocation → **ossification** flakes around radial head
  - Cause unknown; no significance
  - No excision at surgery → **resorbs spontaneously**

Clinical Presentation Ask AI

- Missed fracture, cast removed → sometimes **no pain**
- Complaints with time:
  - Arm hurts → **discomfort**; cannot push with hands
  - Arm "gives way" → **valgus instability** on examination
- Radial head palpable in **cubital fossa**
  - Normal location → empty
  - Sometimes **blocks full flexion**; sometimes flexion normal
- Long-standing valgus instability → nerve palsy
  - **Tardy ulnar nerve palsy**
  - **Posterior interosseous nerve** palsy
- Key findings → valgus instability + **prominent radial head** in cubital fossa

Treatment of the Missed Lesion Ask AI

Principle Ask AI

- Any radial head dislocation → **must be reduced**
- Acute trauma → closed reduction + **ulnar fixation** by any means
- Neglected case, ulna united → **open reduction**, with conditions

	Acute (fresh)	Missed / neglected
Radial head	Closed reduction	Open reduction
Ulna	Fixation (K-wire in young child)	Osteotomy, overcorrection
Time limit	—	None if head shape preserved

Prerequisites for Open Reduction Ask AI

- First → radial head **shape preserved**
  - Very long-standing, deformed head → irreducible
  - Reduced anyway → redislocation, pain or instability → surgery in vain
- Second → **no defect** in radial head from original fracture
- Prerequisites present → reduce **regardless of timing**
  - 3–4 months, intact head → reduce
  - 2–3 years → reduce
  - 5–6 years → reduce
- Head keeping normal shape → **no time limits**

Must-Know

- Missed case → judge the radial head, not the time
- Preserved shape + no defect → reduce
- Deformed head → redislocation, pain, instability

Case: Ten-Year Painful Missed Dislocation Ask AI

Case 10y

- Dr. Hisham Naguib's case → sent for opinion
- Judged by head condition, not time
- Radial head → lost normal **concavity**, flattened
  - Not suitable for reduction
- Painful → wait until **maturity** → radial head excision
- Never rush **radial head excision** before maturity
- Before maturity → usually painless; pain after maturity

Role of Ulnar Osteotomy Ask AI

- Any **anterior angulation** of ulna → prevents reduction
- Ulna remodeled straight after long missed course → still irreducible
  - Required → **ulnar osteotomy** → overcorrect → posterior bowing
- Steps 1 to 90 of open reduction → **ulnar osteotomy,** then head reduction
- Previously → **annular ligament reconstruction**
  - Using a slip of triceps
  - Many drawbacks → now hardly ever done

Open Reduction and Ulnar Osteotomy: Technique Ask AI

Case: 12-Year-Old, Missed Monteggia Ask AI

Case 12y

- Age 12 → **Monteggia fracture-dislocation** missed, undiagnosed
  - Ulna united, strongly remodeled → straight
  - Radial head clearly dislocated
  - Neck axis off capitulum centre → AP and lateral
- Head **maintaining shape**
  - Normal concavity maintained
  - No osteochondral defect
- **CT** confirms:
  - Radial head normal shape; concavity preserved
  - No osteochondral defect in head or capitulum
- Any duration (3, 4, 5 years) → **amenable for reduction**

Positioning and Incision Ask AI

- **Lateral decubitus;** sandbag under forearm; elbow flexed
- Landmarks → medial epicondyle, **olecranon,** lateral epicondyle
- Incision along **posterior ulnar border**
  - Proximally → between olecranon and lateral epicondyle

Exposure Ask AI

- Split part of **triceps**
- Directly onto **subcutaneous border** of ulna
- Muscle mass dissected **subperiosteally**
  - Flap carries whole muscle mass + posterior interosseous nerve
  - → avoids **PIN injury** during surgery
- Incision extends proximally and distally on subcutaneous ulnar border
- All on the lateral side
- Open **joint capsule;** subperiosteal dissection from ulna
- Capsule incised → feel head anteriorly → push to reduce

Abnormal Ulnohumeral–Radial Head Relationship Ask AI

- Reducing head → elbow effectively dislocated
  - Trochlea / proximal ulna → **externally rotated,** supinated
  - **Ulnohumeral joint** → dislocated
- Reduce ulna + flex elbow → head out anteriorly again
- Ulna and head → **move as single mass**
  - Head anterior to ulna
  - Reduce head → pushes ulna out
  - Reduce ulna → pushes head out
- Goal → break abnormal relationship; create **normal relationship**
- Head reduced → ulnohumeral **subluxated** or dislocated
- Ulnohumeral reduced → head anteriorly dislocated
- False reduction → flex elbow → ulna reduces → head out → nothing achieved
- Confirm both → **radial head** reduced + ulnohumeral articulation reduced
  - Not head reduced with ulnohumeral subluxated by supination / external rotation

Ulnar Osteotomy Ask AI

Osteotomy

- First step → **ulnar osteotomy**
- Site → as proximal as possible, at **maximum angulation**
  - Very distal angulation → still keep proximal
- Transverse or **oblique** → preferred: oblique
- After osteotomy → head reducible
  - Overreduce → **posterior angulation** (overcorrection)
- Head no longer related to **proximal ulna**
- Head moved and controlled by **distal ulna** → segment distal to osteotomy
- All manoeuvres → through distal segment
  - Distal segment moves head back
- Final **distal ulnar position** → sets and holds radial head position
- Transverse or oblique? → **oblique**

Temporary K-Wire and Stabilisation Ask AI

- **K-wire** → fixes proximal ulna to distal humerus
  - Ulnohumeral articulation reduced, held → confirmed 100%
- Then reduce radius → adjust **angulation and rotation** of distal ulna
  - Hold position keeping head stable
- Head **stable in all rotations**
  - Supination, pronation, flexion, extension
  - No pressure or manipulation needed
- Osteotomy gap → heals spontaneously; **no bone graft**
- After capsulotomy → clear **sigmoid (radial) notch**
  - Head–coronoid articulation filled with **fibro-fatty tissue**
  - Clean out tissue + cartilage → head moves within notch
- Hold distal ulnar segment in head-stabilising position
  - Usually **posterior angulation** = flexion at osteotomy
  - Plus some supination
- Ulnohumeral reduced + **distal fragment supinated** → head stable

Clinical Pearl

- Osteotomy separates proximal ulna from radius
- Radial head follows the distal ulnar segment
- Confirm ulnohumeral joint and radial head reduced together
- Stable in all rotations without pressure → acceptable

Postoperative Results and Illustrative Cases Ask AI

Results Ask AI

Case 12y

- Postoperative X-ray
- Long-term follow-up → **radial head reduced** in place
- Postoperative X-ray → distal ulnar end adjusted

Second Case: Three Months Missed Ask AI

Case 3m

- 3 months after injury → head clearly **dislocated**
- Same operation, position and approach
- **Oblique osteotomy**
- Reduction requires ulnohumeral reduced + radial head reduced
- Before osteotomy:
  - Head reduced → ulnohumeral **dislocated / subluxated**
  - Ulnohumeral reduced → head dislocates again
- Wrong relationship broken by → **osteotomy**
  - Ulnohumeral reduced; distal ulna controlled → head reduced through it
  - New **ulna–radius relationship**
- Without osteotomy → reduction only forcible
- With osteotomy → both reduced → head does **not dislocate**
  - Ulnohumeral articulation separated from radius
  - Radius related to fixed **distal ulnar segment**

Confirming Stability Ask AI

- Osteotomy gap → fills, heals spontaneously
- After fixation → confirm **ulnohumeral joint** and radial head both reduced
- Attempted dislocation fails → required **intraoperative stability**
- Otherwise → redislocation immediately, soon after, or after cast removal

Plate Position and Screws Ask AI

- Plate position → no difference at all
  - **Direct posterior** or posterolateral
  - Choose → more stability, comfortable screw placement
- Proximal screws → **image intensifier**
  - Avoid penetration into radius
  - Penetration → blocks reduction or fixes radius
- Seen → screw perforated radius → **supination and pronation** fixed

K-Wire Role Ask AI

- Used **temporarily** → placed, then removed
- Not definitive fixation

Remodeling of Overcorrection Ask AI

- Postop X-ray → radius in place; **angulation** starting to remodel
- Posterior ulnar angulation → no fear; **remodels with growth**
  - Especially young child
- After **plate removal** → head in place
- "As proximal as possible" → usually below **coronoid process**
  - Usual site of angulation
- Next case → very proximal angulation → very proximal osteotomy

Case: Three-Year-Old, Very Proximal Angulation Ask AI

Case 3y

- Age 3 → acute trauma image (not presented at this stage)
  - Severe **olecranon fracture**, marked posterior angulation
  - Commonly overlooked → radial head
  - Radial head → **anteriorly dislocated**
  - Neck axis off capitulum centre
- Unacceptable → treating doctor insisted radius normal
- After union → severe **anterior dislocation**
- Must be reduced → operation **very difficult**
  - Angulation very proximal → almost in **olecranon process**
  - Not in shaft, not distal to coronoid
  - → osteotomy very proximal
- Distal osteotomy → fails to restore **proximal ulna–humerus** relationship
- All views → long axis of radius off capitulum centre
- **CT** done for confidence
  - Very young → **cartilage** poorly seen
  - Only 3 months since injury → head reassuringly fine
- Osteotomy **very proximal** → restores proximal ulna–humerus relationship
- Some screws almost in **cartilage**
- Plate here → very small, 1.5
- Most cases → 2.7 **locked plate**
- Locked plate preferred → maintains **osteotomy shape**
- No **bone graft** needed → healed
- **Plate removal** mandatory → blocks proximal ulnar physeal growth

Take-Home Message Ask AI

- Priority → do not miss diagnosis in **acute trauma**
- Acute Monteggia or any forearm trauma → watch the **radial head**
- Long axis of radial neck → **capitulum centre**, AP and lateral
- Missed or late → no such thing as "too long to work"
- Proximal radius shape maintained + no **osteochondral fracture** (head or capitulum) → amenable for reduction, even late
- Step one → **ulnar osteotomy**; no reduction without it
- **Annular ligament reconstruction** → not needed 99.9% of the time

Discussion: Postoperative Care and Nerve Injury Ask AI

Postoperative Regimen Ask AI

- **Cast** → about 6 weeks, until radiological union
- Cast removed → **no physiotherapy**; child moves arm spontaneously
- After 2–3 weeks active mobility → mother does **pronation–supination exercises**
- No **aggressive manipulation**
  - → very major problems
- **Plate removal** → about 9 months to 1 year postop

Radial Osteotomy Ask AI

- Need for **radial osteotomy**? → not widely described
- Ulnar osteotomy → always develops **gapping**
  - Radial head relatively **longer** → reduction opens ulnar gap
- Described option → **radial shortening**; never needed in operated cases
- Gap → closes later, heals spontaneously, no bone graft

Combined Forearm Fractures Ask AI

- Proximal **both-bone fracture** + ipsilateral distal forearm fracture + radial head dislocation
  - Acute trauma → fix everything together at once
  - Intraoperative **closed reduction** of radial head
- Never leave the head; never fix the fracture alone
  - Fixation with head dislocated → **malaligned** fixation

Posterior Interosseous Nerve Ask AI

PIN

- Closed trauma PIN injury → **no indication to explore**
  - Mostly recovers spontaneously
- Real fear → **iatrogenic** PIN injury
- Extensor compartment elevated **subperiosteally** from ulna → nerve lifted with flap
- Never place **Hohmann retractors** / levers / bone levers around radial neck
- To move head → push anteriorly, or reduce while holding ulna
- Lever around radial neck → may injure **PIN**

Pitfall

- Never place a lever around the radial neck
- Lever → posterior interosseous nerve injury
- Subperiosteal extensor flap → protects the nerve

Tardy Ulnar Nerve Palsy Ask AI

- Ulnar palsy after dislocation → not seen personally
- Described in **literature**
  - Valgus instability → ulnar nerve stretch → **tardy ulnar nerve palsy**

Discussion: Technique, Imaging and Congenital Cases Ask AI

K-Wire Entry Point Ask AI

- After osteotomy → reduce **ulnohumeral articulation**, reduce ulna
- Wire → olecranon to distal humerus, in **reduced position**
- Then reduce radius → observe **distal ulna** position
  - = desired osteotomy angulation; plate fixed on this ulna

Congenital: Don't Touch Ask AI

Congenital

- **Congenital dislocation** → irreducible → don't touch
- Malformed head, **capitulum** and distal humerus → never reducible
- Intervention → surgeon may cause **pain**
  - Originally painless → cosmetic appearance / deformity only
- Open reduction in congenital cases → always **failure**

Oblique versus Transverse Osteotomy Ask AI

- Oblique preference → reason, not mere preference

On angulation and distraction	Oblique	Transverse
Bony contact	More contact retained	Bigger gap

Role of MRI Ask AI

- **MRI** → for cartilage visualisation
  - Drawback → no 3D picture
- Indication → doubt about shape of proximal radius / head or **capitulum**
  - Confirms head amenable for reduction
- Coronal and sagittal planes → review **cut by cut**
- Request **thin cuts**
  - Head in only 2–3 cuts → useless

Preventing Recurrence: Key Tip Ask AI

- Most important intraoperative step → **radial head** reduced
  - + ulnohumeral articulation reduced
- Head reduced → ulnohumeral dislocated; ulnohumeral reduced → head dislocated
- Confirm during reduction → **both reduced**
- Achieved by **osteotomy** → separates proximal from distal fragment
- Radial head moves with **distal ulna**
- Radius controlled via distal ulnar orientation → **key tip**
- Never reduce head with **ulnohumeral joint** dislocated
  - Ulnohumeral later reduced → head comes out

Double Approach Ask AI

- Posterior (ulna) + **lateral approach** (head)? → unnecessary
- Lateral approach → poor view of **sigmoid notch**
- One approach enough
  - Field well visualised → no point in two approaches

Discussion: Indications, Timing and Assessment Ask AI

Painful Flattened Head Before Maturity Ask AI

- Head lost concavity + painful + immature → **wait**
- No **radial head excision** in the immature
  - → disruption of inferior (distal) radioulnar joint
  - → **Essex-Lopresti deformity** → proximal migration of radius

Pitfall

- Radial head excision before maturity → avoid
- Risk → distal radioulnar joint disruption
- Result → Essex-Lopresti, proximal migration of radius

Timing versus Fresh Trauma Ask AI

- 3 months, 6 months, 3 years, 8 years → **same procedure**
- Fresh trauma → entirely different
  - **Closed reduction** + fixation of fractured ulna
  - Young child → **K-wire** fixation suffices

Preoperative Assessment of the Head Ask AI

- Key requirement → radial head **maintaining shape**
- Assessed mainly by X-ray and CT

Modality	Role
X-ray	Head shape assessment
CT	Head shape assessment
MRI	Rare cases
Ultrasonography	Described by some authors in very young children

- 3 months, 1 year, 2 years → head usually **preserved**
- Concern → more than ~3–5 years → shape may change → not **amenable**

Radiographic Check After Closed Reduction Ask AI

- Acute closed reduction, ulna fixed → same **X-ray parameters**
- Long axis of radial neck → capitulum centre
- Not through centre → **still dislocated**
- Most important image → radial neck axis
  - → through capitulum centre
- Reduced vs dislocated head → AP or lateral view
- Reduced → axis **through capitulum centre**; dislocated → not through centre

Nail-Patella Syndrome Ask AI

- **Nail-patella syndrome** + congenital dislocation + limited pronation–supination
- Intervention → none; any reduction → **failure**

Annular Ligament Reconstruction Ask AI

- Mostly **not needed** → very rare cases only
- Previously done:
  - Posterior approach → **Bell-Tawse reconstruction**
  - Or anterior approach
- Long-term drawback → **narrowing** of radial neck
  - Impairs proximal radial neck growth → not advisable
- Proper osteotomy + proper head reduction + proper **ulnar fixation**
  - → reconstruction mostly unnecessary

Discussion: Fixation Adjuncts, Healing and Complications Ask AI

Transcapitellar Wire Ask AI

- Capitulum → radial head wire after reduction?
- Drawback → **transarticular** → breaks with movement
- Never use when doubting reduction or its **stability**
- Unstable intraoperatively (dislocates with flexion / rotation)
  - Wire removed at 6 weeks–2 months → **redislocation**
  - Analogy → **DDH**
- Must end with head very stable → no **forceful manipulation**
- Forceful manipulation + cast reliance → self-deception → dislocates on cast removal
- Must be **intraoperatively stable**
- Better alternative → **transverse wire**, head to ulna, after fixation
  - Not transarticular → no wire fracture if child moves or cast breaks
- Still → not advisable; used in few cases, with problems
- Case → **sigmoid notch** convex instead of concave *[C2 · transcript: "sigmoid notch had become concave"]*
  - Head never held on reduction
  - Bone deep to notch hollowed → convex → concave
  - Head placed; worry → transverse wire radial neck
  - → proximal ulna
- Transcapitellar wire → **disliked**

	Transcapitellar wire	Transverse radial neck–ulna wire
Crosses joint	Yes, transarticular	No
With movement	Breaks	Does not fracture
Lecturer's view	Disliked	Few cases; not advisable

Splints and Casts Ask AI

- **Plaster of Paris splint** → not advised
  - Child moves arm while broken and wet → meaningless
- Preferred → **fiberglass cast,** bivalved, anterior part removed
  - Very good splint
  - No problems from swelling or pressure
- **Dependable** → no elbow flexion–extension inside cast

Healing of the Osteotomy Gap Ask AI

- Gap → always healed; no graft needed
- Gap healing → depends on **tissue handling**, not bone
- Osteotomy → **sharp bone saw**
- Most important → preserve **periosteum**; no cutting, no damage
- Spontaneous healing → gap invisible postoperatively
- Periosteum preserved → **bone forms**
- Periosteum damaged → even **bone graft** may resorb; no healing
- **Gentle handling** + preserved vascularity → healing despite gap
  - Applies to children

> **Clinical Pearl**
> - Osteotomy gap → heals without graft
> - Preserved periosteum → bone forms
> - Damaged periosteum → graft may resorb

Plate Screw Crossing the Physis Ask AI

- Screw crossing **physis** → remove later
  - Plate removed **as soon as possible** (last case)
- Removal once healing appears → patient not yet returned
- No alternative → osteotomy very proximal, proximal to coronoid
  - Osteotomy made **through the coronoid**
- Requires **very thin saw** to pass through coronoid
  - Site of original fracture
  - → osteotomy very proximal

Osteotome versus Saw Ask AI

- No small saw blade → osteotomy by **drilling** or osteotome?
- Proper **instruments** → more comfort, better results
- Inadequate tools → risk on surgeon; patient will not excuse
- Patient would ask → why operate without **equipment**
  - Surgeon exposed to more problems than patient
- Good tools → enjoyable work
- Doubtful saw → prepare very sharp **thin osteotome**
- Acceptable → two drill holes + completion with sharp osteotome
- **Osteotome** osteotomy → heals faster than saw osteotomy
- Saw must be **sharp**
  - Blunt saw → burns bone → impaired healing (probable nonunion)

Complications Ask AI

- Most common → limited **supination and pronation**
- Usually **limitation of pronation**
  - Distal ulna used to apply supination → reduces head
- Always some **lag** → mild pronation limitation
- Intraoperative surprises:
  - Irregular head shape
  - **Erosions**
  - Sigmoid notch not concave
- Unpredictable on **X-rays** → must be dealt with
- Poor head shape → expect limited supination–pronation after reduction
- Fix **proximal radius to ulna** rather than transcapitellar wire → maintain reduction until healing
- Tell parents immediately postop → expected limitation; photograph head
- Course → **very variable**
  - Varies with age and rate of growth
  - Children differ completely from adults
  - May improve beyond expectations
- No promise of improvement → **follow up**
- Good radial head → supination–pronation returns
- Analogy → good fabric makes a good suit
- Intact head + intact **sigmoid notch** → full function
- Original trauma + long-standing missed dislocation → outside surgeon's control → affect **outcome**

Longitudinal Instability and Recurrence Ask AI

- **Longitudinal forearm instability** postop → not encountered
- Seen → **recurrent dislocation**
- Radioulnar synostosis (probable term) → not seen

TRASH Lesions Ask AI

- **TRASH lesion** → not radial head–specific
- Meaning → elbow trauma that seems innocent
  - X-ray → nothing; trauma significant
- Possible lesions:
  - **Slipped proximal radial epiphysis**
  - Dislocated radial head → easiest to diagnose
    - Relies on **bony axis** → long axis of radial neck
    - Not on articular surface → very cartilaginous when very young, invisible on X-ray
  - **Monteggia** fractures
  - Capitulum lesions
  - Trochlea lesions
  - **Medial epicondyle** → avulsion and incarceration
- Not clearly visible on X-ray → grouped as TRASH
- Clue → **fat pad sign** (previous lecture)
  - → intra-articular haemorrhage
  - → distension of elbow capsule

- Fracture suspected → **MRI** to define lesion
- Very young → findings unclear → TRASH
- No "TRASH of the radial head" → term = **innocent-looking** X-ray
  - Despite major trauma / injury

Discussion: Acute Reduction and Fixation Choices Ask AI

Contouring the Plate Ask AI

- Ulnohumeral joint fixed reduced; radius reduced
- Ulna held in position → **contour plate** to that shape
  - Plate conforms
- Always contour slightly **more** than visual estimate
- Thumb pushes head in this position → no dislocation

Closed Reduction in the Acute Case Ask AI

- No **isolated** dislocation → traumatic cases include ulnar fracture

Ulnar lesion	Reduction
Complete fracture	Easy → reduce, hold 90 degrees flexion
Greenstick fracture	Easy → reduce, hold 90 degrees flexion
Plastic deformation	Break ulna under anaesthesia

- Problem → **plastic deformation**
  - Under anaesthesia → break ulna; otherwise irreducible
  - Convert to complete fracture → reduce
  - Ulna settles with head reduction → heals in that position

Centre of the Capitulum in Young Children Ask AI

- Identifying **capitulum centre** at 2–3 years?
- Capitulum → among first structures to **ossify**
  - Ossified by 2–3 years
  - One of first distal humeral ossification centres
- Last operated child → 3 years; capitulum visible
- At 3 years → no difficulty seeing **capitulum**

Fixation Method of the Osteotomy Ask AI

- Tension band or wire acceptable?

Fixation	Posterior angulation	Supination / rotation control
Tension band	Never	—
Intramedullary K-wire	No	No
Locked plate	Yes	Yes

- Required → **locked plate,** not ordinary plate → best results

Forearm Position in the Cast Ask AI

- Elbow **flexion** → 90 degrees
- Acute cases → 90–100 degrees, slightly more
  - About 100 degrees
- Avoid excess flexion → **compartment syndrome**
- 90–100 degrees → forearm stable

Discussion: Revision, Physes and Late Indications Ask AI

Why Not Reduce the Congenital Case Ask AI

Congenital

- Not reducible → **no normal relationship**
  - "Square onto round"
- Radial head → very unstable articulation; depends on **bony conformity**
  - Concave head on convex capitulum
  - Both convex → never stable
- Head must rest in ulnar **sigmoid notch**
  - Ulna malformed → no normal notch
- All stabilising factors → **absent**, never formed
- Attempt → loss of **range of motion**; better before surgery than after
- Literature → open reduction not advised; poor results

Contouring Locked Plates Ask AI

- Ordinary locked plates → hard to contour?
- **Locked titanium plates** → contour easily with bender
- Small plate → at most 2.7, not 3.5
- Ulna → 3.5 unusable; 2.7 at most

Ilizarov Ask AI

- **Ilizarov** → no help
  - Not reducible completely closed
- Required → open **capsule**, reduce head, clean sigmoid notch
- Not mere distraction → ulnar **deformity** needs correction
- Head reduced without ulnar osteotomy → **ulnohumeral dislocation**
  - Reduce head → ulnohumeral dislocates; reduce ulnohumeral → head dislocates
- Required → ulnar osteotomy + changed **distal–proximal ulna** relationship
  - Through angulation with supination or rotation
- Distraction alone → very difficult; "not a **magnet**"

Management of Recurrence Ask AI

- Recurrence despite correct technique → **very major problem**
- Preferably done by **experienced** surgeon, or under supervision
  - Second look / second go → not favourable
- Redislocated cases → heavy struggle, extensive **dissection**, damage → poor second results
- Analogy → **DDH:** right first time far better than revision

Arresting the Physes Ask AI

- Arrest radial physes to avoid discrepancy and recurrence?
- No
- Fast-growing end of ulna and radius → **distal end**
  - Proximal growth never equals distal
- Arresting distal radius + proximal ulna → **very short radius**
  - Proximal ulnar growth much less than distal physis
- **Transphyseal screw** → no growth arrest with single entry, fine screw
  - Removal → growth resumes
- Physeal injury during fixation → **iatrogenic**, poor technique
  - Repeated drilling, ~20 holes → physis destroyed
- Repeated in-and-out → **physeal damage** even without crossing
- Last case → screw in **epiphysis** and metaphysis, not crossing
- Crossing with single screw → removal restores growth
- Evidence → **temporary epiphysiodesis** with screws
  - No permanent arrest → single entry, one screw
- Physis 2 cm wide + 4.5 mm screw
  - → ~22–25% of physeal area → **growth arrest**
- Smaller, single entry, minimal damage → injury mostly absent
- Exceptions → **high-energy trauma** (fall from height) or excess manipulation
  - → physeal injury → possible growth arrest

Radial Shortening Ask AI

- Radial **shortening** → not performed, never needed
- Considered only → irreducible despite ulnar osteotomy direction
- Never radial shortening alone instead of ulnar osteotomy
- Actual pathology → **ulnar bowing** and ulnar deformity
  - → ulnar osteotomy required; shortening alone fails

Age Limit Ask AI

- Missed cases → no **closed reduction**; always open
- **No age limit** → head shape maintained + no osteochondral fracture
- Hesitation → 14–15 years with 10-year neglected dislocation
  - Painless for 10 years → leave
  - Painful → **excision**
- Indication for excision → **painful radial head dislocation**
  - Aim → relieve significant radial-side elbow pain
  - Not to restore function or correct anything else
- Mainly → **disabling elbow pain** relief

Pulled Elbow and Acute Monteggia Management Ask AI

Pulled Elbow versus Radial Head Dislocation Ask AI

Pulled

- **Pulled elbow** → different entity; not a true dislocation
- Annular ligament **interposed** between radial head and capitulum
  - Mechanism → traction injury of elbow
- Before 5–6 years → head **rounded**, not dish-shaped
  - Neck same diameter as head
  - → annular ligament slips over easily
- Later → head diameter exceeds neck → slipping stops
- X-ray → long axis of radius with capitulum → **no dislocation**
- Manipulation → ligament returns → child moves arm normally

	Pulled elbow	Radial head dislocation
Lesion	Annular ligament interposition	Head out of capitulum
Radial neck axis	Through capitulum	Off capitulum centre
Treatment	Manipulation	Reduction

Exam Pearl

- Pulled elbow → annular ligament interposition, not dislocation
- Under 5–6 years → rounded head, neck same diameter
- Radiocapitellar line → normal

Acute Bado Type I Monteggia Ask AI

Bado I

- Priority one → reduce **radial head**
- Then assess ulnar fracture
- Wire or plate? → reduced + stable → **cast** suffices
- After casting → **weekly X-ray** → exclude redislocation
- Acute → head, **sigmoid notch** and proximal ulna intact
- Reduced head → aids **ulnar alignment**
- Head well seated in capitulum → dislocates less than expected
  - While ulnar fracture fresh and mobile
- Late presentation → fracture **sticky**
  - Around 2–3 weeks → undo ulnar fracture, reduce radius, fix ulna
  - Osteotomy procedure not needed → treat as acute fresh fracture

Mid-Prone versus Mid-Position Ask AI

- Cast position → mid-prone / approximately **neutral**
- "Mid-prone" → pronation 20–40 degrees
- True middle → **mid-position** between supination and pronation
- "Mid-prone" → to be removed from terminology

Plastic Deformation: Closed or Open Ask AI

- Closed possible → **break ulna** closed
  - Convert plastic deformation → complete fracture → reduce head
- Otherwise → head irreducible
- Forced reduction → deformation recurs → **redislocation** within 1–2 weeks
- Fracture site → apex?
  - Same site as **plastic deformation**
- Osteoclasis at deformation site
- Head reduced as if **overcorrecting** ulna posteriorly
- Won't break → open → drilling and **osteoclasis**

Irregular Radial Head Preoperatively Ask AI

- Irregular head preop → **contraindication** to reduction
  - No intraoperative stage; prerequisites unmet